

Chapter 6

Securing Operating Systems

THE FOLLOWING COMPTIA A+ 220-1202 EXAM OBJECTIVES ARE COVERED IN THIS CHAPTER:

✓ **2.0 Security**

✓ **2.2 Given a scenario, configure and apply basic Microsoft Windows OS security settings.**

- Defender Antivirus
  - Activate/deactivate
  - Updated definitions
- Firewall
  - Activate/deactivate
  - Port security
  - Application security
- Users and groups
  - Local vs. Microsoft account
  - Standard account
  - Administrator
  - Guest user
  - Power user
- Log-in OS options
  - Username and password
  - Personal identification number (PIN)
  - Fingerprint
  - Facial recognition
  - SSO
  - Passwordless/Windows Hello



- NTFS vs. share permissions
  - File and folder attributes
  - Inheritance
- Run as administrator vs. standard user
- User Account Control (UAC)
- BitLocker
- BitLocker-To-Go
- Encrypting File System (EFS)

✓ **2.3 Compare and contrast wireless security protocols and authentication methods.**

- Protocols and encryption
  - Wi-Fi Protected Access 2 (WPA2)
  - WPA3
  - Temporal Key Integrity Protocol (TKIP)
  - Advanced Encryption Standard (AES)
- Authentication
  - Remote Authentication Dial-in User Service (RADIUS)
  - Terminal Access Controller Access-Control System (TACACS+)
  - Kerberos
  - Multifactor

✓ **2.8 Given a scenario, apply common methods for securing mobile devices.**

- Hardening techniques
  - Device encryption
  - Screen locks
    - Facial recognition
    - PIN codes
    - Fingerprint
    - Pattern



        - Swipe
    - Configuration profiles
- Patch management
    - OS updates
    - Application updates
- Endpoint security software
    - Antivirus
    - Anti-malware
    - Content filtering
- Locator applications
- Remote wipes
- Remote backup applications
- Failed log-in attempts restrictions
- Policies and procedures
    - MDM
    - BYOD vs. corporate-owned devices
    - Profile security requirements

✓ **2.10 Given a scenario, apply security settings on SOHO wireless and wired networks.**

- Router settings
    - Change default passwords
    - IP filtering
    - Firmware updates
    - Content filtering
    - Physical placement/secure locations
    - Universal Plug and Play (UPnP)
    - Screened subnet
    - Configure secure management access
- Wireless specific
    - Changing the service set identifier (SSID)
    - Disabling SSID broadcast



- Encryption settings
- Configuring guest access
- Firewall settings
  - Disabling unused ports
  - Port forwarding/mapping

✓ **2.11 Given a scenario, configure relevant security settings in a browser.**

- Browser download/installation
  - Trusted sources
    - Hashing
  - Untrusted sources
- Browser patching
- Extensions and plug-ins
  - Trusted sources
  - Untrusted sources
- Password managers
- Secure connections/sites – valid certificates
- Settings
  - Pop-up blocker
  - Clearing browsing data
  - Clearing cache
  - Private-browsing mode
  - Sign-in/browser data synchronization
  - Ad blockers
  - Proxy
  - Secure DNS
- Browser feature management
  - Enable/disable
    - Plug-ins
    - Extensions
    - Features



This chapter is the second of two chapters that focus primarily on security. Volume 2 Chapter 5, "Security Concepts," covered myriad security concepts, ranging from physical security to the proper destruction of data storage devices in your organization. In this chapter, we will focus on operating system security and mobile security.

Many organizations are adopting a cloud-first initiative for their line-of-business applications. This has further perpetuated the adoption of mobile devices in the workplace. These initiatives, along with the rapid adoption of mobile devices in our personal lives, has created a tremendous need for security. This chapter will address the concerns of operating system security, mobile device security, and best practices.

Working with Windows OS Security Settings

Every operating system offers security features and settings. While you need to know a little about Linux and macOS, the A+ exams focus primarily on Windows and the OS-specific security settings that you need to know to secure them. The following sections will explore some basic Windows OS security features and settings in more detail.



Exam Objective 2.2, "Given a scenario, configure and apply basic Microsoft Windows OS security settings," requires you to know users and groups in the Windows operating system. In addition, you should be familiar with log-in options, NTFS and share permissions, UAC, antivirus, and encryption in the Windows operating system.

Users and Groups

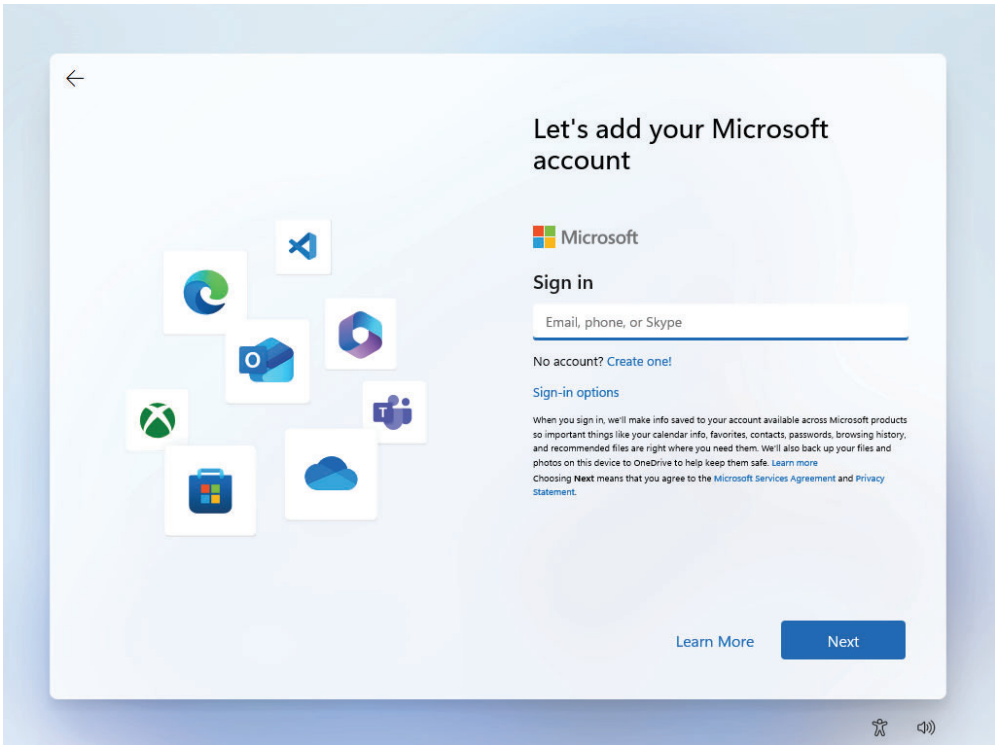
A number of groups are created on the operating system by default. The following sections look at the main ones.

Microsoft vs. Local Accounts

During the initial setup of Windows, Microsoft urges the user to log in with a Microsoft account, as shown in Figure 6.1. You can set up a Microsoft account with an email, a phone,

or a Skype login, which is actually a Microsoft account. This feature was originally introduced with Windows 8.

FIGURE 6.1 Microsoft account screen



Setting up your initial user account is a smart idea, because it allows you to set up tools, such as OneDrive, to back up your files. The most significant feature is it allows setup of synchronization for settings across all of your devices. However, it also allows access to other Microsoft productivity tools and ultimately it is how you access the Microsoft Store and the Store App ecosystem.

You can also select Sign In Options and choose to use an offline account, as shown under the logon name in Figure 6.1. When you use an offline account, many of the features we described earlier need to be set up manually. Also, depending on the feature, it may not work as designed. When you elect to create an offline account, you are actually electing to use local accounts on the operating system. Many corporate-owned devices still use local accounts and connect to a traditional Active Directory (AD) domain.

Choosing to sign in with a Microsoft account or local (offline) account depends on what you are trying to achieve. If the device will be used for personal work, then a Microsoft account is the best option. If the device will be used for an organization, then a local account

might be the best option. A third option exists that is reserved for organizations, which is the use of a corporately owned email address for enrollment into mobile device management (MDM) software, such as Intune.

In the following sections we will cover the basic local accounts and local permissions that you should expect to see on the CompTIA exam.

Administrator

The *Administrator* account is the most powerful of all; it has the power to do everything from the smallest task all the way to removing the operating system. Because of the great power the Administrator account holds, and the fact that it is always created, many who want to do harm target this account as the one that they try to breach. To increase security, during the installation of the Windows operating systems in question, you are prompted for the name of a user who will be designated as the Administrator. The power then comes not from being called "Administrator" (the username might now be "wiley," "sybex," or something similar) but from being a member of the *Administrators group*. (Notice the plural for the group and singular for the user.)

Since members of the Administrators group have such power, they can inadvertently do harm (such as accidentally deleting a file that a regular user could not). To protect against this, the practice of logging in with an Administrators group account for daily interaction is strongly discouraged. Instead, we suggest that system administrators log in with a user account (lesser privileges) and change to the Administrators group account (elevated privileges) only when necessary.

Power User

Originally, Microsoft wanted to create a group in Windows whose members were not as powerful as members of the Administrators group, so they created the *Power Users* group. The idea was that members of this group would be given Read and Write permission to the system, allowing them to install most software but keeping them from changing key operating system files. As such, it would be a good group for those who need to test software (such as programmers) and junior administrators.

The group did not work out as planned, and in Windows 7 up to Windows 11 the group has no more permissions than a standard user. The group is now kept around only for backward compatibility with Windows XP systems.

Guest

The *Guest user* account is created by default (and should be disabled) and is a member of the Guests group. For the most part, members of the Guests group have the same rights as a standard user, except they can't get to log files. The best reason to make users members of the Guests group is that they can access the system only for a limited time.



As part of operating system security, it's usually recommended that you rename the default Administrator and Guest accounts that are created at installation.

Standard User

By default, *standard users* belong to the local Users group. Members of this group have Read and Write permission to their own profile. They cannot modify systemwide Registry settings or do much harm outside of their own accounts. Under the principle of least privilege, users should be made members of the Users group only, unless qualifying circumstances force them to have higher privileges.

If you attempt to run some utilities (such as `sfc.exe`) from a standard command prompt, you will be told that you must be an administrator running a console session in order to continue. If your account is in the Administrators group, then the command prompt must be launched with the elevated permissions of the administrator. To do so, choose Start ➢ All ➢ Windows Tools, and then right-click Command Prompt and choose Run As Administrator. The User Account Control (UAC) will prompt you to continue, and then you can run `sfc.exe` without a problem.

Changing Between Account Types

You can change between account types of standard and administrator for your local user account or for other local user accounts by using the legacy Control Panel applet for User Accounts. You simply open Control Panel, select User Accounts, click Manage User Account, click Properties, and click the Group Membership tab to change your level of access. You can also select another user and then change their group membership. You will be presented with the dialog box in Figure 6.2, where you can change the type of account from Administrator to Standard using the radio buttons.



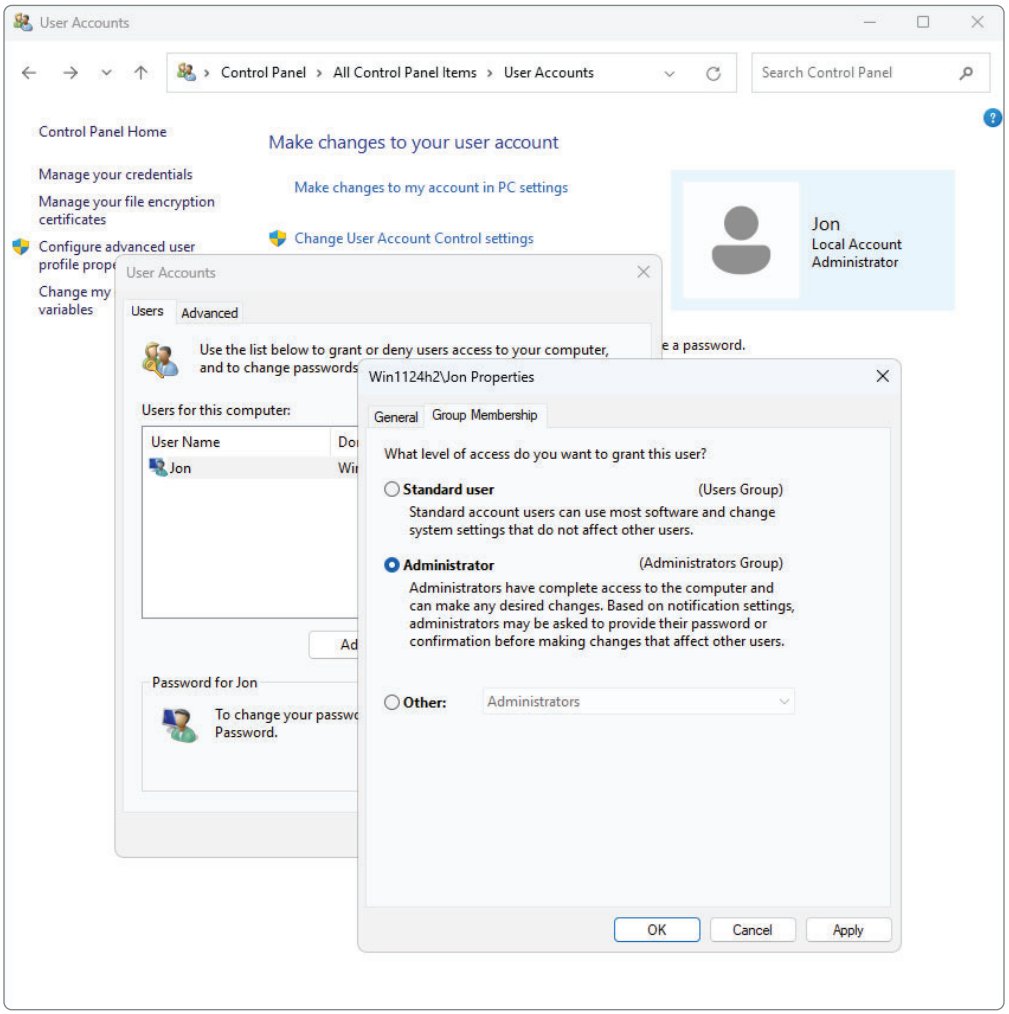
If the Standard option is grayed out and you cannot change the user account to a standard user, then you must make another administrative user. The operating system must have at least one valid administrative account. The operating system will not let you lock yourself out through this set of screens.

User Authentication

Users can log into the local operating system with their username and password, if they have an account, and they will receive a local access token. The *access token* the user is granted is locally significant for the operating system. For example, an administrator (local) who authenticates against the operating system is only an administrator of that operating system and has no further network permissions. Every Windows operating system has a local database and authentication system called the *Security Account Manager (SAM)*, as shown in Figure 6.3.

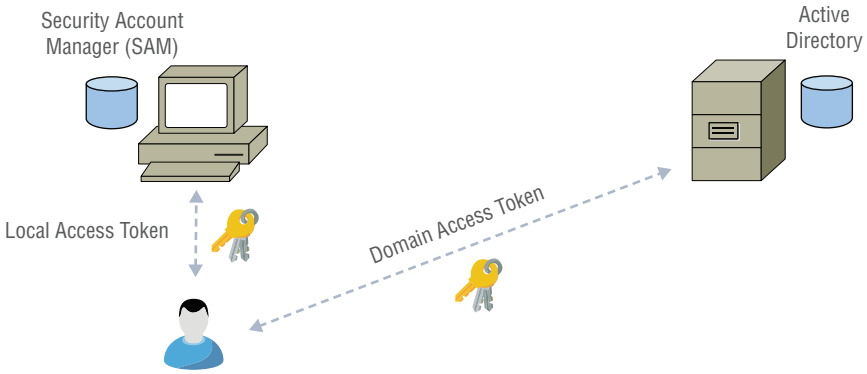
Active Directory simplifies the sign-on process for users and lowers the support requirements for administrators. Access can be established through groups and enforced through group memberships: all users log into the Windows domain using their centrally created Active Directory account. It's important to enforce password changes and make certain that passwords are updated throughout the organization on a frequent basis.

FIGURE 6.2 Changing the account type



Active Directory uses Kerberos v5. A server that runs Active Directory retains information about all access rights for all users and groups in the network. When a user logs into Active Directory, they are granted a *network access token*, also called a *Kerberos token*. This token can be used to authenticate against other servers and workstations in the domain and is accepted network (domain) wide. This token is also referred to as the user's *globally unique identifier (GUID)*. Applications that support Active Directory for authentication can use this GUID to provide access control.

FIGURE 6.3 Windows authentication

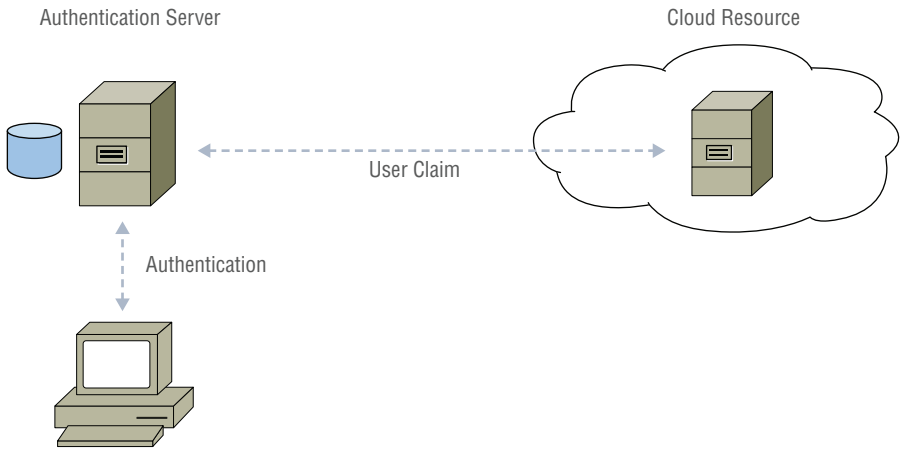


Single Sign-On (SSO)

One of the big problems larger networks must deal with is the need for users to access multiple systems or applications. This may require users to remember multiple accounts and passwords. An alternative to this is that the application must support Active Directory authentication, but that creates other considerations.

The purpose of *single sign-on (SSO)* is to give users access to all the applications and systems that they need when they log in. Single sign-on is often used with cloud-based resources. The principle behind SSO is that the resource will trust that the user has already been authenticated. This allows the user to use one set of credentials for all resources they have access to. The authentication server performs this by sending a claim on behalf of the user, as shown in Figure 6.4. This claim can contain any number of Active Directory user attributes, such as first name, last name, email, and username, just to name a few. It is important to understand that at no time during the authentication process are the username and password sent to the resource that is requesting authentication. The resource must trust that the user has already been authenticated and accept the claim at face value.

FIGURE 6.4 Claims-based authentication



SSO is both a blessing and a curse. It's a blessing in that once users have been authenticated, they can access all the resources on the network and browse multiple folders. Single sign-on is a curse in that it removes the doors that otherwise exist between the user and various resources.

Windows Hello

The problem with traditional usernames and passwords is that they are too complex for the user to consistently type. As administrators we want to make sure that users lock their workstation and type their password when they return to gain access. However, this is a burden on the user, especially if we are constantly changing passwords and making them more complex.

Windows Hello addresses these problems by storing the user's credentials in a secure container called the Credential Manager. The Credential Manager is then locked and unlocked with the authentication of biometrics or a PIN by the user. Once the Credential Manager is unlocked, the credentials can be passed to the operating system to provide the login credentials. Windows Hello can be configured by navigating to Start ➢ Settings ➢ Accounts ➢ Sign-in Options, and you will be brought to the various ways to configure Windows Hello, as shown in Figure 6.5.

There are several different sign-in options for Windows Hello that will allow for easier login of the device. The available options will depend on your hardware connected to the device. As an example, if you want to use your fingerprint, you will need a fingerprint reader. There are several options as shown in Figure 6.5, but they all basically fall into a few categories:

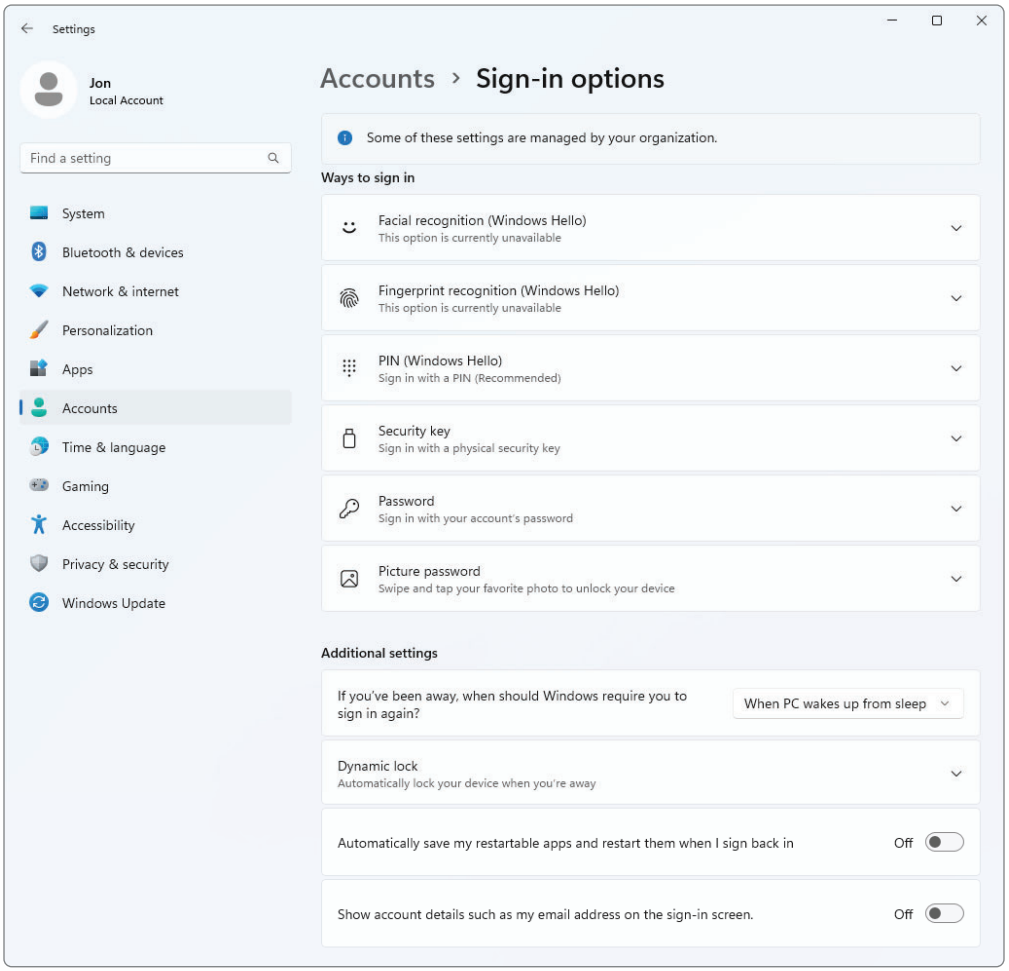
Biometric Sign-In Using the Facial Recognition (Windows Hello) option will allow you to use your face to unlock the operating system. If your device is equipped with a camera, it will sense you entering the scene and verify your face to unlock the device. You can also use your fingerprint to provide authentication, but you will need a fingerprint reader as previously mentioned. Both of these biometric options authenticate you based on something you are. Because it is coupled with something that you have (the device), the process actually provides two-factor authentication/multifactor authentication (2FA/MFA).

PIN The most common option for devices without a camera or fingerprint reader is to use a personal identification number (PIN). The PIN can be a simple password containing numbers such as a PIN you use for your bank. Although you can make this as complex as you wish, its intended purpose is to make it easier to log in for the end user. When you use this method of authentication, it is something that you know. By knowing the PIN and having physical access to the device, the process provides 2FA/MFA.

Security Key Using a hardware security key or token is another way of securing Windows Hello. You can configure Windows to log you in after you present a security key, which is a small USB device that is plugged into the device. This is an authentication method of something that you have. By having both the security key and physical access to the device, the process provides 2FA/MFA.

Picture Password A picture password is similar to a gesture-based password. You can upload a picture, then select points in the picture you must click on to allow authentication. This type of authentication is based on something that you know, and when combined with the physical access to the device, the process provides 2FA/MFA.

FIGURE 6.5 Windows Hello Sign-in options



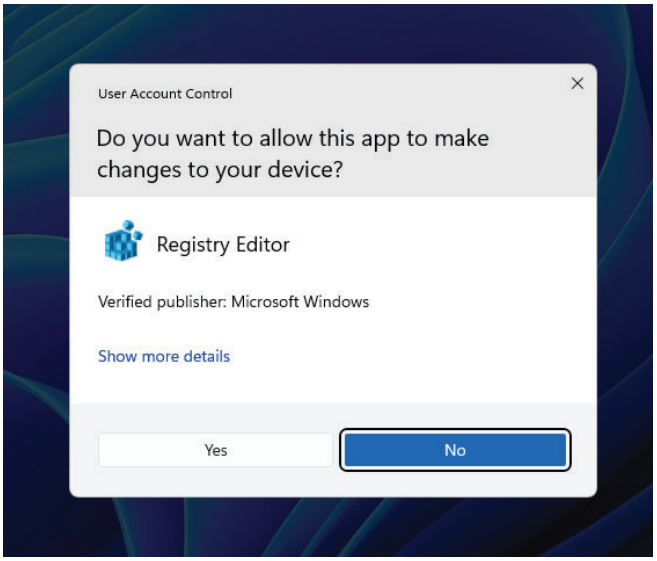
In addition to local logins, Windows Hello can be used to authenticate users for a Microsoft account, Active Directory account, and Azure AD account. It can even be used to authenticate users for identity provider services, which is another way of addressing SSO.

Administrator vs. Standard User

Administrators have rights in the operating system that allows them to change the operating system. Standard users do not have these rights and can only make changes to their environment. Because the administrative rights can cause harm to the operating system or introduce security issues, care should be taken in granting your end users administrative permissions. You should exercise the principle of least privilege with granting end-user rights. This means that unless someone needs administrative privileges, they should always log in with a standard user account. This protects the local operating system and potentially the network (if domain authenticated) from the security threats covered in Volume 2 Chapter 5.

User Account Control (UAC) is a feature that was introduced in Windows Vista. It supports the principle of least privilege by logging an administrator in with minimal permissions. This is extremely handy for users who occasionally need administrative rights, such as your home operating system. For example, if extra privileges were required to modify the operating system, a prompt asking if the user wants to continue would be displayed, as shown in Figure 6.6. If the user answers Yes, then the user will receive the administrative token to complete the task. UAC allows the user to run as a standard user with the ability to escalate privileges.

FIGURE 6.6 UAC prompt



In Exercise 6.1 you will examine the security token for your user account. The exercise assumes that you are the administrator of the operating system.

EXERCISE 6.1

Examining a Security Token

This exercise assumes that you have not turned off the UAC and that you are the administrator of the operating system.

1. Click the Start menu, type **cmd**, and then press Enter.
2. In the command prompt, type **whoami /ALL**, and then press Enter.
3. Record the privileges that you see in the output.
4. Click the Start menu and type **cmd**.
5. Right-click Command Prompt and select Run As Administrator.
6. Answer Yes to the UAC prompt.
7. In the command prompt, type **whoami /ALL**, and then press Enter.
8. Compare the privileges that you see in the output.

When you opened the first command prompt, it was opened with the standard user credentials of your account. When you opened the second command prompt, you selected Run As Administrator. The second command prompt was opened with the administrative security token.

NTFS vs. Share Permissions

The *New Technology File System (NTFS)* was introduced with Windows NT to address security problems. Before Windows NT was released, it had become apparent to Microsoft that a new filesystem was necessary to handle growing disk sizes, security concerns, and the need for more stability. NTFS was created to address those issues.

Although the *File Allocation Table (FAT)* filesystem was relatively stable if the systems that were controlling it kept running, it didn't do well when the power went out or the system crashed unexpectedly. One of the benefits of NTFS was a transaction-tracking system, which made it possible for Windows NT to back out of any disk operations that were in progress when it crashed or lost power.

With NTFS, files, folders, and volumes can each have their own security. NTFS's security is flexible and built in. Not only does NTFS track security in ACLs, which can hold permissions for local users and groups, but each entry in the ACL can specify which type of access is given—such as Read & Execute, List Folder Contents, or Full Control. This allows a great deal of flexibility in setting up a network. In addition, special file-encryption programs were developed to encrypt data while it is stored on the hard disk.

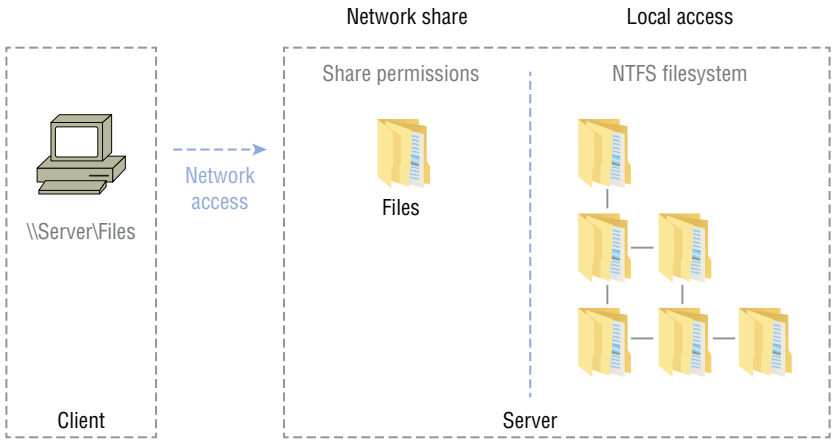
Microsoft strongly recommends that all network shares be established using NTFS. Several current operating systems from Microsoft support both FAT32 and NTFS. It's possible to convert from FAT32 to NTFS without losing data, but you can't do the operation in reverse. (You would need to reformat the drive and install the data again from a backup tape.)



If you're using FAT32 and want to change to NTFS, the `convert` utility will allow you to do so. For example, to change the E: drive to NTFS, the command is `convert E: /FS:NTFS`.

Share permissions apply only when a user is accessing a file or folder through the network, as shown in Figure 6.7. *NTFS permissions* and attributes are used to protect the file when the user is local. With FAT and FAT32, you do not have the ability to assign "extended" or "extensible" permissions, and the user sitting at the console effectively is the owner of all resources on the system. As such, they can add, change, and delete any data or file.

FIGURE 6.7 Network share permissions and NTFS permissions



With NTFS as the filesystem, you are allowed to assign more comprehensive security to your computer system, as shown in Table 6.1. NTFS permissions can protect you at the file level. Share permissions can be applied to the folder level only, as shown in Table 6.2. NTFS permissions can affect users accessing files and folders across a network or logged in locally to the system where the NTFS permissions are applied. Share permissions are in effect only when the user connects to the resource through the network.

TABLE 6.1 NTFS permissions

NTFS permission	Meaning	Object used on
Full Control	Gives the user all the other choices and the ability to change permissions. The user can also take ownership of the folder or any of its contents.	Folder and file objects
Modify	Combines the Read & Execute permission with the Write permission and further allows the user to delete everything, including the folder.	Folder and file objects
Read & Execute	Combines the Read permission with the List Folder Contents permission and adds the ability to run executables.	Folder and file objects
List Folder Contents	The List Folder Contents permission (known simply as List in previous versions) allows the user to view the contents of a folder and to navigate to its subdirectories. It does not grant the user access to the files in these directories unless that is specified in file permissions.	Folder objects
Read	Allows the user to navigate the entire folder structure, view the contents of the folder, view the contents of any files in the folder, and see ownership and attributes.	Folder and file objects
Write	Allows the user to create new entities within a folder.	Folder and file objects

TABLE 6.2 Share permissions

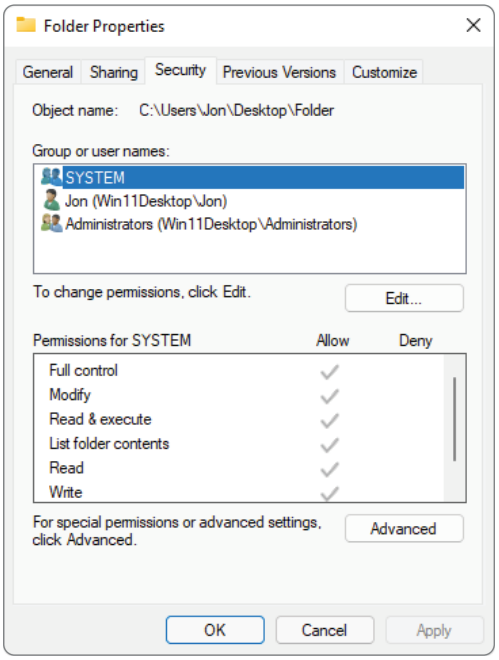
Share permission	Meaning
Full Control	Gives the user all the other permissions as well as permission to take ownership and change permissions.
Change	Allows the user to overwrite, delete, and read files and folders.
Read	Allows the user to view the contents of the file and to see ownership and attributes.

Allow vs. Deny

Within NTFS, permissions for objects fall into one of three categories: Allow, Deny, or not configured. When viewing the permissions for a file or folder, you can select the box for

Allow, which effectively allows the group selected to perform that action. You can also deselect the box for Allow, which does not allow that group that action, as shown in Figure 6.8. Alternatively, you can select the Deny box, which prevents that group from using that action. There is a difference between not allowing (a cleared check box) and Deny (which specifically prohibits), and you tend not to see Deny used often. Deny, when used, trumps other permissions.

FIGURE 6.8 NTFS folder permissions



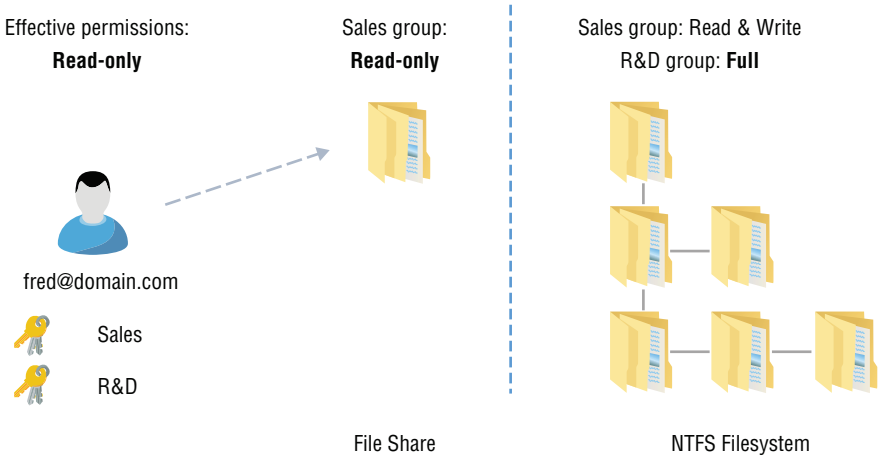
Permissions set on a folder are inherited down through subfolders unless otherwise changed. Permissions are also cumulative; if a user is a member of a group that has Read permission and a member of a group that has Write permission, they effectively have both Read and Write permissions.

Effective Permissions

When a user accesses a file share, both the share permissions and NTFS permissions interact with each other to form the effective permission for the user. Figure 6.9 shows that a user named Fred has logged in and received his access token containing the Sales and R&D groups, since he is a member of both groups. When Fred accesses the Sales file share, the share permissions define that he has read-only access because he is part of the Sales group. You can see that the NTFS permissions are granting him read and write access because of his

Sales group membership, as well as full control because he is also in the R&D group. If Fred were to locally log in to this computer, he would effectively have full control of these files. However, because he is accessing these files from the network, he only has read-only access because of the file-share permissions. The opposite is also true: if he had full permission at the share level and read-only permission at the NTFS level, he would effectively have read-only access.

FIGURE 6.9 Effective permissions



The rule for figuring out effective permissions is simple: If a user is in more than one group for which there are multiple permissions, take the most permissive permission of NTFS and then the most permissive permission of the share; the effective permission is the more restrictive of the two. There are some circumstances that change this rule slightly when the user (or group) is denied. If a user is in any group that is denied permission at the share or the NTFS level, they are denied for that access level. Therefore, when you derive the more restrictive permission, it will always be a deny for the user. A simple way to remember this is that a deny is a deny.

Moving vs. Copying Folders and Files

When you copy a file, you create a new entity. When you move a file, you simply relocate it and still have but one entity. This distinction is important when it comes to understanding permissions. A copy of a file will have the permissions assigned to it that are already in place at the new location of the file, regardless of which permissions were on the original file.

A moved file, on the other hand, will attempt to keep the same permissions as it had in the original location. Differences will occur if the same permissions cannot exist in the new location. For example, if you are moving a file from an NTFS volume to

FAT32, the NTFS permissions will be lost. If, on the other hand, you are moving from a FAT32 volume to an NTFS volume, new permissions will be added that match those for newly created entities.



Folder copy and move operations follow guidelines that are similar to those for files.

File and Folder Attributes

Since the introduction of FAT, file and folder *attributes* have existed in the filesystem. The basic set of attributes—Read-only, Hidden, System, and Archive—can still be found in both FAT and NTFS and are very useful for the operating system, as explained in the following list. You can view the file or folder attributes by right-clicking the file or folder and selecting Properties. From there, you can view the Read-only and Hidden attributes.

Read-Only The Read-only attribute is used to make a file or folder read-only. If the Read-only attribute is present, regardless of the permission of the file or folder, the user cannot write or delete a file or folder. The Read-only attribute is useful for protecting files from being overwritten or deleted.

Hidden The Hidden attribute makes files and folders disappear when using the Windows File Explorer utility, using the defaults or using the `dir` command. The files still function and are still a part of the filesystem, but they just don't appear by default. The Hidden attribute is useful when you want to hide a file or folder from the average user.

System The System attribute marks files and folders as system files. When a file or folder is marked as system, it means the file or folder is critical. With the introduction of NTFS, the System attribute is not really useful anymore, but it is still there for backward compatibility.

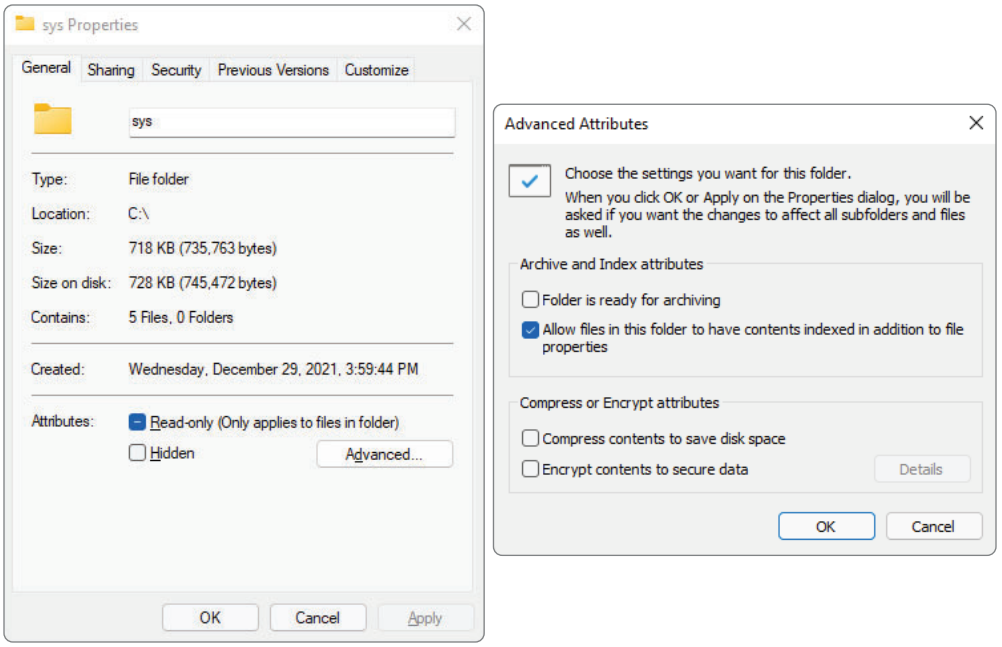
Archive The Archive attribute can be used to tell the system whether the file has changed since the last time it was backed up. Technically, it is known as the Archive Needed attribute. If this bit is on, the file or folder should be backed up. If it is not selected, a current version of the file is already backed up.



Only the Read-only and Hidden attributes can be set or cleared from the GUI. The Archive attribute can be accessed from the Advanced Attributes dialog box. The System attribute can be set or cleared only with the `attrib.exe` command.

Windows uses NTFS, which gives you a number of options that are not available on earlier filesystems, such as FAT and FAT32. A number of these options are implemented through the use of the Advanced Attributes dialog box, as shown in Figure 6.10.

FIGURE 6.10 The Advanced Attributes dialog box





FAT32 does not have as many options as NTFS, such as encryption and compression. These attributes are available only on NTFS partitions.

To reach these options in Windows, right-click the folder or file that you want to modify, and then select Properties from the menu. On the main Properties page of the folder or file, click the Advanced button in the lower-right corner. In the Advanced Attributes window, you have access to the following settings:

Indexing Windows implements a feature called the *Indexing Service* to catalog and improve the search capabilities of your drive. Once files are indexed, you can search for them more quickly by name, date, or other attributes. Setting the index option on a folder causes a prompt to appear, asking whether you want the existing files in the folder to be indexed as well. If you choose to do this, Windows will automatically reset this attribute on subfolders and files. If not, only new files created in the folder will be indexed.

Compression The versions of Windows that you need to know for the exam support advanced *compression* options, which were first introduced in Windows NT. NTFS files and folders can be dynamically compressed and uncompressed, often saving a great deal of space on the drive. As with indexing, when you turn on compression for a folder, you'll be prompted as to whether you want the existing files in the folder to be

compressed. If you choose to do this, Windows automatically compresses the subfolders and files. If not, only new files created in the folder are compressed.



Compression works best on such files as word processing documents and uncompressed images. Microsoft Word files and Microsoft Paint bitmaps can be compressed up to 80 percent. Files that are already packed well do not compress as effectively; EXE and ZIP files generally compress only about 2 percent. Similarly, GIF and JPEG images are already compressed (which is why they are used in Internet web pages), so they compress a little or not at all.

Encryption *Encryption* lets you secure files so that no one else can view them. You encrypt files by encoding them with a key to which only you have access. This can be useful if you're worried about extremely sensitive information, but in general encryption is not necessary on the network. NTFS local file security is usually enough to provide users with access to what they need and prevent others from getting to what they shouldn't. If you want to encrypt a file, go through the same process that you would for indexing or compression.



Encryption and compression are mutually exclusive—you can set one but not both features on a file or folder. Not all features are available in all editions of every operating system. If a user forgets their password or is unable to access the network to authenticate their account, they will not be able to open encrypted files. By default, if the user's account is lost or deleted, the only other user who can decrypt the file is the Administrator account.

In Exercise 6.2 you will have the opportunity to view file permissions for both basic and advanced permissions. Make sure that you don't inadvertently add any deny permissions, as you could be prevented from making any further changes.

EXERCISE 6.2

Examining File Permissions

1. Open Windows File Explorer.
2. Right-click a file or folder and choose Properties.
3. Select and then examine the Security tab.

   You'll see the users and/or groups to which permissions have been assigned.
4. Select a user or group in the list, and examine the list of standard permissions. (To add a new user or group, click Add and follow the prompts.)

EXERCISE 6.2 *(continued)*

Any standard permissions that are checked in the Allow column are applied. If a check box is grayed out, then the permission was inherited.

5. To revoke a set of standard permissions, click the appropriate check box in the Deny column.

   If you click the check box in the Deny column for the Full Control permission, all other standard permissions are also denied.

6. Click Advanced to examine advanced options.

7. Click Cancel twice to close the file or folder's properties.

Be sure that you don't accidentally make any changes that you didn't intend to make. Changing permissions without understanding the ramifications can have negative consequences, such as losing access to files or folders. It is a best practice to assign Deny permissions sparingly. It's better to uncheck Allow. (You may need to turn off Inheritance.)

Shared Files and Folders

You can share folders, and the files beneath them, by right-clicking the file or folder and selecting Show More Options, then choosing Give Access To from the context menu and selecting Specific People. In Windows, the context menu asks you to choose with whom you want to share the folder or file, as shown in Figure 6.11. You can then choose who to share it with along with their respective permissions. It is important to understand that when you use this method to share files and folders, the share permissions are set to Full Control for the Everyone group. The dialog box shown in Figure 6.11 will allow you to manipulate the NTFS permissions via the permission level.

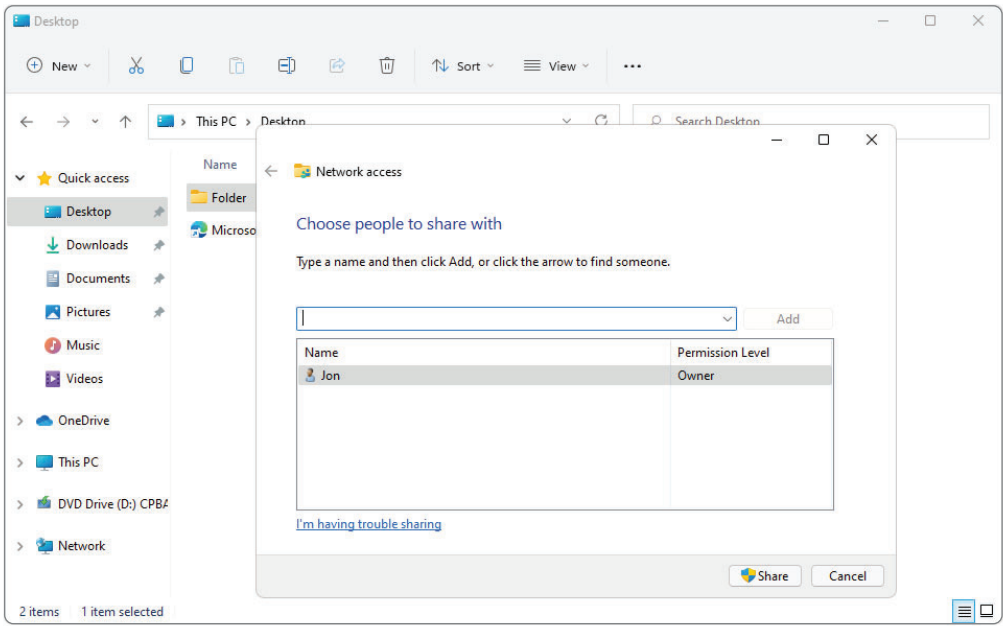
You can access the Advanced Sharing settings by right-clicking the folder you want to share, selecting Show More Options, then Properties, then clicking the Sharing tab, and finally selecting Advanced Sharing, as shown in Figure 6.12. This file-sharing method is more traditional with network administrators because every aspect of the share can be controlled. Using this method, only the share permissions are set from this dialog box. The NTFS security permissions are set on the Security tab. In addition, you can add other share names to the same location, limit the number of simultaneous connections, and add comments.

Administrative Shares vs. Local Shares

Administrative shares are automatically created on all Windows operating systems on the network for administrative purposes. These shares can differ slightly based on which operating system is running, but they always end with a dollar sign (`$`) to make them hidden. There is one share for each volume on a hard drive (`c$`, `d$`, and so on) as well as `admin$`

(the root folder—usually `C:\WINDOWS`) and `print$` (where the print drivers are located). These shares are created for use by administrators and usually require administrator privileges to access.

FIGURE 6.11 Choose People To Share With



Local shares, as the name implies, are shares that are created locally by the administrative user on the operating system. The term *local shares* is used to distinguish between automated administrative shares and manually created shares.

Permission Inheritance and Permissions Propagation

Inheritance is the default throughout the permission structure unless a specific setting is created to override it. A user who has Read and Write permissions in one folder will have them in all the subfolders unless a change has been made specifically to one of the subfolders. If a user has the Write permission, which is inherited from the folder above, the removal of permissions cannot be performed unless inheritance is disabled. Only additional permissions can be added explicitly, since this is actually a new permissions entry and not the removal of an existing permissions entry. You can control NTFS inheritance by right-clicking a folder, selecting Show More Options, then selecting Properties, then choosing the Security tab, selecting Advanced, and clicking the Disable Inheritance button, as shown in Figure 6.13.

FIGURE 6.12 Advanced file and folder sharing

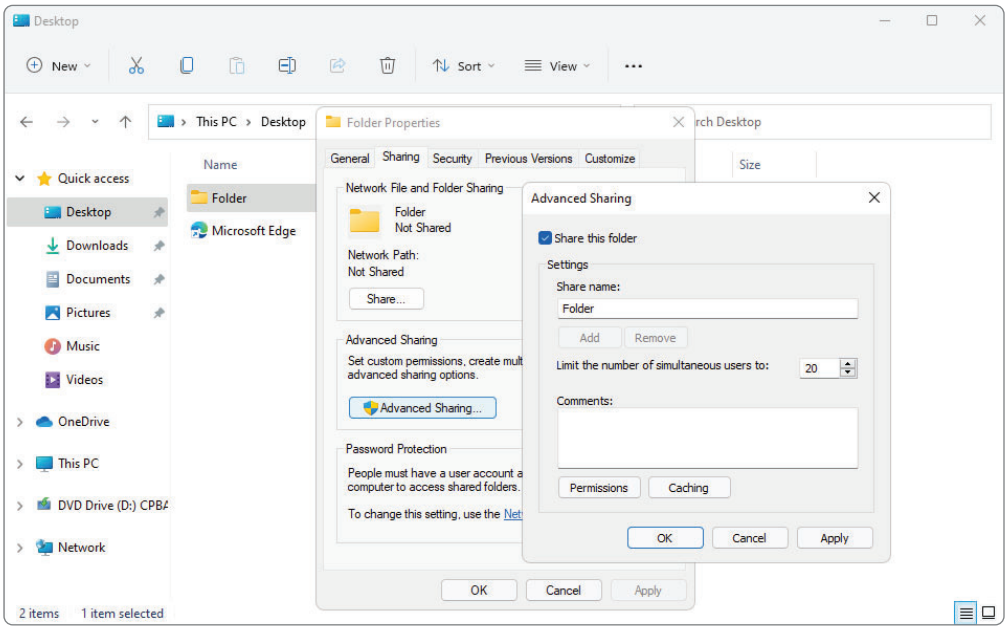
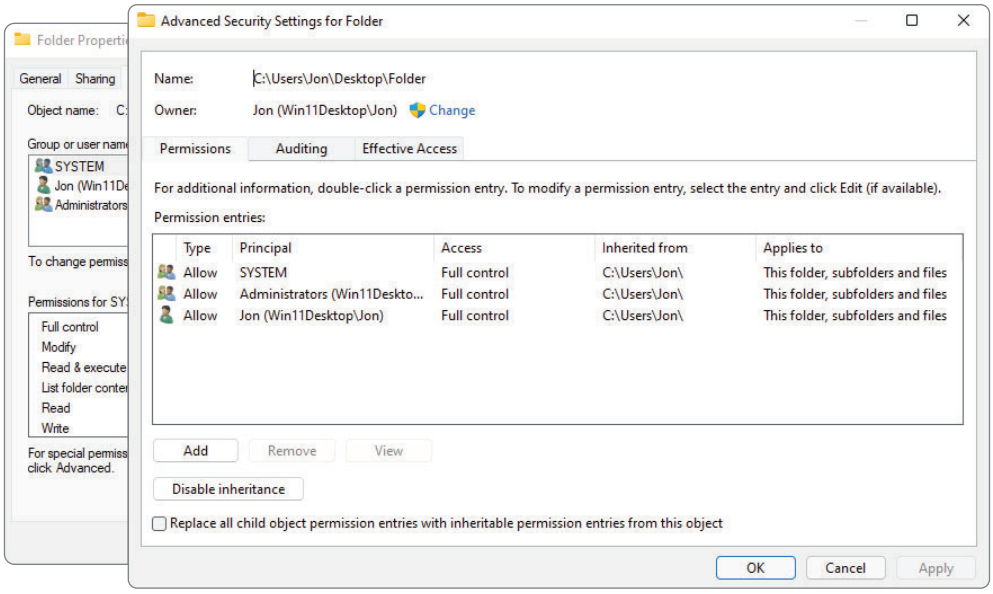


FIGURE 6.13 Disabling inheritance

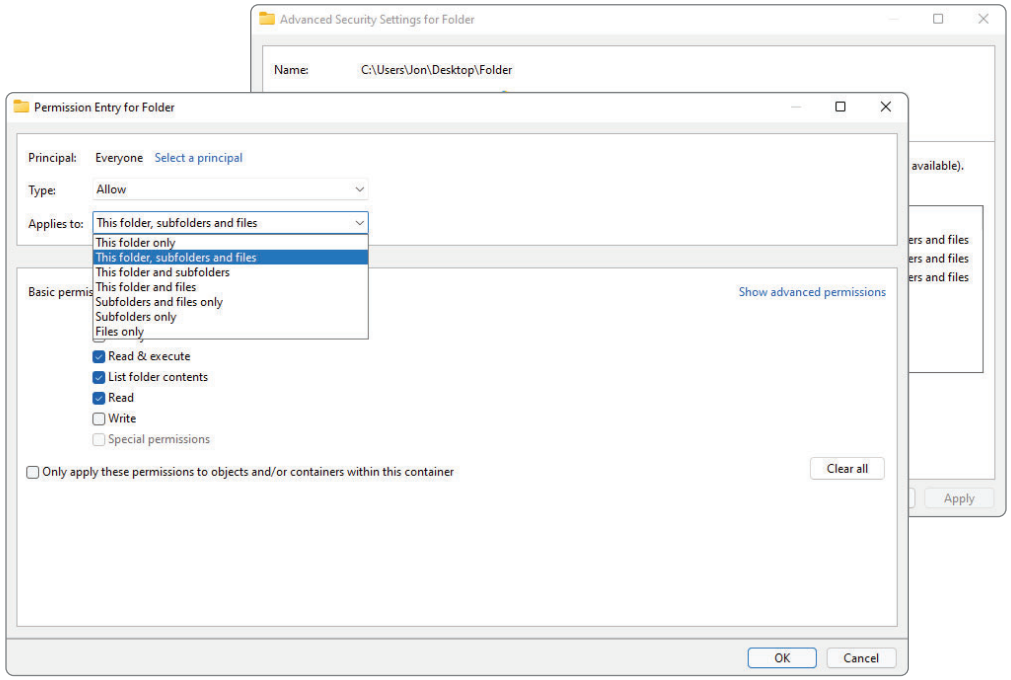


If the Disable Inheritance button is selected, you will be changing the NTFS security settings on the folder. A second dialog box will pop up and you must choose to either keep the existing permissions, by selecting Convert Inherited Permissions To Explicit Permissions On This Object, or to start fresh with no permissions by selecting Remove All Inherited Permissions From This Object. You should use caution when selecting the latter of the two options, because all folders and files below will inherit and propagate the removal of existing permissions.

If you want to make sure that inheritance and permissions for a folder are propagated to all files and folders below, you can use Replace All Child Object Permission Entries With Inheritable Permission Entries From This Object (refer to Figure 6.13). This option will replace every permission in this folder and all the subfolders, regardless of whether explicit permissions were applied further down in the folder structure.

In the Advanced Security Settings, you can also configure permissions entries that only apply to the current folder, current folder and files, all folders and files, or other variations of these, as shown in Figure 6.14. These settings can change the propagation of file permissions to folders and files.

FIGURE 6.14 Permission entry



System Files and Folders

System files are usually flagged with the Hidden attribute, meaning they don't appear when a user displays a folder listing. You should not change this attribute on a system file unless absolutely necessary. System files are required in order for the operating system to function. If they are visible, users might delete them (perhaps thinking that they can clear some disk

space by deleting files that they don't recognize). Needless to say, that would be a bad thing! Most system files and folders are protected by the operating system and won't allow deletion, but better safe than sorry.

Changing File Attributes

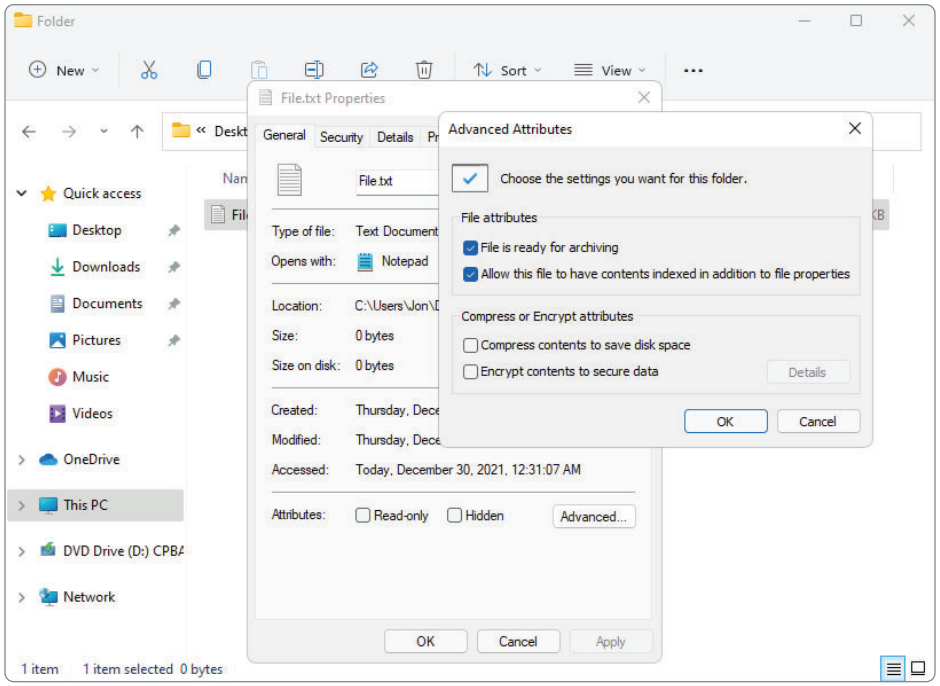
File attributes determine what specific users can do to files or folders. For example, if a file or folder is flagged with the Read-only attribute, then users can read the file or folder but cannot make changes to it or delete it. Attributes include Read-only, Hidden, System, and Archive, as well as Compression, Indexing, and Encryption. Not all attributes are available with all versions of Windows. We'll look at this subject in more detail in a moment.



Some attributes—such as Read-only, Hidden, System, and Archive—date back to DOS. All others—such as Compression, Indexing, and Encryption—are a part of NTFS.

You can view and change file attributes either by entering **attrib** at the command prompt or by changing the properties of a file or folder. To access the properties of a file or folder, right-click the file or folder and select Show More Options, then Properties. You can view and configure the Read-only and Hidden file attributes on the General tab. To view and configure additional attributes, click Advanced, as shown in Figure 6.15.

FIGURE 6.15 Windows file attributes



Windows Security Features

Although everything we've covered so far is a Windows security feature, we have focused on the basic elements of authentication and securing resources. In the following section we will cover the common Windows security features that can protect us from external threats, such as malware, malicious network activity, and data loss.

Microsoft Defender Antivirus

Microsoft Defender, also known as Windows Defender Antivirus, was originally introduced with Windows XP as a downloadable antivirus product. It was later shipped with the Vista operating system, and it has become a pillar of security in the Windows 11 operating system. The addition of Microsoft Defender allows the Windows operating system to be protected right out of the box. A user never needs to install anything else to be protected from malware. That being said, there are lots of products on the market that provide security features above and beyond Microsoft Defender. You can view the Microsoft Defender settings by navigating to Start ➢ Settings ➢ Privacy & Security ➢ Windows Security ➢ Virus & Threat Protection in Windows 11, as shown in Figure 6.16.



You may have noticed that the Microsoft Defender Antivirus branding is slowly disappearing from Windows 11. The Microsoft Security app is slowly replacing the branding of Microsoft Defender Antivirus. There are a few reasons for this; the first is Microsoft Defender Antivirus is being sold as an add-on product for businesses. Another reason is the Microsoft Security app does more than just antivirus, malware control, ransomware mitigation, and a wealth of other security-related features discussed throughout this book. The Microsoft Defender Antivirus product is still under the hood protecting the operating system right out of the box. This book will reference the Microsoft Defender Antivirus product since the CompTIA objectives still reference the product. More information on the Microsoft Security app and Microsoft Defender Antivirus can be found at `https://learn.microsoft.com/en-us/windows/security/operating-system-security/system-security/windows-defender-security-center/windows-defender-security-center`.

On the Virus & Threat Protection screen you can initiate a scan of the computer. In addition, the screen will detail how many threats have been found, when it was last scanned, how many files were scanned, and how long the scan lasted. By clicking Scan Options you can select from Quick Scan, Full Scan, Custom Scan, and Microsoft Defender Offline Scan.

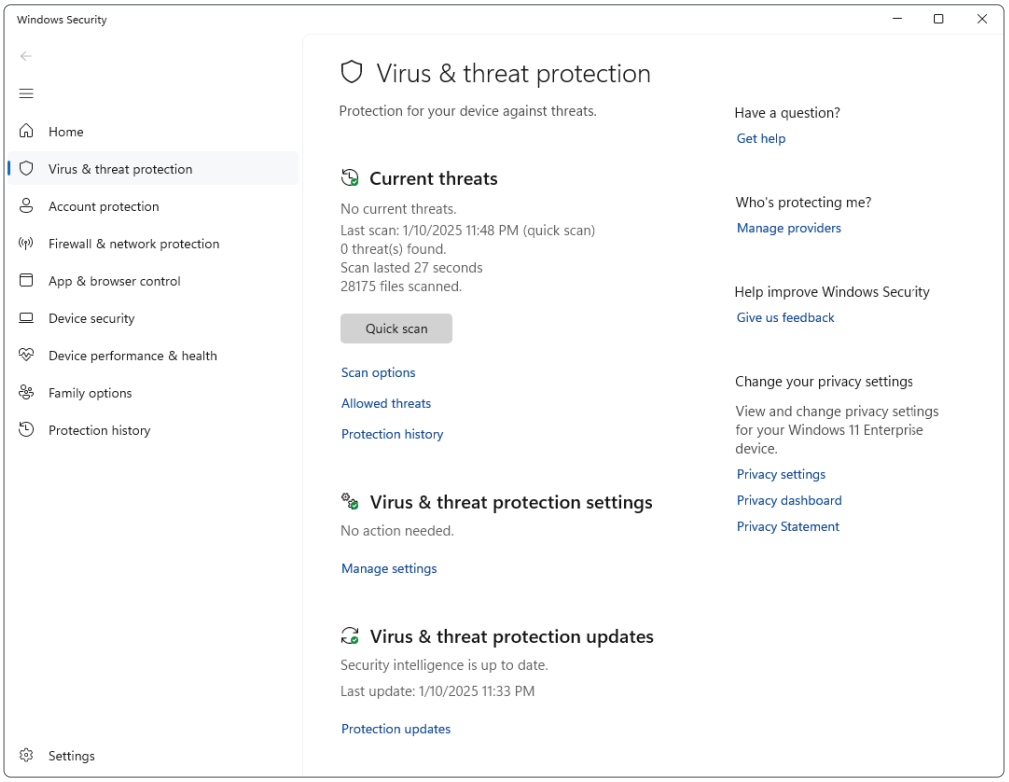
Quick Scan This option will check folders in the system where threats are commonly found.

Full Scan This option will perform a full scan of the entire operating system.

Custom Scan This option will allow you to select the folders on the drive you want to scan.

Microsoft Defender Antivirus (Offline Scan) This option will restart the device, reboot into the Windows Recovery Environment (WinRE), and launch a full Microsoft Defender Scan. It is used to remove malware that is difficult to remove while the operating system is running.

FIGURE 6.16 Microsoft Defender settings

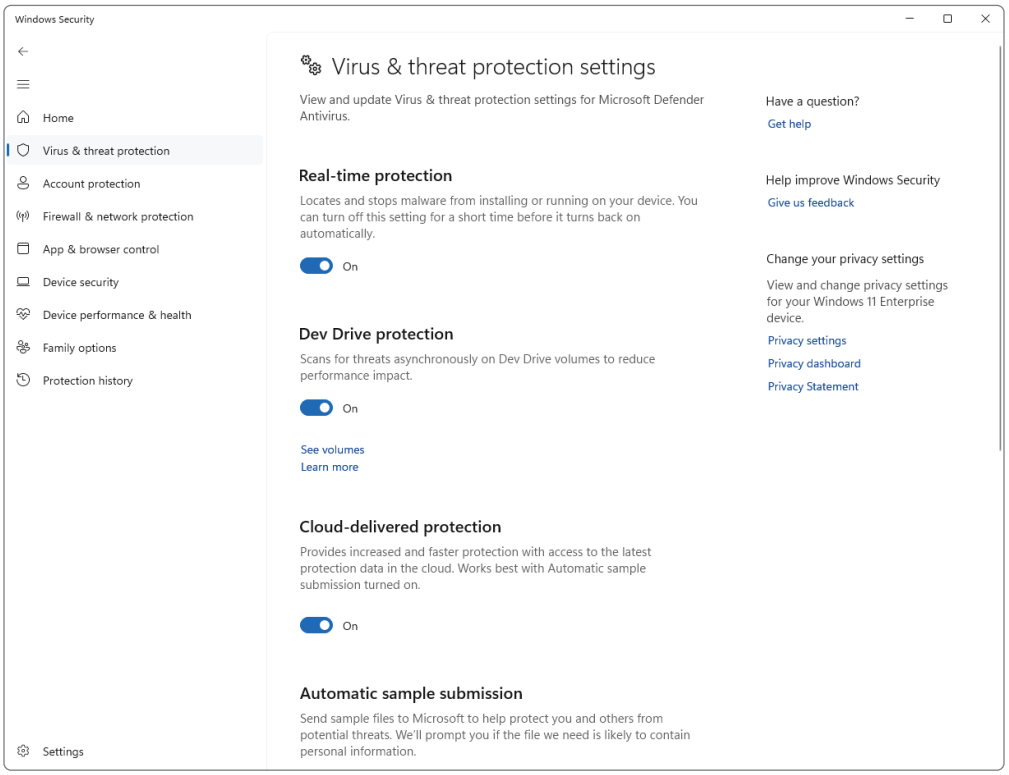


You can also change the Microsoft Defender Virus & Threat Protection Settings by clicking Manage Settings on the Virus & Threat Protection screen. This will allow you to manage a number of settings to change the way Microsoft Defender operates, as shown in Figure 6.17.

Real-Time Protection You can toggle off real-time protection when installing certain applications that require that antivirus be off during installation. However, the real-time protection will turn back on automatically after a period of time.

Dev Drive Protection Dev Drive Protection in Microsoft Defender is a feature designed to enhance performance for developers using Windows 11. When enabled, Dev Drive Protection runs in a special asynchronous performance mode, which defers security scans until after file operations are completed. This mode improves system performance without compromising security.

FIGURE 6.17 This screen provides you with several options to specify how protection is handled by Microsoft Defender Virus & Threat Protection.



Cloud-Delivered Protection You can also toggle Cloud-Delivered Protection, which provides cloud-based data on threats and ultimately faster protection. Turning this setting off might be required for certain regulatory requirements, since it automatically turns on cloud-based sample submission.

Automatic Sample Submission Automatic Sample Submission allows Microsoft Defender to automatically submit samples to Microsoft for analysis. Automatic Sample Submission can be toggled on and off.

Tamper Protection The Tamper Protection security setting prevents malicious applications from tampering with Microsoft Defender settings. Tamper Protection protects against tampering from third-party processes; even Group Policy settings cannot disable Microsoft Defender when Tamper Protection is turned on.

Controlled Folder Access The Controlled Folder Access feature can be accessed in the Virus & Threat Protection Settings. The feature is used to prevent ransomware from harming files, folders, and memory areas on the device. By default, it is turned off, but

it can be easily turned on by clicking the toggle. Once it is set to the default, folders protected are documents, pictures, videos, music, and favorites. Specific folders can be added, and the defaults can also be removed. You can also exempt an application so that it is allowed to modify the files.

Exclusions Although it is not common to need to exempt a folder from the antimalware engine, it can be configured from the Virus & Threat Protection Settings. This option allows you to exempt an entire folder from the real-time protection and scans. One use may be performance-impacted applications such as games, but caution should be used.

Update Definitions An antimalware engine is only as good as its latest definitions, and Microsoft Defender is no different. Therefore, both the Microsoft Defender engine and its definitions are updated quite frequently. Both are updated through the Windows Update process.

Windows Defender Firewall

Windows Defender Firewall is an advanced host-based firewall that was first introduced with Windows XP Service Pack 2. It was integrated and became a security feature with the introduction of Windows Vista. While host-based firewalls are not as secure as other types of firewalls, Windows Defender Firewall provides much better protection than in previous versions of Windows, and it is turned on by default. Windows Defender Firewall is used to block access from the network, which significantly reduces the surface area of attack for the Windows operating system.

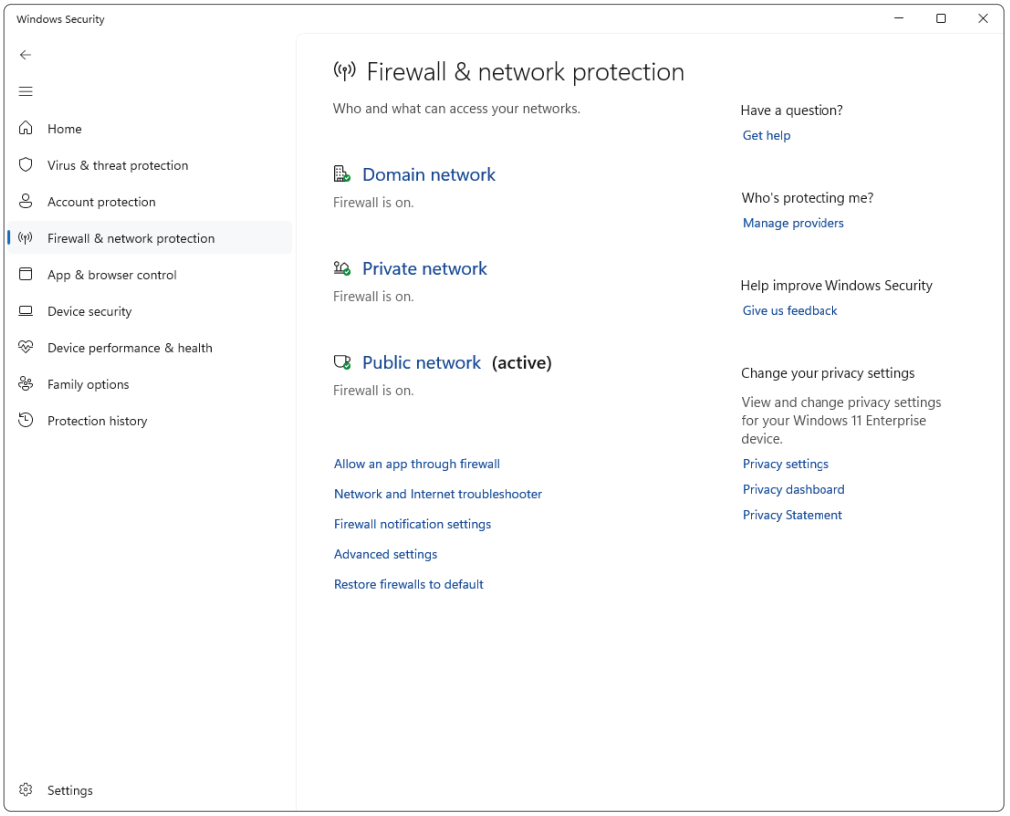
To access Windows Defender Firewall in Windows 11, navigate to Start ➢ Settings ➢ Privacy & Security ➢ Windows Security ➢ Firewall & Network Protection. Windows Defender Firewall is divided into separate profile settings: for domain networks (if you're connected to a domain), private networks, and public networks. In Figure 6.18, you can see the default protection for a Windows client that is not joined to a domain and is active on a public network.

Activate and Deactivate the Firewall By default, the Windows Defender blocks inbound connections, but outbound connections are not blocked by default. You can temporarily deactivate the Windows Defender Firewall for a network profile by clicking on the network profile in the Firewall & Network Protection screen and toggling the Windows Defender Firewall setting to Off, as shown in Figure 6.19. Deactivating the firewall protection is extremely useful if you are trying to troubleshoot a network connectivity problem. Dropping the firewall temporarily and testing the incoming connection will confirm that a firewall adjustment needs to be made.

Allowing Applications When applications attempt to listen on a port for an incoming connection, the Windows operating system will display a notification asking if you want to allow access. However, if the notification does not display and you determine that an application needs to be allowed through the firewall, you can click Allow An

App Through Firewall in the Firewall & Network Protection screen. This will open the Allowed Apps settings page shown in Figure 6.20. Here you can modify the apps that are allowed through the Windows Defender Firewall, as well as add new ones. You will need administrative access to change any network firewall settings.

FIGURE 6.18 Firewall & Network Protection



Windows Defender Firewall with Advanced Security When you allow an application to listen for an incoming connection via the notification dialog box, the operating system creates a rule in the firewall to allow the connection. This is all done for you behind the scenes and it shows up neatly as an allowed application. However, you can also manually create a rule in Windows Defender Firewall with the Advanced Security MMC, as shown in Figure 6.21. You can open the MMC by clicking Advanced Settings on the Firewall & Network Protection screen.

FIGURE 6.19 Deactivating Windows Defender Firewall network protection

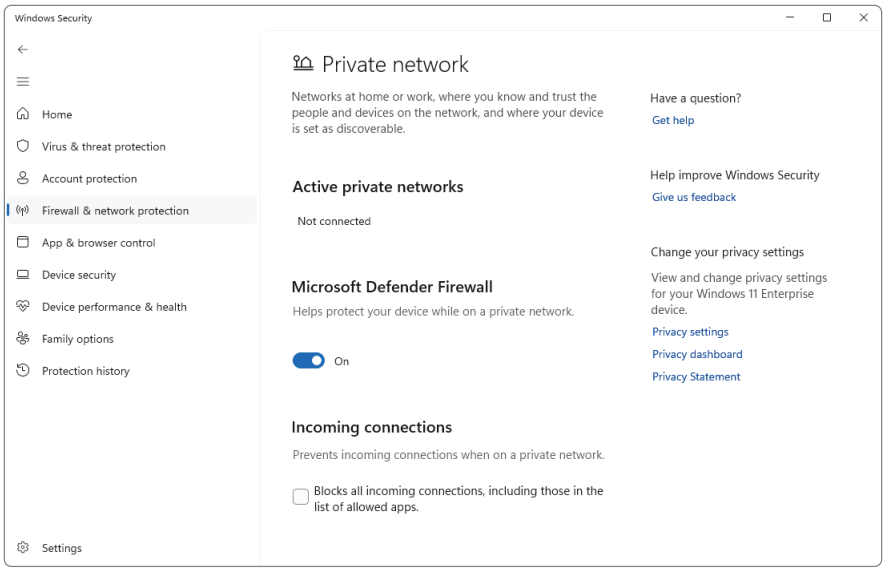


FIGURE 6.20 Windows Defender Firewall Allowed Apps

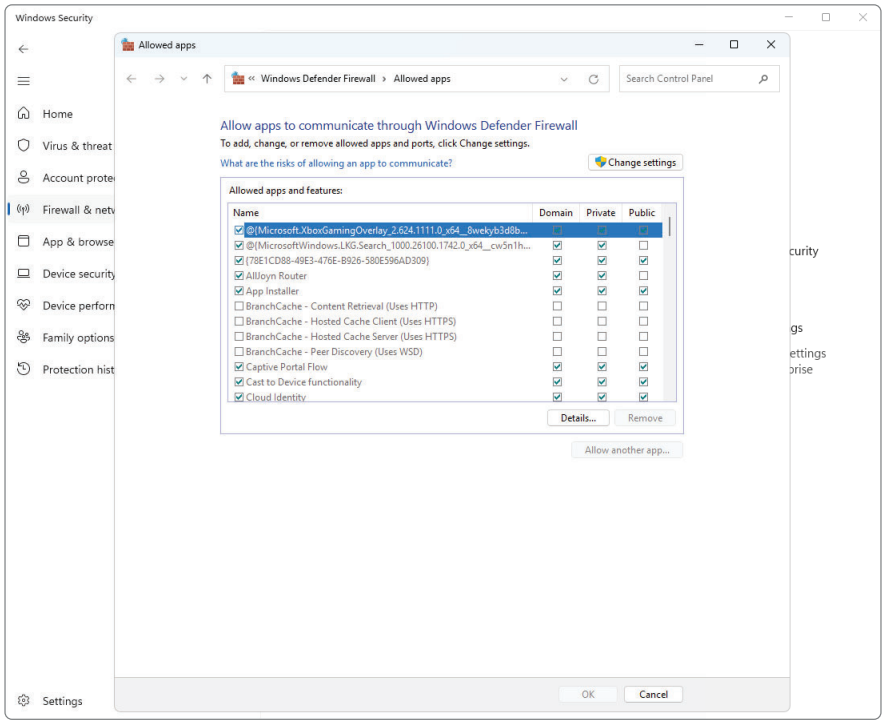
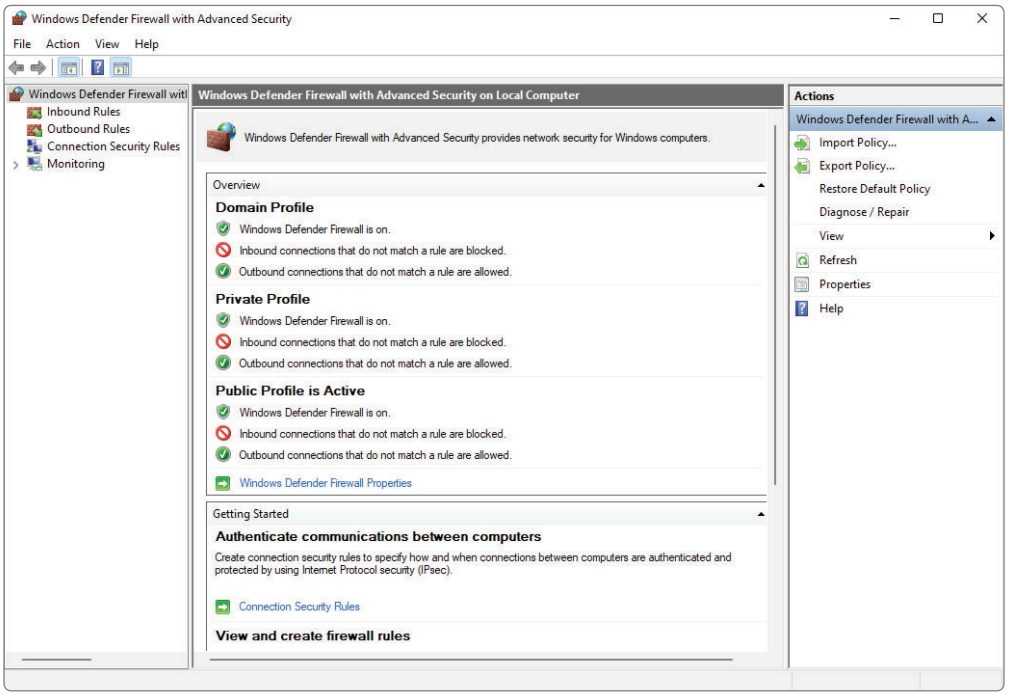


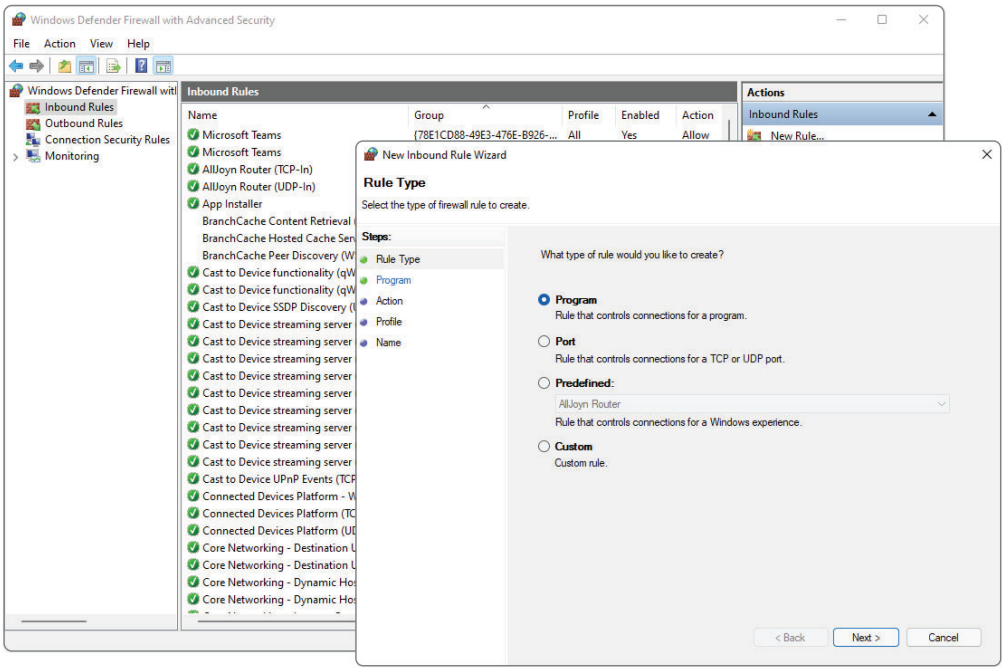
FIGURE 6.21 Windows Firewall with Advanced Security



Here you can configure inbound and outbound rules as well as import and export policies and monitor the security of your system. Monitoring is not confined to the firewall; you can also monitor security associations and connection security rules. In short, Windows Defender Firewall with Advanced Security is an incredibly powerful tool that builds on what Windows Vista introduced. Not only can this MMC snap-in do simple configurations, but it can also configure remote computers and work with Group Policy.

Exceptions Manual exceptions, also known as firewall rules, are configured on the Inbound Rules tab in the Windows Defender Firewall with Advanced Security MMC. You can click New Rule and configure a firewall rule based on a program, port, predefined rule, or totally custom rule. A rule based on a program allows all incoming connections to the application. A rule based on a port allows you to configure a specific rule based on a TCP or UDP connection to a specific port or range of ports. A predefined rule allows you to modify a predefined rule. A custom rule allows you to configure a program and specific ports; you can even scope it down to the incoming range of IP addresses you will allow. Any of these settings can also be configured after running the New Rule Wizard, as shown in Figure 6.22.

FIGURE 6.22 Windows Defender Firewall with Advanced Security inbound rules



BitLocker

You have to be careful, because CompTIA sometimes refers to the utility as "bit-locker" or "Bitlocker," while it is officially known as *BitLocker*. This tool allows you to use drive encryption to protect files—including those needed for startup and login. This is available only with more complete editions of Windows 11 (Pro, Enterprise, Education, Pro for Workstations).

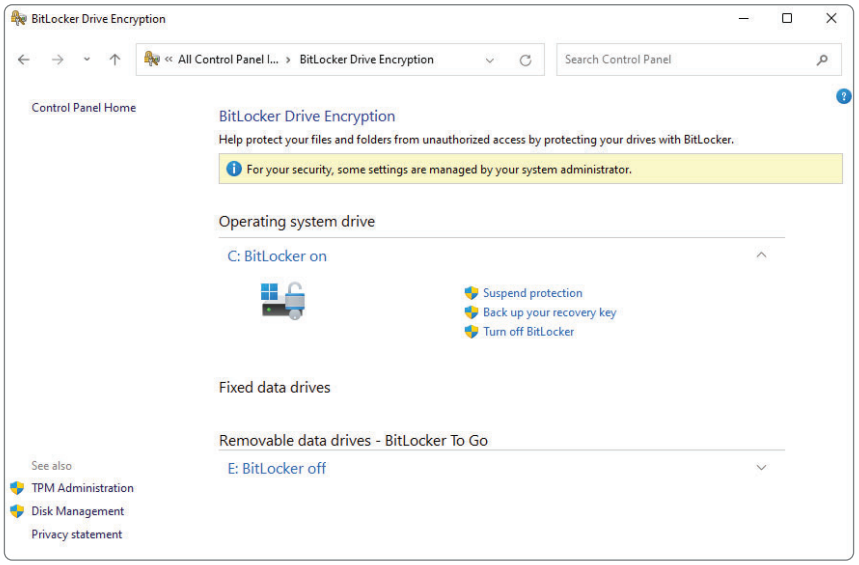
Another requirement is the use of a *Trusted Platform Module (TPM)*. The TPM is a chip on the motherboard that safely stores the encryption key so that the key is not stored on the encrypted disk. BitLocker can mitigate the risk of data loss, because if the disk is separated from the computer, it is still encrypted and only the TPM with the encryption/decrypting key can decrypt the disk. This prevents out-of-band attacks on the hard drive, where it would be mounted and examined on a second system. BitLocker can also sense tampering. If it senses tampering, the recovery key must be reentered. The recovery key is either entered from a printout, loaded from a USB drive in which it was originally saved, or recovered from your Microsoft account. An option of how the recovery key is stored is presented to you when you initially turn on BitLocker.

BitLocker to Go

You can also protect removable drives with *BitLocker to Go*. It provides the same encryption technology BitLocker uses to help prevent unauthorized access to the files stored on them. You can turn on BitLocker to Go by inserting a USB drive into the computer and

opening the BitLocker Drive Encryption Control Panel applet, as shown in Figure 6.23. When a USB drive is inserted into a Windows computer that contains BitLocker to Go encryption, the operating system prompts you for the password to unlock the drive. This password is the one you used originally when you set up BitLocker to Go on the USB drive.

FIGURE 6.23 BitLocker Drive Encryption applet

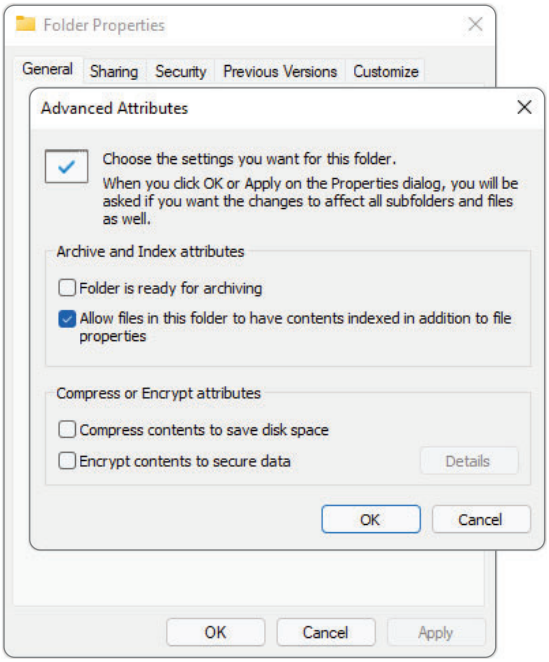


EFS

Encrypting File System (EFS), available in most editions of Windows, allows for the encryption/decryption of files stored in NTFS volumes. EFS uses certificates to encrypt the data, and the private certificate is stored in the user profile. When the first file is encrypted, the operating system automatically generates a key pair. If the computer were joined to an Active Directory domain and a *certificate authority (CA)* existed, the CA would create the key pair. You can encrypt a file or folder by right-clicking the object, selecting Show More Options, then Properties, then Advanced, as shown in Figure 6.24.

All users can use EFS, whereas only administrators can turn on BitLocker. EFS does not require any special hardware, whereas BitLocker benefits from having the TPM. As an additional distinction, EFS can encrypt just one file, if so desired, whereas BitLocker encrypts the whole volume and whatever is stored on it. Finally, EFS can be used in conjunction with BitLocker to further increase security.

FIGURE 6.24 Encrypting a file in Windows 11





Exam Objective 2.11, "Given a scenario, configure relevant security settings in a browser," requires you to know how to configure security settings in a browser. In addition, you should know about browser installation, patching, extensions and plug-ins, password managers, the various settings, and browser feature management.

Web Browser Security

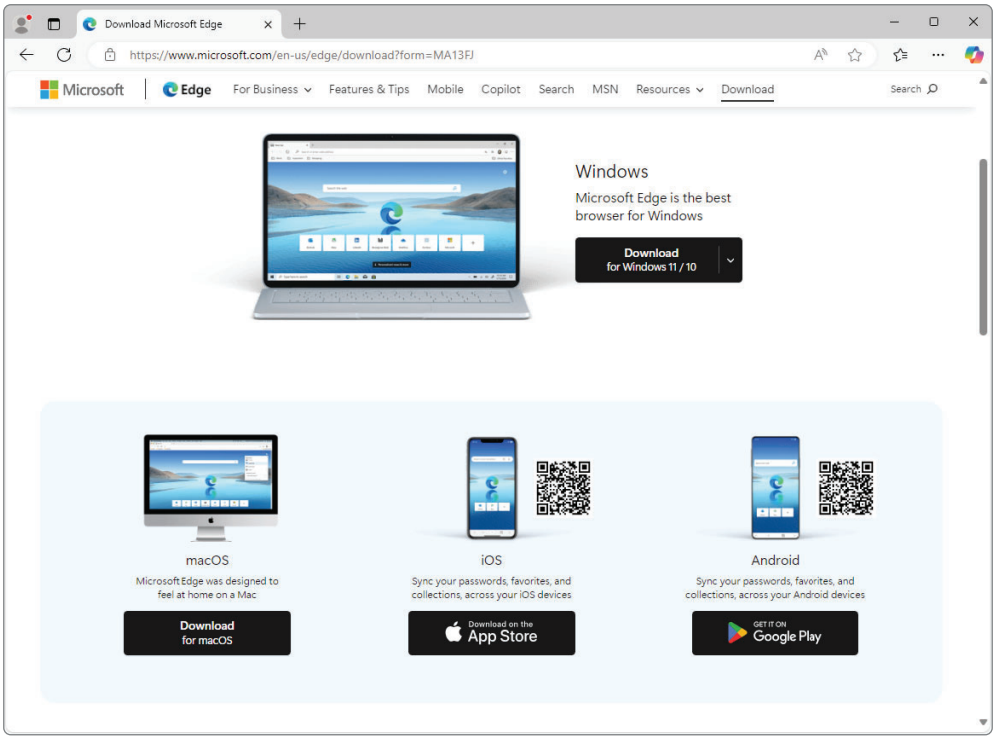
The web browser is arguably the most used application on the Windows operating system. It is also your portal to the Internet and all the bad things that reside outside of your network. Therefore, it makes sense that there is an entire objective dedicated to web browser security for the CompTIA exam. In this section we will explore various browser security topics. We will primarily focus on the Microsoft Edge browser that comes preinstalled in the Windows 11 operating system. However, we will also reference Chrome, as it holds more than 68 percent of the market share worldwide (`https://gs.statcounter.com`) as of this writing.

Browser Download and Installation

Edge is the successor to Internet Explorer and comes preinstalled in the Microsoft Windows 11 operating system. Edge is also the default web browser in the Windows 11 operating system. Although Edge comes preinstalled in Windows 11, there are circumstances when you need to download the Edge web browser and install it in other operating systems. A common scenario is downloading and installing Edge for a server operating system in which it is not preinstalled, such as Windows Server 2019 or another operating system, such as macOS or Linux.

To download the Edge installer, you can often just search for "Edge download." Depending on the search you use, the result can be in the top three results, but malware can also be disguised as the download. Therefore, you need to ensure your download is supplied by a trusted source. You should only download the browser from the vendor, in this case Microsoft. A simple way to check that the source is the vendor is to check the URL in the address bar, as shown in Figure 6.25. If the address bar shows the parent address of `https://www.microsoft.com`, then you are obviously downloading Edge from a trusted source.

FIGURE 6.25 Downloading Microsoft Edge



There are two main ways to download most browsers: online and offline. The online version will initially download a small install application (approximately 2 MB) that will download the rest of the installation and install the remaining download. The online version typically can be downloaded from the Microsoft Edge download page, as previously described. An offline installation can be downloaded from Microsoft as well and offers the benefit on not requiring any Internet connectivity. An offline version can be downloaded from `www.microsoft.com/en-us/edge/business/download`. By downloading the offline version from the Microsoft trusted source, you can be assured that future installations from this install are genuine.

When you store the installation for future installation, you should protect the installer from being tampered with. The best preventive measure is to create a Secure Hash Algorithm (SHA) hash of the executable and store it in another location. Before you run the offline installer, simply run the hashing against the install and check its signature. If the signature matches, then the installer has not been tampered with. If the signature does not match, tampering could have occurred, and it should be treated as being from an untrusted source.

In the following exercise, you will get to use the `Get-FileHash` cmdlet built into PowerShell. If you already have a file hash and it is not in the SHA256 format, you can use the argument of `-Algorithm MD5` or `-Algorithm SHA1` depending on the format you need to verify.

In Exercise 6.3 you will create a hash for a sample file with a PowerShell cmdlet. Then you will slightly modify the file and hash it again and compare the outcomes.

EXERCISE 6.3

Working with File Hashes

1. Open Notepad, type **example**, and save it to the desktop as **`example. txt`**.
2. Navigate to the desktop, hold the Shift key, right-click, and choose Open PowerShell Window Here.
3. Type **`Get-FileHash .\example. txt`**.
4. Examine the file hash and make a note of it.
5. Type **`Get-FileHash .\example. txt`**.
6. Compare the two file hashes; they should be identical.
7. Open the file `example.txt` and make the lowercase `e` in `example` an uppercase `E`; then save the file.
8. Navigate back to the PowerShell prompt.
9. Type **`Get-FileHash .\example. txt`**.
10. Examine the file hash.
11. Change the uppercase `E` back to a lowercase `e`, rerun the `Get-FileHash` cmdlet, and compare the two hashes.

Google Chrome and other third-party browsers require download and installation from the vendor sites, just like Microsoft Edge. These practices can and should be employed with any installation, not just web browsers. You should always download installations from a trusted source, then protect them with file hashing to detect tampering.

Browser Patching

A serious security concern is the number of web browser exploits released every day. A malicious web page could automatically download malware or allow cross-site scripting and compromise your login. These are just a few exploits that commonly need to be patched on a consistent basis. Therefore, constant browser patching is necessary. The version you download today is not going to be the version you will have tomorrow.

Each web browser handles browser patching similarly. Chrome employs a silent update mechanism, which automatically installs updates without user intervention. This method ensures that the browser remains secure and up-to-date. This approach minimizes the risk of vulnerabilities being exploited by attackers. Microsoft Edge, built on the Chromium engine, also utilizes an auto-update feature but offers additional flexibility for enterprise environments. An administrator can use patching tools like Microsoft Configuration Management to deploy constant updates.

When a web browser update is available, the user typically receives a notification indicating that a new version is ready to be installed. Inside Chrome this notification might appear as a subtle change in the browser's menu icon, prompting users to relaunch the browser to apply the update. Users can choose to ignore this notification temporarily, but they will continue to see reminders until they restart the browser. Expecting the user to relaunch the browser is fine in some cases. However, an administrator can enforce a policy that requires users to relaunch the browser within a specified time frame. Once the user decides to proceed with relaunching, the browser will close and reopen, applying the update.

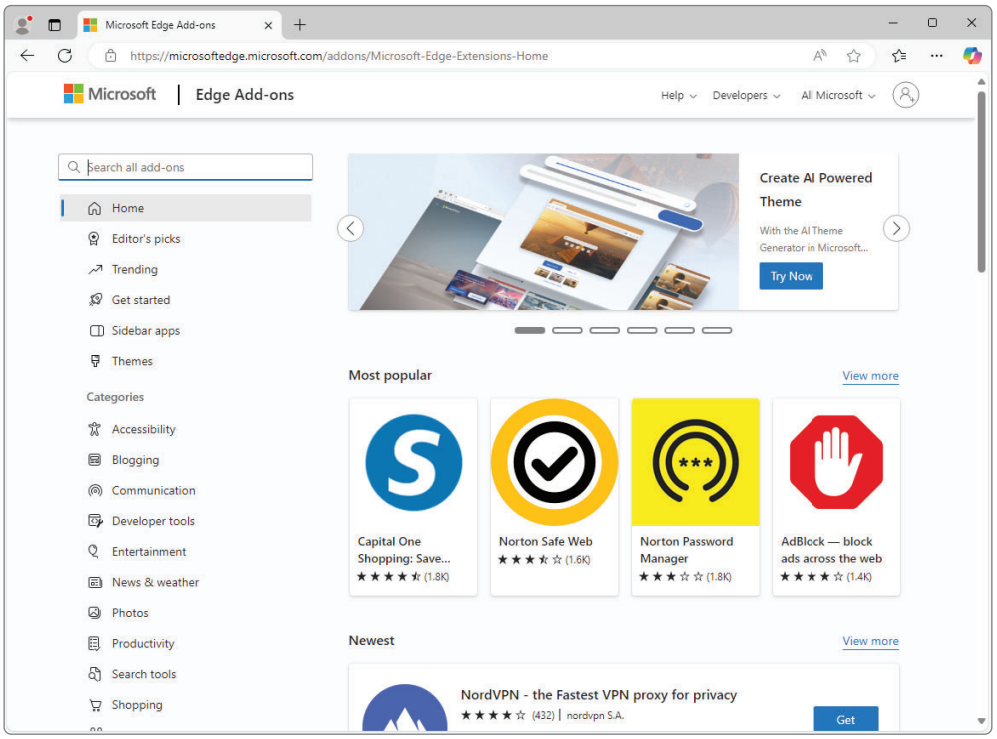
If a system reboot is necessary, the user will be prompted to restart their device to complete the update process. This is typically required when Windows Update performs major upgrades of Edge periodically.

Extensions and Plug-Ins

The best feature in any web browser is the ability to extend its functionality to accommodate your exact needs. By using extensions and plug-ins, you can functionally change the way the web browser works and the web pages it consequently renders. *Extensions* extend the functionality of the web browser that was not originally conceived when it was designed. *Plug-ins* change the way web pages are rendered on the web browser. Each web browser will call these something slightly different—for example, plug-ins might be called add-ons, as in the case of Firefox. Google Chrome refers to both extensions and plug-ins as extensions. Microsoft refers to both extensions and plug-ins as add-ons. For the remainder of this section, we will refer to all plug-ins and extensions as add-ons.

Most modern web browsers have an ecosystem of add-ons. Google Chrome has an ecosystem called the Chrome Web Store that can be accessed at `https://chrome.google.com/webstore`. Microsoft has an ecosystem called Edge Add-ons that can be accessed at `https://microsoftedge.microsoft.com`, as shown in Figure 6.26. An add-on ecosystem is a place where the vendor trusts the publisher of an add-on. The add-on ecosystem in turn distributes the add-on for users, and supplies updates and synchronized installations across multiple devices. The add-on ecosystem should always be considered a trusted source for web browser add-ons.

FIGURE 6.26 Microsoft Edge add-ons



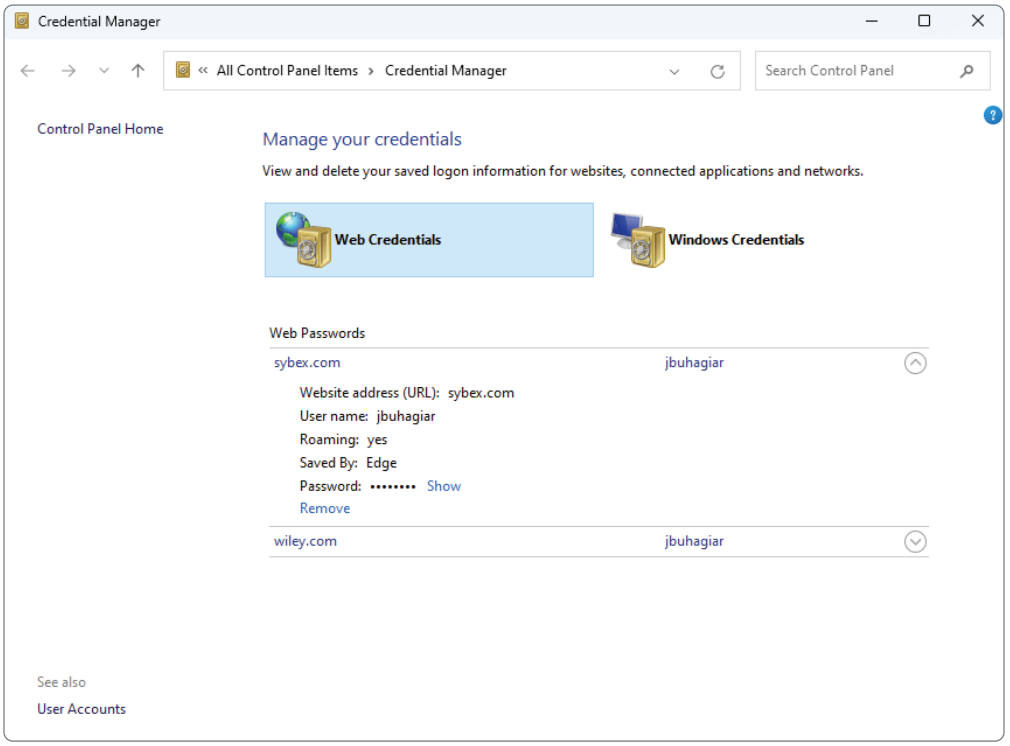
Most web browsers will also allow manual installation of add-ons. This is also called *sideloading*, since you are manually installing the file outside of the ecosystem. This is usually done and allowed by the web browser for development purposes. This type of installation can be found among untrusted applications that the ecosystem doesn't allow. The add-on might look reputable and the web page might explain that manual installation is required, since the developers don't agree to the terms of service (ToS) of the ecosystem. In any case, you should consider these add-ons untrusted and therefore avoid them.

Credentials Managers

As you sign up for websites, they require more complex, lengthier passwords. Sometimes your username might be your favorite nickname, sometimes the nickname might not be available, and sometimes the site might require an email in lieu of a nickname-style username. You should also employ the practice of single-purpose passwords, to prevent credential stuffing attacks, and never reuse a password. So, if your credentials on site A are compromised, then your credentials on site B will not be vulnerable.

No longer can we keep track of all these usernames and passwords. Luckily password managers have come to our rescue. They are built into every operating system and most web browsers. Microsoft Edge and the older Internet Explorer both use the Microsoft Credential Manager, which was originally introduced with Windows XP. It functions as a password manager and is built into the operating system. In Figure 6.27, you can see some web credentials stored by Microsoft Edge. You can access the Credential Manager by navigating to the Start menu, typing **Control Panel** and selecting it in the results, then clicking Credential Manager.

FIGURE 6.27 Microsoft Credential Manager



Credentials are stored by successfully logging into a website with a username and password combination. The web browser will ask if you want to save the credentials. Once the credentials are stored, when a website asks for a username and password matching the site in the Credential Manager, the associated credentials are offered to the user for logging into the site. If you are in the Credential Manager and you want to see the password, click Show and enter your credentials for the currently logged-on user. By entering your credentials for the currently logged-on user, you unlock the Credential Manager and it will let you see the password. You also have the option of deleting the credentials by clicking Remove.

Other web browsers, such as Google Chrome and Mozilla Firefox, have a credential manager built-in. These web browsers do not use the Microsoft Credential Manager. They synchronize their usernames and passwords between installations of the web browser on various devices. You can also download and install a third-party stand-alone credential manager. KeePass is one example, and there are several freely available on the Internet.

Secure Data Transfers

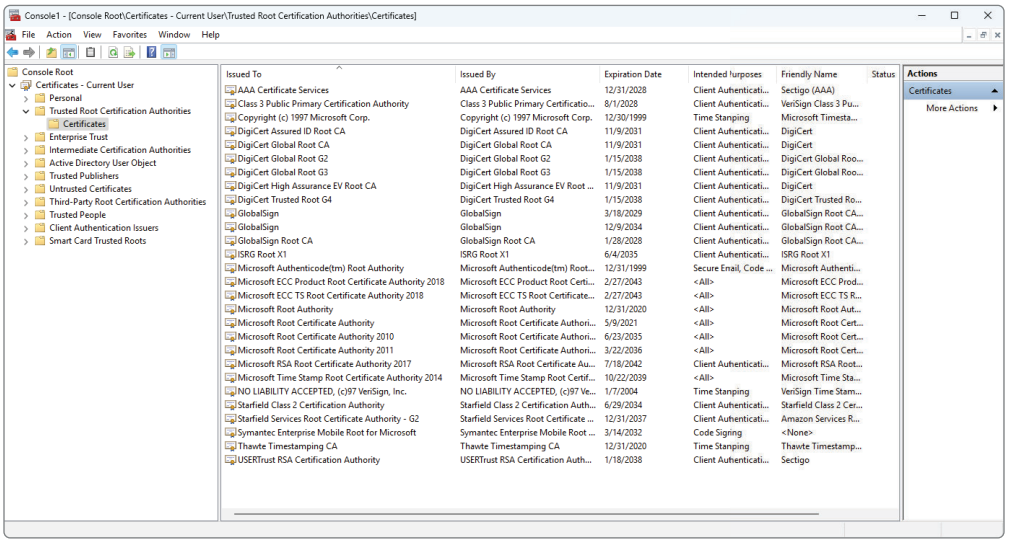
Securing data transfers to and from the web browser is critical for the security of day-to-day operations. We use our web browsers for accessing sensitive information at work, doing our personal banking, accessing social media, and the list goes on. So, securing data in transit is a primary concern to manage the risk of eavesdropping from threat agents. The Hypertext Transfer Protocol over Secure Sockets Layer (SSL), also known as HTTPS, is a method for securing web data communications. SSL is a cryptographic suite of protocols that use public key infrastructure (PKI) to provide secure data transfer.

A PKI is a system that provides the key pair of private and public keys, also known as certificates. The certificates are used to validate communication and encrypt communications between web browsers and web servers. The private key (certificate) from the key pair is installed on the web server. The public key (certificate) is available to anyone who wants to validate the data encrypted with the private key that is installed on the server. The web browser will automatically download the public key for the cryptography process.

Because there is a level of trust involved with the public/private key pair, your web browser must initially trust the publisher of the key pair. The publisher of the key pair is known as the certificate authority (CA). Every web browser comes with an initial list of trusted root certificate authorities, which is a beginning of trust for issuing CAs. The issuing CAs are the CAs that actually issue the key pairs. In most cases the web browser will use the operating system's list of trusted root CAs, as shown in Figure 6.28. However, some web browsers maintain a list of their own trusted root CAs.

Each certificate has an expiration date, and from time to time, certificates expire, get replaced, or in some cases get revoked for various reasons. Therefore, the list of trusted root certificate authorities must be updated now and then. The Microsoft Windows platform does this with routine Windows Updates, but browsers may update their own lists with updates of their own.

FIGURE 6.28 Trusted root CAs





Although the S in HTTPS stands for secure, SSL is the protocol suite used for securing communications. The SSL suite is a suite of protocols that includes the current standard of Transport Layer Security (TLS) 1.3. TLS 1.3 is used for securing websites, as of this writing.

Settings

A lesson on web browser security wouldn't be complete without covering security settings for the web browser. Next, we will explore the various security settings you will find on modern-day web browsers. We discuss the most common topics that you will see on the CompTIA exam, but in no way is this a complete list of security settings. The web browser is the most used application for accessing data, and new settings are introduced all the time to combat threat agents.

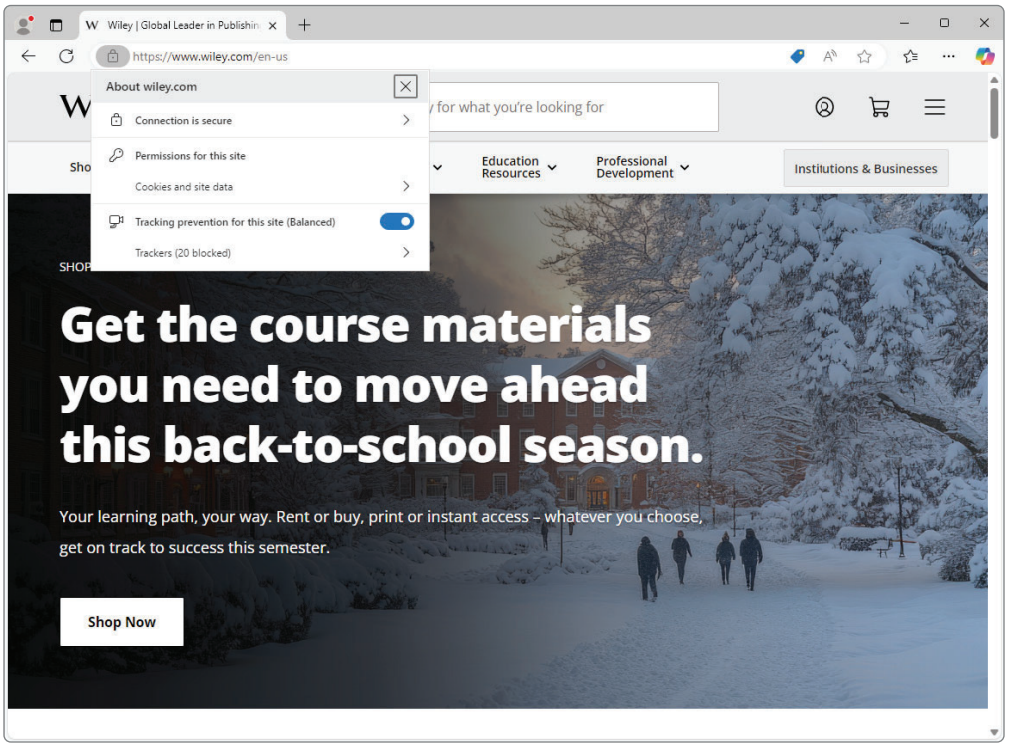
Pop-Up Blocker

Pop-up blockers are used to prevent both pop-ups and pop-unders from appearing when you visit a web page. While older browsers did not incorporate an option to block pop-ups, most current browsers, including the latest versions of Edge and Chrome, have that capability built in.

By default, the pop-up blocker is enabled on all sites globally. However, some sites, such as banking websites, might need to pop up a web page. When this happens, the

address bar will notify you that an attempted pop-up occurred. You can then click and allow the specific site to use pop-ups. If you want to view a site's permission for pop-ups, click the lock at the left of the address bar and choose Permissions For This Site, as shown in Figure 6.29.

FIGURE 6.29 Permissions for a site



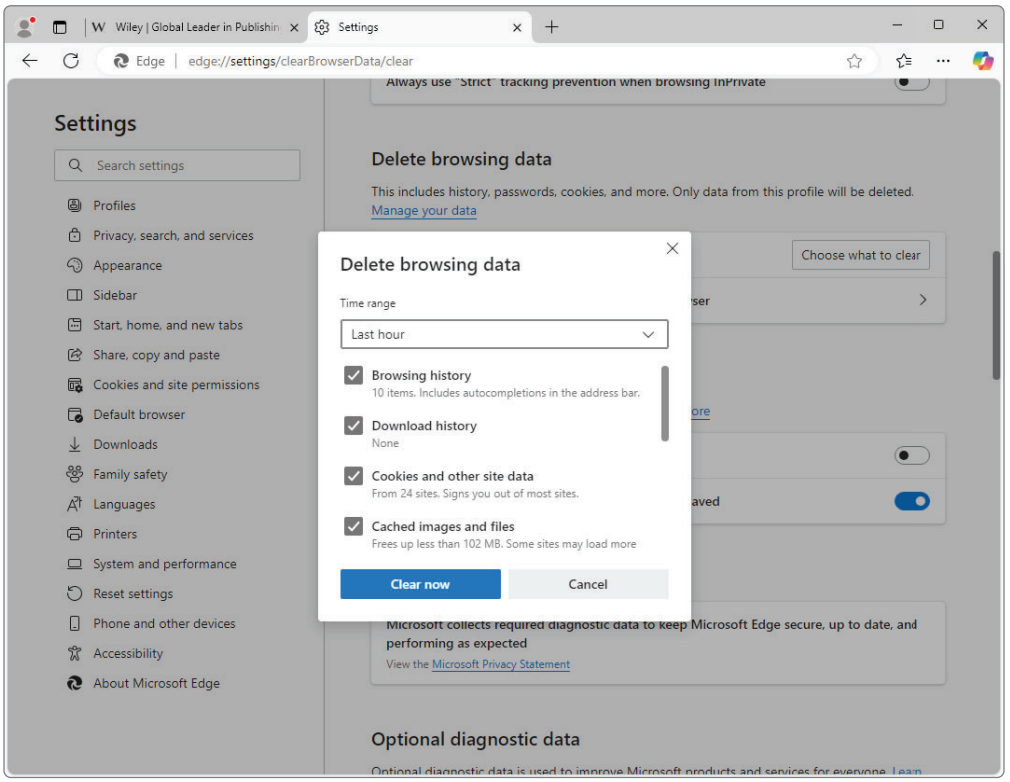
Clearing Browsing Data

Browsing data is a broad term used to describe any data stored while visiting websites. Data that may be considered browsing data is browsing history, download history, cookies, image and file caches, passwords, autofill form data, and site settings. These are just the top categories, so keep in mind that browsing data is a broad term. Browsing history is used by the browser to highlight links you've already visited. Download history allows you to historically see what has been downloaded. Cookies are used to save settings used by the web page, such as themes and login data, to name a few. Passwords are saved for convenience when logging into websites, the same as autofill form data. The site settings are adjustments you've made to the browser to optimally render the web page.

There are various reasons you may want to clear browsing data. The most compelling reason is data privacy. However, it is sometimes necessary to clear browsing data when you are trying to replicate a problem seen with the web browser.

You can clear the browsing data by clicking the three dots in the upper-right corner of the Edge web browser. Then click Settings, choose Privacy, Search, And Services, and scroll down to Delete Browsing Data Now and select Choose What To Clear. You will be presented with a dialog box similar to Figure 6.30. Options for the time range are Last Hour, Last 24 Hours, Last 7 Days, Last 4 Weeks, and All Time. You can also selectively delete the web browsing data that you desire.

FIGURE 6.30 Clearing browsing data



Clearing Cache

When a web browser renders a web page, the files retrieved are cached. This is done so that if you need them again you can quickly retrieve them from storage. This caching mechanism speeds up the web browser and reduces unneeded trips to the Internet. There are times when you need to clear your web browser cache, such as when developing a web page. You will

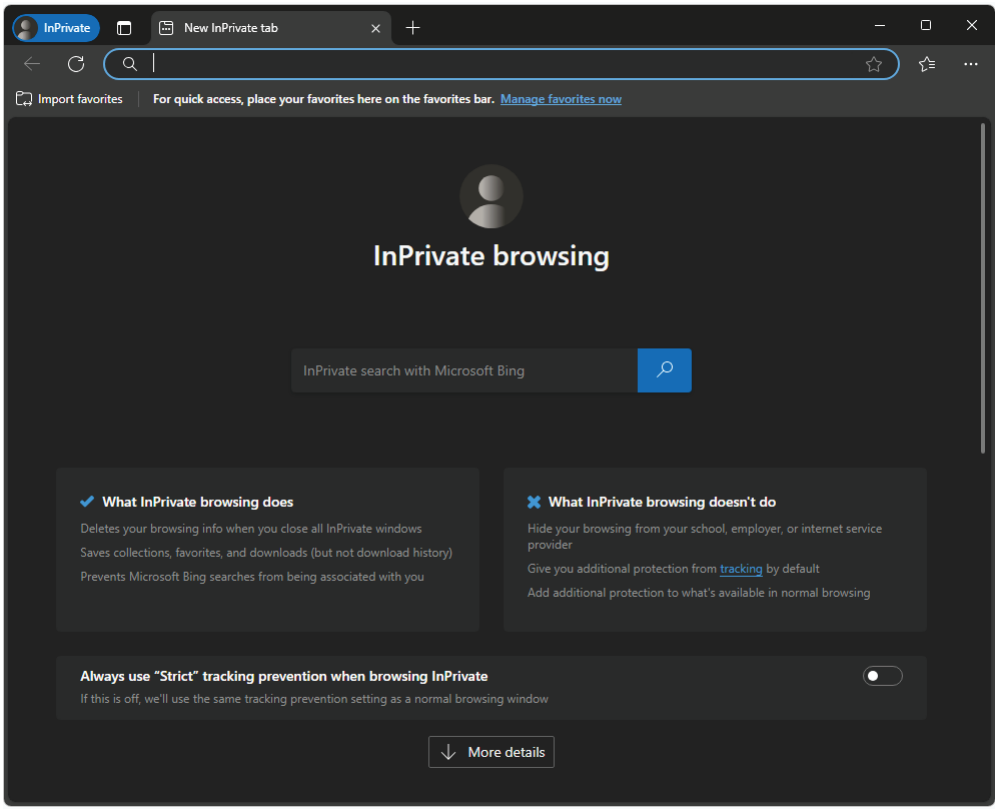
want to retrieve the latest copy of the web page and its assets so that you can verify how it is rendered. The cache images and files are part of the web browsing data that can be cleared. The process is similar to the previously mentioned process for clearing browsing data, except only the cache images and files will be cleared.

Private-Browsing Mode

Private-browsing mode was created primarily to address data privacy while you are web browsing. The most compelling feature of private-browsing mode is that it does not store any web browsing data. Therefore, when you close private-browsing mode, all browsing data is destroyed.

The private-browsing mode on the Edge web browser is called InPrivate browsing; Chrome calls it Incognito mode. On the Edge web browser you can enter InPrivate mode by clicking the three dots in the upper-right corner of the window, then selecting New InPrivate Window. Another way is to press Ctrl+Shift+N to open the InPrivate window. In either case, the window explains the mode, as shown in Figure 6.31. The methods to open a private-browsing window on Chrome are identical, except that it is called Incognito mode.

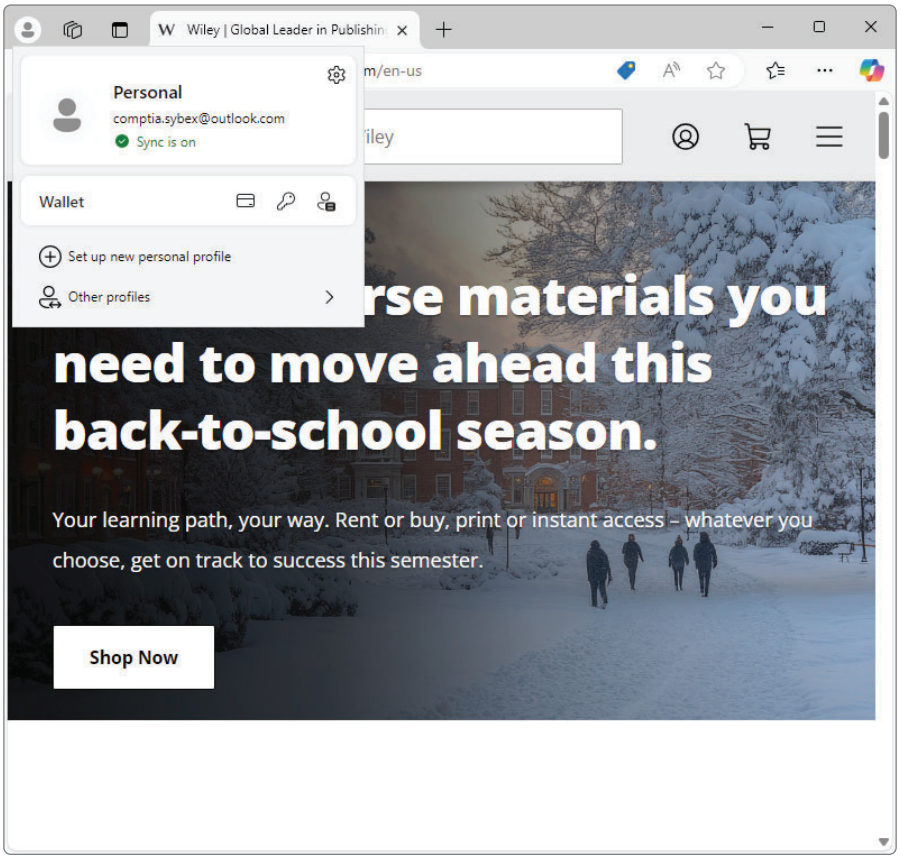
FIGURE 6.31 InPrivate browsing



Browser Data Synchronization

On average, we access our web browsers on several devices. You may access them on your home computer, personal laptop, and mobile device, just to name a few. Therefore, the browsing data should follow you, no matter which device you are using. Modern web browsers fortunately support browser data synchronization, which allows you to share your data across many different devices. To sync your device, you just need to make sure that you have signed in and acknowledged that you want to sync browsing data. You can verify this by clicking on your account in the upper-left corner of the Edge web browser. Your account details will be displayed, as shown in Figure 6.32. If you are not syncing, you can simply sign in.

FIGURE 6.32 Account details



Use caution before syncing your personal web browsing data with your work computer's. You should keep a strict separation between work and leisure. During our leisure time we might search for something that could be misconstrued as mischievous to our coworkers. Therefore, you should maintain different accounts for work and play and never sync your personal account to a work web browser.



Real World Scenario

Leisure vs. Work Browsing

During leisure time you may be watching a movie in which someone picks a lock and curiosity may strike you to search "how to pick a lock." That search will now be in your autofill form data (browsing data). Imagine clicking a search box while giving a presentation in front of coworkers and "how to pick a lock" autofills. You know the context of your search, but do your audience and coworkers know? Think of all the other searches you perform in your leisure time.

Ad Blockers

The Internet was originally embraced as a mechanism to share information and bring people together. However, it was equally embraced by commerce as a mechanism to deliver products and services. One such service is the marketing of other services and products; we get these as marketing ads.

You can install an ad blocker add-on for most web browsers to stop a lot of the spammy ads you might receive on a web page. However, entire websites, such as many news sites, subsidize their income with marketing ads. If you don't have a subscription, then you have to allow marketing ads in order to enjoy their content; it seems a pretty fair trade. Most of these sites allow you to have directions on how to exempt their site from the ad blocker.

You may wonder why you should install an ad blocker at all if every site allows you to exempt it. Ad blockers are extremely useful as an added layer of insulation from threat actors. Although most sites are legitimate and serve relevant ads, there are plenty of sites that serve malicious ads. An ad blocker helps to block all ads and allows you to judge whether the site is worthy of an exemption.

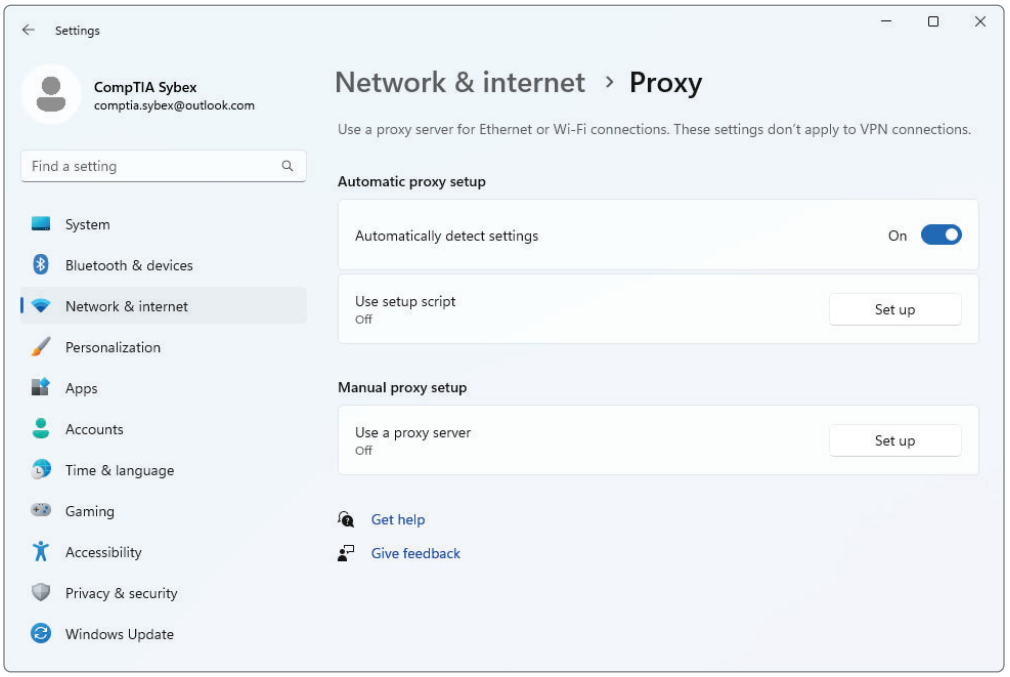
Proxy

A web browser can be configured to use a proxy server, which fetches the request on behalf of your request and returns the data. Proxy servers are also effective for controlling policies of what an employee can view. This provides not only a layer of protection, but also a way to centrally log requests. Filtered access to certain web sites is something that is decided at

the organizational level. It is common to block gambling, pornography, and other illegal or illicit sites.

The proxy setting can be configured for most web browsers at the operating system level. This can be accomplished by opening the Start ➢ Settings ➢ Network & Internet ➢ Proxy, as shown in Figure 6.33. From the proxy screen you can manually configure a proxy server, or allow the automatic detection through the use of JavaScript. It is common for an organization to policy the proxy server settings, so they are automatically applied.

FIGURE 6.33 Proxy Settings



Secure DNS

Secure DNS, also known as DNS over HTTPS (DoH), is a technology designed to enhance privacy and security on the Internet. The need for Secure DNS arises from the vulnerabilities associated with traditional DNS queries, which are typically sent in plain text. This makes them susceptible to eavesdropping, manipulation, and other cyberthreats. By encrypting DNS queries, Secure DNS ensures that these lookups are protected from interception and tampering by malicious actors. This in turn safeguards the user's browsing activities and personal information.

Configuring Secure DNS in Chrome is performed by navigating to the browser's settings, selecting Privacy And Security, and then Security. Here, they can find the Use Secure DNS option, which allows them to choose a preferred DNS provider or enter a custom provider. Similarly, Microsoft Edge offers an easy way to enable Secure DNS. Users can go to Settings, then Privacy, Search, and Services, and under the Security section, they can turn on Use Secure DNS To Specify How To Look Up The Network Address For Websites. Edge also provides options to use the current service provider or select a custom provider. Secure DNS is enabled in both Edge and Chrome by default.

Browser Features

Web browser features can be significantly extended through the installation of extensions, add-ons and plug-ins, which are small software modules designed to add functionality and enhance the user's browsing experience. Extensions, also known as add-ons, modify the core functionality of the browser, allowing users to customize their browsing environment with tools like ad blockers, password managers, and note-taking utilities. Plug-ins, on the other hand, provide additional capabilities without altering the browser's core functions, such as enabling PDF editing or media playback. Their differentiations are so blurred that most web browsers call them add-ons or extensions; these words are interchangeable.

These add-ons are typically available through browser-specific web stores, where users can browse, read reviews, and install them with a simple click. By leveraging these add-ons, users can tailor their browsers to meet their specific needs, improving productivity, security, and overall user experience.

Management

When you want to add an add-on to your web browser, you will typically visit the web browser's add-ons website. This is how you will get new features for your browser. Once you have the features installed, you will need to manage them. In the Edge browser you can manage these add-ons by clicking the three dots in the upper-right corner, then choosing Extensions and then Manage Extensions from the drop-down menu. This will open a page where you can see all the installed extensions for the web browser, as shown in Figure 6.34. In this screen you can flip the toggles for any of the extensions installed. These toggles will control if the extension is enabled or disabled.

Extensions work by interacting with the pages rendered. There could be an instance where an extension effects the page being rendered. This menu allows you to configure if the extension is enabled or disabled. A good rule of thumb when having those problems is to turn all the additions off and test your page.

Settings

All extensions have a Details page, as shown in Figure 6.35, where you can view the Description, Permissions, Site Access, option to allow it to operate with inPrivate mode, Source, and a button to remove it. When you enter inPrivate mode, no extensions are enabled by default. This can be controlled on the extensions detail page.

FIGURE 6.34 Edge Extension Settings

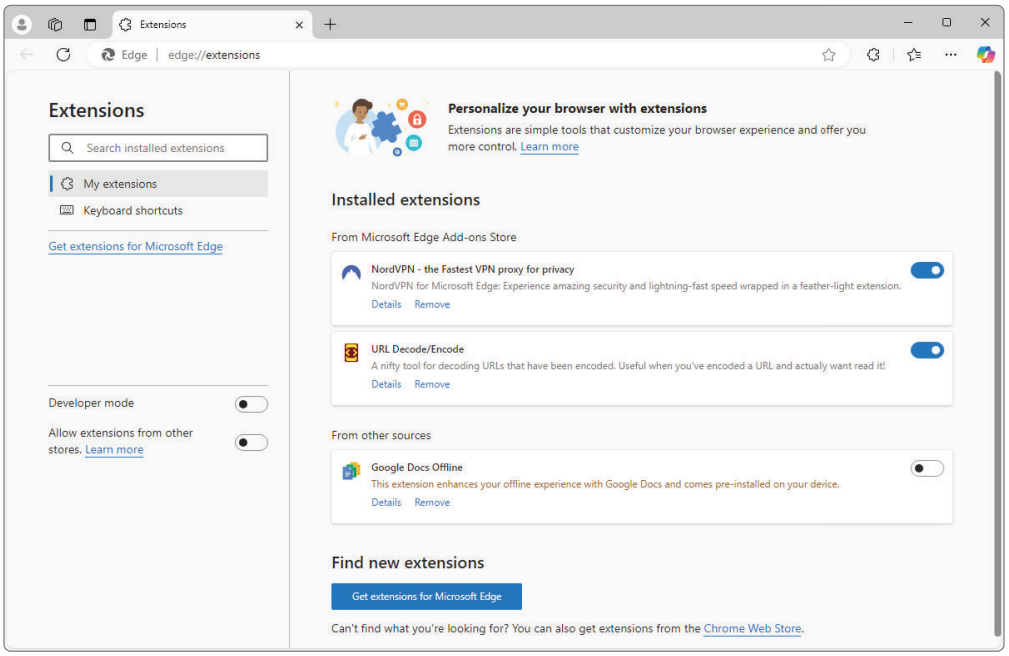
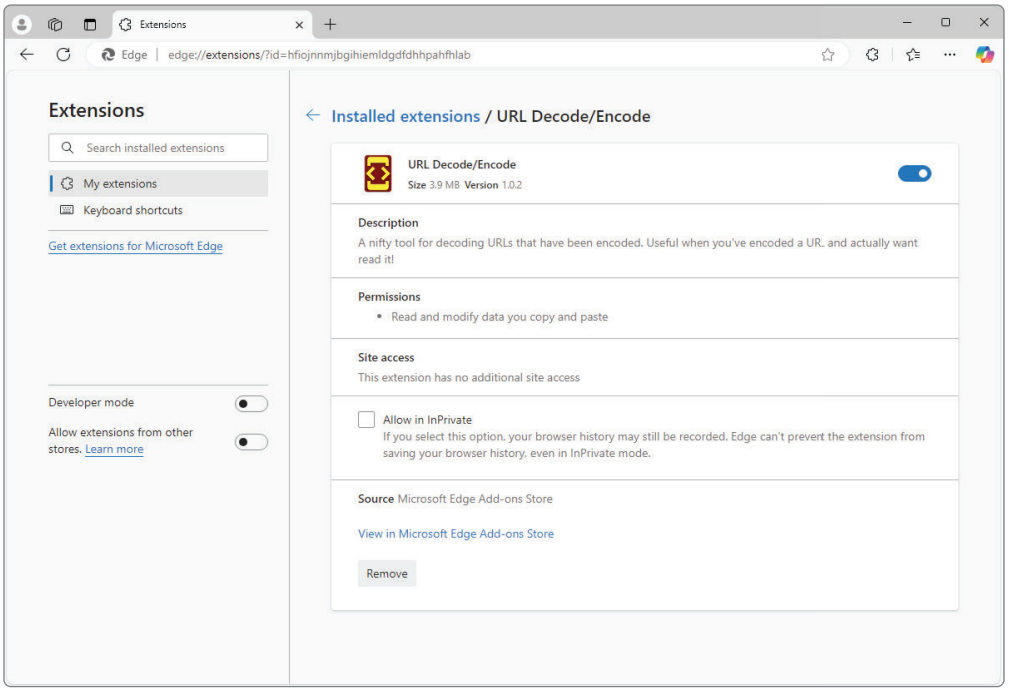


FIGURE 6.35 Edge Extension Details

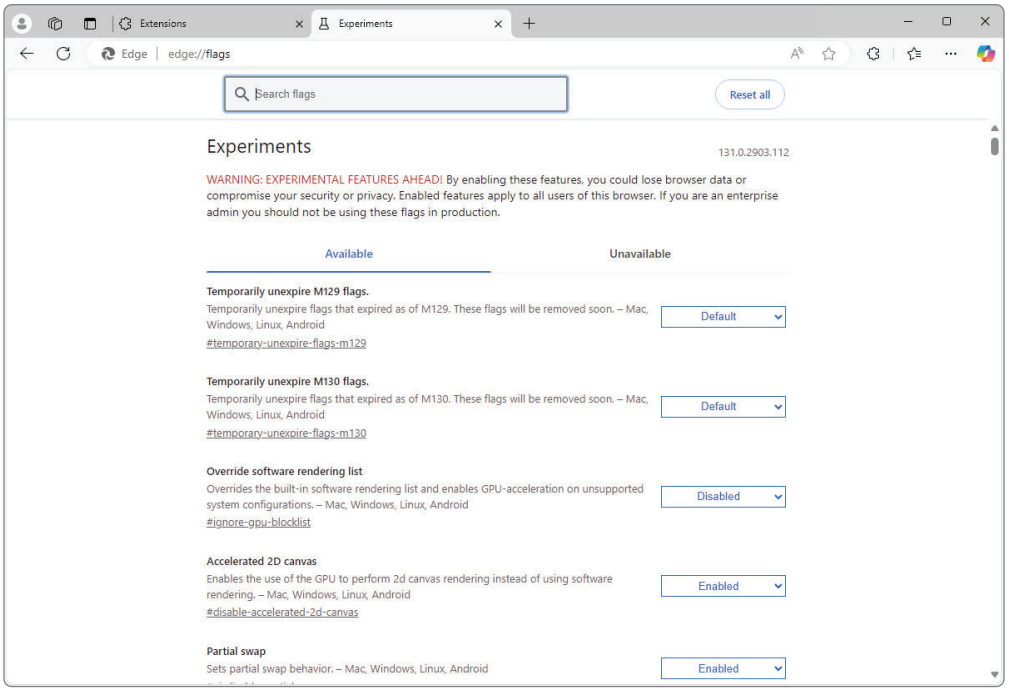


Some extensions have the ability to use their own custom configuration. These extensions will have a link Extension Options, under the View In Microsoft Edge Add-Ons Store link. This link will open a custom web page where you can configure the add-on. Alternately you can click on the puzzle piece icon in the upper right of the web browser, click the three dots next to the add-on, and select Extension Options from the context menu if it provides custom configuration.

Experimental Controls and Features

In most modern web browsers today, there is a hidden menu to control experimental features and control basic features. This is a hidden menu, because clicking around can really cause some issue in your web browsing experience. The Chromium web browsers, such as Chrome and Edge, call this menu Experiments, and it can be accessed at `chrome://flags` or `edge://flags`, respectively, as shown in Figure 6.36.

FIGURE 6.36 Edge Experiments screen



The Experiments screen allows you to turn on experimental features, and control basic features like caching. Typically, unless you know what you are looking to achieve, this hidden menu should be avoided. However, if you wanted to control how cache is handled, or how memory is used, this section is where you would move the levers (so to speak).



Exam Objective 2.10, "Given a scenario, apply security settings on SOHO wireless and wired networks," requires you to know how to apply security settings for small office, home office (SOHO) networks. You should know how to secure both wireless and wired networks.

Securing a SOHO Network (Wireless)

CompTIA wants administrators of small office, home office (SOHO) networks to be able to secure those networks in ways that protect the data stored on them. This objective looks at the security protection that can be added to a wireless SOHO network, while the following section examines similar procedures for a wired network. This knowledge should complement what you have learned in Volume 1 Chapter 9, "Wireless and SOHO Networking." The concepts in this section will help you secure wireless and wired SOHO networks.

A wireless network is not and never will be secure. Use wireless only when absolutely necessary. If you must deploy a wireless network, here are some tips to make improvements to wireless security:

- Change the default SSID.
- Disable SSID broadcasts.
- Disable DHCP or use reservations.
- Use MAC filtering.
- Use IP filtering.
- Use the strongest security available on the wireless access point.
- Change the static security keys every two to four weeks.
- When new wireless protection schemes become available (and are reasonably priced), consider migrating to them.
- Limit the user accounts that can use wireless connectivity.
- Use a preauthentication system, such as RADIUS.
- Use remote access filters against client type, protocols used, time, date, user account, content, and so forth.
- Use IPsec tunnels over the wireless links.
- Turn down the signal strength to the minimum needed to support connectivity.
- Seriously consider removing wireless access from your LAN.

Changing Default Usernames and Passwords

In addition to those created with the installation of the operating system(s), default accounts are also often associated with hardware. Wireless access points, routers, and similar devices

often include accounts for interacting with, and administering, those devices. You should always change the passwords associated with those devices and, where possible, change the usernames.

If there are accounts that are not needed, disable or delete them. Make certain that you use strong password policies and protect the passwords with the same security that you use for users and administrators. (In other words, don't write the router's password on an address label and stick it to the bottom of the router.)

Changing the SSID

All radio frequency (RF) signals can be easily intercepted. To intercept 802.11 wireless traffic, all you need is a PC with an appropriate 802.11 card installed. Many networks regularly broadcast their names (known as an *SSID broadcast*) to announce their presence. Simple software on the PC can capture the link traffic in the wireless AP and then process this data to decrypt account and password information.

You should change the SSID—whether or not you choose to disable its broadcast—to keep it from being a value that many outsiders come to know. If you use the same SSID for years, then the number of individuals who have left the company or otherwise learned of its value will only increase. Changing the variable adds one more level of security.

Guest Network Isolation

Most guests in your network never need to connect to the organization's servers and internal systems. When guests connect to your wireless network, it is usually just to get connectivity to the Internet. Therefore, a guest service set identifier (SSID) should be created that isolates guest traffic from production traffic. These guest network SSIDs are usually created by default on consumer wireless devices. On enterprise wireless LAN controllers, the guest network typically needs to be created.

Some considerations for the guest network are what is open to guests, how long they have access, how much bandwidth, the SSID name, and the list goes on, depending on your organization. Guest networks usually don't give totally unrestricted Internet access; certain sensitive ports like TCP port 25 (SMTP) are normally blocked. The length of time they have access is another concern. Generally, a guest is just that, a guest. So, 4 hours, 8 hours, or 24 hours of access seem responsible. You should give this a lot of thought because too short a time will create administrative overhead and too long a window of access allows for abuse of service. If you don't expect guest access to your wireless network, then it should be disabled.



Exam Objective 2.3, "Compare and contrast wireless security protocols and authentication methods," requires you to know wireless security protocols and authentication methods. Be sure to know the differences between the various security protocols.

Setting Encryption

It's important to remember that you should always enable encryption for any wireless network that you administer. Choose the strongest level of encryption you can work with. The following are some wireless protocols that you might encounter when securing wireless:

Open Open security is just that—open with no passphrase or authentication protocol. Open security was originally how all wireless access points (WAPs) were shipped to the customer. Open security still has its uses when used in conjunction with guest wireless access.

Wired Equivalent Privacy Shared passphrases are used with Wired Equivalent Privacy (WEP). WEP provides 64- or 128-bit encryption via the shared passphrase. The passphrase can easily be cracked with tools and is no longer used to secure wireless.

Wi-Fi Protected Access Wi-Fi Protected Access (WPA) was standardized by the Wi-Fi Alliance in 2003 in response to the vulnerabilities in Wired Equivalent Privacy (WEP). WPA uses 256-bit keys versus the 64-bit and 128-bit keys WEP used previously. WPA operates in two modes for security: preshared key (PSK), also called personal mode, and enterprise mode. PSK is the most common mode, because it can easily be implemented with a mutual agreed-upon passphrase. Enterprise mode, also called WPA-802.1X, requires a certificate server infrastructure. Enterprise mode uses the 802.1X protocol, RADIUS, and EAP; it is often used in corporate environments.

WPA introduced many improved security features over WEP, such as message integrity checks (MICs), which detect packets altered in transit. WPA also introduced Temporal Key Integrity Protocol (TKIP), which uses the RC4 algorithm for encryption. TKIP provides per-packet keying to prevent eavesdropping on wireless conversations. However, despite the improvements in security, WPA is considered exploitable and is no longer used for wireless security. A common exploit used against WPA is an attack on the helper protocol of Wi-Fi Protected Setup (WPS). WPS is used for consumer ease of setup and should be turned off for security purposes.

Temporal Key Integrity Protocol (TKIP) TKIP uses the RC4 encryption algorithm protocol as its cipher. TKIP seeds the RC4 algorithm with a key that is derived from the MAC address and initialization vector. TKIP also works in conjunction with message integrity checks (MICs) to check the integrity of messages received by the access point. The MIC protocol, also called Michael, is a 32-bit cyclic redundancy check (CRC). If two CRC MICs fail within 60 seconds, the access point requires TKIP to rekey the RC4 seed value.

Wi-Fi Protected Access 2 (WPA2) WPA2, also known as 802.11i, is the successor to WPA. WPA was deprecated in 2006 when WPA2 became a wireless security standard. Just like WPA, WPA2 operates in both personal mode (PSK) and enterprise mode.

WPA2 uses the Advanced Encryption Standard (AES) algorithm to protect data. AES is more secure than the RC4 algorithm used with TKIP. WPA2 replaced TKIP with Counter

Cipher Mode (CCM) with Block Chaining Message Authentication Code Protocol (CCMP). However, TKIP can be configured as a fallback for WPA backward compatibility. Like WPA, WPA2 is exploitable if the WPS service is enabled. WPS should be turned off for security purposes.

Counter Cipher Mode with Block Chaining Message Authentication Code Protocol (CCMP) Advanced Encryption Standard (AES) uses a 128-bit key to seed the AES encryption and a 128-bit cipher block. The prior ciphered text is used to encrypt the next block of text; this type of cipher is called code block chaining (CBC). CCMP-AES also uses a MIC to check the integrity of wireless data received. If the MIC fails, the CCMP-AES rekeys the session.

Wi-Fi Protected Access 3 (WPA3) This is the successor to WPA2 and became part of the mandatory Wi-Fi Alliance certification process in July 2020. WPA3 introduced 192-bit cryptographic strength in WPA3-Enterprise mode. WPA3 still requires CCMP-128 as a minimum requirement for WPA3-Personal mode. The most significant improvement is the use of the Simultaneous Authentication of Equals (SAE) exchange that replaces the preshared key (PSK) exchange used with WPA and WPA2-Personal mode. SAE was originally introduced with IEEE 802.11s.

Disabling SSID Broadcast

One method of "protecting" the network that is often recommended is to turn off the SSID broadcast. The access point is still there and can still be accessed by those who know about it, but it prevents those who are looking at a list of available networks from finding it. This should be considered a very weak form of security because there are still ways, albeit a bit more complicated, to discover the presence of the access point besides the SSID broadcast.

Wireless MAC Filtering

Most APs offer the ability to turn on MAC filtering, but it is off by default. In the default state, any wireless client that knows of the existence of the AP can join the network. When MAC filtering is used, the administrator compiles a list of the MAC addresses associated with the users' computers and enters them. When a client attempts to connect, an additional check of the MAC address is performed. If the address appears on the list, the client is allowed to join; otherwise, they are forbidden from so doing. On a number of wireless devices, the term *network lock* is used in place of *MAC filtering*, but the two terms are synonymous.



Adding port authentication to MAC filtering takes security for the network down to the switch port level and increases your security exponentially.

Frequencies

The frequencies used with wireless local area networks (WLANs) vary by standard. The two main frequencies used are 2.4 GHz and 5 GHz. The 2.4 GHz frequencies are governed by the industrial, scientific, and medical (ISM) radio bands. The 5 GHz frequencies are governed by the Unlicensed National Information Infrastructure (U-NII) radio band. The 5 GHz frequencies have been expanded upon into the 6 GHz frequency range with the introduction Wi-Fi 6 and Wi-Fi 7. The 6 GHz frequency range is governed by the U-NII radio band.

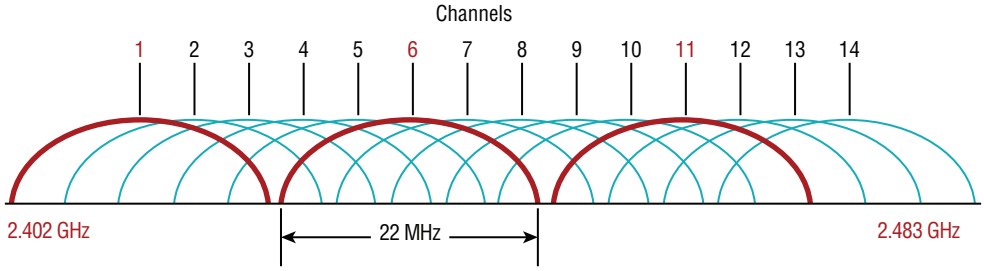


To learn more about Wi-Fi 6 and Wi-Fi 7 refer to Volume 1 Chapter 9, "Wireless and SOHO Networking."

2.4 GHz

The 2.4 GHz spectrum is governed by the ISM radio band. The 802.11b/g/n standards operate on 2.4 GHz frequencies. The band consists of 14 channels 22 MHz wide. In North America only the first 11 of the channels can be used for wireless. In Japan all 14 channels can be used, and almost everywhere else in the world the first 13 channels can be used. Only 3 of the 14 channels are considered nonoverlapping, as seen in Figure 6.37. The channels of 1, 6, and 11 are considered prime channels for WLAN because they do not overlap with the other channels in the channel plan.

FIGURE 6.37 The 2.4 GHz channel plan

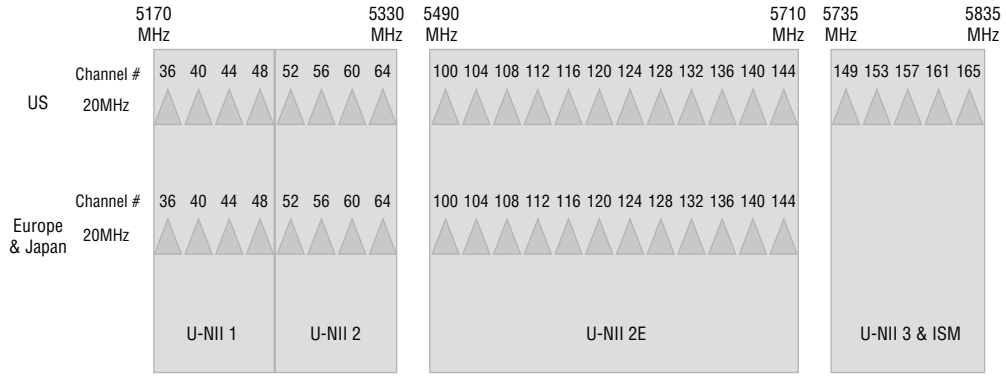


5 GHz

The 5 GHz frequencies are governed by the Unlicensed National Information Infrastructure (U-NII) radio band. The 802.11 a/n/ac/ax/be standards operate on the 5 GHz frequency spectrum.

As seen in Figure 6.38, the band consists of 25 nonoverlapping channels. In North America the 802.11a standard can function on 12 channels consisting of 36, 40, 44, 48, 52, 56, 60, 64, 149, 153, 157, and 161. Each regulatory domain restricts the number of channels and specific channels for the region.

FIGURE 6.38 The 5 GHz channel plan



In North America, the 802.11ac standard can use 25 of the nonoverlapping channels. In Europe and Japan, the channels are limited to the U-NII 1, U-NII 2, and U-NII 2E list of channels. The 802.11n standard only allowed the first 24 channels in North America, because channel 165 is in the ISM band.

6 GHz

The 6 GHz frequencies are also governed by the U-NII radio band. In recent years the U-NII 5, U-NII 6, and U-NII 7 expanded the 5 GHz frequency range into the 6 GHz range. This expansion meant even more channels for 802.11ax and 802.11be operations. The proposed U-NII 8 looks to even expand into the 7 GHz range in the future.

Speed, Coverage, and Security

Speeds will always be higher on 5 GHz and 6 GHz wireless, such as the 802.11 a/n/ac/ax/be standards. The 802.11ac standard can use a maximum of 25 nonoverlapping channels. However, the 802.11 b/g/n standards are restricted to the 2.4 GHz wireless band with three nonoverlapping channels. The 6 GHz wireless band introduced a whopping 60 nonoverlapping channels. You should always be aware of the airspace around your AP. If you are overlapping on a channel, you should make every effort to change the channel on your AP. By changing the channel, you will effectively increase the speed of the connection for all your users within the cell.

Lower frequencies will go further in distance, yet as you go further away from the AP your speed will suffer. High frequencies will tend to go shorter distances, and when you move further away from the AP, speed will sharply decline. Using higher frequencies allows you to lower power and decrease the changes of signals traveling to a public place.



The 802.11 a/n/ac standards operate at 5 GHz exclusively and 802.11 802.11 b/g/n standards operate at 2.4 GHz exclusively. The latest standard of 802.11ax/be operates at 5 GHz, 6 GHz, and 2.4 GHz. 802.11ax/be will steer the client to the best frequency band, also known as band steering. The concept remains the same: the higher the frequency, the shorter the distance; the lower the frequency, the longer the distance.

Radio Power Levels

Consider the radio power level, since the wireless access point has better transmitting power than most mobile devices. Let's use the analogy of two people standing in a field. One person is using a bullhorn to ask the other person the time, and the other person only has their voice to respond. Although they can hear the request, they will not be heard when they answer because they don't have a bullhorn. To fix this problem, the wireless access point should have its power level adjusted so that the client needs to be closer to receive data or associate with the SSID.

From a security standpoint, power levels should be adjusted so that they do not travel past the interior of the organization's building. If they do, then someone sitting in the parking lot or an adjoining building could attempt to infiltrate the wireless network. On the chance that the signal is actually traveling too far, some access points include power level controls that allow you to reduce the amount of output provided.



A great source of information on RF power values and antennas can be found on the Cisco site:

`www.cisco.com/c/en/us/support/docs/wireless-mobility/wireless-lan-wlan/23231-powervalues-23231.html`

Antenna and Access Point Placement

Antenna placement can be crucial in allowing clients to reach the access point. For security reasons, you do not want to overextend the reach of the network so that people can get on to the network from other locations (the parking lot, the building next door, and so on). Balancing security and access is a tricky thing to do.

There isn't any one universal solution to this issue—it depends on the environment in which the access point is placed. As a general rule, the greater the distance the signal must travel, the more it will attenuate; however, you can lose a signal quickly in a short space as well if the building materials reflect or absorb it. You should try to avoid placing access points near metal (which includes appliances) or near the ground. They should be placed in the center of the area to be served and high enough to get around most obstacles.

Wireless site surveys can be performed with specialized software that records the strength of the signal on a map you import of the coverage area. Wireless surveys should be performed in your environment before access point placement to help you determine the best placement. They should also be performed after access point placement for fine-tuning of the location and verification of uniform signal strength.

Assign Static IP Addresses

While DHCP can be a godsend, a SOHO network is small enough that you can get by without it issuing IP addresses to each host. The advantage to assigning the IP addresses

statically is that you can make certain which host is associated with which IP address and then utilize filtering to limit network access to only those hosts.

WPS

WPS (Wi-Fi Protected Setup) can help to secure the network by requiring new machines to do something before they can join the network. This often requires the user to perform an action in order to complete the enrollment process: press a button on the router within a short time period, enter a PIN number, or bring the new device close-by (so that near-field communication can take place). It should be noted that during this brief period the wireless access point is susceptible to an attack. Anyone with the passcode or the ability to guess the passcode can gain access to your wireless network. Therefore, WPS is not used outside of SOHO networks and is not found in corporate networks.

Authentication

When setting up a wireless network, you are extending a wired network to a wireless network. Therefore, you must consider how users will authenticate to the wireless network. The following are various concepts of authentication that can be used with wireless and wired networks (though they are more commonly used with wireless networks):

Single-Factor Authentication As previously mentioned, single-factor authentication is weak authentication. It is based on only one of the following factors:

- Something you know
- Something you have
- Something you are
- Somewhere you are
- Something you do

Setting up a wireless network based on a preshared key limits you to a single authentication factor, which everyone consequently shares. The use of a preshared key is a great example of an authentication factor of something that you know. Unfortunately, a preshared key can be compromised because people must share it, and you only have one preshared key per wireless SSID.

Multifactor Authentication Using single-factor authentication alone allows you to be compromised much more quickly. Therefore, using multifactor authentication is desirable for a higher level of wireless security. *Multifactor authentication* requires two or more single-factor authentication methods to be used to secure the wireless network. A common multifactor authentication implementation is the use of the Extensible Authentication Protocol – Transport Layer Security (EAP-TLS) protocol. EAP-TLS requires that a certificate be installed in the operating system along with a matching password for the computer.

RADIUS *Remote Authentication Dial-In User Service (RADIUS)* was originally proposed as an Internet Engineering Task Force (IETF) standard. It has become a widely adopted industry standard for authenticating users and computers for network systems. RADIUS creates a common authentication system, which allows for centralized authentication and accounting.

The origins of RADIUS are from the original ISP dial-up days, as its acronym describes. Today, RADIUS is commonly used for authentication of virtual private networks (VPNs), wireless systems, and any network system that requires a common authentication system. RADIUS operates as a client-server protocol. The RADIUS server controls authentication, authorization, and accounting (AAA). The RADIUS client can be wireless access points, a VPN, or wired switches. The RADIUS client will communicate with the RADIUS server via UDP port 1812 for authentication and UDP port 1813 for accounting.

The RADIUS server can be installed on many different operating systems, such as Linux and Windows. Microsoft Windows Server includes an installable feature, called the Network Policy Server (NPS), that provides RADIUS functionality.

TACACS+ Terminal Access Controller Access Control System Plus (TACACS+) is a protocol developed by Cisco, from the original dated protocol of TACACS. Although it was developed by Cisco, it was released as an open standard. The protocol is mainly used for authentication of users on routers and switches to allow management access. It is also used to authenticate users connecting to wireless access points via a centralized database. The TACACS+ protocol is declining in popularity and has largely been replaced by RADIUS.

Kerberos Kerberos is an open standard for authentication developed by the Massachusetts Institute of Technology (MIT). It is also the main authentication protocol used by Microsoft Active Directory. Kerberos can be used as the main authentication method for the Microsoft implementation of a RADIUS server, called the Network Policy Server (NPS). Kerberos can also be used with the 802.1X protocol for direct authentication with the Extensible Authentication Protocol (EAP).

Securing a SOHO Network (Wired)

Although a wired network can be more secure than a wireless one, there are still a number of procedures that you should follow to leave as little to chance as possible. Among them, change the default usernames and passwords to different values and secure the physical environment. You should also disable any ports that are not needed, assign static IP addresses, use IP filtering, and use MAC filtering to limit access to hosts that you recognize.

Changing Default Credentials

When installing a network device, the very first thing you must do is log in to the device. There is often a standardized default username and password for each vendor or each vendor's product line. Most devices make you change the default password upon login to the device.

Changing the default password to a complex password is a good start to hardening the device. However, changing the username will also ensure that a brute-force attack cannot be performed against the default username. There are many different websites dedicated to listing the default credentials for network devices, so it doesn't take tremendous skill to obtain the default username and password of a device.

The hosts in the network are no exception to changing default usernames and passwords. In Windows, the Guest account is automatically created with the intent that it is to be used when someone must access a system but lacks a user account on that system. Because the Guest account is so widely known to exist, you should not use this default account but instead create another account for the same purpose if you truly need one. The Guest account leaves a security risk at the workstation and should be disabled to deter anyone attempting to gain unauthorized access.



Change *every* username and password that you can so that they vary from their default settings.

Upgrading Firmware

When you purchase a network device, you don't know how long it's been sitting on the shelf of a warehouse. In that time, several exploits could have been created for vulnerabilities discovered. It is always recommended that a device's *firmware* be upgraded before the device is configured and put into service.

Most hardware vendors will allow downloading of current firmware. However, some vendors require that the device be covered under a maintenance contract before firmware can be downloaded. It is also best practice to read through a vendor's change log to understand the changes that have been made from version to version of firmware.

Filtering

IP filtering, also known as firewall rules, helps secure the internal network from an external network. The external network could be the Internet, or it could be a network less trusted than the internal network, such as a wireless network. In any case, firewall rules help harden the security of an organization because we can restrict activity from the external network to specific applications.

Firewall rules are normally configured with an implicit deny at the end of the rules set. This means that if an application has not explicitly been allowed, it will automatically

(implicitly) be denied. The easy way to remember what *implicit* means is that it implies there is a deny, unless a user or application has been explicitly allowed. This implicit deny operation of a firewall is the default for firewall rule sets.

Although there is an implicit deny at the end of firewall rule sets, you may also need to explicitly deny an application or IP address. An explicit deny is required when another explicit rule follows, allowing access to a wide range of applications or IP addresses. For example, you may want to allow access to all the servers from the client networks. However, you should explicitly deny applications the clients shouldn't have access to, such as Remote Desktop Protocol (RDP) or Secure Shell (SSH).

Disabling Ports

Disable all protocols on the network device that are not required. As an example, many network multifunction printers are preconfigured with a multitude of protocols, such as TCP/IP, Bonjour, and Internet Printing Protocol (IPP), just to name a few. If you don't need them, remove the additional protocols, software, or services, or prevent them (disable them) from loading. Ports on a switch, router, or firewall not in use present an open door for an attacker to enter and should be disabled or disconnected.



Many of the newer SOHO router solutions (and some of the personal firewall solutions on end-user workstations) close down the ICMP ports by default. Keep this in mind, because it can drive you nuts when you are trying to see if a brand-new station, server, or router is up and running.

MAC Address Filtering

Limiting access to the network by employing IP filtering is not the only way to restrict access. *MAC address filtering* can also be employed to restrict traffic to MAC addresses that are known and to filter out those that are not. Even in a home network, you can implement MAC filtering with most routers and typically have an option of choosing to allow only computers with MAC addresses that you list or deny only computers with MAC addresses that you list.



If you don't know a workstation's MAC address, you can find it by using `ipconfig /ALL` (on Windows machines; it is listed as *physical address* and `ifconfig` command on macOS and the `ip` command on Linux operating systems).

Content Filter

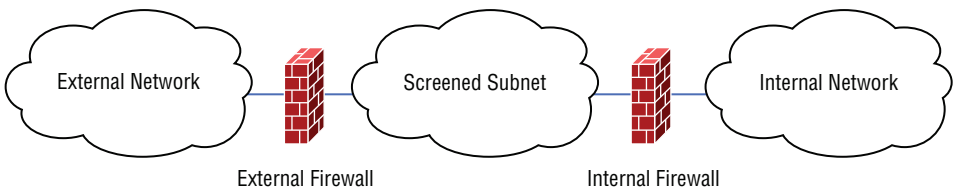
Content filters are useful in networks to restrict users from viewing material that is non–work-related, questionable, or malware. Content filtering is usually dictated by organization policy and management. The content filter operates by watching content and requests from

web browsers and other applications. The content filter functions in two ways: the first is content based; when images and text are requested from a website, the content filter can use heuristic rules to filter the content according to administrator-set policies. The second method is URL based, which is much more common since many websites now use SSL/TLS (encryption) and the traffic is encrypted. Content filters are typically purchased with a subscription that provides updates to the categories of the material administrators' block. Content filters can be hardware solutions or software solutions, although it is common to find them installed as software solutions.

Screened Subnet

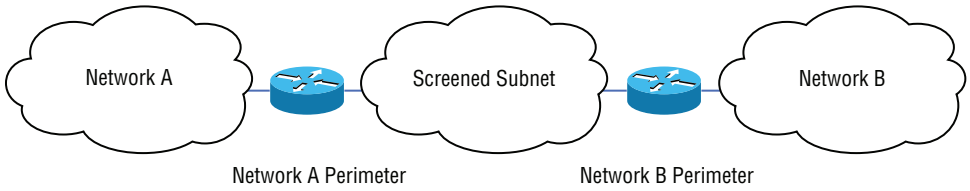
The *screened subnet* is also known as the demilitarized zone (DMZ). The DMZ gets its name from the segmentation that is created between the exterior and the interior of the network. This is similar to where borders of two opposing countries meet with military presence on both sides. Between the two sides there is a neutral segment called the DMZ. As it pertains to a network, hosts that serve Internet clients are placed in the screened subnet. As shown in Figure 6.39, a network segment called the screened subnet sits between an external firewall and the internal firewall. The external firewall contains ACLs to prevent Internet hosts from accessing nonessential services on the server in the screened subnet. The internal firewall restricts which hosts can talk to internal servers. A typical rule on the external firewall would allow HTTP access for a web server in the screened subnet and would restrict all other ports. A typical rule on the internal firewall would allow only the web server to communicate with the SQL backend database in the internal network.

FIGURE 6.39 A typical screened subnet with two firewalls



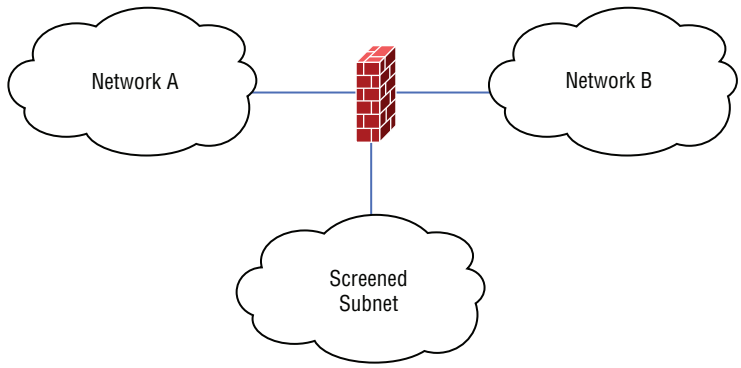
The concept of the screened subnet (DMZ) is still used today in network design; a screened subnet can be created between any two segments in the network. The subnets don't necessarily need to be external and internal in relation to the network. Routers containing ACLs can be implemented in lieu of firewalls to filter traffic to the screened subnet, as shown in Figure 6.40. In the figure, a network called Network A is segmented from the screened subnet by a router with ACLs filtering traffic. On the other side of the screened subnet is another network called Network B, and it too is segmented by a router with ACLs filtering traffic. Each of these two networks have equal access to the hosts in the screened subnet. These two networks, Network A and Network B, could potentially be a wireless network and the wired network, respectively.

FIGURE 6.40 A typical screened subnet with two routers



Some screened subnets are just another interface on a single firewall, as shown in Figure 6.41. In this example, the rules for both the Network A subnet and the Network B subnet would be on the same firewall. The benefit of a single firewall is centralized administration of firewall rules. Each interface is placed into a trust zone, and the firewall rules allow incoming and outgoing connections.

FIGURE 6.41 A typical screened subnet with one firewall



Port Forwarding/Mapping

On the router, the port configuration dictates what traffic is allowed to flow through. The router can be configured to enable individual port traffic in, out, or both; when you implement this type of configuration it is referred to as port forwarding. If a port is blocked (such as port 80 for HTTP or port 21 for FTP), the data will not be allowed through, and users will be affected. *Port forwarding* is also known as port mapping, and both are subsets of what a firewall does, and the amount of tweaking they require to get right is about the same.

Port forwarding is required when you are trying to share a network service with the Internet, such as a web server. It is commonly required when gaming, since some games require you to set up an impromptu game server. The players on the Internet will have to connect to the ports on your local machine behind your router, through port forwarding. There are

websites dedicated to configuring these port forwarding settings manually; an example is `https://portforward.com`.

Universal Plug and Play (UPnP)

If you need to configure port forwarding frequently, then you might want to skip manually port forwarding altogether. The *Universal Plug and Play (UPnP)* is a network protocol that allows for automatic configuration of port forwarding. Most modern-day routers have the feature turned on, which is a security concern if you are not using any port forwarding. The UPnP protocol operates by the clients initiating a connection to the router and communicating the ports needing to be forwarded to the client. The router then opens the ports and forwards them to the client.

DHCP

Dynamic Host Configuration Protocol (DHCP) is responsible for automatic configuration of IPv4 IP addresses and subnet masks for hosts from a pool of IPv4 addresses. It is also responsible for configuration of such options as default gateways, DNS server addresses, and many other IP-based servers. It performs configuration of the host in a series of network broadcasts and unicasts.

Reservations

When a client requests an IP address from a DHCP server, the client's MAC address is transmitted in the DHCP packet. A rule on the DHCP server called a *DHCP reservation* can tie the client's MAC address to a particular IP address. When a reservation is created for a client, the client is guaranteed to obtain the same IP address every time for the DHCP process. When a reservation is created on the DHCP server, no other hosts can obtain the reservation IP address unless they have the MAC address that matches the reservation. This type of assignment is considered a dynamically static–assigned IP address.

Reservations can be very handy when static IP addresses are too troublesome to configure, such as network printers with poor configuration options. It's common to set a reservation on network printers and move on when faced with a finicky static IP address process. You can save an hour of busy work in the right situation. Reservations can also be useful when you need to make specific firewall rules for a client based on its IP address.

Static

Dynamic IP addressing is the standard in small-to-large networks when configuring client computers. Static IP addressing should only be used under certain circumstances for client computers, since it is not very scalable and a nightmare to keep track of manually. DHCP allows for central management of the IP address space versus static assignment of individual hosts (which is decentralized). Static IP addressing should only be used on internal network resources such as routers, network printers, and servers.

Static IP addressing can be useful for wide area network (WAN) connections, also known as your connection to the Internet. If a server is operating at the location, a static IP address is necessary for clients to be able to connect. Name resolution to the IP address is the biggest driver for static IP addressing. There are work-arounds, such as dynamic DNS services, but the best solution is to purchase a static IP address from the Internet provider.

Secure Management Access

Configuring secure management access on both wireless and wired routers is critical to protect your network from unauthorized access and potential threats. Start by changing the default login credentials to strong, unique passwords. You should use secure protocols like Secure Shell (SSH) instead of Telnet for remote access. Configuring the router to use HTTPS vs. HTTP for accessing the router's management interface. HTTPS encrypts the data transmitted between your device and the router, protecting sensitive information from being intercepted by malicious actors. Additionally, consider setting up a dedicated management network to isolate management traffic from regular network traffic, further enhancing security.

Some routers offer an option to restrict management access to only the internal network. This means that management interfaces can only be accessed from devices within the local network, adding an extra layer of security by preventing external access.

Physical Security

Just as you would not park your car in a public garage and leave its doors wide open with the key in the ignition, you should educate users to not leave a workstation that they are logged into when they attend meetings, go to lunch, and so forth. They should log out of the workstation or lock it. "Lock when you leave" should be a mantra they become familiar with. A password (usually the same as their user password) should be required to resume working at the workstation.

Also don't overlook the obvious need for physical security. Adding a cable to lock a laptop to a desk prevents someone from picking it up and walking away with a copy of your customer database. Every laptop case we are aware of includes a built-in security slot in which a cable lock can be added to prevent it from easily being carried off the premises. If this is in your home office, it might deter a burglar from walking away with sensitive information.

When it comes to desktop models, adding a lock to the back cover can prevent an intruder with physical access from grabbing the hard drive or damaging the internal components. You should also physically secure network devices, such as routers, access points, and the like. Place them in locked cabinets, if possible—if they are not physically secured, the opportunity exists for an unauthorized person to steal them or manipulate them to connect to the network.



Exam Objective 2.8, "Given a scenario, apply common methods for securing mobile devices," requires you to be able to identify and apply security to mobile devices.

Mobile Device Security

Apple computers have a pretty decent reputation in the industry for being somewhat resistant to malware. Whether or not this is because of the relatively small installed base or the ease with which hackers penetrate "other" operating systems, this characteristic carries over to Apple's mobile devices. In fact, hackers don't seem to be as interested in attacking the legions of mobile devices as much as they have gone after the Windows operating systems that drive the vast majority of laptops, desktops, and servers in the world. Nevertheless, attacks occur. Coupled with how easy mobile devices are to misplace or steal, it behooves users to have proactive monitoring and contingency plans in place.

The following sections detail the built-in security utilities that are common in today's mobile devices. Furthermore, for threats not covered by the software with which the devices ship, the protection available from third-party utilities is worth discussing.

Hardening Techniques

Hardening techniques for mobile devices enhances security and protecting sensitive data from potential threats and threat actors. By implementing these techniques, users can significantly reduce the risk of unauthorized access, data breaches, and other cyberattacks. In the following sections, you will learn about various hardening methods, including screen locks, which prevent unauthorized access to the device; device encryption, which ensures that data stored on the device is secure; endpoint detection and response (EDR) software, which protect against malicious software; firewalls, which help control incoming and outgoing network traffic; and configuration profiles, which enforce security policies and settings. These measures collectively contribute to a robust security posture, safeguarding mobile devices against a wide range of threats.

Screen Locks

Apple and Android mobile devices include a requisite locking mechanism, which is off by default. The user on the go is encouraged to enable a lock. If your device acts primarily as a home computing device and rarely goes with you out the door, there is very little reason to set a lock. However, knowing how to do so is important. The following are types of locks that you can implement to secure your device:

Fingerprint Lock The *fingerprint lock* is a biometric-type lock that uses your fingerprint to unlock the device. The fingerprint lock is the most secure biometric method for a device. The technology works by placing your finger on a touchpad sensor on the device. On Apple devices, the fingerprint recognition is called *Touch ID*. The technology works exceptionally well, except when your fingers are wet or extremely dirty. When your fingers are wet or extremely dirty, they make different prints from those you originally trained the device to recognize.

Face Lock or Face ID The *face lock* and *face ID* are biometric-type locks that allow you to use your face to unlock your device. There are a high number of false positives that makes the face lock less secure than the fingerprint lock. This means that presumably someone that looks like you could unlock your phone. Almost all mobile devices have a front-facing camera, allowing you to instantly know that you've unlocked the phone seamlessly by looking at your phone. On Apple devices, this technology is called *Face ID*.

Pattern Lock The *pattern lock* is a type of passcode that allows you to unlock your device. Swipe lock works by displaying nine dots in a matrix of 3 × 3. You then swipe with your finger with the registered pattern to unlock the phone. The swipe lock is the least secure of any locking methods. A grease trail from your fingers can allow someone to derive the swipe pattern.

PIN Lock The *PIN lock* is by far the most common way to unlock devices. It is typically a 4- or 6 digit-numeric passcode. However, it can also be alphanumeric, depending on the device. Passcode locks suffer from the same problems that passwords are prone to: people can shoulder surf or learn your passcode over time.

Swipe Lock The s*wipe lock* is not really a lock at all. It allows you to unlock the phone easily by swiping on the screen. This lock method is to really prevent pocket dialing of someone by accident.

Exercise 6.4 outlines the steps for creating a code for your iPhone.

EXERCISE 6.4

Setting the Passcode Lock on an iPhone

1. Tap Settings.
2. Depending on the iOS version, select either Face ID or Passcode.
3. Tap Turn Passcode On.
4. Enter a six-digit passcode.

   If you want to use a four-digit passcode, tap Passcode Options to switch to a four-digit numeric code or a custom alphanumeric passcode.
5. Enter the passcode a second time to confirm. Doing so sets the passcode.

   Optionally, you can set the amount of time that must pass while the phone is asleep before the passcode will be required and whether the Erase Data feature should be enabled. Setting the Required Passcode field to Immediately requires entering the passcode each time the device wakes up.

The same general concept for Android phones is illustrated in Exercise 6.5.

EXERCISE 6.5

Setting the Passcode Lock on an Android Phone

Depending on the Android version, these steps might differ.

1. Swipe up on the home screen.
2. Tap the Settings app.
3. Select Lock Screen and/or Security from the Settings menu.
4. Tap Screen Lock Type on the Lock Screen And Security menu.
5. Select Pattern from the Phone Security list.
6. Use your finger to draw a continuous pattern of four or more dots, and then tap the Continue button.
7. Repeat the same pattern, and then tap the Confirm button.
8. Optionally, tap Secure Lock Settings to lock the phone automatically and adjust the number of wrong swipes before a factory reset occurs.

Full-Device Encryption

As discussed previously, BitLocker and BitLocker to Go greatly enhance security by encrypting the data on drives (installed and removable, respectively) and helping to secure it from prying eyes. At a minimum, the same level of protection that you would apply to a desktop machine should be applied to a mobile device, because it can contain confidential and personally identifiable information (PII), which could cause great harm in the wrong hands.

Full-device encryption should be done on laptops and mobile devices, and you should back up regularly to be able to access a version of your files should something happen to the device itself. When full-device encryption is turned on, both the device and the external storage (SD card) are encrypted. The only way to view the information on the SD card is from the encrypting mobile device.



Android devices do not automatically turn on device encryption and encrypt files upon setting a passcode on the device, like iOS devices do.

Failed Login Attempt Restrictions

After you've set a screen lock, an optional step is to set the device to wipe, or factory reset, after a number of failed attempts. This option will wipe local data on the device if incorrect

passcodes are entered 10 times in a row, or perform a factory reset depending on the device. While this is recommended for users with devices that contain sensitive data and that are frequently taken into public venues or placed in compromising positions, the casual user should not turn this feature on unless they can be sure that a recent backup will always be available in iCloud or Google.

Imagine a user's child or a mischievous, yet harmless, friend poking away at passcodes until the device informs them that it is being wiped clean; it's not for everyone. Restoring from a backup is easy enough, but will a recent backup be available when disaster strikes? Apple performs a backup to the iCloud or the computer running iTunes that the iOS device syncs with.

Apple imposes cooling-off timeout periods of increasing duration, even if the Erase Data feature is disabled and you or someone else repeatedly enters the wrong code over multiple lockouts. The final penalty with the Erase Data feature disabled is that you cannot unlock the device until it is connected to the computer with which it was last synced.

When a passcode is set, Android devices have a similar approach to failed login attempts as their Apple counterparts. A factory reset will occur after 10 failed attempts.

The difference is that if waiting the timeout period won't help because you've forgotten the pattern or code, this device can tie your access back to the Google account you used when setting it up. This is also the account where you receive purchase notifications from Google Play, and it does not have to be a Gmail account (one of the benefits of the open-source nature of Android). If you remember how to log into that account, you can still get into your phone. Or at least you can investigate the credentials to that account on a standard computer and return to the device to enter those credentials.

Endpoint Detection and Response (EDR)

For the most part, mobile devices have been left alone with viruses and malware, as compared with the Windows platform. However, this is not a reason to let your guard down and not worry about viruses and malware on mobile devices. Mobile devices can contract a virus and malware through the installation of a malicious app.

Most malware installed on mobile devices will spam your device with ads from the Internet. However, some malware can expose your personal information and even subject your device to the control of a malicious individual. Viruses, on the other hand, will use your email application to send copies of itself or turn your device into a zombie.

The extent of the damage from malware or a virus is hard to estimate, but one thing is for sure: protection goes a long way. Antivirus and antimalware software should be installed on your mobile device to thwart malicious attempts to infiltrate your device. All the leading vendors, such as AVG, Norton, and Avast (just to name a few), have offerings of antivirus and antimalware. Most installations of these apps are free, and other services or features can be purchased within the apps.

Antivirus software primarily targets known threats such as viruses, worms, and trojans, while antimalware focuses on a broader range of malicious activities, including spyware, ransomware, and adware. Endpoint Detection and Response (EDR) for mobile devices integrates these capabilities to offer advanced threat detection and response. EDR continuously monitors mobile devices for suspicious activities, collects and analyzes data, and provides real-time alerts

to security teams. This enables rapid identification and mitigation of potential threats, reducing the risk of data breaches and ensuring the security of sensitive information on mobile devices.

EDR solutions for mobile devices go beyond traditional antivirus and antimalware by incorporating advanced features such as behavioral analysis, machine learning, and threat intelligence. These technologies enable EDR to detect and respond to sophisticated threats that may evade conventional security measures.

Content Filtering

Mobile content filtering is a technology used in environments such as schools, businesses, and government organizations where it is essential to ensure that users are not exposed to harmful or inappropriate content. By using mobile content filtering, administrators can block access to specific websites, apps, and online services that may pose security risks or distract users from their tasks. Mobile content filtering helps protect against malware infections and phishing attacks by blocking access to malicious websites and links. This is particularly important for maintaining the security of organizational data and ensuring that employees stay on task by preventing access to irrelevant sites that reduce productivity and drain bandwidth. Additionally, mobile content filtering can be configured to comply with legal and regulatory requirements, ensuring that organizations adhere to data protection and privacy laws.

Mobile content filtering works by screening content to restrict or block access to websites and online services that compromise security or breach internal content consumption policies. This is achieved through various techniques such as allowlists and blocklists, keyword filtering, and AI-powered filters that determine allowable data. For instance, allowlists include a small number of allowed sites and block access to everything else, while blocklists apply the opposite principle. AI-powered filters add a valuable layer of subtlety by adapting to different contexts and blocking harmful content without affecting legitimate sites.

Trusted Sources vs. Untrusted Sources

You can limit your exposure to malicious apps by only installing apps from trusted sources, such as the Apple's App Store or Google Play. In most cases, the mobile operating system must specifically be configured to accept installations from untrusted sources. So, it's not likely that you will mistakenly install an app from an untrusted source. Examples of untrusted sources include manual installs of Android APK (Android Package Kit) or untrusted Apple IPA (iOS App Store Package) files. These files can be distributed outside of the Google Play and Apple's App Store ecosystems. When files are distributed outside of these ecosystems, consider them untrusted.

Firewalls

Mobile devices have the same inherent vulnerabilities as desktop operating systems. However, mobile devices are generally not targeted the same as desktop operating systems. When a mobile device starts up, it is configured with an IP address via either the internal cellular radio or the Wi-Fi radio, so the device can be exploited via the network. Therefore, a firewall should be installed and turned on to protect the mobile device.

However, inbound communication is not the only aspect of security you need to be concerned about on mobile devices. Outbound communication also is a primary concern. A mobile device firewall app will allow you to monitor both the inbound and outbound communications on your mobile device. Several third-party firewall apps (both free and for purchase) can be downloaded from either Apple's App Store or Google Play.

Configuration Profiles

Configuration profiles are a key component of mobile device management (MDM). They contribute a major role in maintaining the security and compliance of devices within an organization. These profiles are essentially sets of rules and settings that administrators can deploy to mobile devices to control various aspects of their operation. By using configuration profiles, organizations can ensure that all devices adhere to the same security standards and policies, which is vital for protecting sensitive data and preventing unauthorized access.

For example, Microsoft Intune, a cloud-based MDM solution, allows administrators to create and deploy configuration profiles to manage mobile devices. With Intune, you can configure a wide range of settings, including:

Security Policies: Enforce password requirements, such as complexity and expiration, to ensure that all devices have strong passwords.

Encryption: Require device encryption to protect data at rest, ensuring that even if a device is lost or stolen, the data remains secure.

App Management: Restrict access to certain apps or prevent the installation of unauthorized apps, reducing the risk of malware and other security threats.

Network Settings: Configure Wi-Fi, VPN, and other network settings to ensure secure and reliable connectivity.

For instance, an organization might use Intune to create a configuration profile that enforces encryption on all corporate devices. This profile would automatically apply the encryption settings to any device enrolled in Intune. This would ensure that all data stored on the device is encrypted and protected. Additionally, the profile could include settings to require a complex passcode, further enhancing the security of the device.

Remote Wipes and Locator Applications

Should your work or personal mobile device disappear or fall into the wrong hands, it's always nice to have a backup plan to ensure that no company secrets or personal identifiers get misused by anyone who would use the information with ill will. Apple supplies a free app called *Find My iPhone* that, together with iCloud, allows multiple mobile devices and Macs to be located if powered on and connected to the Internet (via cellular, Wi-Fi, Ethernet,

and so on). The app allows the device to be controlled remotely to lock it, play a sound (even if audio is off), display a message, or wipe it clean.

Within a newer iPhone's Settings screen, you can find an iCloud settings page and select the Find My iPhone switch. If this switch is off, the Find My iPhone app and iCloud web page will be unable to find your device.

On the login screen for the iPhone app, you must enter the iCloud account information that includes the device you are attempting to control remotely.

Note that when you change the password for your Apple ID, it affects your iCloud account but does not transfer automatically within your device. Always remember to update your iCloud account information on each device when you update the associated Apple ID.

The iCloud website's login page (`www.icloud.com`) calls for the same credentials as the app. You are signing in with HTTPS, so your username and password are not traversing the Internet in the clear. With the switch in the iCloud settings screen set to Off for all devices on your account, when you sign in to the app with your iCloud account credentials, you are met with a disabling switch message.

You do not need to go to the website if you have another device with the Find My iPhone app or can borrow one from someone else. The device forgets your credentials when you log out, so the owner will not be able to control your device the next time they use the app.

After logging into the iCloud website, you can click the icon that matches the icon for the Find My iPhone app in iOS. Assuming that you've made it into the app on another device and your Find My iPhone feature is enabled on your missing device, the Info screen tells you that your device has been found and gives you options for the next step you take.

Tapping the Location button in the upper left shows you a map of where your device is currently located. You have three options for how to view the location: a two-dimensional map, a satellite map, and a hybrid version, where the two-dimensional street-name information is laid over the satellite view.

If you tap the Play Sound or Send Message button on the Info screen, instead of the Location button, the screen that pops up allows you to display a message remotely. You might consider first displaying a message without the sound, which is at maximum volume. Ask in the message to be called at another number. If you hear from someone in possession of your device, the hunt is over. Otherwise, send another message with the tone to get the attention of the nearest person. If you are at the reported location when you generate the sound, it can help you home in on the device.

If you do decide to use the remote-lock feature of the app, you'll have the opportunity to reconsider before locking it. You should have no issue with locking the device. Doing so does not prevent you from using the app further; it simply makes sure that the device is harder to break into.

Should you decide to take the sobering step of destroying the contents of the device remotely, you get a solemn notice allowing you the opportunity to reassess the situation before proceeding. Sometimes there's just no other option.

For Android devices, Google's *Find My Device* app performs many of the same functions as Apple's Find My iPhone app, including playing your ring tone for 5 minutes. When you

search Google's web page for "find my device," a map will display the location of your device. Clicking the location will reveal the SSID the phone is connected to, the phone's battery level, and when the phone was last in use at that location. You can then choose to play a sound (ring tone), secure the device, or erase the device. Securing the device will lock the phone and sign you out of your Google account. This option can also allow you to display a message on the lock screen. Erasing the device is the last step when you know it cannot be recovered and you want to be assured your data is removed.



Real World Scenario

Uh-Oh. Where's My Phone?

After a recent visit to a car dealership, a prospective new car owner named Jimmy departed without his iPhone—at least, that's where Find My iPhone said it was. Luckily, Jimmy had the foresight to set up his free iCloud account and log that phone into it. Upon his return to the dealership, Jimmy was temporarily disheartened to find that no one had seen his phone.

It occurred to him that he had laid his phone down on the counter in the men's room as he was washing his hands. He didn't recall having it after that point. A cursory look around the facilities turned up nothing. The app was not accurate enough to tell him where exactly the phone was at the time it was located on the Find My iPhone website, which he checked from home. However, the app did indicate that the iPhone was still somewhere at the dealership.

Jimmy noticed that the salesperson assisting him had an iPhone, so he asked her if she also used the Find My iPhone app. She did. He asked if he could borrow her phone for a brief instant. She obliged, and Jimmy entered his iCloud credentials into the salesperson's app and then entered the message, "I'm in the showroom." He left the Play Sound switch on and tapped the Send button.

In less than five minutes, a manager who had been back at the loading dock came to the showroom with a story. He said he heard a disturbing noise coming from the dumpster, which was quite full and scheduled to be picked up the next morning. The trash bag containing Jimmy's phone was conveniently right at the top of the heap and easily retrieved by the manager.

Jimmy recalled being in a bit of a fluster as he left the men's room. He was about to finalize the terms of the deal he had been working on for more than an hour. In his haste to get back to the table, apparently his phone slipped from his hand as he was disposing of the paper towels he used to dry his hands. So, the cliché "All's well that ends well" has a shred of truth. Jimmy owed this happy ending to the features of the Find My iPhone app and tells his story to anyone who will listen.

Remote Backup

Apple iOS devices automatically back themselves up either to a computer running iTunes that they sync with or to the iCloud account associated with the device. When a mobile device is connected to a computer containing iTunes, the process of backup is called *synchronization*. iCloud Backup is the most common method of backing up Apple iOS devices, since no computer is necessary. iCloud does require a Wi-Fi connection or cell service. To enable iCloud Backup, navigate to Settings, then tap your name, tap iCloud, and finally tap iCloud Backup. You can create an immediate backup by tapping Back Up Now, or you can wait until the next backup interval.

The Android operating system will automatically synchronize the device to Google Drive. Android phones require a Google account during setup. This account is the account that will synchronize to Google Drive. Google Drive is Google's cloud-based storage product. The backup service will back up Wi-Fi passwords, phone logs, app settings, contacts, messages, pictures, and other related files. A multitude of third-party backup apps can be downloaded from Google's Play Store. Each third-party app offers different features over and above the built-in backup service functionality, such as the capability to remove the bloatware apps that come with the phone.



Both Apple iCloud and Google Drive services allot you a certain amount of storage when you sign up for free. The limited cloud storage used for backups of your settings and data is adequate for most people. However, you can always purchase more storage from Apple and Google.

Patch Management

Regularly applying patches helps to fix vulnerabilities that could be exploited by threat actors, thereby protecting sensitive data and preventing unauthorized access. For mobile devices, this means not only updating the operating system but also ensuring that all installed applications receive the latest security patches and feature enhancements. Effective patch management reduces the risk of malware infections, improves device performance, and ensures compliance with regulatory requirements. By keeping both the operating system and applications current, organizations can safeguard their mobile environments against emerging threats and keep them secure.

Operating System Updates

It's easy to forget that these tiny yet powerful mobile devices we've been talking about are running operating systems that play the same role as the operating systems of their larger siblings. As such, users must be careful not to let the operating systems go too long without updates. Occasionally, mobile devices will notify the user of an important update to the operating system. Too often, however, these notifications never come. Therefore, users should develop the habit of checking for updates on a regular basis.

Not keeping up with software updates creates an environment of known weaknesses and unfixed bugs. Mobile devices operate on a very tight tolerance of hardware and software performance. Not maintaining the device for performance at the top of its game will tend to have more pronounced repercussions than those seen in larger systems.

For the iPhone and iPad, you can check for the most important level of updates by tapping Settings ➢ General ➢ Software Update. For the Android operating system, there are multiple updates that can be checked for manually. All of them are accessible by following Menu ➢ Settings ➢ System Updates.

Application Updates

On Android devices, users can enable auto-updates by navigating to the Google Play Store settings and selecting their preferred update method, either over cellular data or Wi-Fi only. This setting allows the device to automatically check for and install updates, ensuring that apps are always current without user intervention. Similarly, Apple devices offer an auto-update feature that can be configured for both cellular data and Wi-Fi only downloads. Navigating to Settings ➢ Apps ➢ App Store, you can configure the application auto-updates. By default, both Android and Apple devices will auto-update applications over Wi-Fi, but not over cellular data.

When auto-updates are enabled on Android devices, the Google Play Store periodically checks for updates and installs them in the background. This process is designed to be seamless, minimizing disruption to the user. On Apple devices, the App Store handles automatic updates similarly. Users can also manually update apps by opening the App Store, tapping on their profile icon, and selecting Update All or updating individual apps as needed.

Multifactor Authentication

Multifactor authentication, mentioned previously in this chapter, involves using more than one item (factor) to authenticate. An example of this would be configuring a BitLocker to a Go–encrypted flash drive so that when it is inserted into your laptop, a password and smartcard value must be given before the data is decrypted and available.

Authenticator Applications

An authenticator app works with mobile devices to generate security codes that can keep accounts secure by requiring two-factor authentication (2FA). Once this is set up, your account will require a code from the app in addition to your account password. Authenticator applications are available for download with Apple, Android, and other mobile device operating systems, as well as desktop operating systems. An account is usually added to the authenticator application by entering a secret key or scanning a QR barcode; this creates the account in the authenticator application. Several different authenticator apps can be downloaded; they vary depending on the mobile operating system.

Policies and Procedures

With the explosive growth of mobile devices in the workplace, there are many different policies and procedures that may be required for your organization to minimize data loss. This section focuses on the policies and procedures specific to mobile devices that you may have in your organization.

As employees are hired in your organization, a certain amount of initial interaction with the information technology (IT) is required. This interaction, called the *onboarding procedure*, is often coordinated with the Human Resources (HR) department. During the onboarding procedure, the IT representative will help the user log in for the first time and set their password. The password policy is often the first policy discussed with the user. Other policies—such as bring your own device (BYOD), acceptable use policy (AUP), nondisclosure agreement (NDA), and information assurance—should also be discussed during the onboarding procedure. The use of email and file storage and policies should also be covered with the user during the onboarding process. Each organization will have a different set of criteria that make up the onboarding procedures.

Eventually, employees will leave your organization. The *offboarding procedure* ensures that information access is terminated when the employment of the user is terminated. The offboarding procedure will be initiated by the HR department and should be immediately performed by the IT department. This process can be automated via the organization's employee-management system. This procedure can also be performed manually if the employee-management system is not automated. However, the procedure must be performed promptly, since the access to the company's information systems is the responsibility of the IT department.

During the offboarding procedure, email access is removed via the *mobile device management (MDM)* software, the user account is disabled, and IT should make sure that the user is not connected to the IT systems remotely. The offboarding procedure may also specify that the user assume ownership of the terminated employee's voicemail, email, and files.

BYOD vs. Corporate-Owned

The traditional workforce is becoming a mobile workforce, with employees working from home, on the go, and in the office. These employees use laptops, tablets, and smartphones to connect their companies' cloud resources. Organizations have embraced BYOD initiatives as a strategy to alleviate the capital expense of equipment by allowing employees to use devices they already own.

Because employees are supplying their own devices, a formal document called the *BYOD policy* should be drafted. The BYOD policy defines a set of minimum requirements for the devices, such as size and type, operating system, connectivity, antivirus, patches, and many other requirements the organization will deem necessary, as well as the level of service your IT department will support for personally owned equipment.

Many organizations use MDM software, which helps enforce the requirements for the BYOD policy. MDM software helps organizations protect their data on devices that are

personally owned by the employees. When employees are terminated or a device is lost, the MDM software allows a secure remote wipe of the company's data on the device. The MDM software can also set policies requiring passwords on the device. All of these requirements should be defined in the organization's BYOD policy.

Corporate-owned mobile devices are also of paramount concern when it comes to security. The equipment is mobile, so these devices sometimes travel and can disappear completely. Luckily, MDM software allows you to not only control the data on these devices, but in many instances even track it via a built-in Global Positioning System (GPS) sensor (if the device supports this functionality). When MDM software is implemented in the capacity of tracing assets, consideration must be given to privacy. Although it may be acceptable to track a corporate-owned mobile device, the end user must be made aware of this policy.

Profile Security Requirements

When you implement an MDM solution to manage mobile devices, part of your implementation is the creation of profile security requirements for the mobile devices you will manage. The profile security requirements allow the management of the mobile devices in a uniform fashion. As an administrator, you can choose settings for mobile devices under your purview and enforce profile security requirements in various ways. In a given scenario, you may want to enforce settings for the entire organization, whereas in other scenarios you may want to differ the settings based on organizational unit, role, or other group type. Among the settings you may want to enforce are those requiring the encryption of drives and the use of complex passwords.

IoT Considerations

The *Internet of Things (IoT)* is an exploding industry. You can now control everything from your lights to your thermostat with a mobile device or computer. The downside to connecting things to the Internet is that they must be patched so that they are not exploitable. Hardware vendors often wire off-the-shelf components into the hardware, and these off-the-shelf components never get frequent updates. So, this is a major security consideration for an organization. Ease of use versus security is the balance beam that you walk when owning IoT devices.

In recent years, attackers have harvested IoT devices for distributed denial-of-service (DDoS) attacks. The Mirai botnet is one such popular botnet that can crush an organization's bandwidth. This can happen when IoT devices in your network are used in a DDoS or are attacked by the botnet. To mitigate inadvertently being used in a DDoS, you can place the IoT devices on an isolated network and police their outbound bandwidth.

Unfortunately, there is not much you can do with IoT devices to prevent being attacked. If an attacker wants to attack your organization with IoT devices, firewalls, ISP controls, and third-party services like Cloudflare can help mitigate these attacks. This is not really an IoT consideration because any botnet can attack you and the mitigations are the same as if you were attacked by an IoT botnet.

Summary

In this chapter, you learned some best practices related to operating system security. We then focused on Windows operating system security settings, such as sensitive accounts and file-system permissions. We concluded with mobile device security for both Apple devices and Google Android. The mobile device operating systems are changing rapidly, as both consumer and enterprise needs change. However, the objectives on the A+ 220–1202 exam are general enough that a mastery of them will enable you to secure a mobile device today and into the future.

Security, as you've already guessed, is a large part of the CompTIA A+ certification. CompTIA expects everyone who is A+ certified to understand security-related best practices and be able to secure both Windows operating systems and mobile devices.

Exam Essentials

Understand Windows operating system security concepts. You should be able to compare and contrast Microsoft Windows operating system security settings. This should include knowing the various accounts that are created by default and why you need to disable or rename these accounts. You should also have an in-depth understanding of NTFS permissions and share permissions, and their interaction, to form the effective permissions the user has when accessing a share remotely. In addition, you should have a good understanding of authentication and the options to encrypt both the operating system and removable drives.

Know how to compare wireless security protocols and authentication methods. Wireless networks can be encrypted through WEP, WPA, WPA2, and WPA3 technologies. Wireless controllers use service-set identifiers (SSIDs)—32-character, case-sensitive strings—that must be configured in the network cards to allow communications. However, using ID string configurations doesn't necessarily prevent wireless networks from being monitored, and there are vulnerabilities specific to wireless devices.

Understand the basics of antenna placement and radio power levels. Antenna placement can be crucial in allowing clients to reach an access point. Place access points near the center of the area to be served and high enough to get around most obstacles. Know that power level controls allow you to reduce the amount of output provided.

Know how to configure security on SOHO wireless and wired networks in various scenarios. Disable all unneeded protocols/ports. If you don't need them, remove them or prevent them from loading. Ports not in use present an open door for an attacker to enter. MAC filtering allows you to limit access to a network to MAC addresses that are known and filter out (deny access to) those that are not.

Understand mobile device security concepts. You should understand how screen locks operate. Explain why we need remote wipe in the event of a device missing or lost. Understand the risks of viruses and malware on mobile devices, as well as measures that can be taken to mitigate these risks. In addition, you should have a good understanding of the importance of full-device encryption, authentication applications, firewalls, and policies for mobile devices.

Review Questions

The answers to the chapter review questions can be found in Appendix A.

1. Which policy would you create to define the minimum specification if an employee wanted to use their own device for email?
   - **A.** MDM
   - **B.** AUP
   - **C.** BYOD
   - **D.** NDA

2. Which term refers to copying data between a mobile device and a computer system in order to mirror such things as contacts, programs, pictures, and music?
   - **A.** Calibration
   - **B.** Remote wipe
   - **C.** Pairing
   - **D.** Synchronization

3. You want to follow the rules of good security administration as set by CompTIA and vendors. To do so, which account should be disabled on most Windows operating systems for security reasons?
   - **A.** Guest
   - **B.** Print Operators
   - **C.** Power Users
   - **D.** Userone

4. What kind of mobile app is being used when the owner's phone displays a message on the screen and emits an extremely loud tone?
   - **A.** Failed login restriction
   - **B.** Antivirus
   - **C.** Locator
   - **D.** Remote wipe

5. As a best practice, after a set period of inactivity on a Windows workstation, what should happen?
   - **A.** The system should shut down.
   - **B.** The system should restart.
   - **C.** A password-enabled screensaver should automatically start.
   - **D.** The system should log out the user.

6. Which Windows feature helps protect against unauthorized changes to system files?
   - **A.** BitLocker
   - **B.** EFS
   - **C.** UAC
   - **D.** Windows Defender

7. Which of the following has the goal of allowing a username/password combination to be entered once and then allowing claims to be used for consecutive logins? (Choose the best answer.)
   - **A.** Tokens
   - **B.** Kerberos
   - **C.** Single sign-on
   - **D.** Multifactor authentication

8. Which of these is a password manager?
   - **A.** Edge
   - **B.** Credential Manager
   - **C.** Internet Explorer 11
   - **D.** Active Directory

9. You have a very small network in a home-based office, and you want to limit network access to only those hosts that you physically own. What should you utilize to make this possible?
   - **A.** Static IP addresses
   - **B.** Disabled DNS
   - **C.** Default subnet mask
   - **D.** Empty default gateway

10. Which wireless encryption protocol provides Advanced Encryption Standard (AES) encryption?
    - **A.** Wired Equivalent Privacy (WEP)
    - **B.** Wi-Fi Protected Access (WPA)
    - **C.** Wi-Fi Protected Access 2 (WPA2)
    - **D.** Temporal Key Integrity Protocol (TKIP)

11. Which type of add-on will extend the functionality of the web browser in a way it wasn't originally designed?
    - **A.** Pop-up blocker
    - **B.** Extensions
    - **C.** Plug-in
    - **D.** Ad blocker

12. What is normally performed when an employee is offboarded?

A. Their user account is deleted.

B. Their user account is unlocked.

C. Their user account is created.

D. Their user account's password is reset.

13. By default, when setting up an Android device, why do you need a Google account?

A. The device requires email setup.

B. The account is used for cloud synchronizations.

C. The account is used for desktop backups.

D. The device requires registration.

14. You need to secure your mobile device's lock screen with the highest level of protection. Which of the following should you use? (Choose the best answer.)

A. Fingerprint lock

B. Face lock

C. Passcode lock

D. Swipe lock

15. You need to encrypt a single file on a Windows desktop. Which technology should you use?

A. EFS

B. BitLocker

C. NTFS

D. BitLocker to Go

16. A user is in both the Sales group and the Marketing group. The Sales group has full permission at the share level, and the Marketing group has Read-only permission. The files on NTFS are secured with the Modify permission for the Sales group and the Read & Execute permission for the Marketing group. Which permissions will the user have?

A. Full

B. Modify

C. Read-only

D. Read & Execute

17. James just moved a folder on the same partition. What will happen with the permissions for the folder?

A. The permissions will be the same as they were before the move.

B. The permissions will be inherited from the new parent folder.

C. The permissions will be configured as the root folder for the drive letter.

D. The permissions will be blank until configured.

18. A user is in the Sales group. The Sales group has no permissions at the share level. The files on NTFS are secured with the Modify permission for the Sales group. What permissions will the user have?

A. The user will have the Modify permission when connecting from the network.

B. The user will have the Modify permission when logged in locally to the computer.

C. The user will have no access when logged in locally to the computer.

D. The user will have Read-only permissions when connecting from the network.

19. You are trying to delete a file on the local filesystem, but the operating system will not let you. What could be the problem? (Choose the best answer.)

A. The NTFS Modify permission is applied to the file.

B. The share permissions are not set to Full Control.

C. The file attributes are set to Read-only.

D. The file attributes are set to System.

20. You need to enforce profile security requirements on mobile devices. Which should you use to achieve this goal?

A. AUP

B. NDA

C. BYOD

D. MDM

Performance-Based Question

You will encounter performance-based questions on the A+ exams. The questions on the exam require you to perform a specific task, and you will be graded on whether you were able to complete the task. The following requires you to think creatively in order to measure how well you understand this chapter's topics. You may or may not see similar questions on the actual A+ exams. To see how your answers compare to the authors', refer to Appendix B.

You have been asked to create a working structure for your organization's network. You have three groups: Sales, Marketing, and R&D. You need to set up a network share and NTFS to allow Sales to access Marketing material but not modify it in any way. R&D must be able to write to marketing files and read Sales information. Marketing must only have read access to R&D and Sales. Each group should have the Modify permission to their respective folder. All permissions should be controlled with share permissions. How will you set up the folders for access, NTFS permissions, and share permissions?



Chapter 7

Troubleshooting Operating Systems and Security

THE FOLLOWING COMPTIA A+ 220-1202 EXAM OBJECTIVES ARE COVERED IN THIS CHAPTER:

- ✓ **2.0 Security**
- ✓ **2.6 Given a scenario, implement procedures for basic small office/home office (SOHO) malware removal.**
  - Investigate and verify malware symptoms.
  - Quarantine the infected systems.
  - Disable System Restore in Windows Home.
  - Remediate infected systems.
    - Update anti-malware software.
    - Scanning and removal techniques (e.g., safe mode, preinstallation environment).
    - Reimage/reinstall.
  - Schedule scans and run updates.
  - Enable System Restore and create a restore point in Windows Home.
  - Educate the end user
- ✓ **3.0 Software Troubleshooting**
- ✓ **3.1 Given a scenario, troubleshoot common Windows OS issues.**
  - Blue screen of death (BSOD)
  - Degraded performance
  - Boot issues
  - Frequent shutdowns
  - Services not starting



- Applications crashing
- Low memory warnings
- USB controller resource warnings
- System instability
- No OS found
- Slow profile load
- Time drift

✓ **3.2 Given a scenario, troubleshoot common mobile OS and application issues.**

- Application fails to launch
- Application fails to close/crashes
- Application fails to update
- Application fails to install
- Slow to respond
- OS fails to update
- Battery life issues
- Randomly reboots
- Connectivity issues
  - Bluetooth
  - Wi-Fi
  - Near-field communication (NFC)
- Screen does not autorotate

✓ **3.3 Given a scenario, troubleshoot common mobile OS and application security issues.**

- Security concerns
  - Application source/unofficial application stores
  - Developer mode
  - Root access/jailbreak
  - Unauthorized/malicious application
  - Application spoofing
- Common symptoms



- High network traffic
- Degraded response time
- Data-usage limit notification
- Limited internet connectivity
- No internet connectivity
- High number of ads
- Fake security warnings
- Unexpected application behavior
- Leaked personal files/data

✓ **3.4 Given a scenario, troubleshoot common personal computer (PC) security issues.**

- Common symptoms
  - Unable to access the network
  - Desktop alerts
  - False alerts regarding antivirus protection
  - Altered system or personal files
    - Missing/renamed files
    - Inability to access files
  - Unwanted notifications within the OS
  - OS update failures
- Browser-related symptoms
  - Random/frequent pop-ups
  - Certificate warnings
  - Redirection
  - Degraded browser performance



Troubleshooting is a major responsibility of an A+ technician's daily job. It may not be as glamorous as we'd like it to be, but it does make up a good percentage of our daily workload. Applying a systematic approach to software troubleshooting is the key to solving all problems. A systematic solution also works well in preventing problems in the first place.

Many of the common software problems that you will spend time solving can be prevented with proper preventive maintenance. Preventive maintenance tends to get neglected at many companies because technicians are too busy fixing problems. Spending some time on keeping those problems from occurring is a good investment of resources.

In this chapter, we'll look at applying the same troubleshooting methodology to common software problems. We'll also apply similar troubleshooting to security issues. First, we'll look at common symptoms of problems and their solutions. We'll then follow up with ways to deal with—and prevent—security-related issues.



Exam Objective 3.1, "Given a scenario, troubleshoot common Windows OS issues," requires you to know the common troubleshooting of the Windows operating system.

Troubleshooting Common Microsoft Windows OS Problems

Windows is mind-bogglingly complex. Other operating systems are complex too, but the mere fact that Windows has nearly 60 million lines of code (and thousands of developers have worked on it!) makes you pause and shake your head. Fortunately, you just need to take a systematic approach to solving software-related issues.

Windows-based issues can be grouped into several categories based on their cause, such as boot problems, missing files (such as system files), configuration files, and virtual memory. If you're troubleshooting a boot problem, it's imperative that you understand the Windows boot process. Some common Windows problems don't fall into any category other than "common Windows problems." We cover those in the following sections, followed by a discussion of the tools that can be used to fix them.

Symptoms/Issues

There are numerous "common symptoms/issues" that CompTIA asks you be familiar with for the exam. They range from the dreaded *Blue Screen of Death (BSOD)* to spontaneous restarts and everything in between. They are discussed in the following section.

Degraded Performance

The performance of your systems will inevitably slow down over time. This could be due to a multitude of causes, ranging from bad Windows Update patches to malware. Sluggish or slow performance is one of the hardest problems to solve on a Windows operating system, because many of the symptoms are related to each other.

The first step to solving the problem is identifying the component that is impacted by the performance issue. The following is a list of critical components that can be affected by slow performance:

CPU A symptom of poor CPU performance is the slow execution of applications. The operating system GUI will be unresponsive and sluggish. You may also hear your CPU fan running higher than normal if overheating starts to occur. CPU problems can be caused by an application that requires high CPU usage, such as movie rendering.

RAM A symptom of the operating system running out of RAM is high disk activity. As physical memory is used up, the less active pages of memory are swapped out to the disk drive. The symptoms of performance closely resemble CPU-related issues, where applications are slow in loading. RAM problems can be caused by too many applications being open at once or an application that has high RAM requirements, such as a database.

Disk A symptom of poor hard disk drive performance is the thrashing of the drive heads on the platters of the drive. Thrashing occurs when there is excessive movement of the drive arm to locate information on the drive. Disk problems can be caused by excessive fragmentation, high RAM usage, or a high volume of drive usage by applications, such as a video capture.

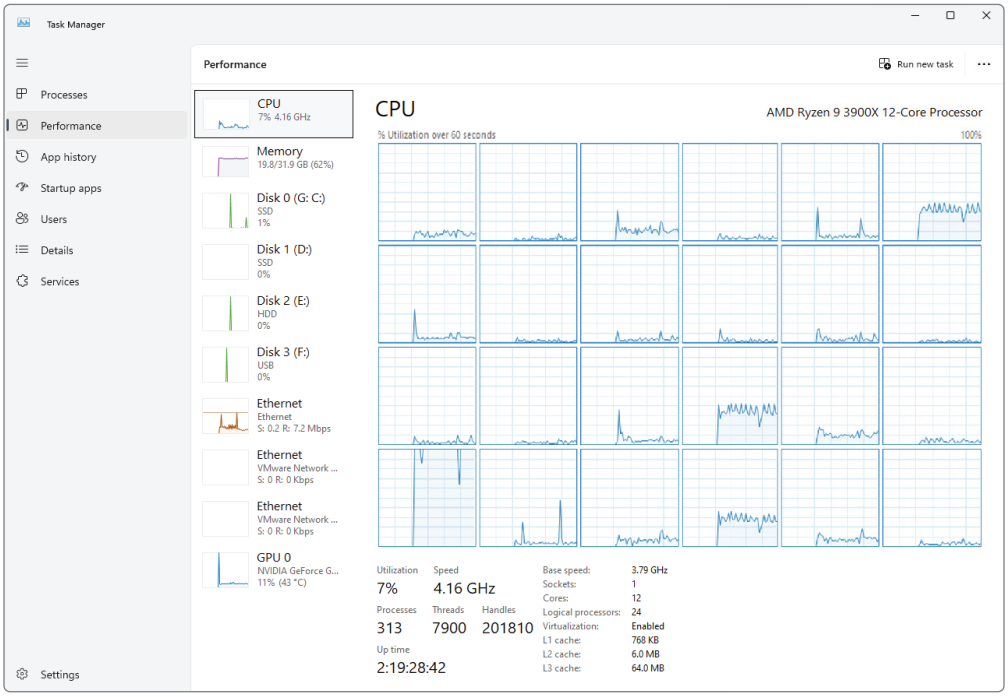
Network Symptoms of poor network performance are slow loading web pages, network applications that load slowly, and even timeouts. If you are using wireless, network issues can be caused by poor signal strength. If you are connected through Ethernet, poor network performance can be directly related to the local area network (LAN). The problems can also be outside of your network.

Graphics Symptoms of poor graphics (GPU) performance are usually related to slow-running video games and playback of videos. The frames per seconds (FPS) will be excessively low as the computer tries to render the screen. It is not common to have graphics-related issues unless you just downloaded the latest shoot-em-up game. Graphics-related problems are usually the hardest to solve because they require third-party tools by the graphics card vendor.

As you can see from the list of possibly affected components, many of the symptoms are closely related, such as RAM, CPU, and disk. The excessive usage of RAM can create performance symptoms with your hard disk drive. If left for a long period of time, these can both lead to an increase in CPU activity.

There are several tools that you can use to identify the problem area so that you can focus your attention on narrowing down the problem. The first tool you should start up is the Task Manager, as shown in Figure 7.1. You can launch Task Manager in several different ways, such as right-clicking the Start menu and selecting Task Manager, right-clicking the Taskbar and selecting Task Manager, or (my personal favorite) pressing Ctrl+Shift+Esc. The Performance tab will show you four of the five critical areas (detailed previously) on the left side. In this example, you can see that the processor is spiked out at almost 100 percent and all other systems are within tolerance.

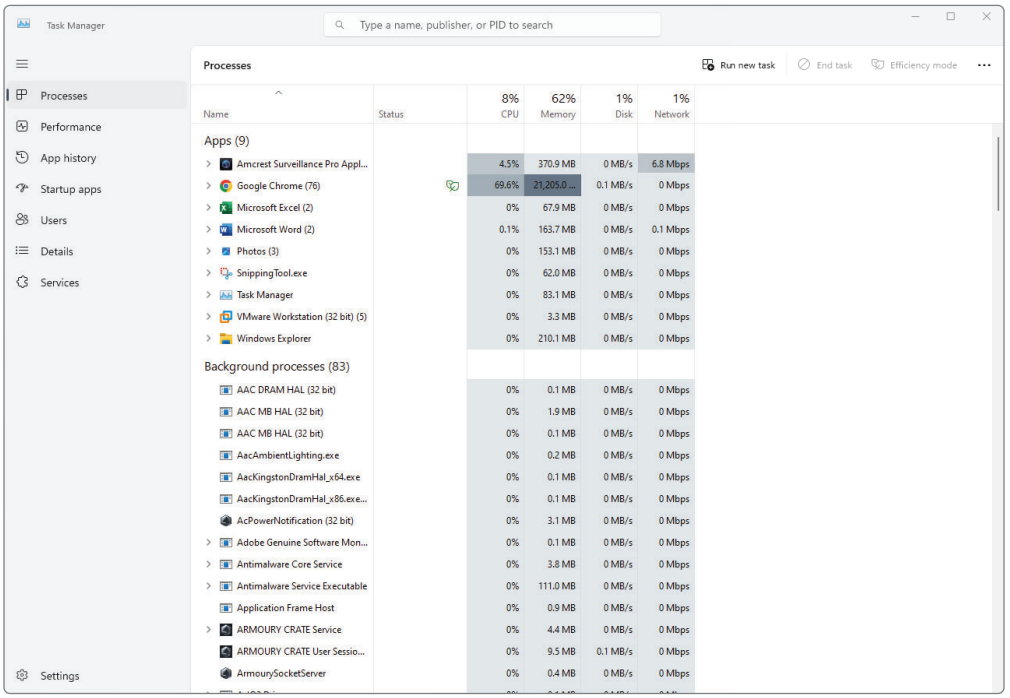
FIGURE 7.1 The Performance tab in Task Manager



Now that you've isolated the problem to the critical area of CPU, you can narrow it down further by looking at the Processes tab, as shown in Figure 7.2. You can see that Google Chrome is using nearly 70 percent of the CPU. The other 30 percent is most likely distributed among other processes. By clicking the core area headings of CPU, Memory, Disk, and Network, you can sort usage from high to low or from low to high. In this

particular instance, the operating system was caught booting up, so that particular process was displaying high CPU.

FIGURE 7.2 The Processes tab in Task Manager





Starting in Windows 8, the Task Manager gained a few tabs. One of those is called Details. The Details tab allows you to see the details of the processes, such as the user who executed the process, the process ID (PID), and the name of the process executable. You can sort by any of the headings by clicking them, as shown in Figure 7.3, just like you can on the Processes tab. You can even right-click the headings and add columns, such as peak memory usage and the exact command line, just to name a few.

Using *Resource Monitor*, you can get a much more detailed view than what is displayed in Task Manager. You can open Resource Monitor by clicking Start ➢ All ➢ Windows Tools ➢ Resource Monitor, or you can click the Start menu and type **Resource Monitor**, and select it when it populates in the search results. Resource Monitor is shown in Figure 7.4. This tool allows you to read real-time performance data on every process on the operating system. Resource Monitor also allows you to sort details, the same as Task Manager. You can click each critical area and drill down to the performance issue.

FIGURE 7.3 The Details tab in Task Manager

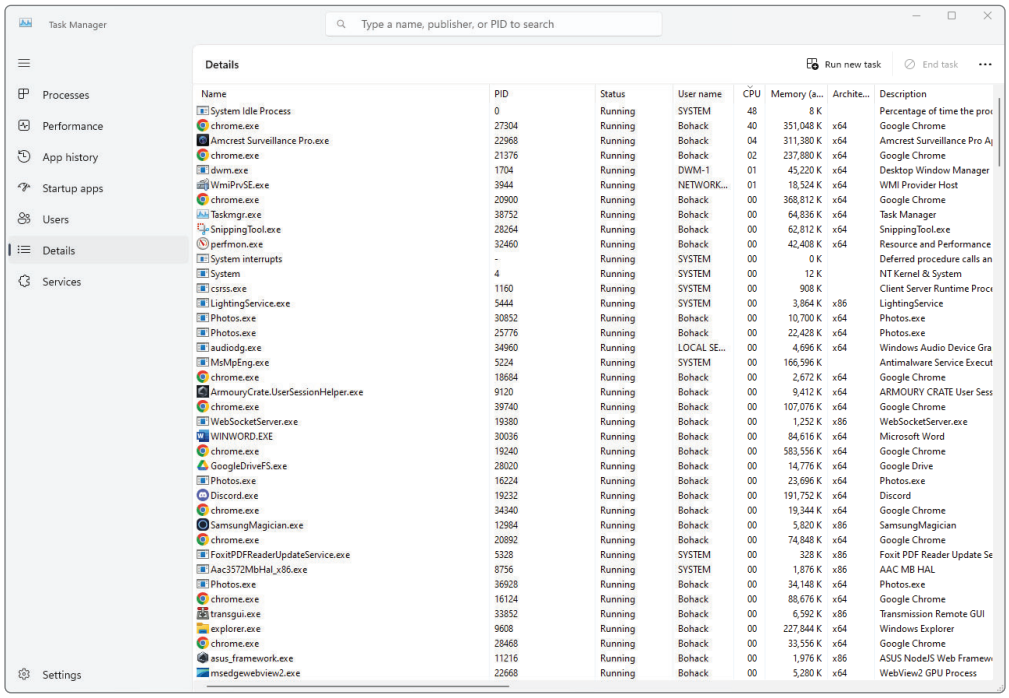
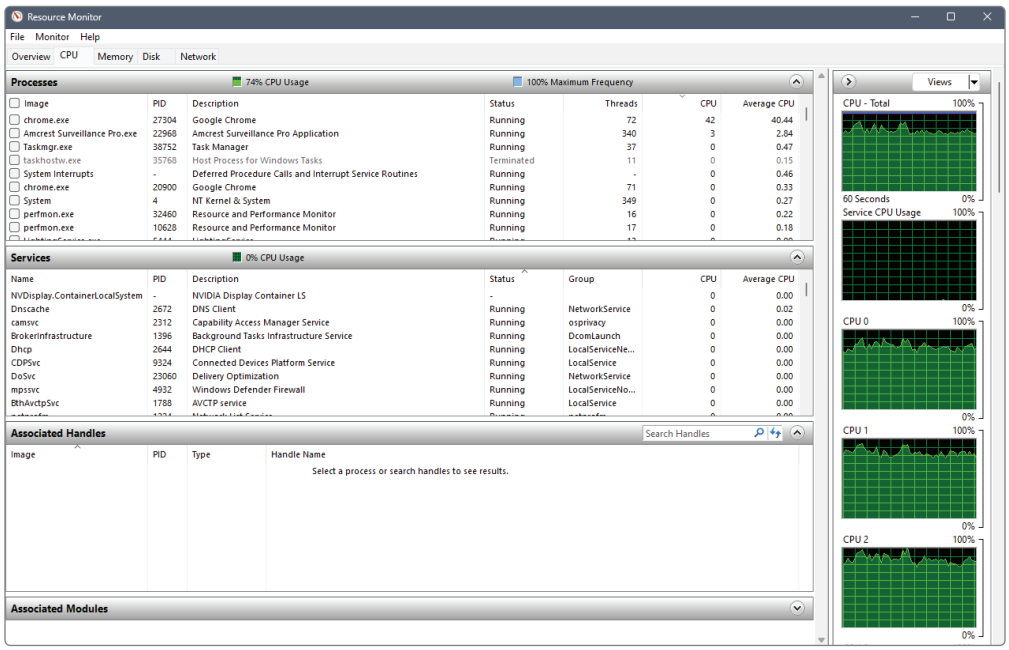
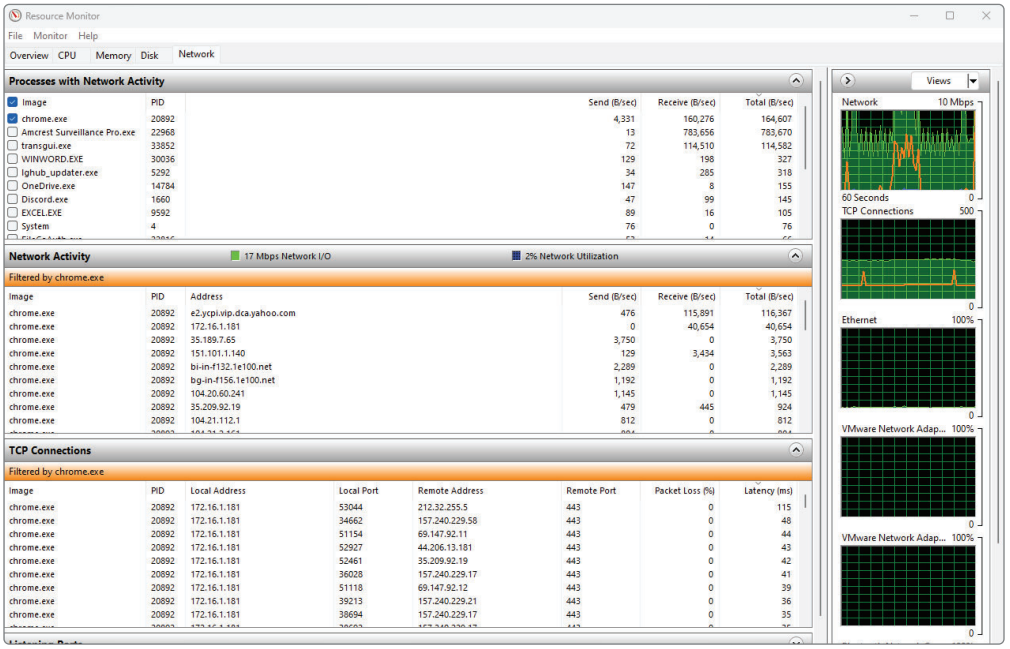


FIGURE 7.4 Resource Monitor



A unique feature of Resource Monitor is the visualization of data. When you select a process on the upper view, Resource Monitor automatically filters the activity of the critical area, as shown in Figure 7.5. As you can see in this example, the Chrome browser processes have been selected and then the Network tab can be chosen to display the network activity and connections. The result is the isolation of network activity for this process. This can be done for any of the critical areas.

FIGURE 7.5 Selective isolation in Resource Monitor



Now that you've isolated the problem to an action or process in the operating system, you need to do the following:

1. Formulate a theory of probable cause.
2. Test the theory to determine the cause.
3. Establish a plan of action to resolve the problem and implement the solution.
4. Verify the full-system functionality and, if applicable, implement preventive measures.
5. Document findings, actions, and outcomes.

The theory of probable cause may be that a hardware upgrade is required due to a new version of the application that demands more resources. Or it could be as simple as the job the application is running is higher than normal in load. Remember to question the obvious and do the simple things, such as rebooting, to see if the problem goes away. It is often joked about that problems always go away after a reboot. It's not too far from the truth.

Sometimes a process is hung up and is affecting another process. A reboot sometimes fixes them both and the symptoms go away. More likely than not, the problem will still be there. This is when you need to start testing your theory of probable cause to determine the cause.

You might find that after running a large query the hard drive is extremely stressed. Your action plan might be to upgrade the hard drive or move the workload to a faster machine. In either case, you need to verify that the process is functioning the way the user expects it to.

If you determine that it's a certain report in the database, remind the user that the report takes time and maybe they should schedule it when the computer is not immediately needed. Or schedule the system for an upgrade of hardware to prevent these problems in the future.

Ultimately, you should document your finding so that other technicians do not waste time with the same issue. The more intricate the problem, the more time is wasted when you forget you've solved it already or don't remember the answer. Always document the actions taken, such as upgrades or changes to the process for the user. You should also note the outcome—whether it was successful or showed no immediate performance increase—so that if another technician is working on the same or a similar issue, they can gauge whether the solution is effective.

Boot Problems

With the introduction of Windows Vista, the boot process had changed from prior operating systems of Windows XP, 2000, and NT 4.0. We've used this current boot process introduced with Windows Vista all the way to today with Windows 11. The current boot process allows for the adoption of UEFI firmware.

In order to troubleshoot a failure to boot, you need to understand the complete boot process, starting with either the BIOS or the UEFI. The process is slightly different depending on which firmware you have on the motherboard. However, the outcome is the same: the hardware hands control over to the operating system so that the operating system can boot.

BIOS Legacy BIOS systems perform a *power-on self-test (POST)*, and then the BIOS bootstrap routine looks at the master boot record (MBR) at the beginning of the disk. The MBR then reads the boot sector on the first primary partition found. This boot sector then instructs the Boot Manager to load.

UEFI UEFI firmware will perform a similar POST. Then the UEFI bootstrap begins by loading drivers for the hardware. One of the differences is that UEFI can contain drivers that allow it to boot across a network or other nonstandard devices. Just like the legacy BIOS, the UEFI firmware looks at the MBR in the GUID Partition Table (GPT). The GPT defines a globally unique identifier (GUID) that points to a partition containing the Boot Manager. Therefore, UEFI firmware requires a partitioning scheme of GPT and cannot use the standard MBR partitioning scheme.

The initial boot sequence from hardware control to software control is almost identical in both BIOS and UEFI firmware. UEFI firmware does give you many more options, because

UEFI drivers can be loaded before control is handed over to the software. This allows UEFI to treat all locations containing an operating system the same. Up to the point at which the hardware hands control over to the software, there is no difference between a network boot and a hardware boot.



Windows 11 requires a UEFI firmware and therefore the boot process will follow the UEFI process. However, legacy MBR booting is still used with operating systems that you put on a USB drive, such as the Windows installation process. UEFI BIOS typically supports legacy BIOS booting, such as physical media, ISOs, or USB devices.

After control is handed over to the software, several files are used to complete the operating system bootup. The most important files are as follows:

Windows Boot Manager The Windows Boot Manager (`BOOTMGR`) bootstraps the system. In other words, this file starts the loading of an operating system on the computer.

BCD The Boot Configuration Data (BCD) holds information about operating systems installed on the computer, such as the location of the operating system files.

`winload.efi` `winload.efi` is the program used to boot Windows. It loads the operating system kernel (`ntoskrnl.exe`).

`winresume.exe` If the system is not starting fresh but resuming a previous session, then `winresume.exe` is called by `BOOTMGR`.

`ntoskrnl.exe` The Windows OS kernel is the heart of the operating system. The kernel is responsible for allowing applications shared access to the hardware through drivers.

`ntbtlog.txt` The Windows boot log stores a log of boot-time events. It is not enabled by default.

System Files In addition to the previously listed files, Windows needs a number of files from its system folders (e.g., `SYSTEM` and `SYSTEM32`), such as the hardware abstraction layer (`hal.dll`), the Session Manager (`smss.exe`), the user session (`winlogon.exe`), and the security subsystem (`lsass.exe`).

Numerous other dynamic link library (DLL) files are also required, but usually the lack of them or corruption of one of these files produces a noncritical error, whereas the absence of `hal.dll` causes the system to be nonfunctional.

We'll now look at the complete Windows boot process. It's a long and complicated process, but keep in mind that these are complex operating systems, providing you with a lot more functionality than older versions of Windows:

1. **Power On:** The system self-checks and enumerates hardware resources. Each machine has a different startup routine, called the POST (power-on self-test), which is executed

by the commands written to the motherboard of the computer. Newer PnP boards not only check memory and processors, but also poll the systems for other devices and peripherals.

2. **UEFI Firmware Initialization:** After the POST, the UEFI firmware continues to initialize the rest of the hardware components. This involves setting up the CPU, initializing the memory, and detecting connected storage devices. The UEFI firmware also prepares the system environment for the operating system to take over.
3. **Boot Manager:** Once the hardware initialization is complete, the UEFI firmware hands control over to the Windows Boot Manager, called `BOOTMGR`. The Boot Manager is responsible for managing the boot process and determining which operating system to load, especially if you have multiple operating systems installed.
4. **Boot Configuration Data (BCD):** `BOOTMGR` reads the boot configuration data (BCD) to get a list of boot options for the next step. The BCD contains multi-boot information or options on how the boot process should continue.
5. **Windows Boot Loader:** The Boot Manager then loads the Windows Boot Loader (`winload.efi`) from the EFI system partition. The Boot Loader is responsible for loading the Windows kernel and essential drivers needed to start the operating system.
6. **Suspended State:** If Windows is returning from a hibernated (suspended) state, `winresume.exe` is responsible for reading the `hiberfil.sys` file into memory and passes control to the kernel after this file is loaded.
7. **Kernel Initialization:** The Windows Boot Loader initializes the Windows kernel (`ntoskrnl.exe`), which is the core of the operating system. The kernel is responsible for managing system resources, such as memory, processes, and hardware interactions. This switches the system from real mode (which lacks multitasking, memory protection, and those things that make Windows so great) to protected mode (which offers memory protection, multitasking, and so on) and enables paging. Protected mode enables the system to address all the available physical memory.
8. **Hardware Abstraction Layer (HAL):** The Windows kernel loads the executive subsystems. *Executive subsystems* are software components that are required for the operating system to communicate with the hardware. One of the subsystems is the HAL, which provides a consistent interface for the operating system to interact with the hardware, ensuring compatibility across different hardware platforms.
9. **Session Manager:** The kernel loads the Session Manager. The Session Manager (`smss.exe`) is then started, which initializes system sessions and loads the Registry. The `HKEY_LOCAL_MACHINE\SYSTEM` Registry hive and device drivers are loaded. The drivers that load at this time serve as boot drivers, using an initial value called a *start value*.
10. **Winlogon:** `Winlogon.exe` loads. At this point, you are presented with the login screen. After you enter a username and password, you're taken to the Windows desktop.

11. **Explorer.exe:** Finally, the Windows Explorer (`explorer.exe`) process starts, providing the graphical user interface (GUI) for Windows 11. This includes the desktop, taskbar, and Start menu, allowing you to interact with the operating system and launch applications.

Now that you understand the boot process, let's look at how you can collect information to identify the problem. We'll consider this in two parts: hardware and software. The hardware process begins with the POST, and the software portion of the bootstrap begins with the `BOOTMGR`.

You can collect information from the BIOS/UEFI firmware boot with third-party system event log (SEL) viewers. However, it is very unlikely that you have a failure to boot because of a BIOS/UEFI firmware issue. It's not impossible, but it is highly unlikely.

To collect information on the software portion of the boot process loads, you can use boot logging. The `ntbtlog.txt` file is located at the base of the `C:\Windows` folder, as shown in Figure 7.6. Boot logging is off by default and needs to be turned on. To enable boot logging, issue the command **`bcdedit /set {current} bootlog Yes`**. You can also use the System Configuration utility (`msconfig.exe`) by selecting the Boot Log option on the Boot tab, as shown in Figure 7.7. Because the BCD is read by `BOOTMGR`, this point of the boot process is where logging would begin and the first entries would be the loading of the kernel.

FIGURE 7.6 The `ntbtlog.txt` file

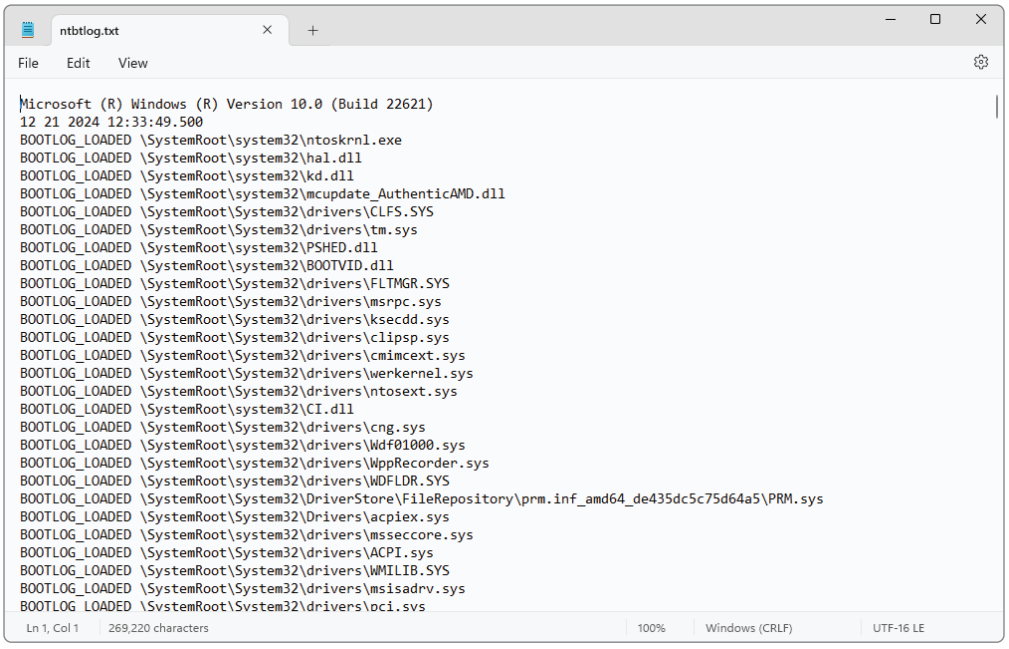
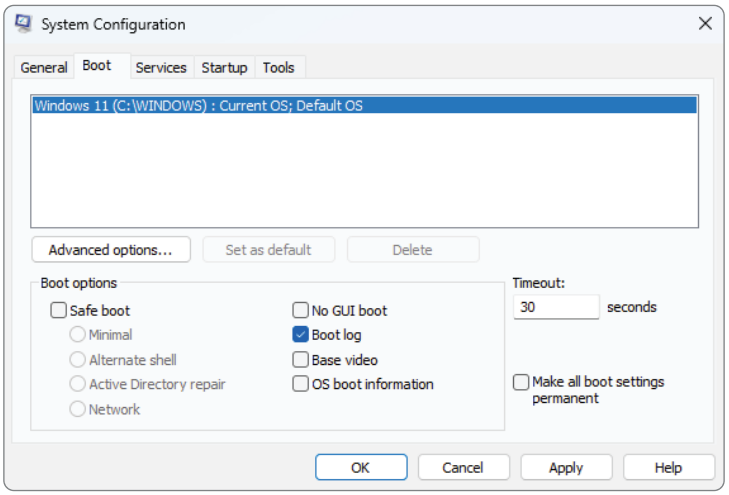


FIGURE 7.7 System Configuration options for boot logging



Chances are, if you're having trouble booting into Windows, you won't be able to access the command prompt to issue `bcdedit` commands, nor will you able to access `msconfig.exe`. Not to worry. You can still access logging by allowing the operating system to fail two times in a row. The third time, the computer will boot into the recovery console, as shown in Figure 7.8. From there, click Troubleshooting ➢ Advanced Options ➢ Startup Settings ➢ Restart. When the computer restarts, it will boot into the Startup Settings menu, as shown in Figure 7.9. Of course, you won't be able to boot the computer to retrieve the files, but you can use the command prompt in the Windows Recovery Environment.

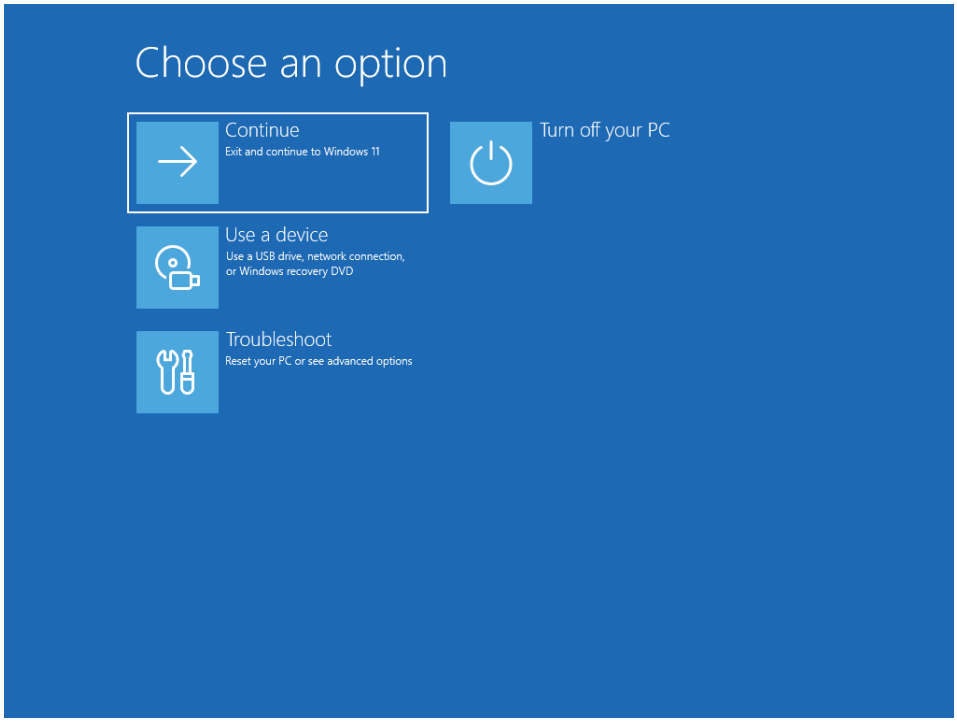
The idea is to collect information to identify the problem and, above all, to fix the problem. Sometimes you need to let Windows repair itself. The Windows Recovery Environment (WinRE) contains a Startup Repair option. Using the Startup Repair option is similar to issuing `bootrec /rebuildbcd` at the command prompt, which will rebuild the BCD.

If that fails, the ultimate solution might be to use the Reset This PC option in the Windows Recovery Environment or to install the operating system from scratch.

No OS Found

When it's reported that an operating system is missing, or "no OS is found," the first thing to check is that no media is in the machine (USB, DVD, CD, and so on). The system may be reading this media during boot before accessing the hard drive. If that is the case, remove the media and reboot. You should also change the BIOS/UEFI settings to boot from the hard drive before any other media to prevent this issue in the future.

FIGURE 7.8 Recovery Environment menu



If the problem is not as simple as removing the non-bootable media, then you may have to boot into the Windows Recovery Environment. This may be a challenge, because if the BIOS/UEFI cannot boot to the Windows Boot Manager, then the Windows Recovery Environment cannot be executed. The Boot Manager is responsible for executing the Windows Recovery Environment. You will have possibly two options to fix this. The first option is to use the vendor's recovery console. This option is dependent on the vendor supplying a recovery console that can be accessed via the BIOS/UEFI; not every vendor supplies this tool. The second option is to boot the installation media and choose Repair My PC when it first boots, as shown in Figure 7.10. Choosing this option will launch the Windows Recovery Environment booted from the installation media. You can then choose to repair the operating system by selecting Troubleshoot ➢ Startup Repair. The Windows Recovery Environment will then attempt to repair the operating system.

System Instability

When an *application crashes*, you want to isolate the cause of the crash and solve it. The cause could be a compatibility issue, a hardware issue, or a host of other problems. One step to take early on is to look for updates/patches/fixes to the application released by the vendor. Be sure to try these updates on a test machine before rolling them out to all machines, and verify that they address the problem and do not introduce new problems.

FIGURE 7.9 Startup Settings menu

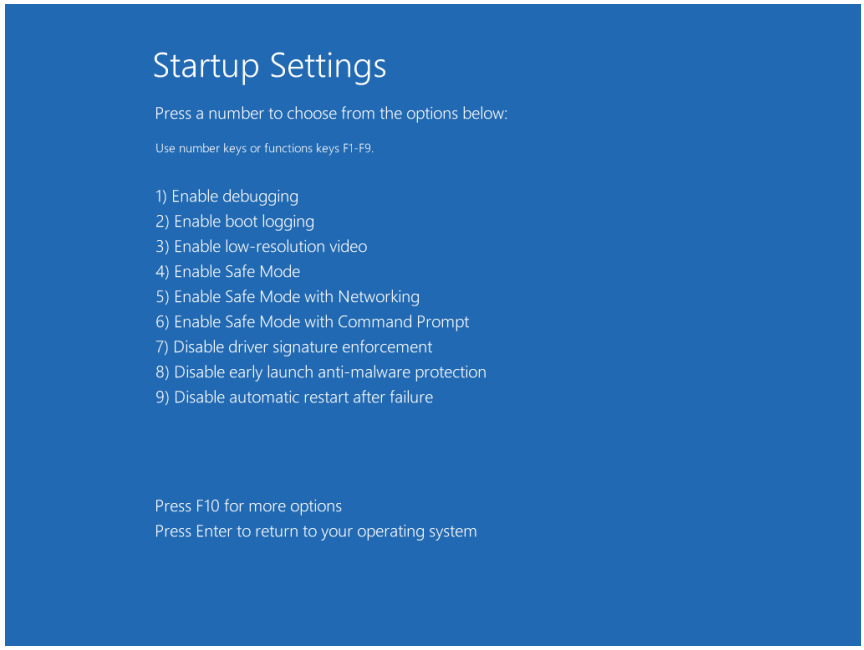
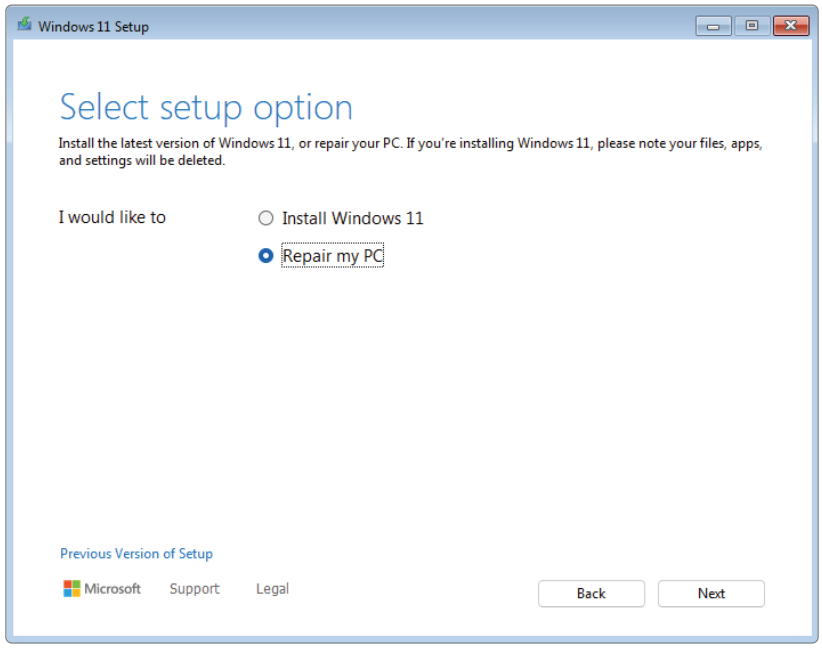
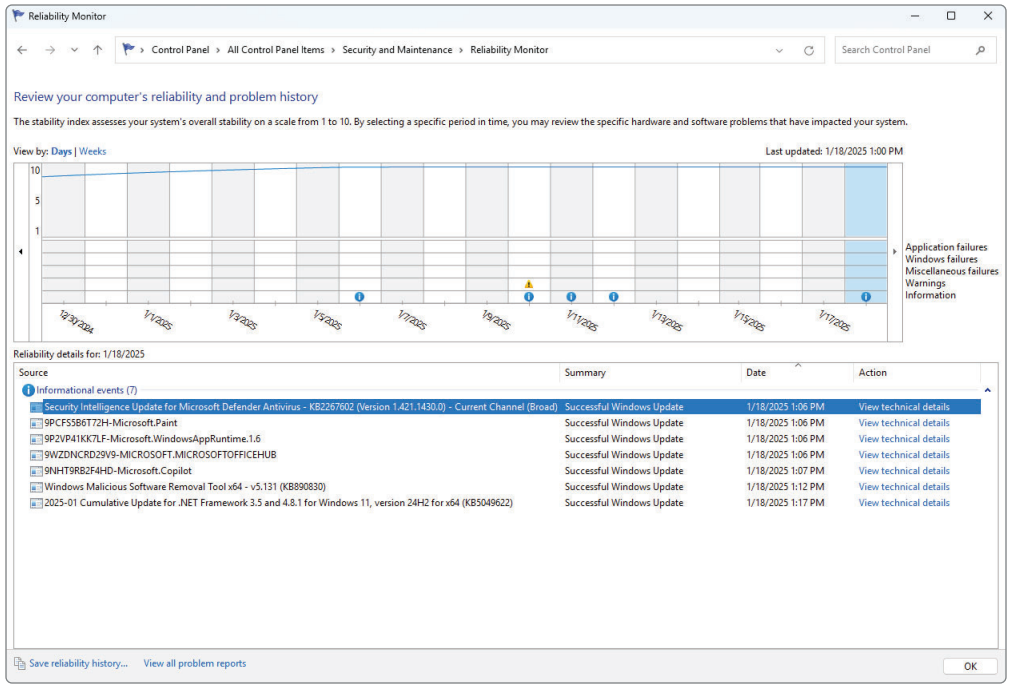


FIGURE 7.10 Windows 11 Repair My PC menu



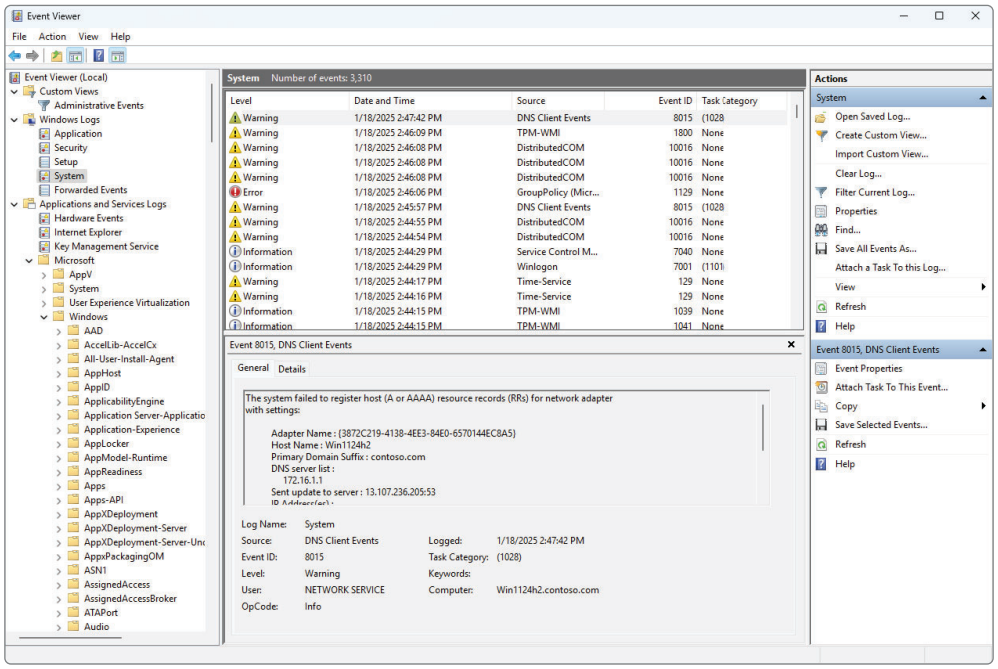
One tool that is extremely helpful in identifying software problems is Reliability Monitor, shown in Figure 7.11. Reliability Monitor allows you to see application crashes and the times and dates they occurred. It also allows you to see which updates were installed before and after the crashes. You can use Reliability Monitor to narrow down whether other software is causing the issues and what led up to the crashes. To open Reliability Monitor click Start ➢ All ➢ Windows Tools ➢ Control Panel ➢ Security And Maintenance, open Maintenance, and click View Reliability History. Alternately, you can click the Start menu and type **View Reliability History**, then select it from the active search results.

FIGURE 7.11 Windows Reliability Monitor



In addition to Reliability Monitor, you can access the Windows event logs in *Event Viewer* for information about Microsoft-based application problems, as shown in Figure 7.12. All third-party vendors should log errors to the Windows event logs, but generally you will only find Microsoft products using these logs. In either case, you might find more information about why an application is crashing by looking at the Application log. You can open the Windows Event Viewer by clicking Start ➢ All ➢ Windows Tools ➢ Event Viewer, or you can click the Start menu and type **Event Viewer**, then select it from the active search results.

FIGURE 7.12 Windows event logs in Event Viewer



The applications on the operating system are not the only elements captured by Reliability Monitor. Reliability Monitor also captures the overall stability of the operating system. It will allow you to see every reboot caused by a problem and even blue screens. The overall stability of the operating system is drawn as a graphical line inside the Reliability Monitor application. This allows you to look back and trace when a problem started.

In Exercise 7.1 you will review the Reliability Monitor built into Windows.

EXERCISE 7.1

Reviewing Reliability Monitor

This exercise is best performed on an operating system that has been running for some time. Although the operating system should be flawless and operate at a reliable level, you should see some fluctuation in reliability. If you don't, then try switching the view from days to weeks in the upper left of the app, above the graph.

1. Click the Start menu, type **View Reliability History**, and then click the Control Panel app result.
2. Make note of your current reliability score by looking at the graph on the rightmost section of the graph and comparing the location with the legend on the left side (1 through 10).

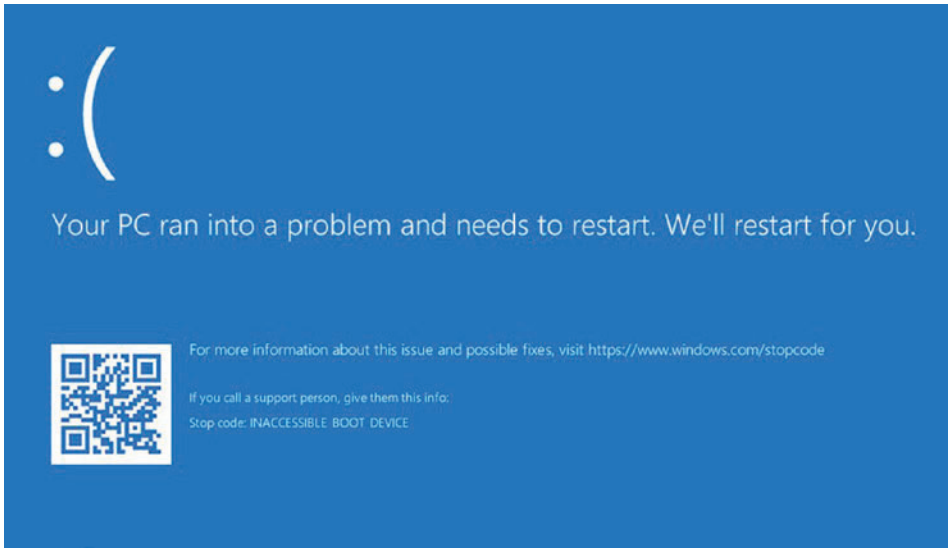
3. Make note of if and why your score has gone down over the days shown.
4. If there is a drop in reliability, click Day, and the lower section of the app will display the reliability details.
5. View the details and make note of the action.

As you click through the reliability events, make note of applications that constantly crash or that are problematic. These are the applications you should investigate further. Some of the solutions in the next section will lend some suggestion on how to fix them.

Blue Screen of Death

The Blue Screen of Death (BSOD)—not a technical term, by the way—is another way of describing the blue-screen error condition that occurs when Windows fails to boot properly or quits unexpectedly, as shown in Figure 7.13. If this happens during a boot, it is at this stage that the device drivers for the various pieces of hardware are installed/loaded. If your Windows GUI fails to start properly, more likely than not the problem is related to a misconfigured driver or misconfigured hardware.

FIGURE 7.13 Blue Screen of Death



You can try a few things if you believe that a driver is causing the problem. One is to try booting Windows into Safe Mode, which you can access via the Startup Settings in the Windows Recovery Environment. In Safe Mode, Windows loads only basic drivers, such as

a standard VGA video driver and the keyboard and mouse. After you've booted into Safe Mode, you can uninstall the driver that you think is causing the problem.

Another option is to boot into the Windows Recovery Environment and use System Restore, which will revert the system drivers back to the state they were in when the restore point was created. Bear in mind that a System Restore will not affect personal files, but it will remove applications, updates, and drivers.

The Windows Recovery Environment will automatically launch if there are two failed attempts to boot the operating system in 2 minutes. You can manually launch the Windows Recovery Environment by holding the Shift key as you reboot the operating system. Another way is to boot the installation media and select Repair My PC.

Frequent Shutdowns

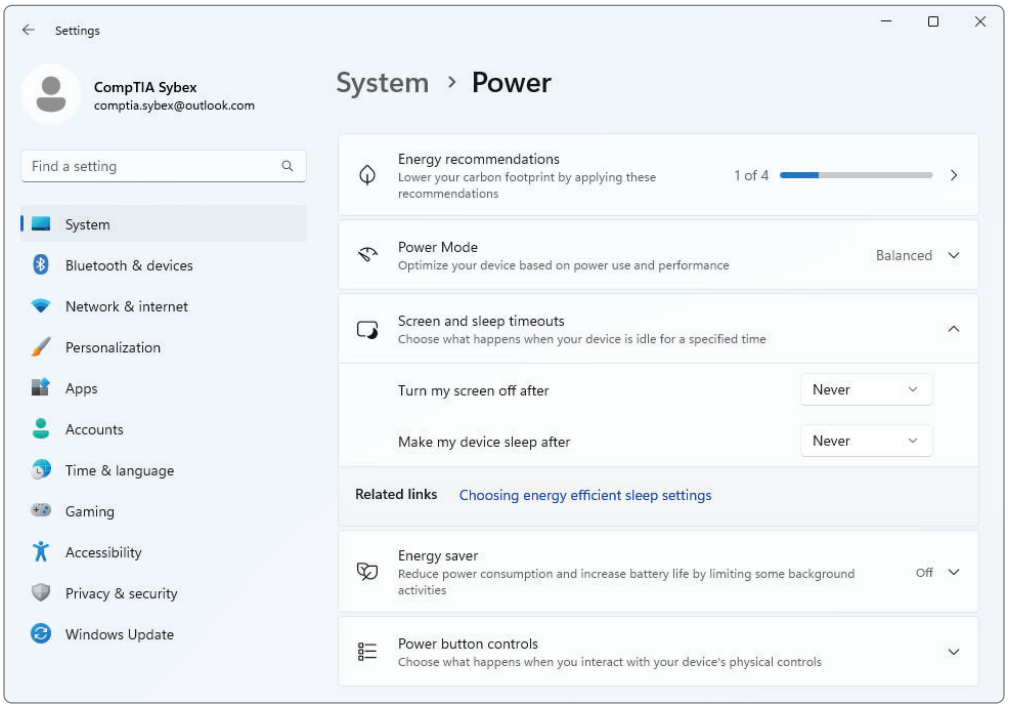
There are many reasons why your computer might randomly turn off or shut down without warning. The problem is almost always related to faulty hardware or a faulty driver, but sometimes it can be as simple as tweaking your advanced power settings.

The first place to check is Event Viewer on the System tab. You should start combing through the logs and looking for the source of Kernel-Boot or Kernel-General. The entries will help you identify if the operating system was shut down properly or if it just suddenly lost power. Any time the operating system is shut down, the kernel will log an entry, and when it is powered back up, it will also log an entry. In addition to the Kernel-Boot and Kernel-General sources, you should also investigate the source of EventLog. These entries will be created if the EventLog service detects a dirty shutdown, such as when the power is removed. By checking the Event Viewer logs, you identify whether the problem is a hardware problem or whether the operating system is actually shutting itself down.

If you identify that the problem is a hardware issue, then the first step to resolving the problem is updating drivers. You should remove any autodetected drivers and reinstall the vendor's driver for the specific hardware. If the problem persists, then swapping known good hardware might help narrow down the issue.

If you determine that the operating system is shutting itself down, then the power settings should be checked, as shown in Figure 7.14. You can access the power settings by navigating to Start ➢ Settings ➢ System ➢ Power ➢ Screen and Sleep Timeouts. Alternately, you can access the Power Control Panel applet by opening Control Panel and selecting Power Options, then selecting Change Plan Settings on the selected power plan. Make sure that the PC is not going into sleep mode when plugged in. From this screen you can access the advanced power settings by clicking Change Plan Settings, then Change Advanced Power Settings. You will need to selectively tweak some of the settings and test your adjustments, such as turning off hard disk sleep, general sleep settings, and processor power management, to name a few. Tweak the timer or adjustment and then monitor the change from the setting to determine the fix.

FIGURE 7.14 Power Options



Services Not Starting

A service's failure to start is directly related either to another application installed with conflicting resources or to a misconfiguration of the service. In either case, the first place to start is the Event Viewer, as shown in Figure 7.15. The System log will display an Event ID of 7000 from the source of the Service Control Manager. The reason for the failure will vary, depending on the problem.

If a service is conflicting with another resource, we recommend that you reinstall the software that installed the service that is failing. Although this might break the conflicting application, it is probably the quickest way to find a conflicting resource.

If the service fails to start because of a misconfiguration, the most likely cause is the user account the service is configured to start with. If a misconfigured user account is the problem, you will see an Event ID of 7000 in Event Viewer, and the description will read that the service failed due to a login failure. You can verify the user configured to start the service in the Services properties, as shown in Figure 7.16. You open the properties of the service by right-clicking the Start menu, selecting Computer Management, then Services, right-clicking the service, selecting Properties, and finally selecting the Log On tab.

FIGURE 7.15 Service Control Manager events

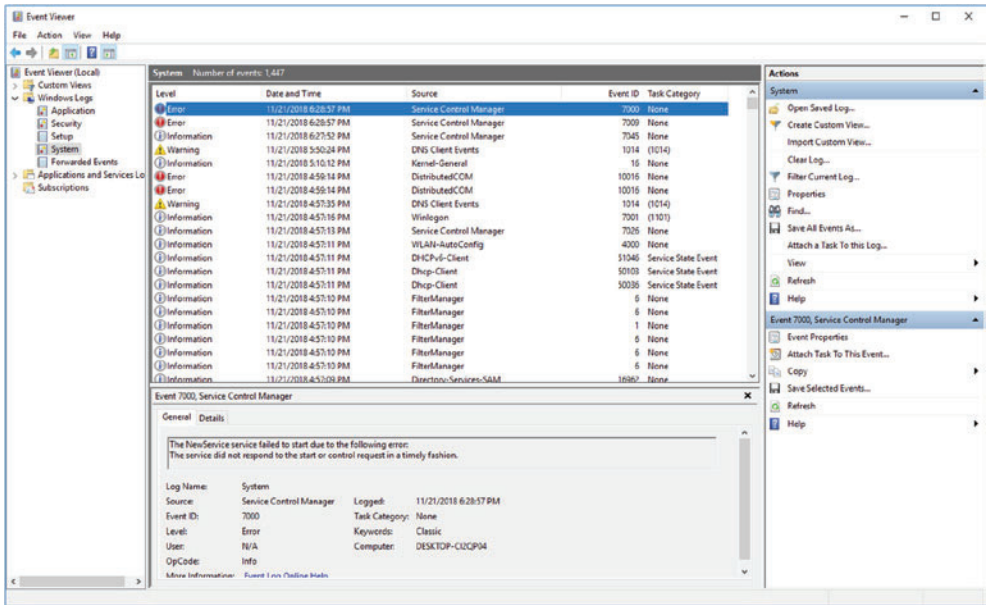
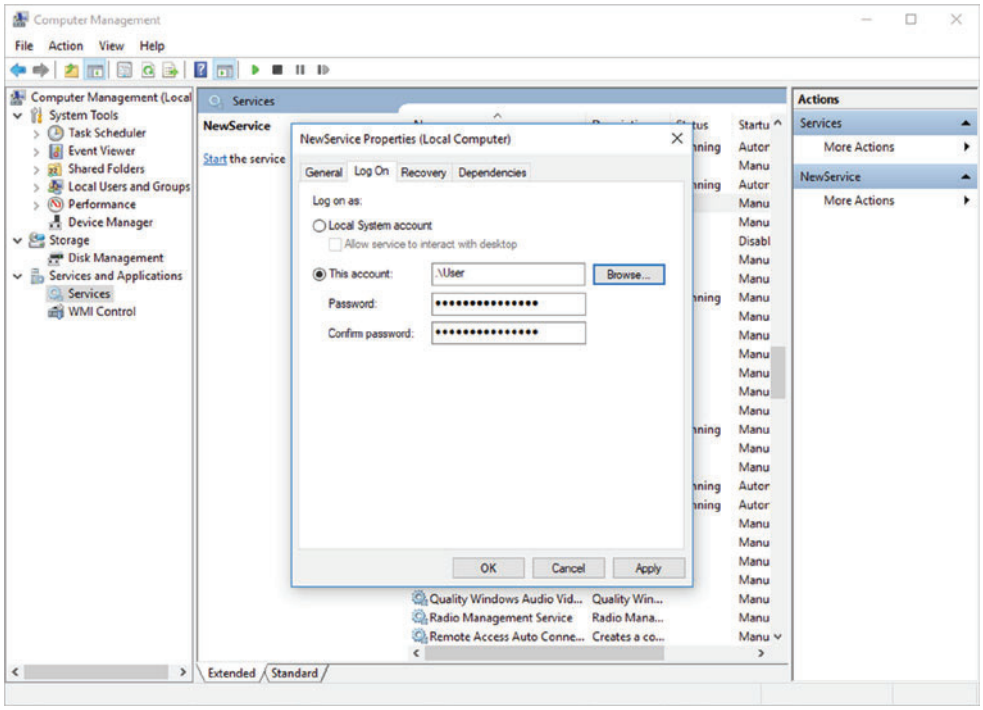


FIGURE 7.16 Services properties



Make sure that the password for the user account has not changed and that the user account is not locked out. You can manually reset the password for the user and reenter the password in the Services properties. Also make sure that the account has the Log On As A Service right.



Real World Scenario

Did You Reboot Your Computer?

Quick quiz: You just got an error in Windows, and it appears that you are on the verge of a crash (of your application or the whole system). What do you do?

The first thing is to write down any error messages that appear. Then save your work (if possible) and reboot your computer.

Anyone who has called tech support, or who has been a tech support person, knows how demeaning the question, "Did you restart your computer?" can seem. Most people respond with an indignant, "Of course!" In reality, they might or might not have actually done it.

Whenever there's a software problem, always, always reboot the computer before trying to troubleshoot. Often, the problem will disappear, and you'll have just saved yourself half an hour of frustration. If the same problem reappears, then you know that you have work to do.

Why does rebooting help? When an application is running, it creates one or more temporary files that it uses to store information. It also stores information in memory (RAM). If a temporary file or information in RAM becomes corrupted (such as by application A writing its information into application B's memory space), the original application can have problems. Rebooting will clear the memory registers and most often remove problematic temporary files, thus eliminating the issue.

It might sound trite, but the first axiom in troubleshooting software really is to reboot. Even if the user says they did, ask them to reboot again. (Tell them you want to see the opening screen for any possible error messages, or make up another good excuse.) If the problem doesn't come back, it's not a problem. If it does, then you can use your software skills to fix it.

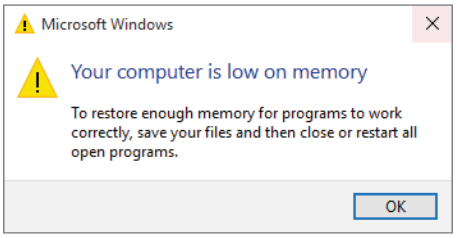
Low Memory Warnings

Random access memory (RAM) is the computer's physical memory. The more RAM you physically have installed in a computer, the more items you can have in the running foreground of the operating system. If you run out of physical memory, then processes that are

backgrounded (minimized) will be loaded into the *page file*, or *paging file*, on the hard drive. This is called paging, and it is totally normal to have a certain amount of paging happen during normal activity. The page file is actually hard drive space into which idle pieces of programs are placed, while other active parts of programs are kept in or swapped into main memory. The programs running in Windows believe that their information is still in RAM, but Windows has moved the data into near-line storage on the solid-state drive (SSD) or hard drive. When the application needs the information again, it is swapped back into RAM so that the processor can use it.

When system processes are at risk of not having enough memory free, you will see a warning message similar to the one shown in Figure 7.17. When this happens it means one of two things; the first is that you simply don't have enough physical RAM in the computer. The second is that a process is using a large amount of RAM that it normally doesn't need. The operating system is letting you know that it can't swap out any more pages of memory to the page file (virtual memory).

FIGURE 7.17 Low memory warning



The larger the page file, the fewer times the machine has swapped out the contents of what it is holding in memory. The maximum possible size of your page file depends on the amount of disk space that you have available on the drive where the page file is placed. Windows configures the minimum and maximum page file size automatically. If you want Windows to handle the size of the page file dynamically, you have to change the default setting by selecting System Managed Size in the Virtual Memory dialog box. We'll show you how to get there in a moment.

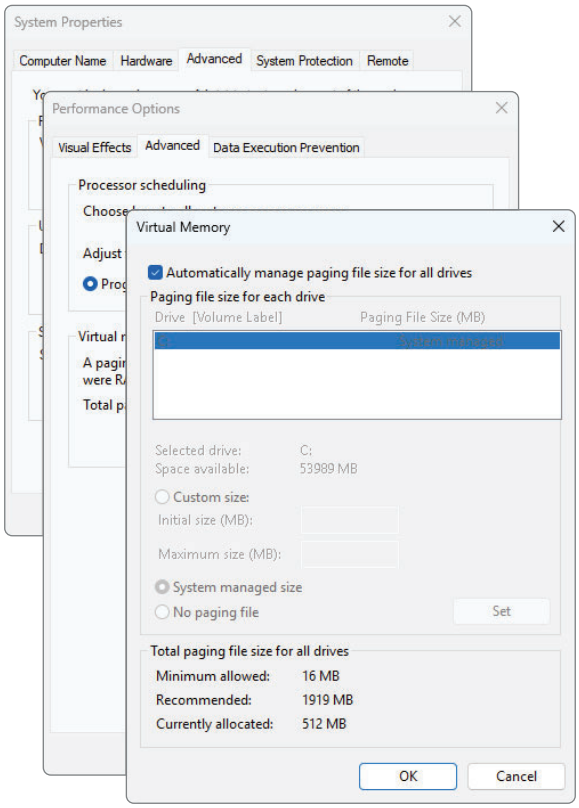
In Windows, the page file is called `pagefile.sys` and is located in the root directory of the drive on which you installed the OS files. The page file is a hidden file; to see the file in Windows File Explorer, you must have the Folder options configured to show hidden files. Typically, there's no reason to view the page file in the filesystem, because you'll use Control Panel to configure it. However, you may want to check its size, and in that case, you'd use Windows File Explorer.



The moral of the story: As with most things virtual, a page file is not nearly as good as actual RAM, but it is better than nothing.

To modify the default virtual memory settings, follow these steps: Click Start menu ➢ Settings ➢ About ➢ Advanced System. In the Performance area, click Settings. Next, select the Advanced tab (yes, another Advanced tab), and then, in the Virtual Memory area, click Change and the Virtual Memory dialog box will open, as shown in Figure 7.18. Note that in addition to changing the page file's size and how Windows handles it, you can specify the drive on which you want to place the file.

FIGURE 7.18 Windows Virtual Memory settings





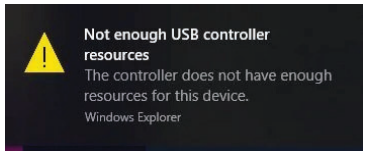
You should place the page file on a drive with plenty of empty space. As a general rule, try to keep 20 percent of your drive space free for the overhead of various elements of the OS, like the page file. Do not set the page file to an extremely small size. If you make the page file too small, the system can become unbootable—or at least unstable. In general, the page file should be at least 1.5× the amount of RAM in the machine.

USB Controller Resource Warnings

The *USB controller* is a hardware component on the motherboard that supplies both power and a data path for the devices connected. It is possible to plug in too many devices and overload the power the port can handle. Most USB 2.0 ports can handle five concurrent loads of 100 mA each, for a total of 500 mA. USB 3.0 can handle six concurrent loads of 150 mA each, for a total of 900 mA. If a device connected draws more than the allotted power, it will malfunction or irradicably disconnect.

The USB controller is also responsible for allotting the number of endpoints for the purpose of accepting data. The endpoints are equivalent to the number of lanes on a highway for cars (data) to travel at any given moment. If you plug in too many devices you can request more ports than are allotted for the USB controller and you will get an error, similar to Figure 7.19.

FIGURE 7.19 USB controller error



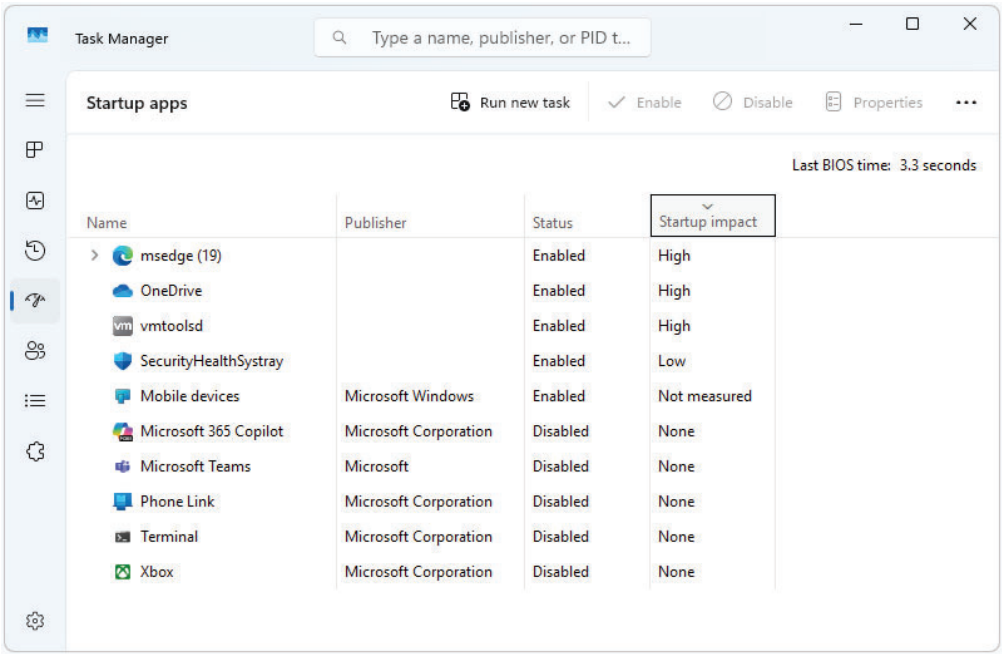
The easiest way to fix this issue is to move some USB devices around on the USB ports. You should move any devices that don't need USB 3.0 to USB 2.0 ports, such as keyboards and mice. Then ensure that your devices that require speed support are connected to USB 3.0 ports. If you are using a USB hub, make sure that you have connected it to a similar port. For example, if you are using a USB 2.0 hub, ensure it is connected to a USB 2.0 port and not a USB 3.0 port. More complex solutions are upgrading the driver to the latest driver the vendor supplies for the USB controller or just simply upgrading the hardware to a newer chipset.

Newer systems will come with a charging downstream port (CDP). These ports can provide higher current (up to 1.5 amps) to charge devices faster while still allowing data transfer. To put things in perspective, USB-C charging bricks can provide up to 100 watts (20 volts at 5 amps); these devices are also considered USB Power Delivery (PD). It is not common to find these types of ports on a laptop or computer, it is usually the interface that powers the device from the wall outlet.

Slow-Loading Profiles

A local profile is a group of settings for the user as well as their personal files. Local profiles can be slow to load because of items set to start when the profile is loaded. You can use Task Manager to selectively disable startup items, as shown in Figure 7.20. By the process of elimination and after several logouts and logins, you can narrow down the performance problem caused by slow-loading local profiles.

FIGURE 7.20 Startup items in Task Manager



Typically, local profiles will not slow down login tremendously. Roaming profiles that are located on a server typically cause slow loading of the profile. The local profiles don't need to traverse a network during login. Roaming profiles, on the other hand, need to traverse the network during login (load from the server) and logout (write back to the server).

There are some things you can do to alleviate the stress on the network and speed up the load time of network profiles. For example, you can save space by deleting temporary Internet files in both Edge and Internet Explorer. You can also save a tremendous amount of space—sometimes gigabytes—by deleting downloaded files. In addition to space traversing the network, login scripts, Group Policy processing, and services starting upon login can also contribute to slow-loading profiles.

Time Drift

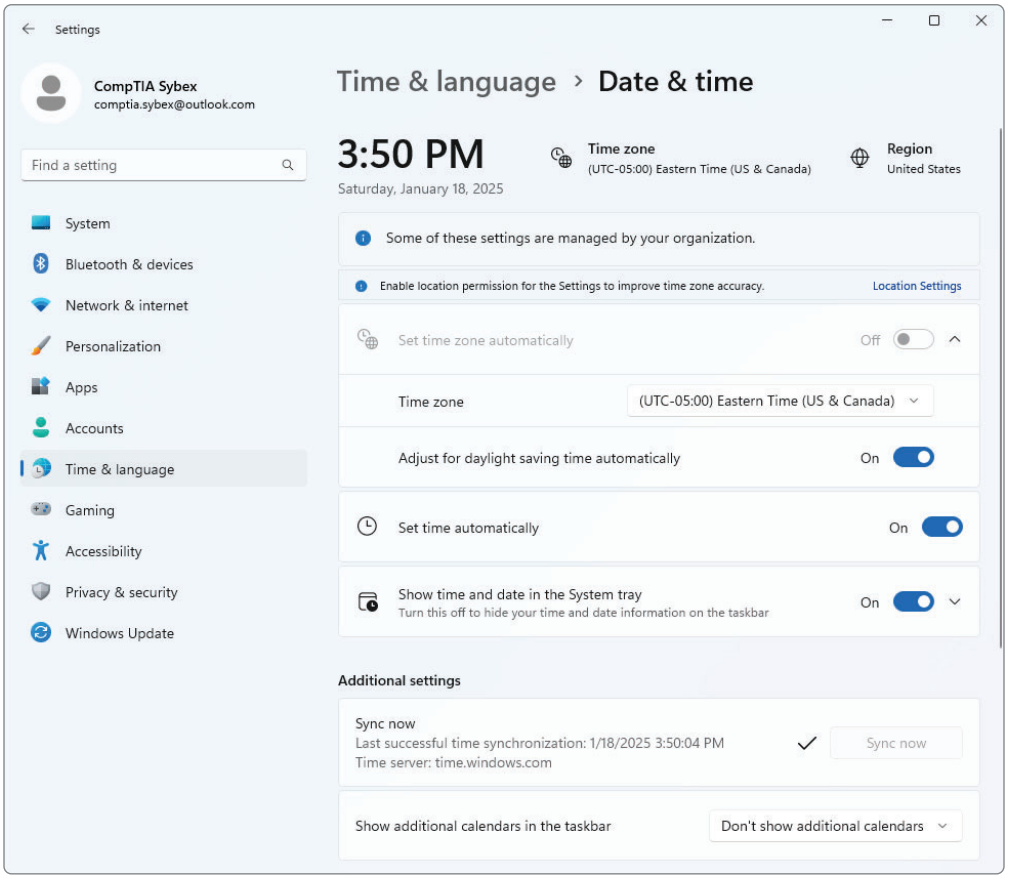
The *real-time clock (RTC)* on the motherboard is responsible for maintaining the correct time. The RTC can drift over time and the computer can become faster or slower. When an operating system is running on a hypervisor, the problem is increased tenfold, since the RTC is actually emulated by the hypervisor. When the *time drifts* too far, you can have authentication problems. If the time drifts too far, certificates can also be invalidated and you'll have problems with web browsers.

Fortunately, the Windows operating system has addressed the problems of time drift by periodically querying a Network Time Protocol (NTP) server. You will need to ensure that

the client has the ability to contact the time server of `time.windows.com`, or you will need to configure a time server the client can reach.

You can verify that the NTP server is reachable by opening the Date & Time Settings screen and trying to update time. This can be performed by clicking the Start menu ➢ Settings ➢ Time & Language ➢ Date & Time. Be sure that the Set Time Automatically option is selected. Then click Sync Now, under the Additional Settings heading. The operating system will attempt to call out to the NTP server and the results will be displayed in the dialog box, as shown in Figure 7.21.

FIGURE 7.21 Configuration of date & time





On virtual machines running on a hypervisor, the VMware guest tools or Hyper-V Integration Services will be responsible for emulating the RTC. These additions will also sync the VM's clock to the hypervisor host's time; this way, you only have to worry about the host's synchronization with an NTP server.

Troubleshooting Steps

Now that we've covered common symptoms of Microsoft Windows OS problems, let's look at solutions that you can implement to solve those problems.

Rebooting

Rebooting a system often takes care of problems, for a multitude of reasons. One of the top reasons is that it allows the operating system to terminate hung processes gracefully. After the operating system reboots, the processes are normally restarted. An added bonus is that only the applications the user requires are relaunched. The goal is to fix the user's problem with minimal disruption. A reboot can't be done every time, especially if the user could lose work as a result.

Depending on the circumstances, when reproducing a problem, one of the first things to do is reboot. It serves an important purpose of isolating the problem so that you can reproduce it. For example, if you've isolated the problem to Excel not scrolling properly when a web browser is open, you should reboot and try to replicate the problem. If you discover the reboot fixed the problem, then you've solved the problem. However, if the problem still exists, you've now isolated the problem further by eliminating other programs that could have been hung in the background affecting this problem. The steps of rebooting and then opening Excel can also be used to verify when you've solved the problem.

Restarting Services

Services normally don't need to be restarted. On occasion, however, a change is made that requires that a service be restarted to reflect the change. Services should be restarted if they crash, of course. Although this is rare, it still happens from time to time. If a service crashes, you can restart the service in the Computer Management MMC by selecting Services, then right-clicking Service and choosing Start, as shown in Figure 7.22. You can use the same method to restart a running service. The Computer Management MMC can be started by right-clicking the Start menu and selecting Computer Management.

Services can be configured to automatically start in the event of failure on the Recovery tab of the Services properties, as shown in Figure 7.23. For example, by default; the Print Spooler service is set to restart on the first and second failure, but after that it will remain stopped. The reset counter can be set for a number of days, and the service can be started after a specific number of minutes after its failure. You can even have the computer restart or run a program in the event a service fails.

Fixing Application Issues

If an application is crashing and acting erratically, it may be due to another application that has overwritten critical files used by the application, or the files may have become corrupted. In either case, choosing to repair an application will validate that it is installed properly and the process will replace any missing critical files for the application. Data files and configuration files will not be touched while the application is being repaired; only critical files (such as DLLs) will be checked and repaired.

FIGURE 7.22 Manually starting a service

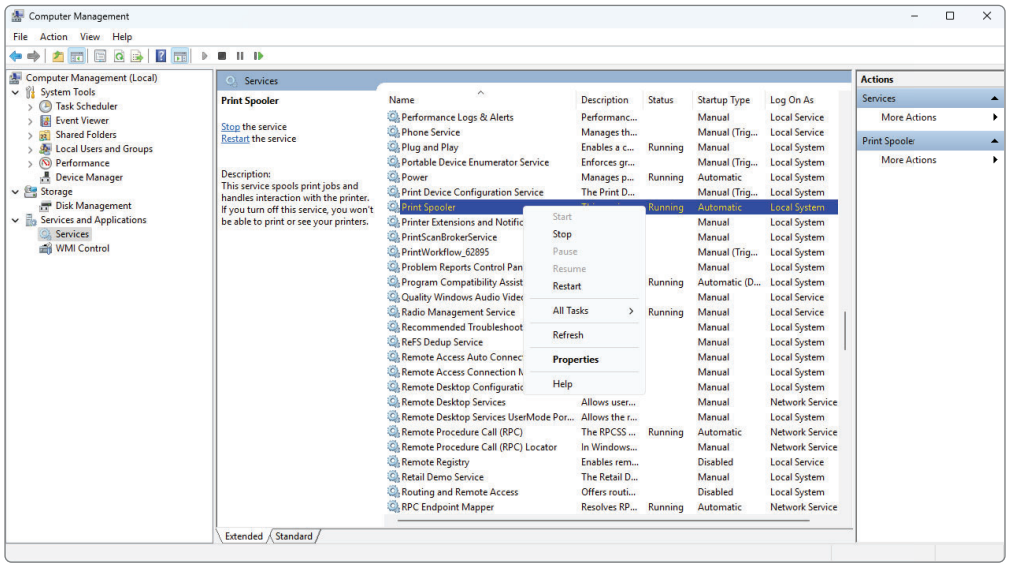
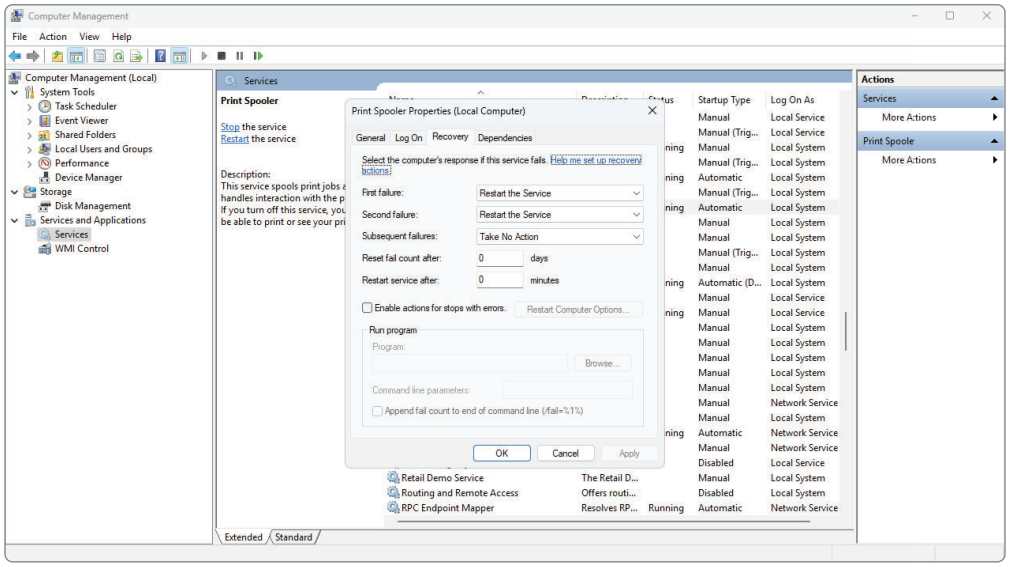


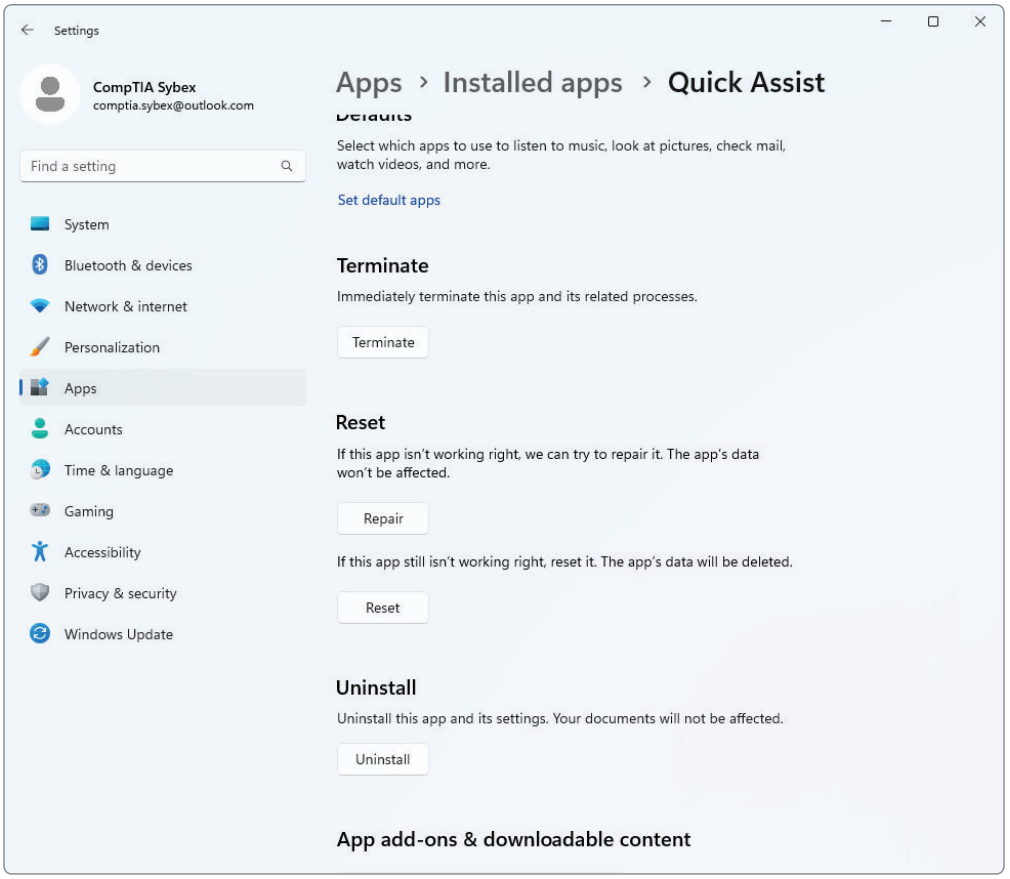
FIGURE 7.23 Service recovery



You can repair an application by right-clicking the Start menu, selecting Installed Apps, finding the application, clicking the three dots to the right of the application, selecting

Advanced Options, then selecting Repair under the Reset heading, as shown in Figure 7.24. The application's installer will launch and start to repair the application. Older applications might not have the Advanced Options menu; in that case, after clicking the three dots, select Modify. If the installer supports a repair, you will be presented with the option to repair the installation or remove it.

FIGURE 7.24 Repairing an application



If a repair does not fix the application, then you should perform a complete uninstall and reinstallation of the application. When you uninstall the application, the uninstaller should remove configuration files that could be causing the issue. Some applications require you to manually remove the configuration files, which are often in a folder of your profile, such as `AppData`. You should always contact the vendor for the specific

folders to remove to ensure the application has been completely uninstalled. Once you've completely uninstalled the application, you can reinstall the application and test to see if the problem is fixed.

If after a repair or uninstall and reinstall, the application is still broken, it could be caused by a bug. You can update the application to get the latest fixes. However, it is always recommended to identify the problem and then cross-reference it with the vendor's change log for the application.



Using the attitude "I can't break it any more than it is broke" might work for most applications. However, just blindly updating an application may cause other problems and complicate the original issue. Always try to verify if the problem is a known bug and update the application accordingly.

Application Requirements and Resources

Applications require a certain amount of RAM, storage space, and CPU speed. Some applications may also require an SSD hard drive, or a GPU with a specific speed and amount of VRAM. When searching for a solution for a problem, you should first verify the requirements for the application based on the vendor's requirements. This will establish an expectation of performance for the application on the given hardware. If the requirement is higher than the given hardware, then you will need to scale the hardware up by adding resources.

Although the hardware might meet the requirements of the application, most often the application is not the only application running on the hardware. This must be taken into consideration when trying to solve a problem.

The performance of the application might also need to be addressed. Often vendors will publish the minimum specification for the application requirements. However, when you speak with support their recommendations for your workload might be very different. This is another aspect of the solution that must be taken into consideration.

Computers are built with a finite amount of resources, such as RAM, CPU, and storage. After reviewing the application requirements and weighing the considerations of application coexistence with other applications and the current load of the application in question, you may decide to add more resources. Adding resources such as RAM, CPU, and storage is considered to be scaling up the hardware. Before adding resources, you should document the performance and the utilization of the resources. Then after you've added the resources, you should compare the current performance and utilization of resources by the application.



The System File Checker (`sfc.exe`) was covered in Volume 2 Chapter 2, "Windows Configuration" and maps to Exam Pbjective 1.5, "Given a scenario, use the appropriate Microsoft command-line tools."

System File Checks

When a feature of the operating system stops functioning or errs in a manner that makes you suspect corruption, the System File Checker tool can scan and replace critical files. The System File Checker is launched from the command line with the command `sfc.exe` and performs a myriad of functions. For example, you can execute the command `sfc.exe /verifyonly`, and the System File Checker will inspect all the critical files and verify integrity only. You can also supply the command `sfc.exe /scannow` and the tool will scan and repair any files that fail the integrity check. You can perform the same task on individual files such as `kernel32.dll`. The System File Checker tool will also allow offline repair and checks.

Using the System File Checker tool on the operating system to be inspected is pretty straightforward, as described previously. However, if the operating system can't boot, you can use the command in offline mode as well. You simply boot the Windows Recovery Environment locally from installation media or some other source. Once the Windows Recovery Environment is booted, select Advanced Options and open a Command Prompt as Administrator. Then enter **`sfc.exe /scannow /offbootdir=C:\ /offwindir=C:\Windows`**, assuming that Windows is installed on the C: drive.

Resetting Windows

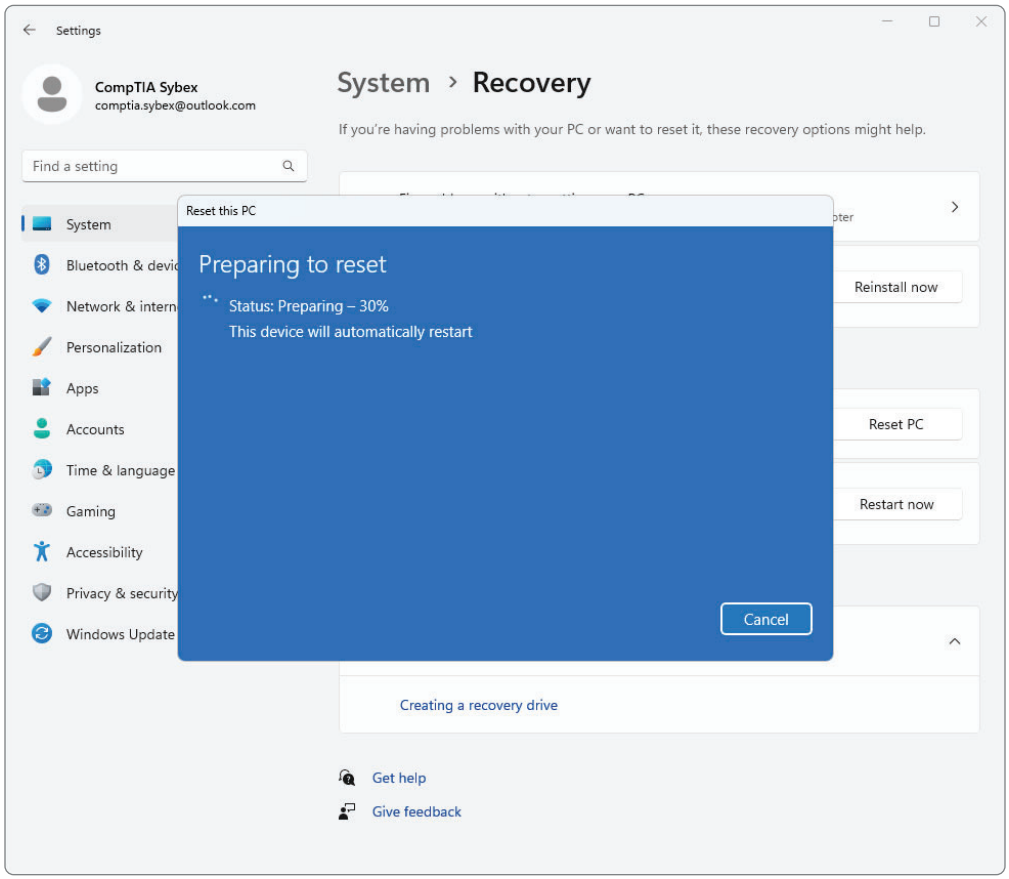
As previously discussed, the System File Checker utility will verify the integrity and replace any corrupted files. However, the utility will only replace files if they fail an integrity check. An alternate method to ensure the operating system is properly installed is to perform a repair installation of Windows. The repair installation will reinstall all files from source media regardless of their integrity. The repair installation will leave all applications and user files in place.

To initiate a repair installation of Windows, click Start ➢ System ➢ Recovery ➢ Reset This PC. You can then select to Keep My Files or Remove Everything. If you select Keep My Files, then the operating system will reinstall itself not touching any of your data. The latter of the two will completely reset the PC and all data. The next option you have is to perform the reset from a Cloud Download or Local Reinstall. The Cloud Download option will download and reinstall Windows from the cloud. The Local Reinstall option will be used to locate files to reinstall the operating system. You will then be prompted to confirm your settings and clicking Reset will begin the reset (reinstallation) of the operating system, as shown in Figure 7.25.

Restoring Windows

Almost everyone, no matter how hard they've tried to keep their computer running properly, will experience a computer crash at some point. Many of the ways to get your computer back up and running (such as reinstalling the operating system) take a lot of time. In Windows, System Restore allows you to create restore points to make recovery of the operating system easier.

FIGURE 7.25 Preparing to reset



A *restore point* is a copy of your system configuration at a given point in time. Restore points are created a couple different ways:

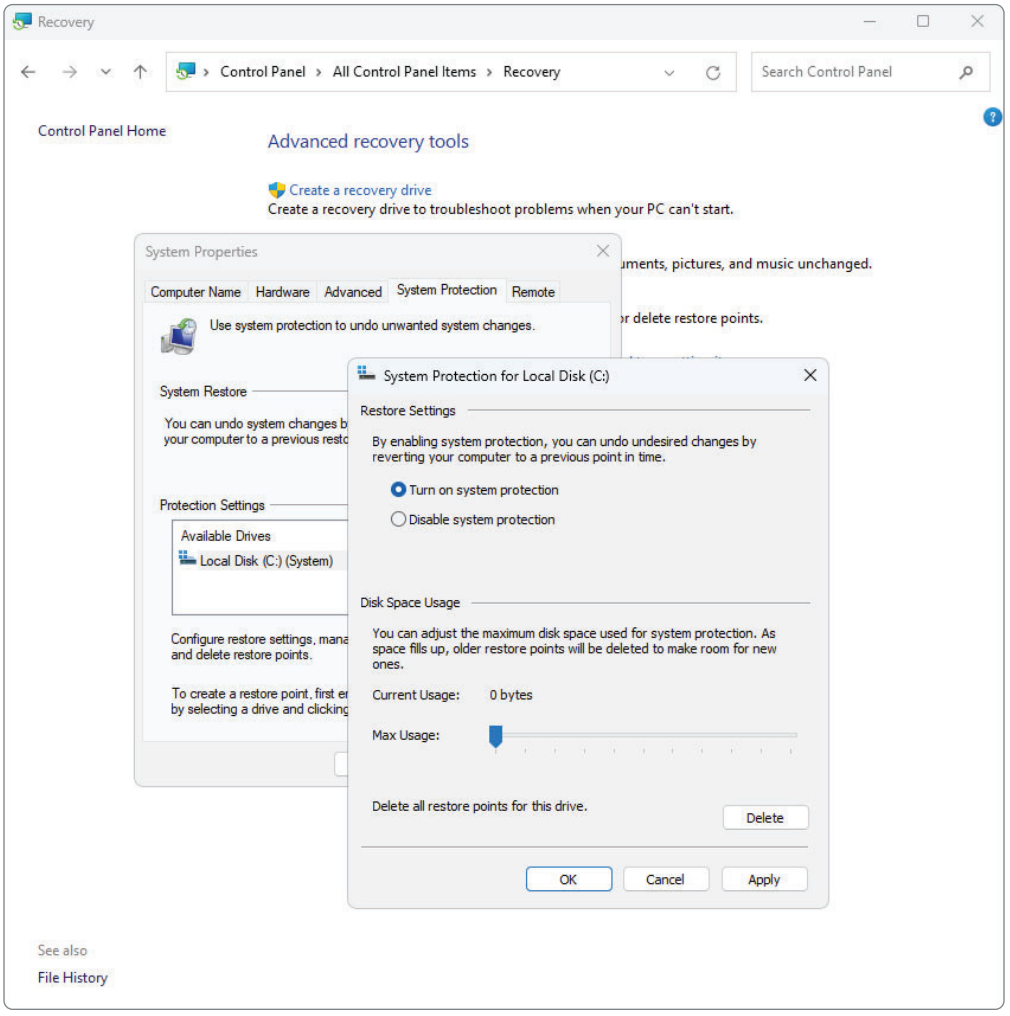
- You can manually create them yourself (which is highly recommended before you make any significant changes to the system, such as installing new drivers).
- During the installation of some programs, a restore point is created before the installation; that way, if the installation fails, you can "roll back" the system to a preinstallation configuration.

Restore points are useful for when Windows fails to boot but the computer appears to be fine otherwise, or if Windows doesn't seem to be acting right and you think it was because of a recent configuration change.

It's important to note that in Windows 11, the automatic system recovery option is disabled by default. It must be turned on manually. We will cover how to do so in Exercise 7.2.

Microsoft is moving toward unifying all settings under the Settings app for Windows 11. This differs from prior operating systems and the legacy Control Panel app. System Restore is one of those settings that can be opened only from the legacy Control Panel app. To open System Restore, click Start, type **Control Panel** and select it from the results, and then click Recovery. This will open to a list of Control Panel choices. Select Configure System Restore. This will open the System Properties, and you can then select Configure and select Turn On System Protection, as shown in Figure 7.26.

FIGURE 7.26 System Protection





If you need to use a restore point and Windows won't boot, the system will boot in the *Windows Recovery Environment*, in which a full system restore can be performed.

The other option is to select how much disk space is available for System Restore. The less disk space you make available, the fewer restore points you will be able to retain. If you have multiple hard drives, you can allocate a different amount of space to each drive.

Exercise 7.2 demonstrates how to create a restore point manually in Windows.

EXERCISE 7.2

Manually Creating a Restore Point in Windows

1. Click the Start menu, type **Control Panel**, and then click the Control Panel app result.
2. Click Recovery, if the menu item does not appear, adjust your View by from Category to Large Icons.
3. Click Configure System Restore. The System Properties dialog box will open.
4. Select Local Disk (C:) (System), and then click Configure.
5. Click the Turn On System Protection radio button, and then click OK.
6. Click Create below the Configure button.
7. Type a name for the restore point.
8. When the process is finished, click OK, and then exit out of the Control Panel windows.

Reimaging/Reinstalling the Operating System

In certain situations, a problem may require you to reinstall software. The time required to uninstall and reinstall the software can sometimes exceed the time it takes to reimage the operating system with the software preinstalled. Reimaging the computer will depend on whether you use operating system images or load each computer by hand.

If your organization does not use a standardized image for its computers, you can use the Windows Recovery Environment and select the Reset This PC option. If the computers have a preinstalled image, you can use the System Image Recovery option to reload the operating system. You can select this option by holding down the Shift key as you reboot the operating system, then choosing Advanced Options after the reboot, then selecting System Image Recovery. Depending on the type of computer you have, it may have a proprietary process for recovering system images.



A factory recovery partition may exist on a machine. A vendor has a great deal of freedom with what it puts here, but usually it contains copies of drivers and preinstalled programs. This partition—if it exists—may serve as a last resort for stabilizing a system if all else has failed.

Windows Updates

Occasionally, applying an update will fix a problem, mainly because that is what updates do: they fix problems. Once you've identified that applying an update is the solution, you need to download, distribute, and install the update. Luckily, by default Windows 10/11 automatically installs updates for the operating system to keep you up to date and problem-free.

In large-scale networks, the organization may employ a corporate patch-management solution. Microsoft offers a free patch-management solution called Windows Server Update Services (WSUS). Microsoft also sells a licensed solution called Microsoft Endpoint Configuration Manager (MECM), which performs many other functions in addition to patch management. If an update is required and your organization uses one of these products, the patch must be approved, downloaded, and deployed. Third-party patch-management solutions may also be used in your organization. Third-party solutions are usually specific to an application or suite of applications, such as Adobe or Autodesk.

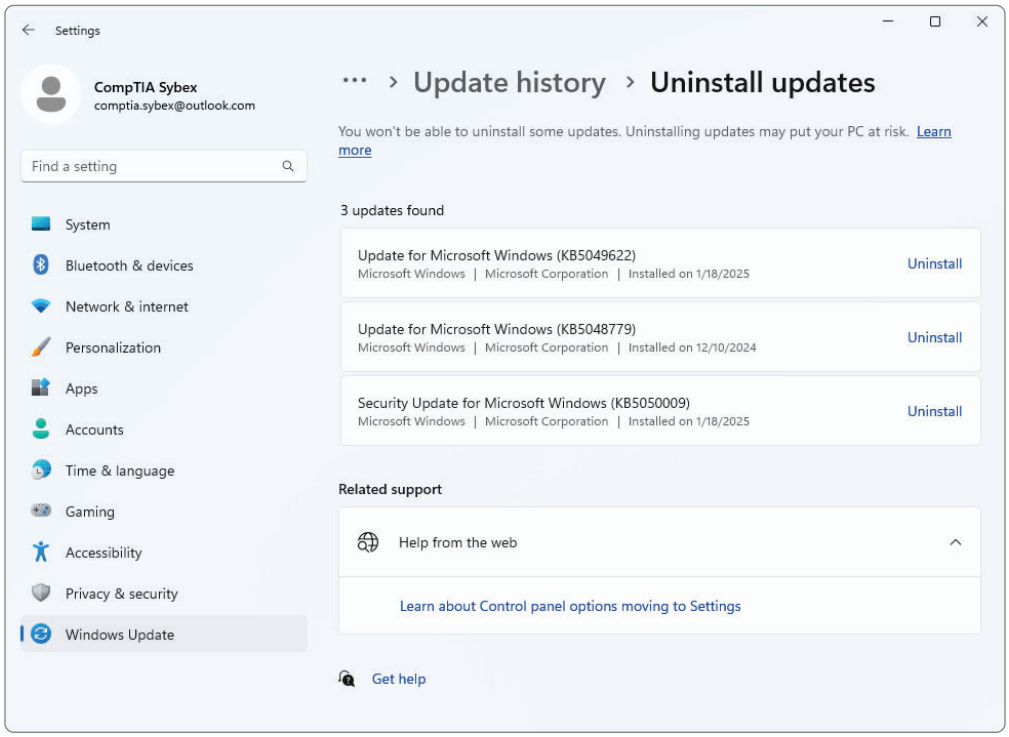
In small office, home office (SOHO) environments and small network environments, the update may be a one-off installation for a specific application. In this case, the update just needs to be downloaded and installed, per the vendor instructions. Always make sure to have a plan to roll back from a bad update. Turning on System Protection before the update is a good idea. If an update fails, you can simply use System Restore to restore the operating system to a prior point in time.

Very rarely you will find that a Microsoft or third-party update has created a problem on the operating system. When this happens, it's pretty easy to roll back updates by uninstalling them. Simply open the Settings app, select Windows Update, then Update History, then Uninstall Updates, and finally select the update and choose Uninstall, as shown in Figure 7.27.



An alternative to uninstalling an update is to use System Restore to revert to an earlier time before the update. However, System Protection must have been turned on before the update was installed. System Protection in Windows 11 is turned off by default. You can turn it on by clicking the Start menu, then typing **Recovery** and selecting it, then clicking Configure System Restore, selecting the System drive, clicking Configure, selecting Turn On System Protection, and finally clicking OK.

FIGURE 7.27 Uninstalling an update



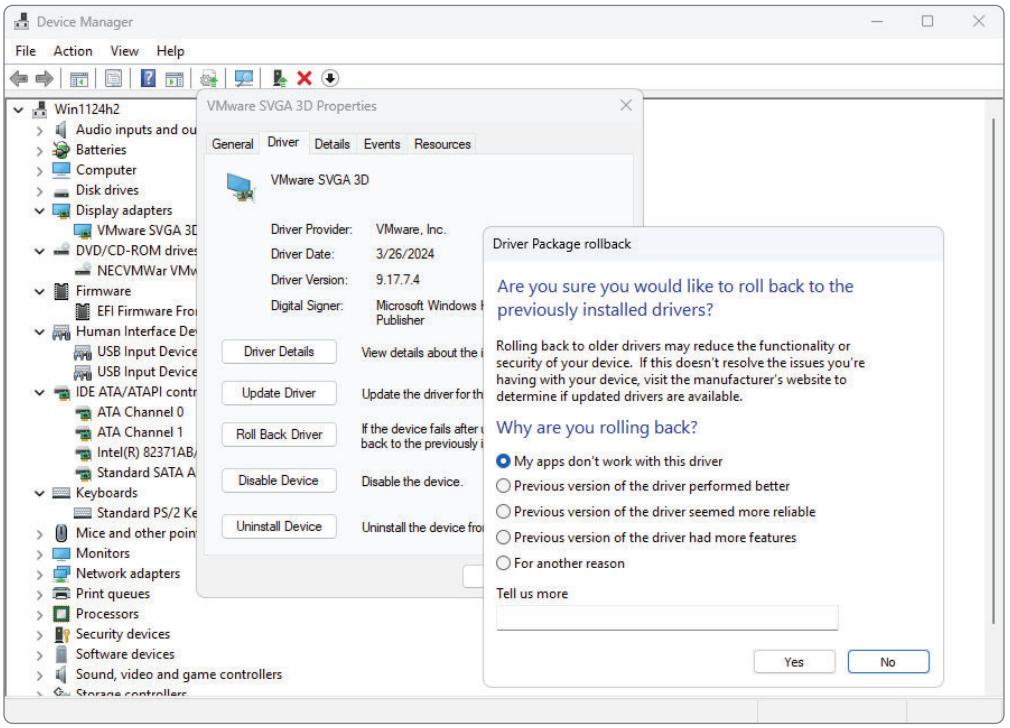
Rolling Back Device Drivers

When you isolate a hardware problem to a faulty device driver, it is sometimes necessary to roll back the current driver to a prior version. This action will roll back the driver to the original version detected by Windows, also called the *out-of-box driver*. In some cases, it may roll back to a generic driver, which reduces functionality until a proper driver is installed.

This process can be completed with these steps:

1. Right-click the Start menu.
2. Select Device Manager.
3. Select the device.
4. Right-click and select Properties from the context menu.
5. Select the Driver tab.
6. Click Roll Back Driver.
7. Provide the reason for rolling back the driver.
8. Click Yes, as shown in Figure 7.28.

FIGURE 7.28 Rolling back a driver



When the rollback is complete, you should reboot the computer before testing to see if it fixed the issue.

Rebuilding Windows Profiles

When a problem has been determined to be a profile-related issue, it is necessary to reset the Windows profile. An example of a profile-related issue is a slow loading profile. This is often related to the accumulation of non-critical files that bog down the login process. A profile reset allows the user to start with a clean profile without all the bloat of these files. When performing this action, ensure that the user's data is backed up. It is best to keep an entire copy of the profile before resetting it. The following are the most common places data is kept by the operating system:

- Contacts
- Desktop Items
- Documents
- Favorites
- Links

- Music
- Pictures
- Videos
- Saved Games (optionally)
- Downloads (optionally)
- 3D Objects (optionally)

To back up a local profile, log into an administrative account (other than the one you are backing up), and then copy the profile under `C:\Users` to a new location. Do not move the profile, because the operating system references it in the Registry.

You can then reset a local profile on the Advanced tab of System Properties, as shown in Figure 7.29. You can access the User Profiles dialog box by following these steps:

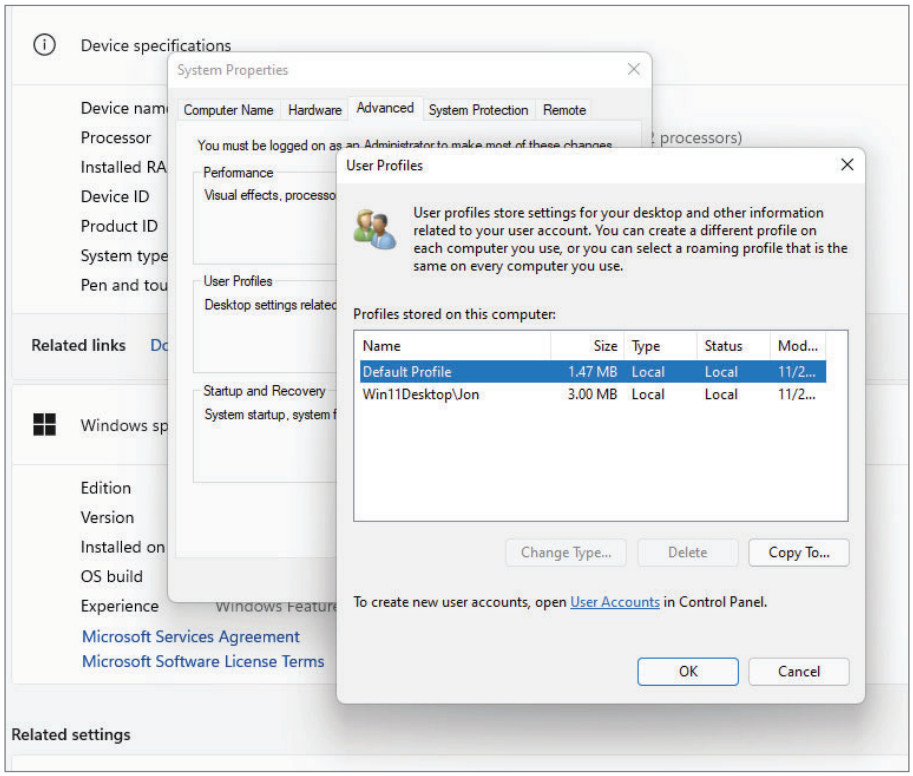
1. Click the Start menu.
2. Type **System**.
3. Click Advanced System Settings in the search results.
4. Select the Advanced tab.
5. Select Settings (User Profiles).
6. Select the profile you want to reset.
7. Click Delete, and answer Yes to confirm the deletion.
8. Log in as the user. Windows will create a new profile.

The user's files can then be manually copied over. The Profile dialog box also allows you to view the overall size of a user's local or remote profile, so it also helps in the troubleshooting process.

You can also use that procedure to delete a roaming user profile that has been left on the Windows operating system. However, performing the procedure on a roaming profile will not reset the profile. You will only remove the profile to clear space. To reset a network-based roaming profile, perform the following steps:

1. Ensure the user is logged out completely.
2. Delete all local copies of the user's profile left on any machine.
3. Navigate to the network location containing the user's profile and rename the folder.
4. Log the user into the machine on which you have deleted the locally cached copy.
5. Copy any useful items back to the user's profile.
6. Log the user out to ensure the roaming profile is saved back to the user's profile.

FIGURE 7.29 Deleting a local user profile





User profile management can become complicated, depending on how it is implemented. Exam Objective 3.1, "Given a scenario, troubleshoot common Windows OS issues," requires you to have a basic understanding of how to remedy a slow profile load. For additional information on user profiles outside of the objective, visit:

`https://learn.microsoft.com/en-us/previous-versions/windows/desktop/legacy/bb776892(v=vs.85)`



Exam Objective 3.4, "Given a scenario, troubleshoot common personal computer (PC) security issues," requires you to have a basic understanding of common operating system symptoms and browser-related symptoms as they relate to security.

Troubleshooting Security Issues

There are a number of topics CompTIA expects you to know for the 220–1202 exam as it pertains to security issues. Many of these issues also appear in other CompTIA certification exams, such as the CompTIA Security+ and other exams that have a security component. Rest assured that for the 220–1202 exam, you do not need to know the depth of content as if you were preparing for the CompTIA Security+ exam. However, you should be familiar with the following symptoms when determining if a security-related problem has occurred:

- You are unable to access the network.
- Desktop alerts mysteriously appear.
- You receive false alerts regarding antivirus protection.
- Altered system or personal files are discovered.
- Missing or renamed files are discovered.
- Unwanted notifications appear within the operating system.
- Operating systems updates fail.
- You receive random or frequent pop-ups.
- You receive certificate warnings.
- Your browser is redirected to websites.

This list is by no means exhaustive. What is an absolute is the fact that you should immediately rectify the security issue or quarantine the system if you experience even one of these symptoms. In the following sections we will cover the aforementioned security-related symptoms, as well as explore some causes for these symptoms.



It cannot be overstated that establishing security policies and procedures, updating your operating systems, updating your applications, and updating your network devices are all good measures to help eliminate potential security problems.

Common Symptoms

The operating system is by far the largest attack vector for threat agents. The software installed on the operating system and the files that the operating system stores are the perfect mixture of targets. Threat agents can target an unpatched application that contains a vulnerability. A threat agent can also sneak a file that is infected with malware onto the operating system. In the following section we will identify symptoms of common security issues along with their possible causes.

The solutions for the symptoms vary significantly. However, the overall goal is to keep your operating system protected. Protecting the operating system can be achieved by keeping

the OS current on all patches. The applications that are installed on the operating system should be kept current with patches as well. Antimalware software should be installed, and best practices should be employed, as previously discussed in past chapters.

Network Connectivity Issues

If your computer is hooked up to a network, you need to know when your computer is not functioning properly on the network and what to do about it. In most cases, the problem can be attributed to either a malfunctioning network interface card (NIC) or improperly installed network software. The biggest indicator in Windows that some component of the network software is nonfunctional is that you can't log in to the network or access any network service. To fix this problem, you must first fix the underlying hardware problem (if one exists) and then properly install or configure the network software.

In some situations, network connectivity issues can be related to security threats. Although you might not seem to have network connectivity, the NIC might be working fine, while the real problem is a malicious program that has crashed or that it is not operating as the creator of the malicious program intends. The malware will act as a *proxy* for the network traffic. This type of malware is usually intent on stealing credentials or banking information. However, it can also be used to inject ads and cause *browser redirection*.

Not all malware that causes network connectivity issues acts as a proxy. Some malware changes network settings, such as your DNS servers. This type of malware will cause browser redirections by controlling what you resolve through its DNS. It is also common for malware to change your system proxy so that all requests go through the threat agent's remote proxy.

False Alerts and Hoaxes

Users have plenty of real viruses and other issues to worry about, yet some people find it entertaining to issue phony threats disguised as security alerts to keep people on their toes. Some of the more popular *hoaxes* that have been passed around are the Goodtimes and the Irina viruses. Millions of users received emails about these two viruses, and the symptoms sounded awful. The mention of these two hoaxes serves to outline the most well-known hoaxes. Since these came out, there have been many different hoaxes, most of which were not as well known as these.

Both of these warnings claimed that the viruses would do things that are impossible to accomplish with a virus. When you receive a virus warning, you can verify its authenticity by looking on the website of the antivirus software you use, or you can go to several public systems. One of the most helpful sites to visit to get the status of the latest threats is the website for the *Cybersecurity & Infrastructure Secure Agency (CISA)*. CISA monitors and tracks threats and provides regular reports on this site.

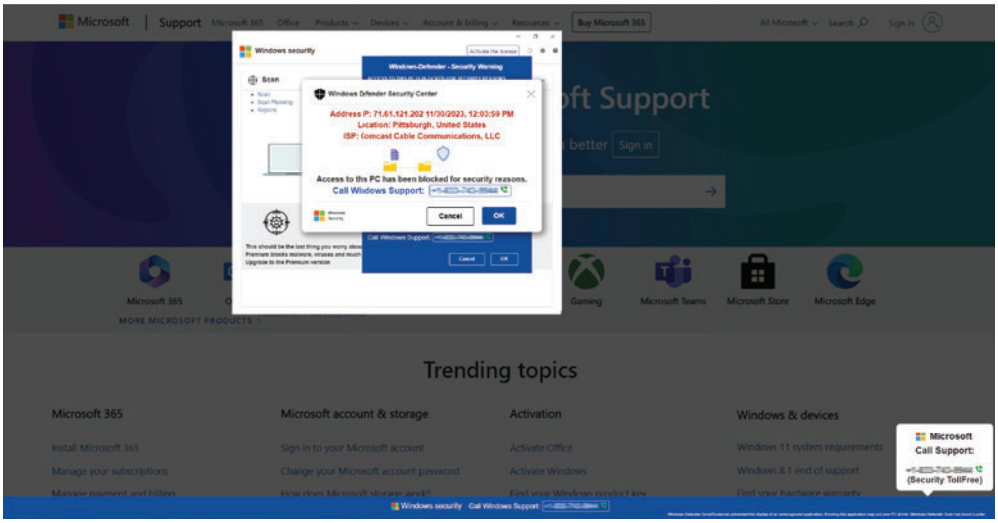
When you receive an email that you suspect is a hoax, check the CISA site before forwarding the message to anyone else. The creator of the hoax wants to create widespread panic, and if you blindly forward the message to coworkers and acquaintances, you're helping the creator accomplish this task. For example, any email that includes "forward to all your friends" is a candidate for research. Disregarding the hoax allows it to die a

quick death and keeps users focused on productive tasks. Any concept that spreads quickly through the Internet is referred to as a *meme*.

Desktop Alerts

A desktop alert is a notification or dialog box that is crafted to look like it was generated by the operating system. This is a crafty way of social engineering the user into becoming a victim. The malware is crafted to generate a pop-up box that states there is a security error detected and that you should call Microsoft or Windows Support right away, as shown in Figure 7.30. When you call the number given, you are calling scammers who will try to sell you software you don't need.

FIGURE 7.30 Malware-generated call-in alert



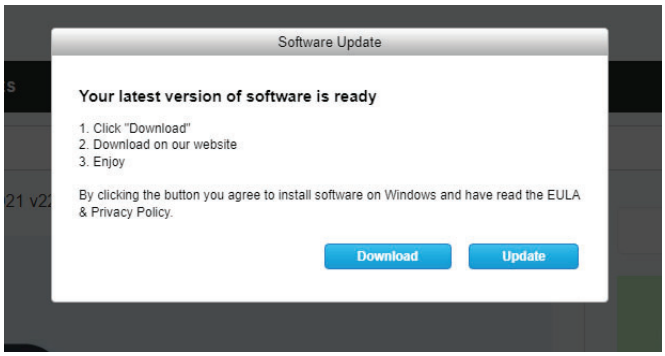
Social engineering is not the only method of a threat agent. The threat agent can generate realistic operating system dialog boxes that coax you into downloading and installing malware, such as shown in Figure 7.31. The average person might just figure it's time to update their software—it even states that, by downloading the software (malware), you agree to the EULA.

OS Notifications

Another really popular method of distributing malware is by using browser push notification messages. The user will browse to a malicious site and then the user will be coaxed into allowing push notifications for the site. Once this is allowed, the site can push notifications to the operating system and spawn a notification that looks like it's coming from the

operating system. Use of the operating system notifications is a well-known attack aimed at coaxing the user into installing malware or pushing advertising to the user.

FIGURE 7.31 Malware-generated download alert



In some cases, the initial installation of the malware is prompted by a browser push notification. After the user installs the malware, it might start prompting deals of the day or other ads. This type of malware is considered adware, and it is becoming rare compared to other types of malware covered in this chapter.

Outside of user education, antimalware software can be used to prevent this type of threat. However, user education is much more effective. A routine review of sites allowed to send notifications should be performed periodically. A routine review of installed applications should be performed as well to ensure that malware has not been installed at some point.

Rogue Antivirus

One clever way of spreading a virus is to disguise it so that it looks like an antivirus program. When it alerts the user to a fictitious problem, the user then begins interacting with the program and allowing the rogue program to do all sorts of damage. One of the trickiest things for threat agents to do is to make the program look as if it came from a trusted source—such as Microsoft—and mimic the Windows Notification Center interface enough to fool an unsuspecting user. The notification might show that a new download or update is waiting for you to install it. It may even notify you that your antivirus software is disabled and needs attention.

Education is the only way to combat a rogue antivirus. You should arm yourself with the knowledge of current antivirus programs. You can achieve this knowledge by reading consumer articles on the latest and greatest antivirus and antimalware applications. You should also pass that education to others in your family and organization. This can easily be achieved by detailing which antivirus and antimalware you have installed. Then there will be no confusion when a notification or pop-up occurs stating it's time to install the rogue antivirus malware.



Threat agents are designing malware to look very realistic and mimic big-name software. If you are unsure of the source and do not trust it, visit the site and redownload the software. Do your homework, as the threat agents are doing their homework.

Renamed System Files/Disappearing Files/Permission Changes/Access Denied

Threat actors that create malware have a number of methods by which they can wreak havoc on a system. One of the simplest ways is to delete key system files and replace them with malicious copies. When this occurs, the user can no longer perform the operation associated with the file, such as printing, saving, and so on. When malware is embedded on an operating system and gains privilege level access, it is known as a *root kit*. Once the operating system is infected with the malware, the threat actors will comb through files looking for sensitive information that they can ransom.

Just as harmful as deleting files, the malware can rename the files or change the permissions associated with them. This could prevent the user from accessing the files or even copying them off to an uninfected system. When an operating system is infected with *ransomware* the malware will encrypt the files. The files might also disappear from the user's normal view and the ransom request may be placed in the parent directory. The mode of operation for most ransomware is to rename the files with a unique extension as the malware encrypts the files. The ransom note is then placed in every folder so the user has instructions on payment for decrypting the files.

Starting with Windows Vista, Microsoft enabled the User Account Control (UAC) by default. This change to the operating system greatly reduced the number of attempts to use elevated privileges and definitely made it more difficult to change system files. In addition to enabling the UAC, Microsoft removed the Modify NTFS permission from system files for the Administrator account. Only the *Trusted Installer* (Windows Update) has access to modify these files; even the System (operating system) permissions are Read and Execute. If that wasn't enough, a self-healing service watches for files changed and replaces them with trusted versions. The System File Checker (SFC) is a user tool that can be used to manually heal missing or modified system files. Malware can maliciously modify files and, in some cases, cause them to go missing. The System File Checker was covered in Volume 2 Chapter 3, "Windows Administration," and was also discussed earlier in this chapter.

OS Update Failures

Failed updates for Windows—assuming they aren't caused by connectivity issues—can often be traced to misconfigured settings. These settings can also cause the operating system to report that an update needs to be installed when it has already been installed. The best solution is to find the error code being reported in Windows Update Troubleshooter, solve the problem, and download the update.

Recent versions of Windows 11 include a troubleshooting utility. To access the utility, click the Start menu and select the Settings app. Once the Setting app opens, select System, then click Troubleshoot, then Other Troubleshooters, and finally choose Windows Update.

Microsoft has also published some common problems that should be checked. You can access the document at `https://support.microsoft.com/en-us/windows/troubleshoot-problems-updating-windows-188c2b0f-10a7-d72f-65b8-32d177eb136c`. The solutions Microsoft recommends in the documents are as follows:

- Reboot.
- Verify free space.
- Run Windows Update again.
- Update third-party drivers.
- Disconnect external hardware.
- Check Device Manager for errors.
- Perform a system restore or repair installation.

Browser-Related Symptoms

The web browser is the most used application on the operating system. It's so popular that Google has made an operating system around their Chrome web browser called the ChromeOS. On Windows, the web browser is just an application like any other application on the operating system. However, it is the easiest way a threat agent can access your operating system. Therefore, in this section we will cover some of the most common symptoms you may observe related to the web browser.

Pop-Ups

Pop-ups (also commonly known as popups) are both frustrating and chancy. When a user visits a website and another instance (either another tab or another browser window) opens in the foreground, it is called a pop-up; if it opens in the background, it is called a *pop-under*. Both pop-ups and pop-unders are pages or sites that you did not specifically request and that may only display ads or bring up applets that should be avoided.

Most modern web browsers come standard with a pop-up blocker and by default they block all pop-ups. If you visit a link that contains a pop-up, the browser will notify you that it has blocked it. If the pop-up is on a trusted website, you will have the option to allow pop-ups for the site.

Threat agents have found other creative ways to pop up ads or referred content. Through the use of JavaScript they can serve overlays over the original web page. In older web browsers, the JavaScript could even hold you hostage at the malicious page. However, most newer browsers limit the access of JavaScript, and you can always close the web page.

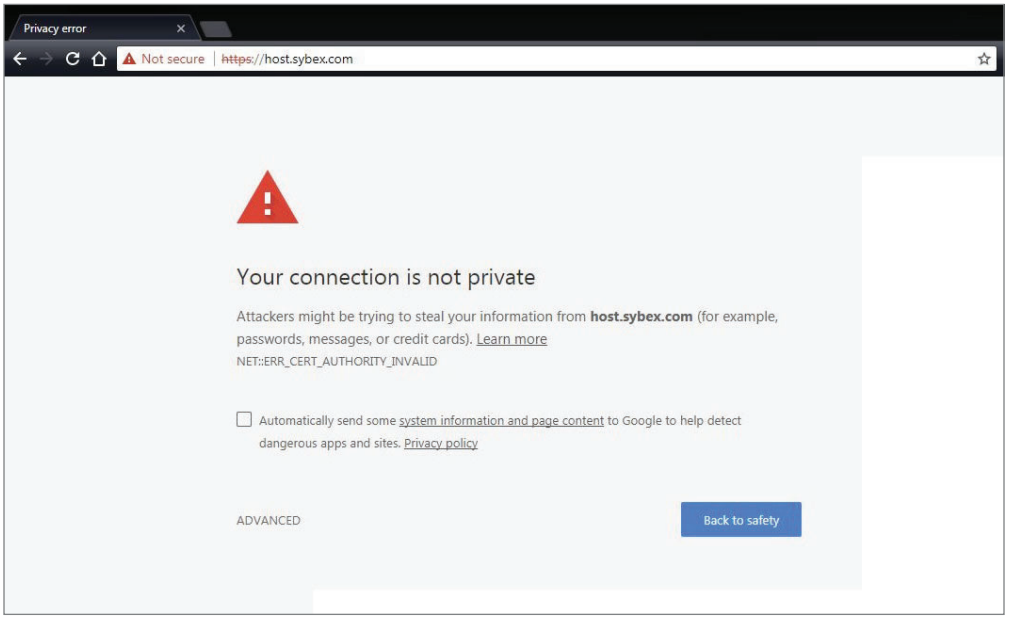
If you continually receive pop-ups or overlays, then you may be infected with malware or a rogue page is minimized serving the pop-ups/overlays. A reboot should clear the problem, but it is also best to scan your operating system with antivirus/antimalware software.

Certificate Issues

There are several problems that plague *digital certificates*. Of the two major problems, one is related to the proper setting of time and date and the other is trust related. The time on the host should always be checked along with the expiration of the SSL certificate. If the certificate is expired, this will cause problems.

On the other hand, when an untrusted SSL certificate is encountered, the web browser will alert you that the SSL certificate is not valid, as shown in Figure 7.32. Every web browser comes with a list of trusted certificate publishers. If a certificate is issued to a website or is not trusted, a warning box will come up preventing you from visiting the site. You can click through the warning prompt and visit the site anyway, but the address bar will still read "Not secure" or display an unlocked lock icon during your visit.

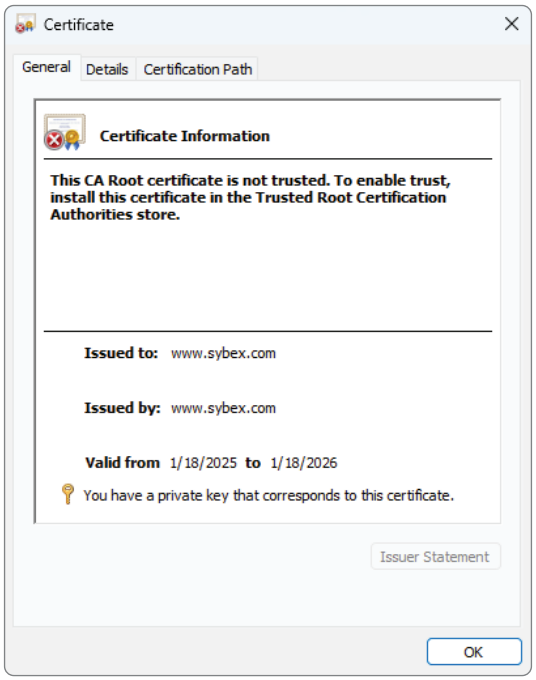
FIGURE 7.32 An untrusted SSL certificate warning



The problem should always be investigated further, since information entered in the site could be intercepted if the site was hacked. The first step to diagnose is checking the hostname in the URL. All certificates must match the hostname in the URL that they are issued for. If you tried accessing the site by the IP address, this warning is benign and can

be disregarded. However, if you entered the correct hostname, then the certificate should be inspected. Every web browser is different, but every web browser will let you view the certificate. In Figure 7.33 you can see that the certificate has been self-signed.

FIGURE 7.33 A self-signed certificate



Both the Issued To and Issued By fields in the certificate are the same. This is common when the website is in development, but it is not normal once the website has been placed into production. It is also common on network management equipment that allows configuration through a web page. Often the management web page will use a self-signed certificate. For this purpose, the certificate can be imported into your trusted publisher certificate store so that it can be trusted in the future.

Browser Redirection

Pharming is a form of redirection in which traffic intended for one host is sent to another. This can be accomplished on a small scale by changing entries in the hosts file and on a large scale by changing entries in a DNS server, also known as *DNS poisoning*. In either case, when a user attempts to go to a site, they are redirected to another site. For example, suppose Illegitimate Company ABC creates a site to look exactly like the site for Giant Bank XYZ. The pharming is done (using either redirect method) and users trying to reach Giant

Bank XYZ are tricked into going to Illegitimate Company ABC's site, which looks enough like what they are used to seeing that they give their username and password.

As soon as Giant Bank XYZ realizes that the traffic is being redirected, it will immediately move to stop it. But while Illegitimate Company ABC will be shut down, it was able to collect data for the length of time that the redirection occurred, which could vary from minutes to days.

Another form of browser redirection is called affiliate redirection. This type of browser redirection can be very subtle. For example, when you search for a product and click the link in the results, the malware will redirect your browser to the intended site with an affiliate link attached. Now anything you purchase will credit a commission to the person who redirected the browser with the affiliate link. This malware is usually related to an unscrupulous plug-in in the browser.

Because an attacker can use many different tactics to launch browser redirection, the mitigation is not straightforward. However, implementing end-user education, maintaining updates for browsers and operating systems, and ensuring that your antimalware/antivirus software is up to date are best practices to protect against browser redirection.

Degraded Browser Performance

Degraded browser performance is a frustrating experience. The symptoms are often slow loading times, frequent crashes, and unresponsive tabs. This can be caused by a number of factors, including an excessive number of open tabs, an outdated browser, or the browser needs a restart to complete the upgrade process. However, there are more insidious causes of degraded performance—for example, the presence of malicious software, such as malware, cryptominers, and web pages running malicious JavaScript, and malicious plug-ins or add-ons.

Malicious proxies are a common cause of degraded browser performance. These proxies intercept and redirect your Internet traffic through a server controlled by a threat actor. This can lead to slower browsing speeds, as your data takes a longer route to reach its destination. Additionally, malicious proxies can be used to inject unwanted ads, steal sensitive information, or even install further malware on your system. You should regularly monitor your proxy settings and ensure they haven't been altered without your knowledge.

Malware, can significantly impact browser performance. Cryptominers use your computer's resources to mine cryptocurrencies without your consent, leading to high CPU usage, increased power consumption, and overall system slowdowns. This not only affects your browsing experience but can also shorten the lifespan of your hardware, running it 100% all the time without your knowledge. Malware also comes in the form of adware, which generates intrusive ads and pop-ups. This of course results in degrading performance and potentially exposes you to more malicious content.

Malicious web pages running JavaScript are another culprit behind degraded browser performance. These web pages are often in the form of pop-unders that open behind your main browser window. The malicious JavaScript web pages often run scripts that consume system resources. They can be used to track your browsing habits, display unwanted ads, or

even execute malicious code. Because they run in the background, you might not immediately notice their presence, but their impact on performance can be significant. You should use an ad blocker and antimalware software to mitigate these issues. It's also important to keep your browser and security software up to date, to combat these shady attempts by threat actors.

Malicious plug-ins are another common cause of degraded browser performance. These plug-ins are often disguised as useful tools or extensions. Once installed, they can hijack your browser, redirecting you to unwanted websites, displaying intrusive ads, or even stealing your personal information. Just like any browser-related malware, they can also consume significant system resources, leading to slowdowns and crashes. To protect yourself, install plug-ins only from trusted sources, regularly review and remove any unnecessary extensions, and use security software to detect and block malicious plug-ins. In addition, you should monitor the permissions plug-ins have. It's a red flag for a plug-in to have complete access to see all web activity or even access to your filesystem.



Exam Objective 2.6, "Given a scenario, implement procedures for basic small office/home office (SOHO) malware removal," requires you to know how to remove malware on a small network such as a small office or home office.

Best Practices for Malware Removal

Best practices for malware removal are a key objective for the 220-1202 exam. The best way to think about this is as a seven-item list of what CompTIA wants you to consider when approaching a possible malware infestation. The following discussion presents the information that you need to know.

1. Investigate and Verify Malware Symptoms

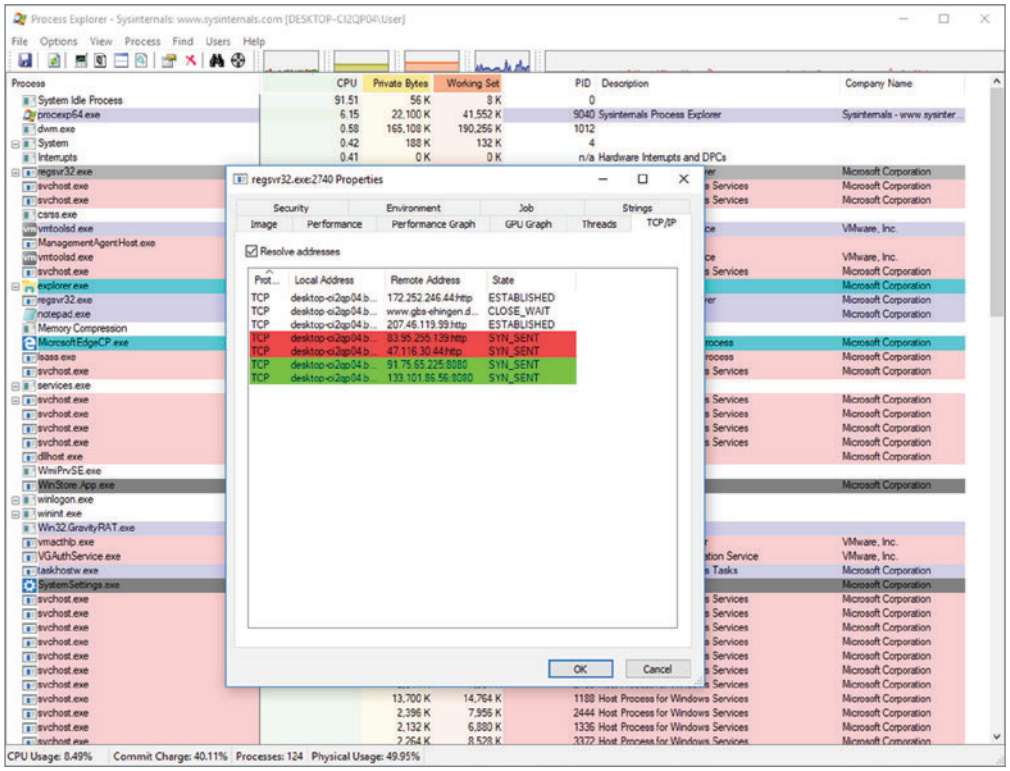
Before doing anything major, it is imperative first to be sure that you are dealing with the right issue. If you suspect malware, try to investigate the type (spyware, virus, and so on) and look for the proof needed to substantiate that it is indeed the culprit.

You first need to identify the problem. This can be done with a multitude of tools, but hopefully your antivirus/antimalware software will be the first tool that helps to identify the problem. If the antivirus/antimalware software fails to identify the problem, then other third-party tools must be used.

Earlier in this chapter, in the section "Troubleshooting Common Microsoft Windows OS Problems," we introduced you to Resource Monitor to isolate performance problems. A similar tool, called Process Explorer, can be downloaded from Microsoft Sysinternals. This tool allows a different visualization from what Resource Monitor provides, as

shown in Figure 7.34. You can see the process list on the operating system; in this case, there is a process called `regsvr32.exe`. When you look more closely, you can see that it is creating network traffic and is very active on the operating system. The process is actually a ransomware application calling out to command-and-control servers. It is sneakily disguising itself as the `regsvr32.exe` utility, which is normally used to register DLLs.

FIGURE 7.34 Process Explorer



Unfortunately, this lone example will not give you the expertise of a professional virus/malware hunter. However, it provides just one of many examples of third-party software that can help you detect and identify viruses and malware running on a computer.

Many built-in tools, such as `netstat.exe`, can also provide assistance. For example, the `netstat -nab` command enables you to view all the processes on the operating system and their network connections. Using the `netstat -nab` command is how it was identified that something looked wrong with the `regsvr32.exe` process; otherwise, the command would have looked like any other process on the operating system.

In addition to applications that can identify viruses and malware, third-party websites can aid in detection. One such website is VirusTotal (`www.virustotal.com`). VirusTotal allows users to upload potentially unsafe applications. Their service will scan the applications against more than 70 antivirus engines and report if the signature is found. It's a valuable tool to validate that you've found an application on your operating system that is malicious. Many tools, such as Process Explorer, can even check against the VirusTotal database.

2. Quarantine Infected Systems

Once you have confirmed that a virus or malware is at hand, then quarantine the infected system to prevent it from spreading the virus or malware to other systems. Bear in mind that the virus or malware can spread in any number of ways, including through a network connection, email, and so on. The quarantine needs to be complete enough to prevent any spread.

Ransomware is probably the biggest risk, since it will spread through a network rapidly and encrypt files in its path. The ransom is usually equivalent to the number of files or the total size of files. In either case, over the past eight years it has made headline news, as it has taken down extremely large companies. In one instance, the Petya ransomware even took down most of the computers in Ukraine, along with those of several other countries.

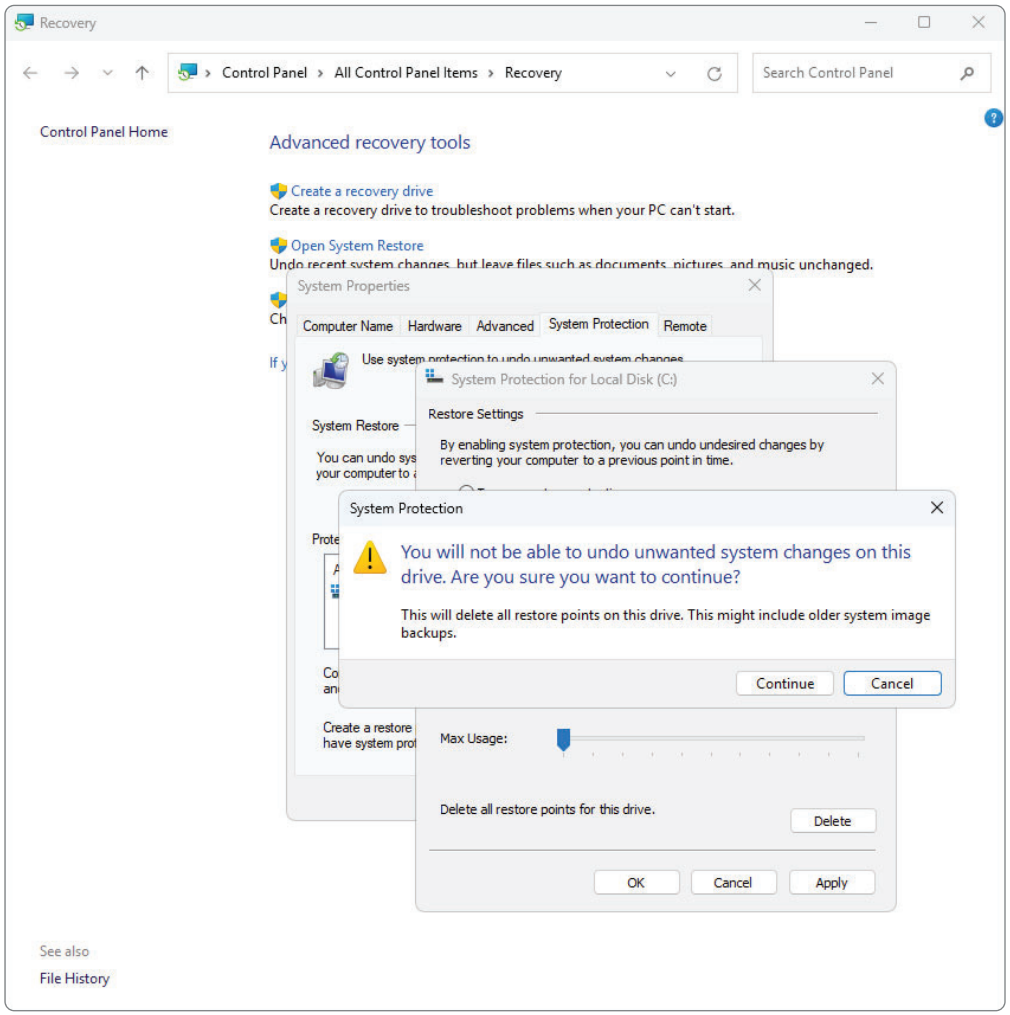
If an infected system is discovered and needs further analysis, it should be quarantined from the network and put into an isolated network. This hot network is a place where it can be studied further without repercussions to the operational network.

3. Disable System Restore in Windows

This is a necessary step because you do not want to have the infected system create a restore point—or return to one—where the infection exists. System Protection in Windows 10/11 is turned off by default. However, threat actors often turn System Protection on, so they can automatically restore your system back to when it was infected. It is best practice to disable System Protection, just in case the threat actor turned it on. You can disable System Protection with these steps:

1. Click the Start menu.
2. Type **Recovery** and select it from the results.
3. Choose Configure System Restore.
4. Select the system drive and click Configure.
5. Disable System Protection.
6. Click Delete (Disk Space Usage).
7. Click Continue (confirmation), then Close, OK, and finally Yes (confirmation), as shown in Figure 7.35.

FIGURE 7.35 System Protection





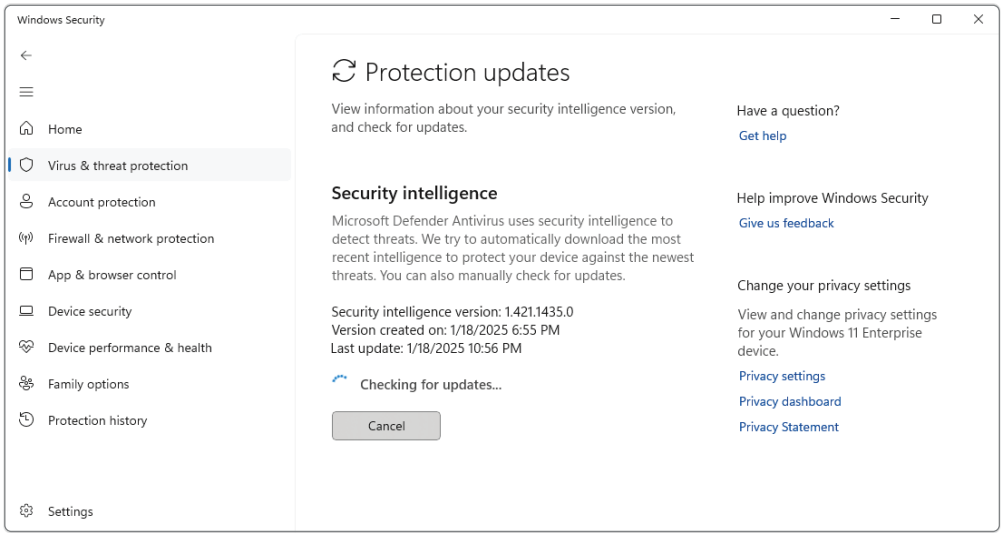
Most ransomware will dump your restore points for you, since you could potentially recover the operating system and then recover files using your previous versions.

4. Remediate Infected Systems

The steps taken here depend on the type of virus or malware with which you're dealing, but they should include updating antivirus and antimalware software with the latest definitions

and using the appropriate scan and removal techniques. You can update *Microsoft Defender* from the Windows Security screen; to access it, click Start ➢ Settings ➢ Privacy & Security ➢ Windows Security ➢ Open Windows Security ➢ Virus & Threat Protection, and select Protection Updates under the Virus & Threat Protections Updates heading. You can also click the blue shield in the task tray (Notification area) in the lower-right corner of the desktop, which will open the Windows Security screen. You can then click Virus & Threat Protection, then Protection Updates under the Virus & Threat Protections Updates heading. Then click Check For Updates, as shown in Figure 7.36.

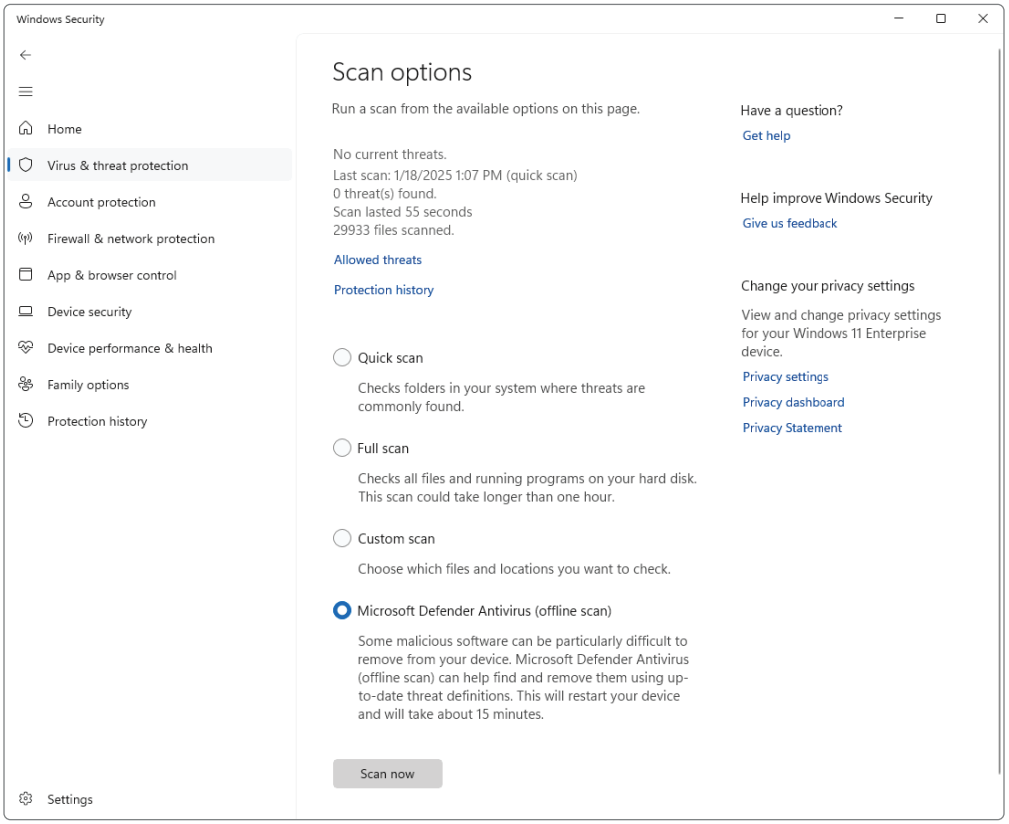
FIGURE 7.36 Microsoft Defender Antivirus Security updates



Depending on the type of virus or malware, you may need to boot into Safe Mode or the Windows Recovery Environment (as discussed earlier in this chapter). However, the remediation of the virus or malware will be different for each situation. Microsoft Defender Security can automatically perform an offline scan. To perform an offline scan, inside of the Windows Security screen click Virus & Threat Protection, click Scan Options, and select Microsoft Defender Offline Scan, as shown in Figure 7.37.

After you confirm that you will save your work by clicking Scan in the confirmation dialog box, the UAC will prompt you to answer Yes, and then Windows will reboot. The Windows Recovery Environment will boot, and Windows Defender Antivirus will run, as shown in Figure 7.38.

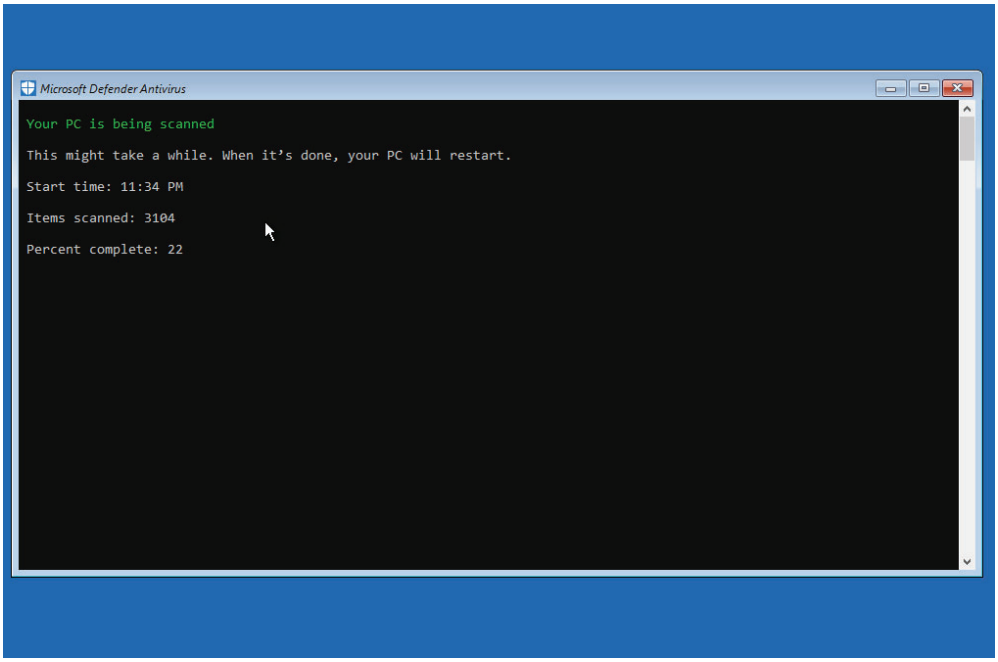
FIGURE 7.37 Microsoft Defender Antivirus (offline scan)



In some situations, such as in a ransomware attack, no remediation can be performed because the user files are encrypted. In these cases, the malware should be removed from the operating system, and then the user data must be restored from a backup. The unfortunate and terrifying fact when it comes to ransomware is that there will be loss of work.

In many instances, remediating the virus or malware is impossible because no one knows for sure what the virus or malware actually does. Antivirus researchers can document the delivery system that a virus or malware uses to enter your system. You can then patch the vulnerability, which is part of the remediation process. What antivirus research cannot do most of the time is document the payload of a virus or malware. This is because most of the time the payload is encrypted and changed, depending on the need of its creator. In these cases, the remediation might be to sanitize the drive and reinstall the operating system from an image or manually install it.

FIGURE 7.38 An offline Microsoft Defender Antivirus scan



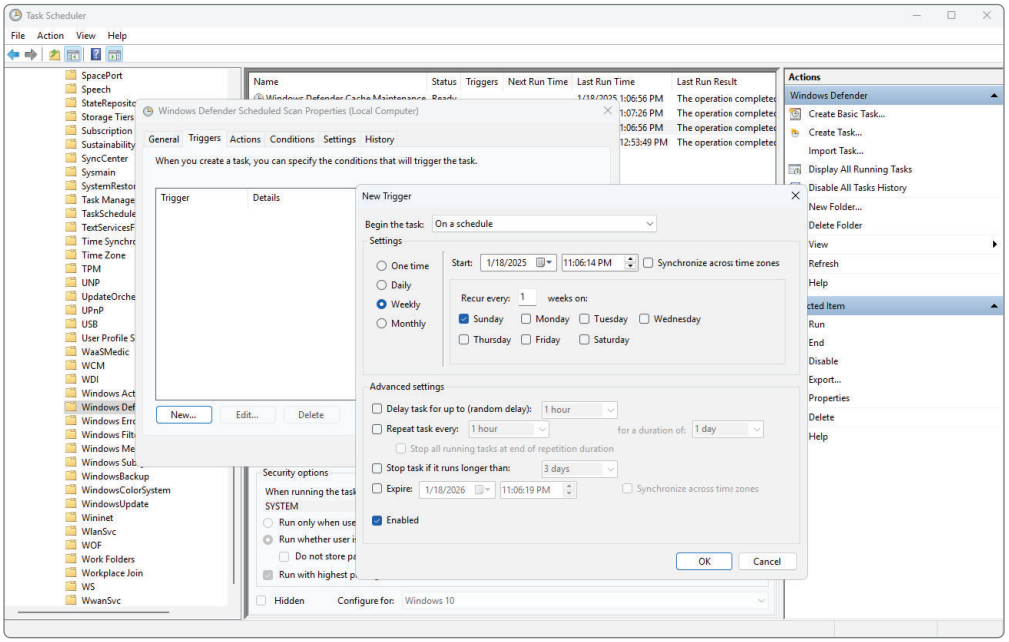
5. Schedule Scans and Run Updates

The odds of the system never being confronted by malware again are slim. To reduce the chances of it being infected again, schedule scans and updates to run regularly. Most antimalware programs can be configured to run automatically at specific intervals; however, should you encounter one that does not have such a feature, you can run it through Task Scheduler.

Microsoft Defender Security is scheduled to automatically scan the operating system during idle times. However, if you want to schedule a scan, you can use Task Scheduler:

1. Click the Start menu.
2. Type **Task Scheduler** and then select Task Scheduler from the results.
3. Open the Task Scheduler Library.
4. Select Microsoft, then Windows.
5. Select Windows Defender, and double-click Windows Defender Scheduled Scan.
6. Select the Triggers tab.
7. Click New on the Triggers tab, then select Weekly and choose the day of the week in the New Trigger dialog box, as shown in Figure 7.39.
8. Click OK, You'll need to click OK again after the New Trigger dialog box closes.

FIGURE 7.39 Creating a Windows Defender Security scheduled scan



Windows Defender Security is scheduled to automatically download updates during the Windows Update check, which is daily. If you require the latest updates, use either the Check For Updates option in the Windows Update settings or the Check For Updates option in the Microsoft Defender Security Center.

6. Enable System Restore and Create a Restore Point in Windows

Once everything is working properly, it is important to create restore points again, should a future problem occur and you need to revert back. You can enable System Protection by following these steps:

1. Click the Start menu.
2. Type **Restore** and select it from the results.
3. Click Create A Restore Point.
4. Select the System drive and click Configure.
5. Click Turn On System Protection, and click OK.

You can then manually create a restore point by clicking Create in the System Protection dialog box, typing a description (such as **after remediation - date**), clicking Close (in the confirmation dialog box), and clicking OK to close the System Properties.

7. Educate the End User

Education should always be viewed as the final step. The end user needs to understand what led to the malware infestation and what to avoid, or look for, in the future to keep it from happening again. This training can be formal training in a classroom setting, or it can be an online training in which the user must participate and answer questions.

It is common for large companies to require annual or biannual end-user training for threats. It is becoming more common for training to be done online, and a number of companies offer this as a service. It is not uncommon for a company to send a phishing attempt to their employees. When an employee falls for the phishing attempt, they are automatically signed up for mandatory training. Incentives are also common, such as the first employee who notifies the IT department of the phishing attempt gets a gift card.



Exam Objective 3.2, "Given a scenario, troubleshoot common mobile OS and application issues," requires you to know how to troubleshoot mobile OS issues. These include issues of an application failing to launch, an application failing to close or crashes, an application failing to update, an application failing to install, is slow to respond, the OS failing to update, battery life issues, random reboots, connectivity issues, near-field communication (NFC), and the screen does not autorotate.

Troubleshooting Mobile OS Issues

As mobile devices have been rapidly replacing the desktop and laptop machines that used to rule the workplace, the equipment an administrator must maintain has now evolved to cover a plethora of options. This section focuses on common mobile OS and application issues and some of the tools that can be used to work with them. A subsequent section will look at the same topics with more of a focus on security.

In the following sections, we will cover many common symptoms of problems that are common with mobile OSs and applications. In this section, we cover how to identify application symptoms as they appear in the objectives.

Application Problems

Mobile devices are generally error free and function fine. However, as we install applications to the mobile OS, we introduce potential problems. This happens because a mobile application is generally developed for the nominal platform. The application is also usually only tested on one or two devices as the developer sees fit. The developer can't account for every make and model of mobile device, and this is generally why we see application problems. In this section we will cover the most common application problems you may encounter with mobile devices. You will notice that they are all somewhat connected and usually have the same steps to rectify the problem.

Application Launch Issues

If an application (app) does not load (fails to launch), it could be attributed to a multitude of reasons. One common reason is that the application is still running in the background and is not really loading; it is just becoming a foreground application and becoming the application in focus. When you close an application, it sometimes doesn't close all the way down and free up memory. Instead, it gets moved to the background and is technically still running.

The first thing you should try if an application is not loading is to force-quit the application. To force-quit an application on an Android device, press the tab view (usually the leftmost soft button) and then swipe the application left or right to close. On Apple devices, double-tap the Home button or swipe up on the application you want to close.

Another common problem related to applications not loading is that sometimes the cache associated with the application is corrupted. This usually happens right after an application upgrades itself, which is all the time on mobile devices. On an Android phone, you can clear the application's cache by tapping Settings, tapping Apps, choosing the application, tapping Storage, and finally tapping Clear Cache. On Apple devices, tap Settings, tap General, tap iPhone Storage, choose the application, and finally tap Offload App and confirm it by tapping Offload App again. Clearing the cache will not affect the majority of the application's storage.

In many cases, an application will not allow you to clear its cache. If the option is not there, or after you've cleared the cache the application is still not loading, uninstall and reinstall the application. This option will (should) remove any data associated with the app.

You can remove Android applications by tapping Settings, tapping Apps, choosing the application, and tapping Uninstall. You can then visit the Google Play store and reinstall the application.

Apple is even simpler. All you need to do is tap and hold on an icon until all the icons dance back and forth. An *X* will be displayed in the upper-left corner of the application icon. Simply tap the *X* to uninstall the application. You can then visit the App Store and reinstall the application.

Application Fails to Close/Crashes

Another less common problem with applications is that they crash or fail to close out unexpectedly. This issue is less common since most applications run fine, but occasionally you will have a mobile application just close or crash out on you. This is frustrating, because it usually happens when you most need it, and it is generally an intermittent problem. This is how this particular problem differentiates itself from an app not loading; it doesn't crash every time.

The recommendations are the same for a crashing or closing application. Ultimately, you need to find the series of events or steps to trigger the bug and crash or close the application. Once you've reproduced the problem, it's time to try to fix the issue by doing one or more of the following:

- Force-quit the application.
- Clear the application cache.

- Clear the application data.
- Uninstall the application.
- Reboot the device.
- Reinstall the application.

If none of these solutions works, then it may be time to check the vendor's site for any similar problems (and solutions) encountered by others. Support for applications on mobile devices is normally forum or community support. However, some paid mobile applications have email-based support. Describe the issue, the device make and model, steps to reproduce the problem, and other applications installed on the device.

Application Fails to Update

You may have one or more applications on your mobile device that fail to update. Generally, the developers will backward support an old application until it doesn't make sense anymore, because of new features. The developer's expectation is that their application will be updated on all the mobile devices so that features still operate as expected. So the user will most likely see crashes, closes, or other erratic behavior if the application is not up to date.

The Google Play store and the Apple App Store manage the purchase, installation, and upgrade of applications. This upgrade maintenance on installed applications happens in the background without the user ever knowing it even happens. However, from time to time you may encounter an application that does not want to upgrade automatically. The first troubleshooting step should be to try to manually upgrade it from the Play Store or the App Store.

If manually updating the application does not work, then there are a few other steps you can take to troubleshoot the application, such as force-quitting the application and rebooting the phone to close any applications that may be stuck in memory. The next step is to temporarily disable any antivirus or antimalware software installed on the device. Then try to upgrade the application manually from the Play Store or the App Store.

Another consideration is to make sure that you are connected to the Internet via Wi-Fi. Most app store applications will treat cellular data as a metered connection and will not automatically update applications. Also make sure that the app store is configured to automatically update applications.

If all else fails and the application still doesn't want to upgrade, you can try to uninstall the application and reinstall it. You can follow the same guidelines for a crashing or closing application. Before uninstalling the application, make sure you check the compatibility for the latest version of the application. You could uninstall the application and find you can't reinstall it, because your device does not meet the minimum specifications. This could also be the original problem, which would explain why the application won't update.

Application Fails to Install

There are several reasons why a mobile app fails to install, and understanding them can help you troubleshoot the issue effectively. One of the most common reasons is insufficient storage space on your device. Mobile apps require a certain amount of free space to install.

The make and model of newer phones have a fixed amount of storage. If your device is running low on storage, the installation process may fail. To fix this problem, you can simply free up space by deleting unnecessary files, apps, or media. Additionally, moving some of your data to cloud storage can help create the needed space for the new application.

Another reason for app installation failure could be compatibility issues. Not all applications are designed to work on every device or operating system version. Typically, the mobile app ecosystem is responsible for weeding out (so-of-speak) apps that are not compatible with your mobile device. However, if the app shows up in the app store and the installation fails, it might not be compatible with your device's hardware or software. To resolve this, check the app's requirements in the app store and ensure your device meets them. Updating your device's operating system to the latest version can also help, as it may include necessary updates that improve compatibility with newer apps.

Network connectivity problems can also cause application installation failures. A stable Internet connection is required for downloading and installing apps, especially larger ones. If your connection is weak or intermittent, the download might be interrupted, leading to a failed installation. To address this, ensure you have a strong and stable Wi-Fi connection. If you're using mobile data, make sure you have sufficient data allowance and a good signal.

Performance Issues

There are a number of reasons you can have performance issues with a mobile device. Most performance issues are directly related to the applications on the device. For example, an application may use too much processing time and it could cause poor battery life and performance. A group of applications can use all available RAM and starve the unit for processing space. In the following we cover the various performance problems you may encounter with a mobile device.

Slow Performance

Slow performance is almost always related to RAM usage. Mobile operating systems operate the same as conventional desktop or laptop operating systems. The only real difference between traditional operating systems and mobile devices is the default action for an application is not to close it but to put it into the background. As programs are loaded into RAM, they allocate a percentage for their variables and inner workings. When RAM is filled up, the mobile device will swap background memory pages onto the built-in storage; this process is similar to the page file process. This slows down the device because now its focus is on clearing up memory for competing applications.

Fortunately, most mobile devices allow you to see the RAM usage at a glance and over a longer period of time. On an Android device, tap Settings ➢ Battery And Device Care ➢ Memory. You will see the memory usage for the device, along with each application and its own usage. You can also clear up memory from here, which basically just closes the applications. Unfortunately, you cannot monitor RAM usage on an Apple mobile device, but a simple soft reset works just as well.

It is uncommon to have a performance problem attributed to high CPU on a mobile device. That is not to say it doesn't happen; it's just uncommon. To narrow down problems with a particular application performing slowly, reboot the device and launch only that particular application to isolate and monitor its performance.

Frozen System

Frozen is a silly term that technicians use when something is not functioning or responding. It doesn't really describe the temperature, just the functionality—a block of ice.

If the system is frozen (not responding to a single thing), it will appear with the same symptoms as a nonresponsive touchscreen. One of the ways you can differentiate between a frozen system/lockup and a nonresponsive touchscreen is if the device will soft-reset. If the device will not soft-reset, then a hard reset might need to be performed. You should research the hard-reset procedure for the type of device you are trying to hard-reset. For example, Apple has several different types of procedures depending on the model and generation of the device. Samsung also has hard-reset procedures that differ based on model of device.

If the restart does not work, plug in the device, let it charge (an hour or more is recommended), and then try to restart it. The power level in the battery can sometimes be so low that performance is turned all the way down, and the device will appear unresponsive and frozen.

Random Reboots

Random reboots and restarts could be a symptom of a hardware issue. They can also be related to a problem in the operating system. When a problem is intermittent or random, it is very hard to diagnose. However, by checking the following you can at least rule out some of the most common culprits of random reboots:

Battery Health The most obvious culprit is the battery that powers the phone. Check to make sure that your phone has a good charge and the battery is not swollen. If your phone allows for the battery to be replaced, you should clean the contacts and replace the battery, then monitor for reboots.

Update The operating system should be on the current revision of software. If it isn't, then it should be updated to the latest. Stability issues are often addressed in updates for the operating system.

Storage Check the storage on the device and make sure that it is not over 90 percent. If it is over 90 percent, then you should clean up some space and monitor the device. You can do this by clearing data from applications, clearing application cache, and uninstalling unused applications.

Running Applications Check the applications running on the operating system. This should be checked 10 minutes after a reboot so that you can see the applications that are set to automatically start on boot. You should disable or uninstall any application

that is rarely used and automatically starts on boot.

Auto Restart On some Android phones there is an Auto Restart feature that should be turned off by default. This feature should be checked because it could have been erroneously turned on and might causing the random reboots. Apple mobile devices do not have this feature.

Factory Reset The dreaded factory reset is a last-ditch effort. It will, however, rule out the device's operating system from being the culprit of random reboots. You should also be selective in the software that you reinstall. A cyclical process of install and monitor, then install more and monitor, may help identify an application that is affecting the stability of the device.

OS Fails to Update

When an OS fails to update for a mobile device, there can be a number of reasons for the issue. However, as you will see, the common troubleshooting steps are no different than those for any other application on the mobile device.

Reboot A reboot of the mobile device is always recommended, since a process could be preventing the update from installing. A reboot of the operating system also allows for memory to be freed up since insufficient memory could be preventing the update.

Compatibility You should check to make sure that the operating system update is compatible with the mobile device hardware. It is common to find that a hardware device is only supported for 5 to 7 years, at which time it is no longer compatible with future operating system upgrades.

Storage Make sure that there is enough storage space on the mobile device to accommodate the update. When an update is to be applied to the device, at some point both operating systems will need to exist: the upgrade and the current operating system.

Connectivity Be sure that you have a Wi-Fi connection and that you are currently connected. Many operating systems will not download the update over a metered connection, such as cellular. A bad Internet connection can also prevent an OS update from completing.

If none of these suggestions allow you to successfully update the mobile device, then you can try to manually install the operating system update via the over-the-air (OTA) update. This will require a computer, and depending on the vendor, this option may not work. The vendor needs to support manual installation of updates with their third-party utility. Apple supports the iTunes application, which will work with the device to upgrade the operating system. For more information, visit `https://support.apple.com/en-us/108964`.

Extremely Short Battery Life

Batteries never last as long as you would like. Apple defines battery life as the amount of time a device runs before it needs to be recharged (as opposed to battery life span, which is the amount of time a battery lasts before it needs to be replaced). Tips for increasing battery life include keeping OS updates applied (they may include energy-saving patches), avoiding ambient temperatures that are too high or too low, letting the screen automatically dim, and turning off *location-based services*. You should also disconnect peripherals and quit applications not in use. Wi-Fi, for example, uses power when enabled, even if you are not using it to connect to the network. The following is a list of tips to conserve battery life:

- **Enable Power Saving Mode:** Most devices have a power-saving mode that reduces performance and limits background activity to extend battery life.
- **Turn Off Unused Features:** Disable Bluetooth, Wi-Fi, GPS, and NFC when not in use. These features consume a lot of power even when idle.
- **Use Airplane Mode:** When you don't need to be connected, turning on Airplane Mode can save a lot of battery by cutting off all radio signals.
- **Update Apps Regularly:** Developers often release updates to optimize battery usage. Make sure your apps are up to date.
- **Limit Background Processes:** Restrict background data usage for apps that don't need to be constantly updated.
- **Charge Smartly:** Try to keep your battery level between 40% and 80%. Avoid letting it drop to 0% or charging it to 100% frequently.
- **Use Dark Mode:** If your device has an OLED or AMOLED screen, using dark mode can save battery life as these screens use less power to display black pixels.
- **Close Unnecessary Apps:** Make sure to close apps that you are not using to prevent them from running in the background.

Outside of the preceding usage tips, sometimes battery life is attributed to a performance problem. High RAM usage can shorten battery life because power is expended on moving pages of memory in and out of RAM. This is done to keep the foreground applications running.

Overheating

When most mobile devices get too warm, they will tell you that they need to cool down before they can continue to be used, and they will automatically take measures to protect themselves (turning off features, closing apps, and so on).

One of the most concerning reasons is the lithium-ion (Li-ion) battery contained inside these devices. When a Li-ion battery gets too hot, you risk explosion or fire from a situation called *thermal runaway*. This is where the battery starts to get so hot that the separator inside the battery melts and causes a chain reaction.

Luckily, mobile devices shut down when they get too hot from ambient temperatures or internal temperature from the CPU. One of the best ways to prevent overheating is to avoid ambient temperatures that are too hot. Avoid having the device in direct sunlight for extended time periods, in a hot car on a summer day, or on top of a heat source. When the device does overheat, you can often help it cool down quicker by removing any protective case that may be there—and putting it back on later.

Mobile Device Overheating

When a mobile device overheats, be sure to understand why it is overheating. Was the ambient temperature too hot? Was the CPU usage too high and indicative of another problem? Was it a combination of the two, or did you just leave it in the direct hot sun?

When a lithium-ion battery is exposed to heat (even in small amounts), its useful life span will degrade. Most phones today have a built-in battery, which is not user-replaceable. When the battery goes bad, either you are tied to a charger every hour or it's time for a new phone.

Connectivity Issues

We rely heavily on our mobile devices, and the day-to-day functionality of our mobile devices relies heavily on connectivity to the outside world. The connectivity to the provider network (cellular) is generally supported by the provider, such as Verizon, T-Mobile, or AT&T, just to name a few. When you have a problem with the cellular network, make sure that you have coverage and a fresh reboot. However, outside of the obvious you are best to escalate the problem to your provider. The day-to-day connectivity we are responsible for as A+ technicians will be covered in the following section.

Intermittent Wireless

There are a number of reasons why intermittent wireless connections can occur, but the two most common are lack of a good signal and interference. Increasing the number of *wireless access points (WAPs)* for coverage, or being closer to them, can address the lack of a good signal.

Interference can be addressed by reducing the number of devices competing for the same channel. In many instances, however, the interference may be coming from an external source, such as a microwave oven or even a Bluetooth device on the 2.4 GHz band. To avoid common interference of this nature, use an SSID that is dedicated to the 5 GHz band. Using the 5 GHz band won't guarantee you an interference-free connection, since radar operates in this band. However, you will have better odds of selecting a channel without interference. In an effort to reduce interference and speed up wireless connectivity, 802.11ax/802.11be has been developed to use a 6 GHz band. The Wi-Fi Alliance officially released Wi-Fi 7 in 2024.

Another common problem with intermittent wireless is the auto-reconnection feature for the SSID. Your phone normally goes into a sleep mode every so often. This is normal and saves battery life. One of the first devices to sleep for battery conservation is the wireless circuitry. When you power on your phone, the wireless circuitry needs to associate with your WAP, which will happen unless you do not tell it to automatically reconnect. You can verify your wireless SSID and the auto-reconnection settings on Android by tapping Settings, tapping Connections, tapping Wi-Fi, tapping the current SSID, tapping Edit, and then making sure Auto Reconnect is selected. On an Apple device, tap Settings on your Home screen, tap Wi-Fi, tap the blue circled I next to your current SSID, and make sure that Auto-Join is on.



The auto-reconnect features can open you up to security concerns, such as the evil twin attack. The evil twin attack involves two access points; one is operated by an organization, and the other is operated by the threat agent. The threat agent will assign the same SSID to their AP, then they will send disassociation frames to the organization's AP and its clients. This forces the clients to disconnect, and if auto-reconnect is on they will reconnect to the evil twin operated by the threat agent. Information is then relayed to the network and actively sniffed for usernames and passwords.

No Wireless Connectivity

A common cause of a lack of wireless connectivity is that the wireless radio has been turned off. It happens from time to time, when an application that controls the Wi-Fi doesn't turn it back on. On an Android phone, swipe down from the status bar, then tap the wireless icon to make sure it is lit up. On Apple devices tap Settings, then tap Wi-Fi, and make sure that the slider is turned to the right and lit up in green.

Another common cause of a lack of wireless connectivity is for a device to be in *Airplane mode*. When a mobile device is in Airplane mode, all the radios for the cellular network of the provider, Wi-Fi, Bluetooth, and *near-field communication (NFC)* will be turned off. This function was created so that in one tap you could comply with the Federal Aviation Administration (FAA) or the European Aviation Safety Agency (EASA).



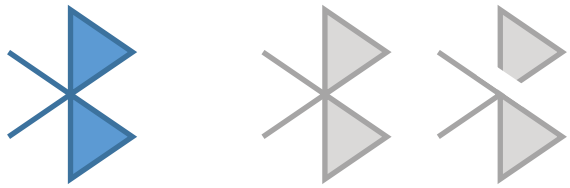
On newer iOS devices, such as iPhones, the Bluetooth is not turned off when you enter Airplane mode.

To make sure that your device is not in Airplane mode, look on the upper status bar that displays your cellular strength. If a plane appears there, then your phone is in Airplane mode. On Android, you can swipe down, then tap the icon of the airplane so that it's no longer lit up. On Apple devices, tap Settings, then Airplane Mode, and then tap the slider so that it is no longer lit up. On both Android and Apple devices, both the cellular service and wireless networks will be restored after Airplane mode is disabled.

Bluetooth Connectivity

The lack of Bluetooth connectivity can also be attributed to the use of Airplane mode, or Bluetooth can just be turned off. So, be sure to check this setting in addition to Airplane mode. On Android devices, you can swipe down, then tap the Bluetooth icon if it is not lit up. On Apple devices, go to Settings, then tap Bluetooth if it is not lit up. Depending on the phone and version of the operating system, the Bluetooth icons will be displayed on your upper status bar and will look similar to Figure 7.40.

FIGURE 7.40 Bluetooth status icons



Lack of Bluetooth connectivity can also be caused when a device is not turned on and/or has an improper setting for pairing. A common pairing issue is not having the proper Bluetooth passcode entered for the device. Each device, when paired, has a specific code from the vendor. Most vendors use a common code, such as 1234, but the code could also be 0000, or any combination, so it's best to check the vendor's documentation for Bluetooth pairing information.

To pair or re-pair a device, first ensure that the device is turned on and that it's discoverable. (Consult the vendor's documentation, as necessary.) On Android devices, tap Settings, then tap Connections, then tap Bluetooth (the phone will immediately start scanning for discoverable devices), select the available device, and enter the passcode from the vendor's documentation. On Apple devices, pairing can be performed by tapping Settings, tapping Bluetooth, tapping the device name, and entering the passcode from the vendor's documentation.

Near-Field Communication (NFC)

Near-field communication (NFC) is a short-distance wireless communication protocol. NFC is built into many mobile devices for the application of payment systems, such as Google Pay and Apple Pay. In addition, NFC is used for data exchange between mobile devices. A use case for this application is the transfer process when a new mobile device is purchased. You can simply tap the two devices together, and the new mobile device pulls the information from the existing mobile device. NFC nominally requires a distance of 4 centimeters or less to operate.

The first thing to check on the mobile device is that Airplane mode is not on. Airplane mode will impede the functionality of NFC, because NFC uses electromagnetic radio fields to enable communications between the phone and the NFC device.

The next thing to check is that it is not the reader. However, when attempting to pay for something and Google Pay or Apple Pay isn't working, anxiety often builds. Troubleshooting between you and the cashier is usually the last thing that comes to mind. However, briefly asking the cashier if anyone else has had an issue today can rule out the reader.

The case on the mobile device can also interfere with NFC communications. If the case is a ruggedized case and has an aluminum back, it could impede the NFC signal. Simply popping it out of the case can rule out the phone case as part of the problem.

Signing out of the mobile payment system will sometimes rectify the problem. In the process of signing back in, it will also validate if the provider's network is down, as this can often be a problem for mobile payment systems. Another troubleshooting step is selecting another credit card in the mobile payment system if you have multiple cards available in the app.



You can learn more about all of the various indicators for Apple devices at `https://support.apple.com/en-us/118203`. Android is an open operating system and customized by the phone vendor. Samsung has a similar article at `www.samsung.com/uk/support/mobile-devices/what-do-the-different-indicator-icons-mean-on-your-galaxy-phone`.

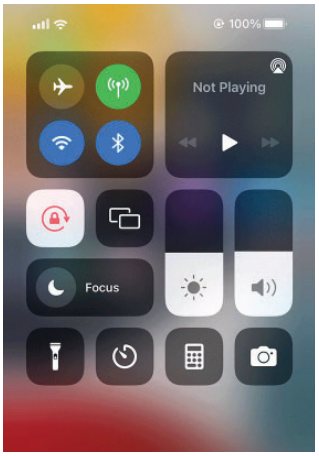
Autorotate Issues

The autorotate function allows a phone to switch between portrait mode and landscape mode by sensing how you are holding the phone. Autorotate is a feature of convenience, because no matter which way you are holding the phone, you can read the information displayed. This of course is assuming you have the screen facing you.

The first item to check is that you do not have autorotate turned off or locked. On the Android operating system there are several different ways to check this, depending on the vendor and the Android version, so it is best to check your specific model of phone. On Apple devices, this can be checked by swiping down from the top-right corner of your screen. When the Control Center opens, look for the circle with the lock in the middle and make sure it is not enabled. This is the Portrait Orientation Lock button, and if it is set, as shown in Figure 7.41, the screen will not autorotate.

If the autorotate function is not turned off or locked to a specific orientation, then you should suspect an application has possibly locked the orientation. A quick reboot will close out all running applications that could have a lock on the autorotate function. The reboot will also reset the autorotate service, in case it has crashed.

FIGURE 7.41 The Portrait Orientation Lock button



If closing the applications and rebooting the device does not remedy the issue, then you should suspect a hardware issue. There are third-party tools available that allow you to test the sensors. Ultimately, the service center can verify that a sensor is bad and malfunctioning.



Exam Objective 3.3, "Given a scenario, troubleshoot common mobile OS and application security issues," requires you to know how to troubleshoot mobile OS security issues. These include issues of security concerns and common symptoms of security issues.

Troubleshooting Mobile Security Issues

The preceding section—and its corresponding objectives—looked at mobile devices and focused on common OS and application issues; this section builds on that and focuses on security-related issues. Once again, it looks at security concerns and common symptoms, differing only in that there is more of a focus on security. It needs to be pointed out, though, that CompTIA is stretching the definition of the word *security* to include more scenarios than many would typically consider. A fair number of the issues that appear in this section would have fit easily in the preceding section.

Security Concerns

As it pertains to mobile devices there are a number of security concerns that you should be aware of. These concerns are the same for personal devices as they are for organizationally owned devices. Understanding these concerns will help you secure mobile devices and allow you to be more knowledgeable about the consequences.

Application Unofficial Sources

Unofficial sources of applications for Apple devices, referred to as third-party app stores provide an alternative to the official Apple App Store. Examples of these platforms are AltStore, TutuApp, and AppValley, just to name a few. Unofficial sources for Android devices are Aptoide, APKPure, and F-Droid, among others. These third-party app stores allow you to download and install apps that are not available through the official channels. These apps can include modified versions of popular games, apps that have been removed from the Google Play Store, or Apple App Store, or even entirely new applications that haven't been approved by Google and Apple. While these sources can offer a wider variety of apps, they come with significant risks.

One of the primary concerns with using unofficial sources is the potential for security vulnerabilities. Apps from these sources do not undergo the same rigorous review process as those on the official Play Store and App Store. This means they could contain malware, spyware, or other malicious code that can compromise your device's security and steal your personal information. Additionally, these apps might not receive regular updates, leaving them susceptible to bugs and compatibility issues with newer versions of Android and iOS.

Despite the risks, some developers choose to distribute their apps through unofficial sources to reach a broader audience or to bypass the strict guidelines imposed by Google and Apple. For example, developers might use these platforms to test new features or gather user feedback before releasing an official version. However, you should exercise caution and thoroughly research any third-party app store and the apps they offer before downloading.

Sideloading

Sideloading apps involves installing software on your device from sources other than the official app stores, such as the Apple App Store or Google Play Store. This process bypasses the standard security and vetting procedures that official stores enforce, potentially exposing your device to unverified and potentially harmful applications.

The risks are identical to using a third-party store. While sideloading can offer access to a wider range of apps, including those not available in your region or apps that provide additional features, they also carry significant risks. These include the possibility of downloading malware, missing out on automatic updates, and compromising your privacy due to excessive permissions requested by unofficial apps. Therefore, while sideloading can be tempting for its flexibility, it requires careful consideration and caution to avoid compromising your device's security and performance.

An Android package (APK) is a developer file format for installation of Android applications. When developing an Android application, the developer will sideload the application, usually using an Android tool such as Android Debug Bridge (ADB). The ADB will allow the developer to install the APK directly onto the device they are testing with.

An iOS App Store Package (IPA) is a file format used for the installation of iOS applications. When developing an iOS application, developers typically use Xcode, Apple's integrated development environment, to build and test their apps. The IPA file can be sideloaded onto an iOS device using tools like Xcode, or third-party software such as Cydia Impactor. This process allows developers to install and test their applications directly on their iPhones or iPads before submitting them to the App Store.

Developer Mode

The *developer mode* on Android and Apple devices allows a developer to connect to the device via a USB connection. A developer can then create a bridge from a computer to sideload applications as well as debug. The developer mode on Android offers myriad settings that can be changed, such as viewing running services, staying awake, setting a mock GPS location, and USB debugging, just to name a few. The Android operating system allows you to change and tweak settings that can be security concerns. Because of all these tweaks and settings changes, the developer mode is a security concern. By using the mock GPS locations, you can make your phone think that it is within a geolocation security perimeter and possibly circumvent security.

Apple's latest iOS does not have a developer mode like Android. However, you can perform development functions through the Xcode application on a Mac to help you develop iOS applications.

The development mode on Android can be accessed by navigating to Settings ➢ About Phone ➢ Software Information, then tapping Build Information seven times. The Development menu will be on the parent menu under About Phone.

Root Access/Jailbreak

The terms *root access* and *jailbreak* are synonymous with each other. However, *root access* is normally associated with Android and *jailbreak* is normally associated with Apple iOS. When you attain root access for an Android device, you literally have access as the super user named root. This access allows you to change various aspects of the operating system, such as turning on premium features like hotspot and tethering, and changing the operating system on the device by flashing new firmware.

When you root an Android phone and flash a new firmware, you no longer have the patch management from the parent vendor of the phone. For example, if you root a Samsung phone and install Havoc OS, then you no longer receive the Samsung security policies and updates. The Google Play store will also be affected, since only older versions of the applications will be available for download. This means that patch management will lag behind.

When jailbreaking the Apple iOS, you attain a higher level of access to the iOS, just as you do when you root an Android device. The motivation to jailbreak an iOS device is to access premium features, such as adding photo modes, hiding apps, and installing newer features on old devices. Jailbreaking is a security concern because you are modifying the operating system of the phone. Malicious software can easily be installed, and with the new level of access, it can hide itself.

Many organizations that employ mobile device management (MDM) create policies to prevent rooted and jailbroken devices from attaining access to organization information. This policy might restrict getting email from the device, or it could restrict access to the network in your organization.

It is best to use the operating system that has shipped with your mobile device. In many cases, once a device has been rooted or jailbroken, it cannot be reverted back to stock.

Malicious Applications

A *malicious application* is any application with malicious intent for the user or the user's device. You can find malicious applications on both Android and Apple mobile platforms. You can identify a malicious app by reviewing its permissions and contrasting it with its function—for example, if you download a camera application and it asks to record calls. There should be some suspicion and you should revoke the permissions and potentially uninstall the software.

The obvious security concern for a malicious application is that it has excessive permissions to your data or your organization's data. Periodically you should review the permissions each application installation has on your device. Another rule of thumb is to look before you click and think of the security concerns before installing software.

Bootleg Applications

A *bootleg application* is a premium application that has been cracked or nullified to remove the digital rights management (DRM). Bootleg applications can be found for a number of premium mobile apps; they generally are in the form of an APK. Bootleg applications usually contain malicious software, because that is how the bootlegger makes their money. This obviously goes back to the discussion of verifying the source of the application and being cautious with APK installations.

Application Spoofing

Application spoofing is the act of a malicious application spoofing a legitimate application. Application spoofing is much more prevalent in the mobile application marketplace and can be observed on both Apple and Google mobile platforms.

The security concern for application spoofing is the possibility of installing malicious software on your device. Disguising itself as a legitimate application, it has the same security concern of access to personal data and organizational data. You can prevent application spoofing by verifying the name of the publisher, the icon for the application, and the number of installations. If you are downloading a social media platform and it has only 100 downloads, this should send up a red flag. Another method of validating the application is to read reviews for the application. Again, think before you click and download the software.

Common Symptoms

The following sections discuss common symptoms of problems with mobile operating systems and application-related security issues. As with so many issues involving troubleshooting, common sense is most important. Using logic and a systematic approach, you can often identify and correct small problems before they become large ones.

High Resource Utilization

High resource utilization can be a telltale sign that a device is running more than you think it should be—perhaps the drives are being searched or the camera is recording your every move. Monitor for high resource usage. If you discover it, find out what is causing it and

respond appropriately. In this section I will cover some basic components and what you should look out for.

High Network Traffic

A higher than normal amount of traffic can be a symptom of a security issue. Spikes in traffic for extended periods of time can mean that data is being stolen from your device or relayed through your device. You should have an idea of the volume of traffic you would normally expect on your device.

You can start closing applications as you watch the volume of traffic. When the volume of traffic subsides, you probably have your culprit. Then, as covered previously, check the application's permissions to see if the application is malicious or is compromised in some way. Clear cache and data, then uninstall and reinstall the application from a known good source. This procedure may identify the issue or verify that you had a malicious application installed. A telltale sign is if the application is no longer available.

Data-Usage Limit Notification

Exceeding the limits on data plans can also be symptomatic of a security issue. Data usage coincides with the volume of traffic previously discussed. A malicious application running on the device could be used to send *spam* or malware, or to conduct a multitude of other malicious activities from your device. A malicious application can also continually spy on you and your data. All of these activities can rob you of precious data in your data plan, pushing you over your contracted limits.

Excessive malicious use of data on a mobile device can be mitigated with two methods:

- The first method is watching the normal usage of data from month to month. Identifying a normal baseline of usage can alert you when data usage is abnormal.
- The second method is to use a mobile firewall, which limits the traffic leaving the mobile device.

Sluggish Performance

While applications, normal usage, and so on can contribute to *degraded performance*, another offender could be malware or a virus. When you observe degraded response time on your device, you need to investigate the symptom, as it could indicate a security issue. Check RAM and CPU usage, and if an application is out of control, it could be infected with malware. It is best to run an antivirus/antimalware scan on the device to check it thoroughly.

Not every problem is related to a possible security threat. The normal search for a cellular signal can be just as taxing on the device. However, if you are in the normal locations in your day-to-day travels, such as work and home, and still experience degraded performance, you may have an application problem (out of memory) or a security threat. In either case, the issue needs to be checked out quickly.

Limited Internet Connectivity

When you have limited Internet connectivity on your mobile device, you should not immediately think that the limited connectivity is a security symptom. Limited Internet connectivity can be a result of many different problems. Mobile devices are very susceptible to limited

Internet connectivity because they contain small transceivers for wireless and cellular communications. The radio firmware also plays a big role in choosing the right radio frequency and is often the problem with connectivity.

Taking everything into consideration, if you know nothing has changed in the wireless environment, your firmware is the same, you've rebooted, and no one else is having a problem, you can suspect this is a security symptom. Malicious applications will often monopolize your connection or proxy the connection in an effort to sniff usernames and passwords. Both monopolization of the connection and proxying of the connection can create intermittent Internet connections. If your connection is being monopolized, then you will see high network bandwidth as previously discussed. There are a number of ways that a network connection can be proxied for malicious purposes, such as DNS proxy, network transmission, and wireless, just to name a few. The ideal way to combat this issue is with a mobile device firewall and antimalware software.

No Internet Connectivity

If all the usual causes have been reviewed, then you should suspect no Internet connectivity to be a security-related symptom. In some rare instances, malware will cause the mobile device to have no Internet connectivity. This generally happens because a DNS server that the mobile device is pointed to for malicious reasons has ceased to function or the relay server that is proxying the connection has ceased to function. There are a multitude of reasons why no Internet connectivity would be experienced. As previously recommended, a good mobile device firewall and antimalware should be employed.

Complete failure of an Internet connection is easier to diagnose than an intermittent problem. Therefore, a factory reset of the device should aid in figuring out whether the hardware is bad or the problem is software, such as malware, triggering the problem.

High Number of Ads

When a mobile device is experiencing a high number of ads, this is a security-related symptom of adware. Adware is a type of malware that pops up ads for malicious purposes, usually to entice the user to buy something. Adware is usually the result of installing a malicious application on the mobile device. There are two ways to diagnose the problem of adware. The first depends on the number of ads and the frequency of the ads. You can start by uninstalling applications until the ads stop popping up. This method is preferable because it is probably the quickest.

The second method, a factory reset, is much more effective, but it will not identify the malicious application. A factory reset will remove any malware along with all the applications. Of course, you can start installing applications until the ads start. However, resetting a phone is a pretty anxiety-producing process for avid mobile device users frantically trying to sign back into applications. Therefore, the first method may be preferred. Once the application is uninstalled and the device is factory reset, be sure to install antimalware software on the device.

Fake Security Warnings

A *fake security warning* on any system is a big red flag and a security symptom. Mobile devices are not exempt from fake security warnings, although these warnings on full operating systems are more common. Regardless, when a fake security warning is

discovered, it should be treated as if malware is installed on the device. The device should be factory reset and antimalware should be installed prior to reinstalling the applications.

Unexpected Application Behavior

Unexpected application behavior is not always an indication that you have been infected with malware or have a security symptom. Applications have unexpected behavior all the time. However, when an untrusted and newly installed application behaves in an unexpected way, this could be a symptom of a security problem with the application.

When you experience unexpected application behavior, you should immediately question the trust of the application. This can be done by reading reviews for the application to determine if others have run into similar problems. Also judge the application by its installer base, which is proportional to the reviews. For example, if an application has 100 installs and only has 5 people commenting that it's a great application, then this app should fall under suspicion.

The first step to be taken is to scan the device for malware. If the application is flagged as malware, then a factory reset should be performed. Then install only the trusted applications that you use daily.

Leaked Personal Files/Data

When authorized users access devices through unintended connections or unauthorized users access stolen devices, they can access the data on the device. Outside of these risks, there is always the risk of loss or theft of the device itself.

Therefore, security for mobile devices should be applied in a layered approach. Antivirus and antimalware software should be installed on the device to protect it from malicious applications. In addition, a mobile firewall should be installed along with the antivirus and antimalware software. Fortunately, there are third-party security suites that can protect you from all these threats.

Mobile device management (MDM) software should also be employed. This software is like the Swiss army knife of security for mobile devices. It can require passcodes, the installation of antivirus, antimalware software, mobile firewalls, current updates, and so much more. One of the most notable features is the ability to remotely wipe the device in the event it is stolen or lost.

In addition, there should be a firm policy that details the encryption of data in use, at rest, and in transit. A written policy should be drafted along with procedures on how to deal with leaks when they occur. These policies are usually drafted with an insurance company in order to protect an organization in case of a data leak of personal information.

Summary

This chapter addressed systematic approaches to working with computer problems as well as troubleshooting operating systems and resolving security-related issues. In our discussion of troubleshooting theory, you learned that you need to take a systematic approach to problem solving. Both art and science are involved, and experience in troubleshooting is helpful but not a prerequisite to being a good troubleshooter. You learned that in troubleshooting, the first objective is to identify the problem. Many times, this can be the most time-consuming task.

Once you've identified the problem, you need to establish a theory of why the problem is happening, test your theory, establish a plan of action, verify full functionality, and then document your work. Documentation is frequently the most overlooked aspect of working with computers, but it's an absolutely critical step.

Next, we discussed operating system–related troubleshooting issues. First, we looked at common symptoms, and then we discussed some tools that can be helpful in solving problems.

Finally, we looked at security-related troubleshooting issues as well as best practices for removing malware. Again, we started by looking at common issues and then how to solve them.

Exam Essentials

Know the common symptoms and solutions for operating system problems. Be able to identify the five critical systems of CPU, RAM, network, disk, and graphics that interact with an operating system. Know the various symptoms that will lead to operating system problems. Also, be able to identify solutions to operating system problems.

Understand the various security issues related to operating systems. Be able to identify common security issues that you will see affecting operating systems. You should also understand the impact of these security issues on both the operating system and the organization, as well as common ways to mitigate and solve these security issues.

Know the process for removing malware. Know the seven steps to remove malware. Be able to identify how to perform these seven steps during the malware-removal process. You should be familiar with the various tools used to identify and remove malware.

Know how to troubleshoot mobile OS and application issues. Be able to identify the common symptoms of application-based issues for mobile devices. Know how to solve mobile issues for applications on both Android and Apple devices.

Understand mobile device security issues. You should have a good understanding of the various mobile device security symptoms, problems, and solutions. Be able to articulate how these issues pose a risk to the organization, as well as how these issues can be mitigated and solved.

Review Questions

The answers to the chapter review questions can be found in Appendix A.

1. In Windows, which utility is responsible for finding, downloading, and installing Windows patches?
   - **A.** Device Manager
   - **B.** Microsoft Management Console
   - **C.** Download Manager
   - **D.** Windows Update
2. Which Startup Setting option allows you to boot with basic drivers?
   - **A.** Enable Debugging
   - **B.** Enable Safe Boot
   - **C.** Disable Driver Signature Enforcement
   - **D.** Enable Low-Resolution Video
3. Which `bootrec` option can be used in Windows to rebuild the boot configuration file?
   - **A.** `/fixboot`
   - **B.** `/rebuildbcd`
   - **C.** `/scanos`
   - **D.** `/fixmbr`
4. What is the first step in malware removal?
   - **A.** Quarantine the infected system.
   - **B.** Identify and verify the malware symptoms.
   - **C.** Remediate the infected system.
   - **D.** Educate the end user.
5. Which tool will allow you to troubleshoot a slow-loading profile?
   - **A.** Profile tab of the Advanced System Properties
   - **B.** Regedit
   - **C.** Windows Recovery Environment
   - **D.** Windows Preinstallation Environment
6. Which of the following components are only used to restore Windows from a suspended state?
   - **A.** BCD
   - **B.** `ntoskrnl.exe`
   - **C.** `winload.efi`
   - **D.** `winresume.exe`

7. One of the users you support has a Windows 11 laptop that will not boot up. The user just installed brand-new drivers for a graphics card. They need to access a tax application and their data files. What should you try first?
   - **A.** Use System Restore.
   - **B.** Use Reset This PC.
   - **C.** Reimage the laptop.
   - **D.** Manually reinstall Windows 10.

8. Which partitioning type is required when you have UEFI firmware?
   - **A.** GPT
   - **B.** MBR
   - **C.** POST
   - **D.** Boot Sector

9. Which of the following are used to prevent pop-unders from appearing?
   - **A.** Antimalware utilities
   - **B.** Pop-up blockers
   - **C.** Phishing sites
   - **D.** Antivirus software

10. In general, how often should you update your antivirus definitions?
    - **A.** Weekly.
    - **B.** Monthly.
    - **C.** Daily.
    - **D.** Antivirus definitions do not need to be updated.

11. Which tool can be used to diagnose why Windows 11 is slow and sluggish?
    - **A.** Resource Monitor
    - **B.** `msconfig.exe`
    - **C.** Device Manager
    - **D.** Reliability Monitor

12. Which tool will allow you to diagnose why Windows Update keeps failing?
    - **A.** `ntbtlog.txt`
    - **B.** Windows Update Troubleshooter
    - **C.** Windows Recovery Environment
    - **D.** Safe mode

13. Which of the following programs could be considered antimalware?
    - **A.** Windows Defender Security
    - **B.** MDM
    - **C.** Windows Action Center
    - **D.** VirusTotal

14. Which of the following tools allows you to manually fix maliciously modified system files?

A. `regedit`

B. SFC

C. `bootrec`

D. UAC

15. Which of the following can you do to help eliminate security problems? (Choose the best answer.)

A. Establish security policies and procedures.

B. Optimize drives.

C. Prevent booting into Safe Mode.

D. Prevent booting into Windows Recovery Environment.

16. A mobile device is running out of RAM. What could be the most likely problem?

A. The device is not charged to capacity.

B. The digitizer is not functioning properly.

C. The device is in DND mode.

D. The device has background applications open.

17. What is a risk of using the auto-reconnect feature on a mobile device?

A. The device will reconnect to any SSID.

B. The device could be exploited by an evil twin attack.

C. The device's battery life could be shortened.

D. You may exceed your cellular data plan's limits.

18. You notice that the reliability of the operating system has diminished in Reliability Monitor. Where can you find more details on why applications are failing?

A. Device Manager

B. Event Viewer

C. Windows Recovery Environment

D. `msconfig.exe`

19. Why would the operating system write out large amounts of RAM to the page file?

A. The CPU is running high on utilization.

B. This is a normal process of the operating system.

C. The amount of physical RAM is low.

D. The page file is faster than conventional RAM.

20. What is one consequence of an overheating mobile device?

A. Higher RAM usage

B. Degraded battery life

C. Inaccurate touchscreen response

D. Inability to decrypt emails

Performance-Based Question

You will encounter performance-based questions on the A+ exams. The questions on the exam require you to perform a specific task, and you will be graded on whether or not you were able to complete the task. The following requires you to think creatively in order to measure how well you understand this chapter's topics. You may or may not see similar questions on the actual A+ exams. To see how your answers compare to the authors', refer to Appendix B.

List, in order, the seven best practice steps associated with malware removal.



Chapter

8

Scripting, Remote Access, and Artificial Intelligence

THE FOLLOWING COMPTIA A+ 220-1202 EXAM OBJECTIVES ARE COVERED IN THIS CHAPTER:

✓ **4.0 Operational Procedures**

✓ **4.8 Explain the basics of scripting.**

- Script file types
  - .bat
  - .ps1
  - .vbs
  - .sh
  - .js
  - .py
- Use cases for scripting
  - Basic automation
  - Restarting machines
  - Remapping network drives
  - Installation of applications
  - Automated backups
  - Gathering of information/data
  - Initiating updates
- Other considerations when using scripts
  - Unintentionally introducing malware
  - Inadvertently changing system settings
  - Browser or system crashes due to mishandling of resources



✓ **4.9 Given a scenario, use remote access technologies.**

- Methods/tools
  - RDP
  - VPN
  - Virtual network computer (VNC)
  - Secure Shell (SSH)
  - Remote monitoring and management (RMM)
  - Simple Protocol for Independent Computing Environments (SPICE)
  - Windows Remote Management (WinRM)
  - Third-party tools
    - Screen-sharing software
    - Videoconferencing software
    - File transfer software
    - Desktop management software
- Security considerations of each access method

✓ **4.10 Explain basic concepts related to artificial intelligence (AI).**

- Application integration
- Policy
  - Appropriate use
  - Plagiarism
- Limitations
  - Bias
  - Hallucinations
  - Accuracy
- Private vs. public
  - Data security
  - Data source
  - Data privacy



CompTIA has identified that with the rapid adoption of cloud-based services, system administrators have a need for scripting and remote access now more than ever. Scripting allows you to manage a system as if you have logged in and performed the task yourself, such as purging old log files from a web server to make space. Remote access technologies allow you to connect to these remote systems so that you can manually administer them.

This chapter looks at a variety of scripting languages. After completing this chapter, you may not be an expert at writing scripts—the goal is to familiarize you with the various scripting languages so that you can learn the basics of scripting and their purposes. We'll also look at various remote access methods and their security considerations.

A new addition to the CompTIA objectives is artificial intelligence (AI). The subject is an inch deep and 5 miles wide. So, we will cover some a myriad of topics that will make you more knowledgeable with AI, while covering the objectives for the exam.



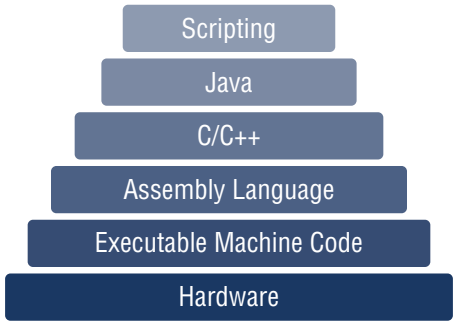
Exam Objective 4.8, "Explain the basics of scripting," requires you to be able to identify and read basic script types of Windows batch scripts, PowerShell, Visual Basic Script, Bash, JavaScript, and Python. You should also be able to apply scripting concepts to use cases. In addition, you should know the considerations when scripting.

Scripting

Before diving into scripting, we'll begin by discussing the differences between a programming language and a scripting language. When you write a script, you are basically using a high-level programming language. An example of a low-level programming language is assembler, also known as *assembly language*. Figure 8.1 shows what we call the "programming pyramid."

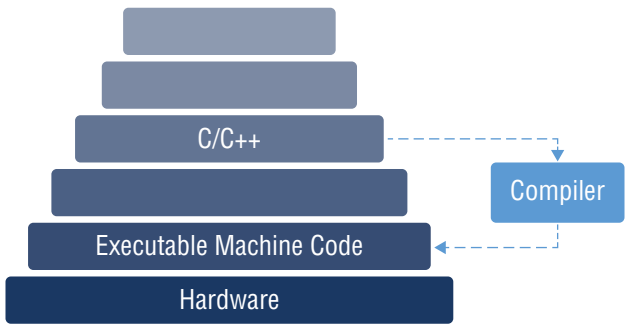
The lowest layer is the actual hardware, which is the central processing unit (CPU). Directly above the hardware layer is the *executable machine code* that interacts with the hardware to make it perform some useful function or process. The operating system is usually programmed in a low- to mid-level programming language, such as C/C++ or even assembly language. It is then *compiled* into executable machine code. Applications are often programmed in high-level languages, such as Java, C#, or VB.NET, and are compiled to executable machine code or an intermediate code. Scripts, however, are not compiled; they are interpreted, as we discuss in this section.

FIGURE 8.1 Programming pyramid



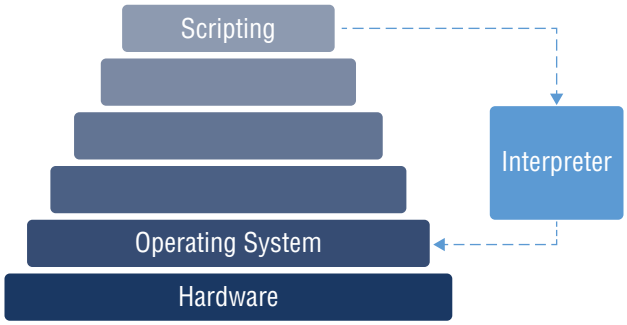
Depending on which layer you program on, you gain some advantages, but at the same time you are also capped by some of the limitations of each layer. For example, a program created in assembly language will be quite complex because you will need to perform low-level functions just to add two numbers together. However, because you are at such a low level, you have direct access to the hardware, so you are limitless in your control of the hardware. If you were to use a higher-level language, such as C/C++, the application would be relatively easy to write; adding two numbers is as simple as `c = a + b;`. Because it is a higher-level language, however, you do not have the same lower-level control over the hardware as with assembly language. You have access only to what the compiler will understand and compile to executable machine code, as shown in Figure 8.2.

FIGURE 8.2 Compiling a programming language



Scripting languages do not need to be compiled. They are interpreted by the shell, command line, or external interpreter, as shown in Figure 8.3. The interpreter reads the script and executes the instructions in the operating system. The big difference is that you do not need to compile scripts to executable machine code, as in the previous example of a C/C++ program. Unfortunately, the higher the level, the less control you have over the process. The benefit is you can create a script rather quickly, and you don't need to compile it.

FIGURE 8.3 Interpreting a scripting language



Another big difference between applications and scripts is that scripts require applications to complete their purpose. If an application doesn't exist for a function in your script, then you should evaluate whether a script is the right course of action. For example, if you need to resize pictures, an application must exist that can be called via the command line that will take the appropriate input and resize the pictures. If an application doesn't exist that can resize the picture, then you might need to write an application that resizes a picture in lieu of a script. A script cannot normally create functions of this nature; it can only call on them.



Java is a compiled language that requires the Java Development Kit (JDK) in order to be compiled to an intermediate code called bytecode. The bytecode is then run on top of the Java runtime, known as the Java Runtime Environment (JRE). This is why you need to install Java to run Java programs on your operating system. JavaScript, on the other hand, is a scripting language that many different web browsers support. JavaScript does not need a JRE since it is interpreted inside the web browser.

Scripting Basics

There are several different types of scripting languages. Each scripting language has its own nuances and syntax. This section discusses some scripting basics that you can apply to any scripting language. You'll only have to adjust the syntax to the scripting language you've selected to get the desired outcome.

Variables

A *variable* is a symbolic word or combination of letters that can be used to hold a value. The value can be defined as a number, text, Boolean, or array. Numbers can be either *integers* or *floating-point*, but the scripting languages you choose must support floating-point math. Floating-point math is a value that contains a decimal number, whereas integers are whole numbers. Text is also called a *string* and has no numeric equivalent. A string containing `30`

will not be a value of `30`; it will be only be a string of characters containing a `3` and a `0`. Mathematical computations cannot be performed on a string. Boolean values are true or false values, and true values and arrays are collections of strings, numbers, and Boolean values.

You might see a programming language described as a strongly typed language. That means that the variable must be defined as to what type of value it will hold and how long the value will be. This is yet another difference between scripting languages and programming languages: scripting languages do not need variables to be defined and are normally not typed. Most of the time, you can just load a variable with a value and it is dynamically typed. This is not the most economical use of memory, but that shouldn't be an issue, because scripts are often simple ways to complete a simple task; therefore, it doesn't matter that loading a variable with a value and having it dynamically typed isn't the most economical use of memory.

The name of a variable should have some meaning and cannot be a reserved word. So if you had a variable that needed to hold a counter, you could simply create a variable called `count`. However, you would need to ensure that `count` was not used elsewhere in the program for a function; otherwise, it's a *reserved word* in the scripting language and cannot be used as a variable.

The following is an example of a variable named `count` being created in the *PowerShell* scripting language and loaded with the value `1`. Generally, you load a value by specifying the variable, followed by an equal sign, and then the value. In PowerShell, the syntax dictates that you place a `$` in front of the variable in order to notify the interpreter that you are addressing a variable. Each scripting language is slightly different, but the concept remains the same. You can also view the contents of the variable either by typing the variable name and pressing Enter or by using the `echo` command before the variable.

```
PS C:\Users> $count = 1
PS C:\Users> $count
1
PS C:\Users> echo $count
1
PS C:\Users>
```

Environment Variables

As previously mentioned, variables hold values in scripts so that we can do things such as counting. Environment variables also hold values, but they are used for the environment of either the system or the current user. Environment variables often hold values like the path to an executable or the location of the temporary folder.

Environment variables are inherited in a structured fashion: system, user, and then program. They can be overwritten at any underlying level, but only for that entity. For example, if a script changes the user environment variable for the temporary folder, the change is applicable only for that script unless the user environment is changed—then it will affect all scripts run by a user. Here are the types of variables you will encounter in the operating system:

System Variable Defined for use by the system, or any user or program

User Variable Defined for use by a specific user, or any program executing under that specific user

Program Variable Defined for use inside a script or application and can only be accessed from inside the script or application

Comment Syntax

Your scripts should have a level of end-user readability. The code should describe what is going to happen without external documentation. Although we strive to create eloquent scripts that read like computer poetry, sometimes we need to make comments inside the code.

Each language has a different way to make comments. The following example uses the PowerShell syntax. As you progress through the next section, "Scripting Languages," you will see the way to comment in the various languages. You'll notice in the following code that we've added some commenting that is generally used for script creation. The three basic components of the comments are author name, authored date, and the purpose of the script. Although the following comment block is best practice, it is not required; one-line comments inside the code are allowed also. In this scripting language, PowerShell, the comment is preceded by the # symbol. Other scripting languages have their own syntax, and you'll see the various examples throughout this chapter.

```
# Sybex
# 4/20/25
# This script will output a directory listing of the C drive
Get-ChildItem C:\
```

Loops

When writing a script, you may need to create a controlled loop inside the script. An example of a controlled loop is to read a file and then do some things for each line contained in the file. You can use two basic loops in your scripts: `for` loops and `while` loops.

A `for` loop is a stepped loop with a defined beginning and a defined end; each step is defined as well. The following example shows a `for` loop for PowerShell that starts at 1 and counts to 10. The output is the word *Number* and an incrementing number. You can see that the variable is initially set to `1` with (`$count=1`), and a test is done to check if the variable is less than or equal to 10 with (`$count -le 10`). The `for` loop is stepped by adding 1 to the existing number with (`$count++`). This is just one example of a `for` loop, and every scripting language is a bit different in syntax, but the concept is the same.

```
For($count=1; $count -le 10; $count++) {
    Write-Host "Number $count"
    }
[ Output of Script ]
Number 1
```

```
Number 2
Number 3
Number 4
Number 5
Number 6
Number 7
Number 8
Number 9
Number 10
```

A `while` loop continues to loop until either it is exited or a condition is met. The `while` loop has no defined beginning, only a defined end, and it can be exited without consequence to the function. In the following code, the variable `count` is set to `0`, and then the `while` loop begins. Inside the `while` loop, the `count` variable is incremented by `1` each time it loops with (`$count++`). The loop will continue as long as `count` is less than `11` with (`$count -lt 11`). The output will be identical to the prior example.

```
$count = 1
While ($count -lt 11) {
    Write-Host "Number $count"
    $count++
    }
[ Output of Script ]
Number 1
Number 2
Number 3
Number 4
Number 5
Number 6
Number 7
Number 8
Number 9
Number 10
```



A `while` loop and a `do while` loop are identical in functionality; they both loop until a condition is no longer true. The difference between the two is that a `while` loop puts the condition before the code in the loop, whereas a `do while` has the condition after the code in the loop. Each language supports either one loop or the other, and sometimes both types of loops. However, the concept remains the same: The loop will continue until the condition is no longer true.

Branch Logic

Branch logic enables code to deviate, or branch, depending on a condition. The `if` statement is the most common conditional branch logic found in scripts. The `if` statement is usually followed by a condition and a `then` clause. It can even use an `else` clause. The `then` clause is executed only if the condition is true; if the condition is not true, the `else` clause is executed.

In the following example, the string `str` is set to `1`. The `if` statement then checks the condition of `str` being equal to `1` with `($str -eq 1)`. It is true, so the next statement of `Write-Host "Yes"` is executed. It's important to note that PowerShell implies the `then` clause; only the `else` clause needs to be spelled out. If `str` were anything other than `1`, the condition would be false, executing the `else` clause of `Write-Host "No"`. This is an example for PowerShell, but in the next section, "Scripting Languages," you will find many different examples for each scripting language.

```
$str=1
If ($str -eq 1) { Write-Host "Yes" } Else { Write-Host "No" }
[ Output of Script ]
Yes
```

Scripting Languages

Now that you understand the basic differences between a programming language and a scripting language and some scripting basics, let's look at several different scripting languages you are likely to encounter as a technician. As we cover the various languages, we'll highlight which operating systems they are common to or natively supported on—which is often the biggest deciding factor when you choose a scripting language.

Windows Batch Scripts

Windows batch scripts have been around since the release of Microsoft's Disk Operating System (DOS) back in 1981. The original script interpreter was `command.com`, and since then, Windows NT was released, which included an updated command-line interpreter called *Command Prompt*, or `cmd.exe`. The original file extension used with batch scripts was `.bat`, but today both `.bat` and `.cmd` can be used to initiate a batch script because they are both associated with the command-line interpreter `cmd.exe`.

A Windows batch script is probably the fastest way to get something done when all you need is a list of commands run one after the other. For example, say you need to create new user accounts for a school. You can get the usernames in an Excel sheet, and you can create a script from the entries by using an Excel formula, adding in the column of the username. As an example, the formula `="NET USER" & A1 & "PassW0rd /ADD /DOMAIN"` copied into cell B1 will produce the line you need to execute. Then you just need to drag the formula down, and the script will be built. A quick copy and paste, and the script will look similar to the following output. It's a quick and dirty way to create user accounts. Using the

combination of an Excel sheet and a copy-and-paste into a batch script is the fastest way to build and execute a laundry list of commands.

```
NET USER UserOne PassW0rd /ADD /DOMAIN
NET USER UserOne PassW0rd /ADD /DOMAIN
NET USER UserOne PassW0rd /ADD /DOMAIN
NET USER UserOne PassW0rd /ADD /DOMAIN
[ Output Cut ]
```

Batch scripts can also contain logic. The following is a simple batch script that tests whether a variable of `FLIPFLOP` is equal to `0` using an `if` statement. If `FLIPFLOP` is equal to `0`, the script will proceed to write to the screen the word *Zero*, set `FLIPFLOP` to a value of `1`, and then jump to `:LOOP`. Because `FLIPFLOP` is set to `1`, it is not equal to `0`; so, the `else` clause in the `if` statement will be processed, and the word *One* will be printed to the screen, and `FLIPFLOP` will then be set to `0`, and the script will jump to `:LOOP` again. This will proceed until Ctrl+C is pressed to stop the processing of the script.

```
@ECHO OFF
REM FlipFlop Script
SET /A FLIPFLOP=0
:LOOP
IF %FLIPFLOP% EQU 0 (ECHO Zero && SET /A FLIPFLOP=1) ELSE (ECHO One && SET /A
FLIPFLOP=0)
GOTO :LOOP
```

In Exercise 8.1 you will create a simple Windows batch script from the preceding script. The end result will print out Zero and One in an alternating pattern.

EXERCISE 8.1

Creating and Running a Windows Batch Script

1. Click the Start menu and type **Notepad**. Select Notepad from the search results.
2. Type the preceding batch script example into Notepad.
3. Choose File ➢ Save As, and then save the batch script to your desktop.
4. Name the file **`flipflop.cmd`**, and then click Save.
5. Press the Windows key and the R key at the same time, type **`cmd.exe`**, and then press Enter.
6. Type **`cd %userprofile%\desktop`** and press Enter.
7. Type **`flipflop.cmd`**, and then press Enter.

   Notice that your screen display will alternate between *One* and *Zero*.
8. Press Ctrl+C.

This will display the question `Terminate batch job (Y/N)?`.

9. Answer Y, and then press Enter.
10. Switch to Notepad and add **`REM`** in front of the `@echo off` command.
11. Save the file by choosing File ➢ Save.
12. Switch to the command prompt and rerun `flipflop.cmd`.

    You will now see the commands as well as the variable contents while the script is running.

13. Press Ctrl+C.

    This will display the question `Terminate batch job (Y/N)?`.

14. Answer Y, and then press Enter.



Another way to open a command prompt to a specific folder is to first browse to the folder, and then hold the Shift key and right-click an empty space in the folder. Select Open PowerShell Window Here on Windows 11. After opening the PowerShell window, type **`cmd`** and then press Enter. This will change your interpreter to `cmd.exe`.

PowerShell Scripts

PowerShell allows for the automation and management of the Windows operating systems, as well as cloud-related services such as Microsoft Azure and Microsoft 365. One of the limiting features of any scripting language is its ability to perform a needed task. PowerShell was created to be totally extensible. It was built on the .NET Framework *Common Language Runtime (CLR)*. Any programmable library a .NET application has access to, PowerShell can use, which is what makes it so extensible. It has been used since Windows Server 2008 as a configuration tool for the operating system. In fact, most of the time when you configure a service in the Server Manager tool, you actually run a PowerShell command in the background. Many of the GUI wizards allow you to see the PowerShell script that will be executed so that you can reuse the line in a script of your own.

PowerShell introduced the concept of *cmdlets*. PowerShell has over 100 cmdlets installed, called the core cmdlets. You can always add your own cmdlet by creating a PS1 script and installing it into the PowerShell cmdlet store in the operating system, as you will do in Activity 8.2. A cmdlet is simply a verb and a noun separated by a dash. Here are a few examples:

`Get-Item` Gets an item such as a directory listing, environment variable, or Registry key

`Set-Item` Changes the value of an item, such as creating an alias or setting an environment variable

`Copy-Item` Copies an item, such as a file or folder

`Remove-Item` Deletes an item, such as a file, folder, or Registry key

`Move-Item` Moves an item, such as a file, folder, or Registry key

These are just a few of the built-in core cmdlets for PowerShell. Others exist for `-Item`, such as `Rename-Item`, `New-Item`, `Invoke-Item`, and `Clear-Item`. Each one performs a corresponding action on the noun following the dash. You can even extend the functionality of a command with your own PS1 cmdlet. There is a `Get-Verb` command so that you can see all the appropriate verbs that you can use for your own command.



To learn more about creating your own cmdlets, visit:

`https://docs.microsoft.com/en-us/powershell/scripting/developer/cmdlet/how-to-write-a-simple-cmdlet?view=powershell-7.2`

If you use the `Get-Item` cmdlet and specify a folder, information about that folder will be returned. If you want to see all the other folders contained within that folder, you can use a `*` wildcard. Or you can use the `Get-ChildItem` cmdlet and specify the directory, as follows:

```
PS C:\Users\UserOne> Get-item c:\*
    Directory: C:\
Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----       11/28/2024   9:52 PM                Dell
d-----        5/16/2024   9:32 PM                NVIDIA
d-----        4/11/2024   7:38 PM                PerfLogs
d-r---        5/28/2024  10:04 PM                Program Files
d-r---        8/12/2024   5:31 PM                Program Files (x86)
d-r---        5/28/2024   6:11 PM                Users
d-----       10/18/2024  10:17 PM                Windows
PS C:\Users\UserOne> Get-ChildItem c:\
    Directory: C:\
Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----       11/28/2024   9:52 PM                dell
d-----        5/16/2024   9:32 PM                NVIDIA
```

```
d-----        4/11/2024   7:38 PM                PerfLogs
d-r---        5/28/2024  10:04 PM                Program Files
d-r---        8/12/2024   5:31 PM                Program Files (x86)
d-r---        5/28/2024   6:11 PM                Users
d-----       10/18/2024  10:17 PM                Windows
```

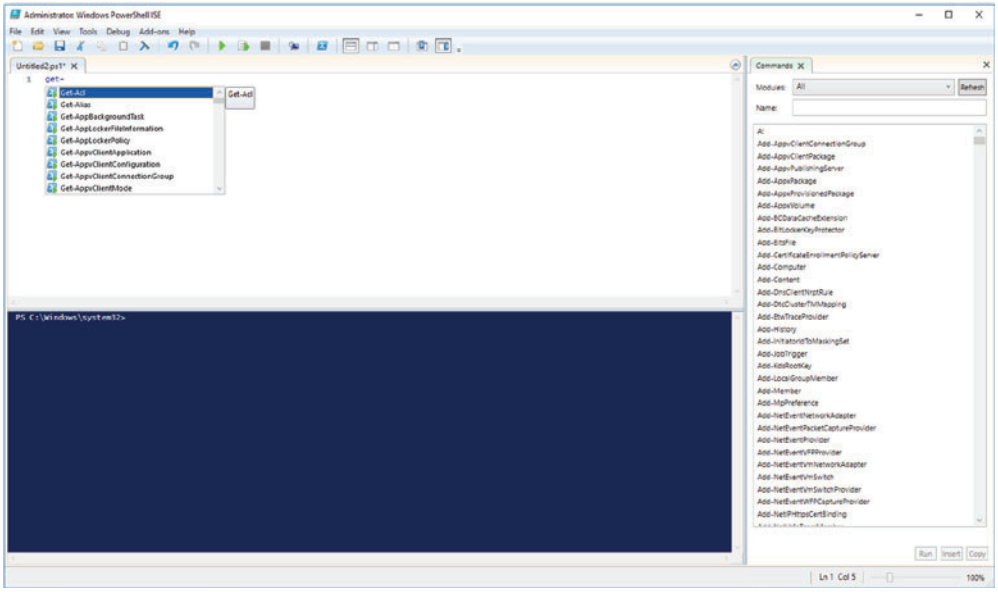
When you use the `dir` command in PowerShell to view a directory listing of files, you are actually using something called an *alias*. The alias then calls the `Get-ChildItem` cmdlet. To see all the aliases on the operating system, you can use the `Get-Alias` cmdlet. You can see all the commands mapped over to PowerShell cmdlets, as follows:

```
PS C:\Users\UserOne> get-alias
CommandType     Name
-----------     ----
Alias           % -> ForEach-Object
Alias           ? -> Where-Object
Alias           ac -> Add-Content
Alias           asnp -> Add-PSSnapin
Alias           cat -> Get-Content
Alias           cd -> Set-Location
Alias           CFS -> ConvertFrom-String
Alias           chdir -> Set-Location
Alias           clc -> Clear-Content
Alias           clear -> Clear-Host
Alias           clhy -> Clear-History
Alias           cli -> Clear-Item
Alias           clp -> Clear-ItemProperty
Alias           cls -> Clear-Host
[ Output Cut ]
```

PowerShell also has a great way to develop scripts in what is called an *Integrated Scripting Environment (ISE)*. The ISE allows you to write a script and test it without having to switch back and forth between a text editor and the execution environment. In addition to the writing and execution environment in the same window, there is a type-ahead feature that allows you to pick a command if you remember only the first few letters, as shown in Figure 8.4. You can also use the Tab key to complete a command, which makes writing scripts easy when you know the first couple of letters. Formatting is also automated and makes for easy-to-read scripts. The formatting highlights variables and commands in different colors so that you can differentiate between the two.

Before any script can be executed on the Windows operating system, you must first allow scripts to run. By default, any PowerShell scripts will be blocked. Exercise 8.2 shows you how to "unrestrict" PowerShell scripts using the `Set-ExecutionPolicy` cmdlet.

FIGURE 8.4 PowerShell ISE



EXERCISE 8.2

Creating Your First PowerShell Script

1. Click the Start menu, type **PowerShell**, right-click Windows PowerShell ISE in the search results, and select Run As Administrator.
2. Click Yes in the User Account Control dialog box when prompted.
3. In the lower portion of the ISE, type **`get-executionpolicy`**, and then press Enter.

   You'll notice that it returns `Restricted`.
4. Type **`set-executionpolicy unrestricted`**, and then press Enter.
5. Answer Yes to the prompt.
6. Type the following into the top portion of the ISE:

```
#FlipFlop Script
While ($true) {
  If ($FLIPFLOP -eq 0) {
    Write-Host "Zero"
    $ FLIPFLOP=1
    }  Else {
```

```
        Write-Host "One"
        $ FLIPFLOP=0
    }
}
```

7. Click File ➢ Save.

   The save dialog box will default to your `Documents` folder.

8. Rename the file **`flipflop.ps1`**, and then click Save.

9. In the lower portion of the ISE, type **`cd ~\Documents`**, and then press Enter.

   Note that you used the ~ character, which represents your home directory, as previously covered in Volume 2 Chapter 4, "Working with macOS and Linux."

10. Type **`.\flipflop.ps1`**, and then press Enter.

    This will run the script and display the alternating words *Zero* and *One*. To stop it, press Ctrl+C. Alternatively, you can click the red stop button on the ribbon at the top of the window. You can execute the script by clicking the green play button.

11. Type **`set-executionpolicy default`**, and then press Enter to change the execution policy back to the default setting.



Microsoft has created a cross-platform version of PowerShell, originally called *PowerShell Core*, that runs on Linux and macOS. This version will run nearly all the commands that the Windows version of PowerShell offers, with a few platform-specific nuances. You can find more information on this topic at `https://learn.microsoft.com/en-us/powershell`.

Visual Basic Scripts

Visual Basic scripts—also known as *VBScripts*—are scripts based on Microsoft's Visual Basic language. VBScript technology has been around since Windows 98 and Windows NT, so it is very mature, but it is slowly being replaced by PowerShell, as we'll explain. VBScripts also run only on Windows operating systems, unlike PowerShell scripts.

VBScript technology is based on the *Component Object Model (COM)* to allow interaction with the operating system. Any COM object that can be instantiated can be accessed through a VBScript. As you learned in the preceding section, PowerShell is based on the .NET Framework. The Component Object Model predates the .NET Framework as a way to register programming libraries with the operating system. By default, VBScripts cannot access programming libraries that have been written for the .NET Framework, which is why VBScripts are slowly losing popularity and support by Microsoft.

VBScripts are still extremely useful when it comes to a structured language for creating login scripts. As previously mentioned, scripts are an interpreted language and are not compiled. VBScripts are no different; they require an interpreter to process. There are three main interpreters that can process VBScripts: *Windows Scripting Host (WSH)*, Internet Information Services (IIS) Active Server Pages (ASP), and Internet Explorer. ASP and Internet Explorer are deprecated, so we will focus on WSH.

The Windows Scripting Host is an environment that allows you to run VBScripts from the command line. By default, when a *VBS* script is run, a program called `wscript.exe` processes the script. Any output will be sent to a Windows message box that you must close by clicking OK. This can be quite annoying if you have multiple lines of output, as each line will pop up a message box you have to close. A VBScript can also be executed with the `cscript.exe` program. This version of the VBScript processor outputs to a console window—the name stands for console script.

VBScripts are normally edited in Notepad and saved with a `.vbs` extension. A number of third-party editors allow type-ahead. As previously mentioned, type-ahead is a feature that allows for the completion of code after a few letters have been entered. Microsoft supports an editor called Visual Studio Code, which you can download from `https://code.visualstudio.com`.

The following VBScript will perform the same output as the two preceding examples. As you can tell, the syntax is a little different.

```
'FlipFlop Script
Do Until FLIPFLOP> 2
  If FLIPFLOP = 0 Then
    Wscript.Echo "Zero"
    FLIPFLOP = 1
  Else
    Wscript.Echo "One"
    FLIPFLOP = 0
  End If
Loop
```



Because of the popularity of Windows PowerShell and its access to the .NET Framework, there will be no future releases of the VBScript processor. Microsoft will maintain the VBScript processor until it is retired. However, Windows batch scripts would probably be deprecated before the VBScript processor.

Shell Scripts

Bash stands for the Bourne Again Shell. Bash is backward compatible with its predecessor, called the *Bourne shell* (or *sh*). Whether you're using the Bash shell or the sh shell, either will perform similar functions for scripting purposes. The shell itself interprets commands

for scripting, similar to the Windows DOS shell. However, the Bash and sh shells are much more advanced in their structure and functionality than DOS. Linux/UNIX shell scripts are basically how the operating system boots itself up and starts all the necessary services.

Linux and UNIX scripts often end with the `.sh` extension to signify to the end user that the file contains a script (text) and not an executable code. Shell scripts do not require the `.sh` extension in order to be executed. They do, however, require execute permissions to be applied. This can be done with the `chmod` command, as you learned in Volume 2 Chapter 4. Another nuance is that you must specify the relative path using `./` and then the script—for example, `./script.sh`.

Shell scripts can be edited in any text-based editor. The vi editor was probably the first editor—and on older systems, your only choice. Today, there is a multitude of editors from which to choose, such as pico/nano, jed, Gedit, and Kate/Kwrite, just to name a few. Any of these editors will do a fine job for the basic editing of scripts. Some people find that opening two consoles is the fastest way to write a script on Linux/UNIX. One console serves as the editing console, with the text editor loaded and saving changes that are made, while the other console serves as the execution environment.

The following is a simple shell script that performs identical functionality to the scripts you have seen throughout this chapter. There are two major things to note with this example. The first is the syntax of the script, which differs slightly from the other examples but basically looks similar. The second thing to note is the directive on the first line. The directive tells the operating system which shell to process the script in. The `#!` is called a *hashbang*. The hashbang lets the operating system know the path of the script interpreter that will be used to process the script—in this case, it is `/bin/bash`.

```
#!/bin/bash
#FlipFlop Script
while [ !$FLIPFLOP ]; do
  if [ $FLIPFLOP -eq 0 ]
  then
    echo Zero
    FLIPFLOP =1
  else
    echo One
    FLIPFLOP =0
  fi
done
```



The best way to start scripting with Linux/UNIX shell scripts is to load up Linux and try it out. You can find a great guide to Linux scripting here:

`https://help.ubuntu.com/community/Beginners/BashScripting`

Python Scripts

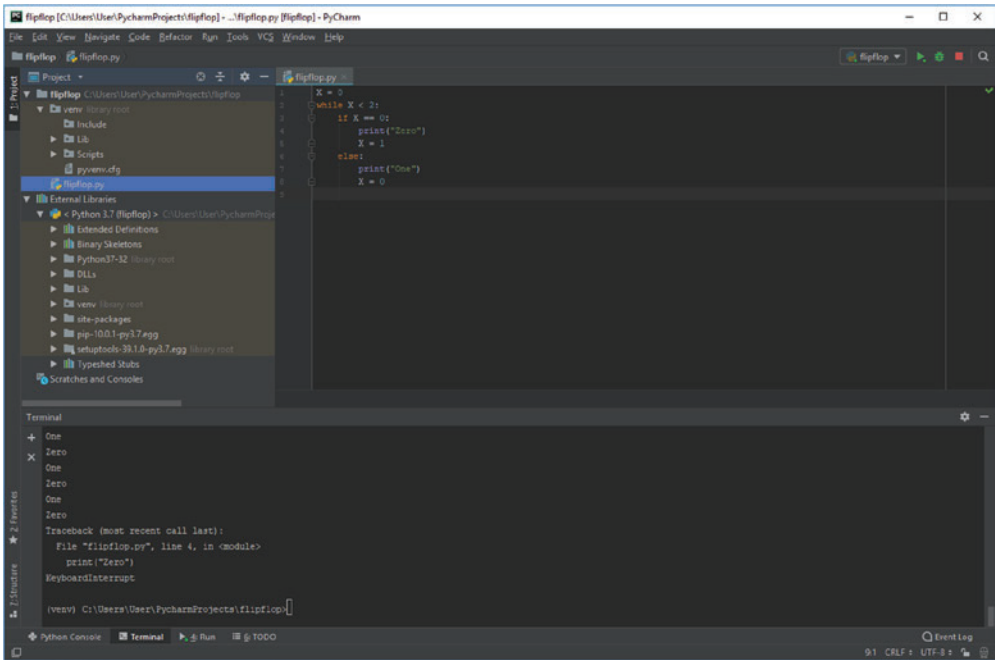
The *Python* scripting language was first released in 1991, and it has started to gain popularity over the past 10 years or so. It is not normally installed by default in any operating system. If you want to use Python on a Windows or macOS operating system, you need to visit `www.python.org` to download the latest version, and then install it. If you are running a Linux operating system, you will install Python through the package management system of the operating system.



If you install Python from `www.python.org`, you will only be prompted to download Python 3.x. However, when installing on Linux, if you do not specify Python 3, you will install the older version, Python 2.x. Python 2 is no longer being supported and has fewer features than the current version, Python 3.x.

The Python installation does not include an *integrated development environment (IDE)*, also known as an *Integrated Scripting Environment (ISE)*. The installation contains only the interpreter and some documentation to get you started. You can use any text editor to create and edit scripts. An extremely popular Python IDE is the Free Community version of PyCharm, by JetBrains. The IDE allows for script development similar to the PowerShell ISE but with many more features, as shown in Figure 8.5.

FIGURE 8.5 The PyCharm IDE



Python was created to be an easy scripting language to learn, and it is very forgiving with syntax. It is probably one of the best languages that we can recommend to start scripting in because it is so forgiving, unlike Bash or Windows batch scripting. Python also doesn't lack in features. Like VBScript and PowerShell, Python is extensible and can use external libraries. One disadvantage to using Python is that it's not been widely adopted in enterprise environments. This means you will not be able to build login scripts with it, because the interpreter does not have direct ties into the operating system. You'll often find that your flexibility is limited when you try to integrate a Python script into an enterprise process.

Python scripts normally end with the `.py` extension so that the end user can identify the scripting language used within the file. The `.py` extension also allows the script to be launched on Windows operating systems. With Linux/UNIX and macOS systems, however, the hashbang defines the interpreter to use and generally looks something like `#! /usr/bin/python3`.

This next script was written in Python and follows the same functionality as the previous examples in this chapter. As you can see, the program's readability is similar to the earlier examples and the syntax differs slightly.

```
#FlipFlop Script
FLIPFLOP = 0
while FLIPFLOP < 2:
  if FLIPFLOP == 0:
    print("Zero")
    FLIPFLOP = 1
  else:
    print("One")
    FLIPFLOP = 0
```

JavaScript

JavaScript and the Java programming language have several similarities—for example, they are both object-oriented languages, and their syntax is similar. However, the similarities end there, because Java is a programming language and JavaScript is a scripting language, with the difference being that Java is compiled whereas JavaScript is a scripting language.

JavaScript is mainly interpreted in web browsers to allow for interactive web pages. JavaScript is one of the three core web technologies; the other two are *Hypertext Markup Language (HTML)* and *Cascading Style Sheets (CSS)*. JavaScript can also be adapted to run outside the web browser with a runtime called `Node.js`. Both web browser JavaScript and `Node.js` scripts end in the `.js` extension, which identifies the contents of the file as JavaScript code.

JavaScript can be edited with any text-based editor. Microsoft Visual Studio Code does an excellent job of formatting for Windows-based editing of JavaScript. Brackets is another popular editor for JavaScript and runs on a variety of platforms.

The following is an example of JavaScript code. The example is similar to the previous examples in relation to functionality. This particular script was coded to run in `Node.js`, since JavaScript normally outputs to HTML. The structure is similar to the previous examples. As always, every language differs slightly in syntax.

```
//FlipFlop Script
FLIPFLOP = 0;
while (FLIPFLOP < 2) {
  if (FLIPFLOP == 0) {
    console.log ('Zero');
    FLIPFLOP = 1;
  } else {
    console.log ('One');
    FLIPFLOP = 0;
  }
}
```



For the 220-1202 exam, CompTIA does not expect you to write scripts. However, you should be able to identify scripts and know their associated extensions.

Scripting Use Cases

Scripting is a fairly new objective for the CompTIA A+ exam, and it may look overwhelming to you. You are not expected to have mastered the skills of writing scripts, such as those included in this chapter. They have been included so that you can visualize the various languages and understand the syntax. You will be required to read and understand what a script is doing, as well as identify the various elements of a script, such as variables, branch logic, basic loops, and the type of variables.

In addition to basic comprehension of scripts, you will need to know the various use cases where you may find yourself scripting something together. The basic rule of thumb should always apply: If the task is repetitious and needs to be completed several times, then a script should be developed. For example, if you needed to create 20 users, then a script is your best choice. Time versus reward should be calculated based on the time it takes to develop the script compared to the time it will take to complete the job. One consideration is how often you must complete the task; if you have three users you need to create every week, then developing a script is a good investment of time.

In this section, we'll examine several different scenarios in which you might find yourself developing scripts to complete.

Automation

The most compelling situation for the use of a script is one that requires some form of automation for a task. Scripts are perfect for *basic automation* tasks, such as creating users, adding users to groups, or even more intricate and sophisticated tasks. When a task is automated with a script, it guarantees that the task will flow the same every time it is executed.

There are two types of scripts that you will most likely create: scripts for automating your own tasks and scripts that automate tasks for others. During your career you will most likely find tasks that you have to do over and over. These tasks should be automated as much as possible, and each repetitive task should have its own script. As a best practice, you should create a folder that contains all the scripts you use on a daily basis. This way, you always know where they are, and when you move to a new computer, you can simply copy them over. Obviously, when you create scripts for others, you won't use them on a daily basis. However, these scripts should also be grouped together in a common folder, since you will probably reuse a part of one script or the entire script for another user.

The following is an example of a task that should be automated. It assumes that you have the Remote Server Administration Tools (RSAT) installed on your system, which includes the `dsquery` and `dsget` commands. The command line will query Active Directory for users with *test* in their name. The output from the initial command of `dsquery user -name *test*` accomplished that task and outputs the distinguished name (DN) of the user. We then pipe that output to the `dsget user -samid -ln -fn -email` command line and that retrieves the username, first name, last name, and email address of each user.



Remote Server Administration Tools (RSAT) is a feature that can be added to the Windows operating system to allow you to administer Windows Server roles such as Active Directory. These tools are not an objective of the CompTIA A+ exam but are useful to know. To learn more about RSAT, visit:

`https://learn.microsoft.com/en-us/troubleshoot/windows-server/system-management-components/remote-server-administration-tools`

```
C:\sys>dsquery user -name *test* | dsget user -samid -ln -fn -email
  samid            fn           ln       email
  testuser1        user1        test     test.user1@wiley.com
  testuser2        user2        test     test.user2@wiley.com
  testuser3        user3        test     test.user3@wiley.com
C:\sys>
```

This is a handy command line, but typing it in every time you need an answer like this is tedious. Luckily, we can write a simple script where we just need to change one part of the script. Where we had `test`, we simply replace it with `%1` to capture the first argument from the Windows batch script. Then we just start Notepad and copy the line in and save it as `lookup.cmd`.

```
dsquery user -name *%1* | dsget user -samid -ln -fn -email
```

Then when we run it, we get the following output:

```
C:\sys>lookup test
C:\sys>dsquery user -name *test* | dsget user -samid -ln -fn -email
```

```
  samid            fn           ln       email
  testuser1        user1        test     test.user1@wiley.com
  testuser2        user2        test     test.user2@wiley.com
  testuser3        user3        test     test.user3@wiley.com
C:\sys>
```

The script works just like before, but each command is echoed to the console. This is easily fixed by editing the file `lookup.cmd` with the following additional lines:

```
@echo off
dsquery user -name *%1* | dsget user -samid -ln -fn -email
```

Now when the file is executed, you won't see the echo of the command actually executed, but only the output of the command, as shown here:

```
C:\sys>lookup test
  samid            fn           ln       email
  testuser1        user1        test     test.user1@wiley.com
  testuser2        user2        test     test.user2@wiley.com
  testuser3        user3        test     test.user3@wiley.com
C:\sys>
```

We can refine the script further to add some branch logic, so if the user doesn't supply an argument, it explains the argument required:

```
@echo off
if {%1}=={} (
echo.
echo You must supply the following.
echo ex. %0 {name of user}
echo.
goto :END
)
dsquery user -name *%1* | dsget user -samid -ln -fn -tel -email
:END
```

The process of refining the script is called the development process. The first script you build will have common parts that you will reuse, such as error handling when no argument is supplied. In this example we used a Windows batch script, but you should use whichever scripting language(s) you are comfortable with. There is no right way to develop a script, and the purpose of this example is to show you the thought process involved in writing a script.

Restarting Machines

Another common task might be to restart multiple machines. You can solve this problem in several different ways, and it's all about the end goal of the task. If this is a one-time task, then you could write a simple script, such as the following:

```
shutdown /s /m \\computer1 /t 0
shutdown /s /m \\computer2 /t 0
shutdown /s /m \\computer3 /t 0
shutdown /s /m \\computer4 /t 0
```

If this is a recurring task, then you could develop a script that reads a text file into a variable and restarts each computer. It sounds complex, but it really isn't when you break down the task into smaller pieces. For example, say you need to read a file of computer names, and then walk through each line in the files and restart the computer. For this example, let's switch to PowerShell, since it allows reading of files and parsing them. Start by creating a text file named **`list.txt`** with four computer names in it: `computer1` through `computer4`. Now we'll test the script by saving it as `restartcomps.ps1`.

```
$list = Get-Content .\list.txt
ForEach($line in $list) {
  Echo $line
}
```

The script will read the contents of the `list.txt` file into the variable `$list`. Then the script will load the variable `$line` for each entry in the `$list` collection. When we run the script, we get the following output:

```
PS C:\sys> Set-ExecutionPolicy Unrestricted
Execution Policy Change
The execution policy helps protect you from scripts that you do not trust.
Changing the execution policy might expose you to the security risks described
in the about_Execution_Policies help topic at
https:/go.microsoft.com/fwlink/?LinkID=135170. Do you want to change the
execution policy?
[Y] Yes Yes to All [N] No [L] No to All [S] Suspend [?] Help (default
is "N"): y
PS C:\sys> ./restartcomps.ps1
computer1
computer2
computer3
computer4
PS C:\sys>
```

Now that we have the first two parts of the script working, we just need to add the functional piece. While we are refining the script, let's also remove the variable and read the lines directly:

```
foreach($line in $(Get-Content .\list.txt)) {
  Write-Output $line
  Restart-Computer -Computer $line
}
```

When the script is run, the output will remain the same, but you'll also be restarting the computers that are written as output to the console. We can make the script fancier by allowing arguments for the file to read, but we think you have the idea behind the script development process. It is a work in progress, and you always have some outstanding code to automate your job.

Remapping Network Drives

Remapping network drives can be done in a multitude of ways with VBScript, PowerShell, Window batch scripts, or some other favorite language. However, Windows batch script and PowerShell are the most common. In the following, we see a script that maps a few drives using a Windows batch script. When we map a drive, we are mounting a remote filesystem through to a drive letter. For example, we can mount the remote filesystem of `\\server1\files` to a local drive letter of m:

```
net use m: \\server1\files
net use n: \\server2\files
net use o: \\server3\files
```

The same script can also be developed in PowerShell:

```
New-PSDrive -Name "m" -PSProvider FileSystem -Root "\\server1\files"
New-PSDrive -Name "n" -PSProvider FileSystem -Root "\\server2\files"
New-PSDrive -Name "o" -PSProvider FileSystem -Root "\\server3\files"
```

These scripts don't need to be complex like the previous scripts. We just want to obtain a reproducible result every time the script runs.

Installation of Applications

When scripting is combined with the installation of applications, you can perform a number of functions that are not possible on their own. For example, you can write a script that installs the prerequisites for an installation and then only succeeds if the subsequent installations are successful. A lot of these scenarios are going to be custom to your specific needs and environment. The following is an example of a PowerShell command that will install an application called `App.msi`:

```
Invoke-CimMethod -ClassName Win32_Product -MethodName Install -Arguments
 @{PackageLocation='\\server\installs\App.msi'}
```

A few assumptions are made with the PowerShell example. The first assumption is that you are calling an MSI installer. If you aren't, the code will not work, since every installer has its own methods for invoking an installation. The second assumption is that you are an administrator of the operating system. Writing a PowerShell script will not circumvent Windows security.

Creating a scripted installation of an application is rewarding, but it is also time consuming. You will most likely have to refine your script several times before it works as expected. This means that you will need to install the application many times to get it right. However, if you have an application that requires installation across a number of computers, you can easily reclaim the time spent on the script.

Automated Backups

Backups should be trusted to back up software that is engineered to expire media, rotate media, and generally backup and restore the data and systems that the organization depends on. This type of software is considered off-the-shelf backup software and scripts are not expected to replace this software. However, by using scripts you can automate pieces of the backup process to make the process much more reliable.

A common example of automating backups with scripts is the backup of SQL databases. You can use off-the-shelf backup software to back up the database. However, one common problem is that the agent installed on the SQL server that facilitates the backup will take a snapshot of the database and back it up in whole. This might seem like what we are trying to achieve as an end goal—except when you try to restore it, you'll quickly find that you need to restore the database in whole, even though you only need one table of records.

If you preprocess the database with a script, you can export it to a file. This file can then be backed up and restored to any SQL server, even down to the record level. The following is a maintenance script used with Microsoft SQL to back up the database to a file. The script is written in the SQL scripting language.

```
USE TestRecords
GO
BACKUP DATABASE [TestRecords]
TO  DISK = N'D:\DBBackups\TestRecords.bak'
WITH CHECKSUM;
```

You can automate many other types of backups, such as custom application data, Registry settings, email data, and any other type of data you can address with a script. However, one of the most common backups you will automate with scripts are SQL database backups.

Gathering of Information/Data

Although you can use applications such as Device Manager to gather information about a computer, using the GUI is not scalable when you need to gather information from a large group of computers. This is where scripts come in handy to gather information and even export the information to a file for you. In the following examples, we will look at some

simple PowerShell commands for gathering information and exporting the information. However, you can use any scripting language you like and Windows batch scripting is very commonly used as well.

```
PS C:\sys> Get-Service
Status   Name               DisplayName
------   ----               -----------
Running  agent_ovpnconnect  OpenVPN Agent agent_ovpnconnect
Stopped  AJRouter           AllJoyn Router Service
Stopped  ALG                Application Layer Gateway Service
Stopped  AppIDSvc           Application Identity
Stopped  Appinfo            Application Information
[ Output Cut ]
```

The `Get-Service` cmdlet will show you all the services running on the operating system. It will output a long list and will display each service's Status, Name, and DisplayName. By piping the output to the `Export-Csv` cmdlet, we can output a lot more detail and send it directly to a comma-separated values (CSV) file. An example of this is in the following command string:

```
PS C:\sys> Get-Service | Export-Csv .\Services.csv
```

The `Get-Service` cmdlet isn't the only command you can use to gather data; there are various other commands that let you gather data and export it. Obviously, you can chain these commands together in a script and gather a large amount of data. You can then use scripts to mine the data for specific information, such as a service state or free space.

Initiating Updates

The Windows platform has been automatically patching itself since the release of Microsoft Update (now called Windows Update) and Windows XP. The feature of Microsoft Update and the Windows platform have evolved since the early release. It has turned into a robust feature to keep Windows up to date. However, if you need an immediate result to patch a security hole, then scripting a solution is the best remedy.

There are a number of ways to get patches to immediately install on Windows. The approach you choose depends on your patch management solution, such as Windows Update, Windows Server Update Services (WSUS), Microsoft Endpoint Configuration Manager (MECM), or a third-party patch management solution. For the remainder of this section, we will use the Windows Update solutions for examples.

If you want to script Windows Updates to initiate patching via Windows batch scripting, then the utility of choice is `wuauclt.exe`. The command can be directed to detect patches with the `/detectnow` argument. However, don't expect anything elaborate; the utility will not notify you that it is doing anything. If you want to watch the progress, then you'll need to keep an eye on the log file, `C:\Windows\WindowsUpdate.log`.

An alternative is to use PowerShell and a `PSGallery` module call `PSWindowsUpdate`. This module allows you to fully automate the patch management and obtain a great level

of detail. For example, after installing the module you can execute the command `Get-WindowsUpdate` to obtain the pending list of available updates, as shown here:

```
PS C:\sys> Install-Module -Name PSWindowsUpdate
NuGet provider is required to continue
PowerShellGet requires NuGet provider version '2.8.5.201' or newer to interact
with NuGet-based repositories. The NuGet provider must be available in
'C:\Program Files\PackageManagement\
ProviderAssemblies' or'C:\Users\bohack\AppData\Local\PackageManagement\
ProviderAssemblies'. You can also install the NuGet provider by running
'Install-PackageProvider -Name NuGet
-MinimumVersion 2.8.5.201
-Force'.
Do you want PowerShellGet to install
and import the NuGet provider now?
[Y] Yes  [N] No  [S] Suspend  [?] Help (default is "Y"): y
Untrusted repository
You are installing the modules from an untrusted repository. If you trust this
repository, change its InstallationPolicy value by running the Set-
PSRepository cmdlet. Are you sure
you want to install the modules from 'PSGallery'?
[Y] Yes Yes to All [N] No [L] No to All [S] Suspend [?] Help (default
is "N"): y
PS C:\sys> Get-WindowsUpdate
ComputerName Status     KB          Size Title
------------ ------     --          ---- -----
CERES        -------    KB5007406   80MB 2021-11 Cumulative Update Preview for
.NET Framework 3.5, 4.7.2 and 4.8 for…
CERES        -------    KB2267602    2GB Security Intelligence Update for
Microsoft Defender Antivirus - KB2267602 (...
PS C:\sys>
```

You can then execute the command `Get-WUInstall -AcceptAll -AutoReboot` and the operating system will begin to install the updates and automatically reboot. These are just a few ways you can script the installation of Windows Updates.

Other Scripting Considerations

Along with the knowledge of scripting comes great power and even greater responsibility. You should consider several key points before scripting a solution and during the development of the script. In the following section we will cover some of these key considerations.

Of course, there may be factors outside of these CompTIA objectives. The one consideration that has resonated throughout the previous section is the decision between investing time to develop a script or just completing the task. This is something that you will need to take into account before you even begin scripting.

Introducing Security

A common pitfall with scripting is inadvertently introducing a security issue. Security issues come in all different forms when scripting. The most common security issue is the embedding of security credentials in scripts. Regardless of how secure you think the script will be, it's a bad habit and should be avoided at all costs.

That being said, there are instances where embedding a password cannot be avoided. In these situations, you should use mechanisms that are supported in the scripting language to act as a digital locker for your password or methods that encrypt the password. One such mechanism is the `ConvertTo-SecureString` cmdlet. It means more lines of code, more time, and sometimes more aggravation in getting it to work, but the benefit is a secure system.

Another potential problem is the unintentionally introduction of malware with scripting. This usually happens when you need additional functionality, such as email capability or changing a system setting. Windows batch scripting has its limitations, and it is easy to use a third-party tool to get the last bit of functionality out of the script. However, that third-party tool may have malicious intent, and you could introduce that malicious code to the system in which the script is executed. PowerShell has its own potential malware risks; when you add an untrusted module, the same could happen.

The best way to avoid these pitfalls is to not shortcut the solution. Do your homework and avoid embedding passwords or using untrusted modules or third-party utilities in your scripts. It means more work, but this should be factored into your cost-benefit analysis when taking on developing a script.

Changing System Settings

Scripts are generally created to change various user and system settings. However, sometimes you can inadvertently change a system setting that is not meant to be changed. If this happens, it can adversely affect the system it is executed on. Therefore, it is always best to test the script on a sacrificial computer. A virtual machine is the best option, since it can be reverted to a prior snapshot if something unexpected happens.

An example of a simple mistake is using the `setx` command incorrectly. The `setx` command modifies the system environment variables and is used by Windows batch scripting. By using the statement `setx path "$path;c:\sys" /m`, you could cause irreversible problems. This statement will set the `path` variable to `$path;c:\sys`, because the incorrect syntax was used to address the variable of `path`. The correct statement is actually `setx path "%path%;c:\sys" /m`. The statement now sets the `path` variable to the current path variable contents (`%path%`) and the addition of `c:\sys`.

The mistake can be as simple as mixing up syntax from one scripting language with syntax from another scripting language. However, some commands are not as forgiving, such

as the previous example. When using PowerShell the mistake can be very subtle, since many commands cover both user and system settings. Always test your script before you execute it on a live system. Otherwise, you might be reinstalling an operating system in addition to completing your scripting task.

System Crashes

Scripting allows you to automate processes, which is the main reason we are developing the script in the first place. However, automation can sometimes create problems that we can't foresee during the development of the script. As an example, you can quickly gobble up all the usable RAM resources with the following Windows batch script. The script will launch instances of Notepad until the operating system runs out of memory.

```
:loop
start notepad.exe
goto :loop
```

Although this script is obvious in its intent, it is an extreme example of automation that if left unchecked will crash the system. Windows batch scripting is not the only scripting language where things can go awry; you can do the same with PowerShell. This example will create an HTML-formatted file of the directory structure:

```
Get-ChildItem c:\ -Recurse | ConvertTo-Html | Out-File -FilePath .\output.html
```

The problem with this example is the sheer size of the resulting file. This statement will recursively list all the files from the C: drive down. When the file is launched in the web browser, the browser will quickly run out of memory trying to display the large file. More elaborate scripts may automatically open the web browser and immediately crash it.

To prevent similar problems from happening in your environment, you should test and monitor your scripts. By testing your script solution for errors or conditions that can run the system out of resources unintentionally, you identify and correct problems that would otherwise cripple the system. Monitoring should be performed after the script is in place, such as a script that is started via the Task Scheduler. You want to identify any spikes in resource usage after the solution is in place.



Exam Objective 4.9, "Given a scenario, use remote access technologies," requires you to know the various methods and tools used for remote access. This includes third-party tools like screen-sharing, videoconferencing, file transfer, and desktop management software.

Remote Access

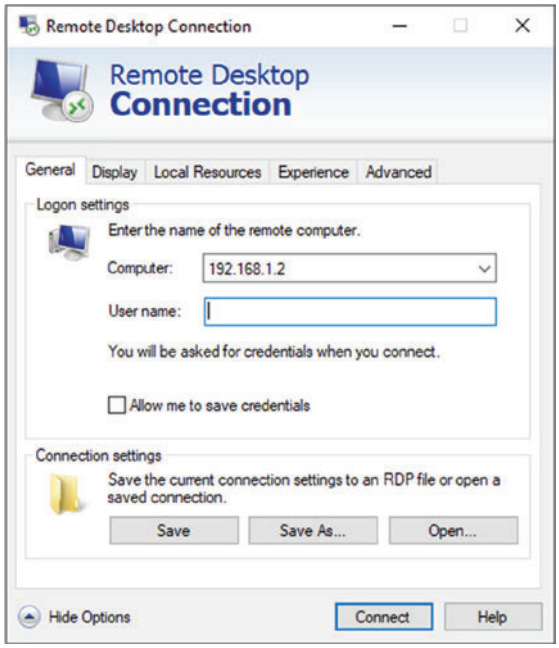
Remote access technologies have been around for as long as networks have been in existence. As a technician, you can't always be in all the places you are needed.

Remote access technologies allow you to do just that. You can create a remote connection to a distant computer on the network and change its configuration. In this section, we will discuss several different types of remote access technologies that are covered on the CompTIA 220-1202 exam, as well as their security implications for a network.

Remote Desktop Protocol

Remote Desktop Protocol (RDP) is used exclusively with Microsoft operating systems. An RDP client—the *Remote Desktop Connection client*, also known as the `mstsc.exe` utility—is built into the Microsoft operating system, as shown in Figure 8.6. The Remote Desktop Connection client can provide remote access as though you were sitting in front of the keyboard, monitor, and mouse. The Remote Desktop Connection client and RDP can transport several other resources, such as remote audio, printers, the clipboard, local disks, and video capture devices, as well as any other Plug and Play (PnP) devices.

FIGURE 8.6 The Remote Desktop Connection client



RDP communicates over TCP port 3389 to deliver the remote screen and connect the local mouse and keyboard for the RDP session. RDP uses Transport Layer Security (TLS) encryption by default, and it provides 128-bit encryption. Microsoft allows one remote user connection or a local connection on desktop operating systems via RDP, but not both

at the same time. On server operating systems, Microsoft allows two administrative connections, which can be a combination of local or remote access but cannot exceed two connections.

Microsoft also uses RDP to deliver user desktops and applications via terminal services. When RDP is used in this fashion, a centralized gateway brokers the connections to each RDP client desktop session. Terminal services require terminal services licensing for either each user connecting or each desktop served. RDP can also be used to deliver applications to end users using Microsoft RemoteApp on terminal services. When RemoteApp is used, the server still requires a terminal services license. However, just the application is delivered to the user rather than the entire desktop.

Virtual Private Network (VPN)

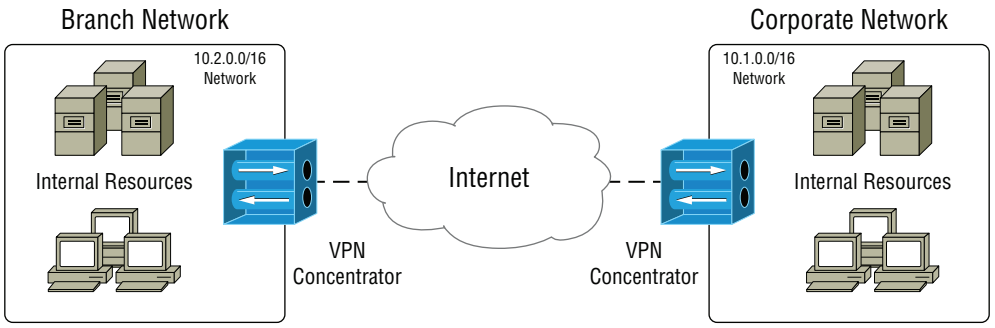
A virtual private network (VPN) extends your company's internal network across the Internet or other unsecured public networks. This remote access technology allows clients and branch networks to be connected securely and privately with the company's network. There are several different ways a VPN can be used in your network architecture, and we will cover them in the following sections. A VPN achieves this private connection across a public network by creating a secure tunnel from end to end through the process of encryption and encapsulation. The encryption protocols used vary, and we will cover them as well. Since a tunnel is created from end to end, your local host becomes part of the company's internal network along with an IP address that matches the company's internal network. We don't have to be bound to only TCP/IP across a VPN, since this technology can encapsulate any protocol and carry it through the tunnel.

Site-to-Site VPN

Over the past 10 to 15 years, using high-bandwidth connections to the Internet has become cheaper than purchasing dedicated leased lines. So, companies have opted to install Internet connections at branch offices for Internet usage. These lines can serve a dual purpose: connecting users to the Internet and connecting branch offices to the main office. However, the Internet is a public network and unsecured, but site-to-site VPN connections can fix that. Companies with multiple locations have reaped the benefits of creating VPN tunnels from site to site over the Internet by ditching their leased lines, installing VPN concentrators at each location, and creating VPN tunnels. Site-to-site VPN is also much more scalable than leased lines because locations only need a connection to the Internet and a VPN concentrator to be tied together. Figure 8.7 details two locations tied together with a VPN tunnel. The magic happens all in the VPN concentrator. Since VPN concentrators also have a routing function, when a tunnel is established, a route entry is created in the VPN concentrator for the remote network. When traffic is destined for the branch office with a destination network of 10.2.0.0/16, the router encrypts and encapsulates the information as data and sends it to the other side of the tunnel over the Internet. This is similar to a host-to-site VPN, the difference being the routing is performed in the VPN concentrator. When the packet is received on the other side of the tunnel, the VPN concentrator decapsulates the data, decrypts the packet, and sends the packet to its destination inside the branch network.

It is common to find that the appliance performing VPN is also the firewall and router. Firewalls today are sold with VPN software built in and licensed accordingly.

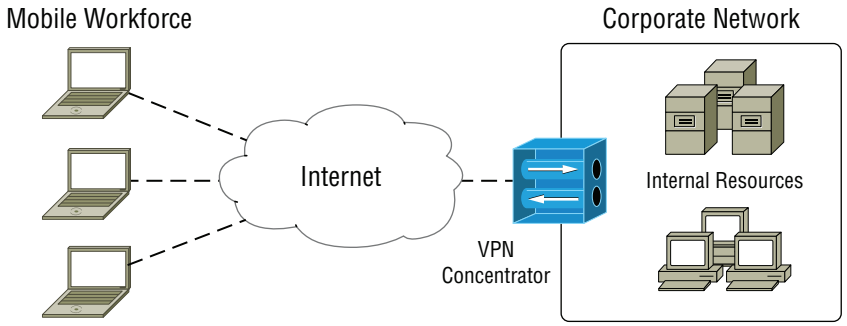
FIGURE 8.7 A typical site-to-site VPN



Client-to-Site VPN

Client-to-site VPN connectivity is a remote access strategy for mobile access. It can be used for telecommuters, salespeople, partners, and administrative access to the internal network resources. The key concept is that VPN access is granted on an individual or a group basis for the mobile users. Using the example in Figure 8.8, you can allow salespeople to connect to the corporate network so they can update sales figures or process orders. This can all be done securely over the Internet while the users are mobile and have access to a network connection.

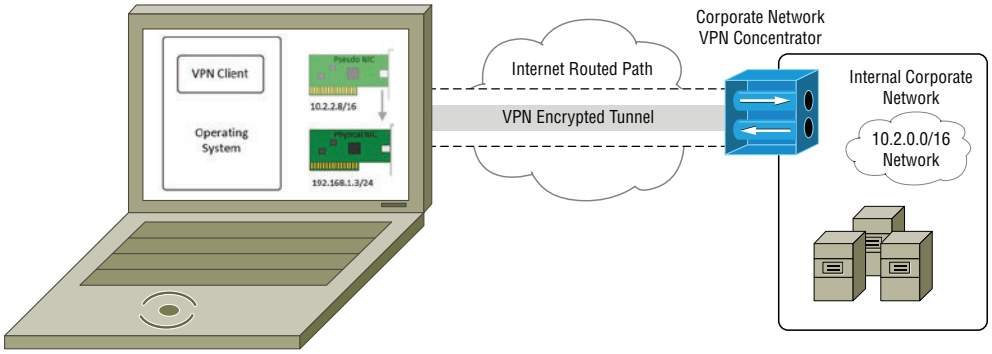
FIGURE 8.8 A typical host-to-site VPN



When a client computer establishes a VPN connection, it becomes part of the internal corporate network. This happens by assignment of an IP address from the internal corporate network. In Figure 8.9, you can see a mobile device such as a laptop with a VPN client

installed in the operating system. When the connection is established with the VPN concentrator over the Internet, a pseudo network adapter is created by the VPN client. In this example, the pseudo network adapter is assigned an internal IP address of 10.2.2.8/16 from the VPN concentrator. The laptop also has its own IP address of 192.168.1.3/24, which it uses to access the Internet. A routing table entry is created in the operating system for the 10.2.0.0/16 network and through the pseudo network adapter. When traffic is generated for the corporate network, it is sent to the pseudo adapter, where it is encrypted and then sent to the physical NIC and sent through the Internet to the VPN concentrator as data. When it arrives at the VPN concentrator, the IP header is stripped from the packet, the data is decrypted, and it is sent to its internal corporate network resource.

FIGURE 8.9 Client-to-site VPN connection



Clientless VPN

There are many different VPN solutions on the market. Each one of them traditionally requires the installation of a VPN client. However, there are a growing number of products that do not require the installation of a client; these products are called clientless VPN solutions. The VPN client is the web browser on the mobile device requiring connectivity back to the corporate network. The VPN appliance acts as a reverse proxy to the various resources internal to the organization.

Virtual Network Computing

Virtual Network Computing (VNC) is a remote-control tool for the sharing of desktops. The VNC client normally operates on TCP port 5900. VNC is similar to Microsoft RDP, with the exception that VNC is an open source protocol and typically allows only one console session per operating system. It supports encryption via plug-ins, but it is not encrypted by default.

VNC operates in a client-server model. The server allows for the remote control of the host on which it is installed. It is normally configured with a simple shared password, but it can also be configured with Windows groups. Several different clients can be used, such as RealVNC, TightVNC, and many others, but they all perform similarly.



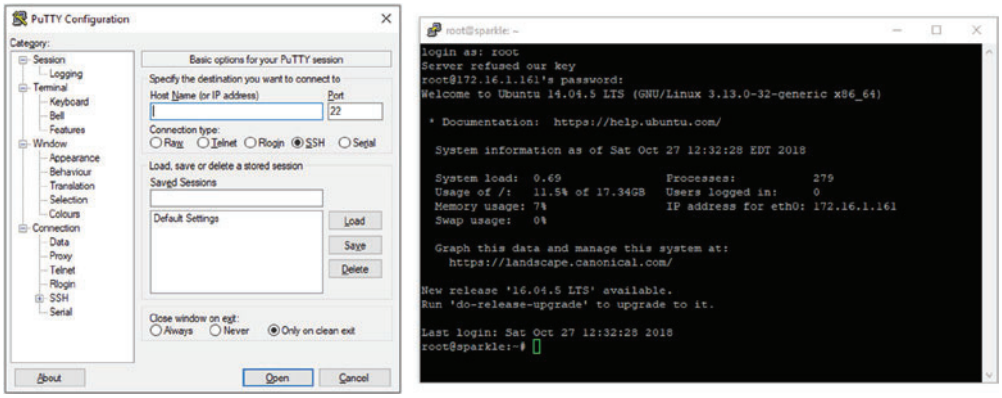
Many different VNC clients are available for download. To learn more, visit `www.realvnc.com` and `www.tightvnc.com`. A simple web search for "VNC clients" will produce a staggering number of results. Each client has its own unique feature set.

Secure Shell

Secure Shell (SSH) is commonly used for remote access via a text console for Linux and UNIX operating systems. The SSH protocol encrypts all communications between the SSH client and the SSH server using TCP port 22. The SSH server is also known as the SSH daemon. SSH uses public-private key pair cryptography to provide authentication between the SSH client and server. SSH can also use a key pair to authenticate users connecting to the SSH server for the session, or a simple username and password can be provided.

It is important to understand that both the user and their host computer are authenticated when the user attempts to connect to the server. During the initial connection between the user's host computer and the SSH server, the encryption protocol is negotiated for the user's login, and the cryptography keys are verified. PuTTY, shown in Figure 8.10, is a common SSH client that is free to download and use. PuTTY provides various methods of connecting to a remote device, including Telnet and SSH.

FIGURE 8.10 PuTTY SSH Client





Microsoft included the OpenSSH client by default in Windows 10 version 1803 and later. The OpenSSH client is located in the `C:\WINDOWS\SYSTEM32\OpenSSH` folder.

Beyond logging into a Linux or UNIX server for remote access, SSH can provide remote access for applications. Through the use of SSH port forwarding, the application can be directed across the SSH connection to the far end. This allows applications to tunnel through

the SSH session. It also encrypts application traffic from the client to the server, because it is carried over the SSH encrypted tunnel. SSH can behave similarly to a VPN connection, but it is more complex to set up.

Remote Monitoring and Management (RMM)

In the early days of your organization, it may have been simple to monitor and manage all the various systems from one location. However, as your organization's footprint grew across different sites and many employees now work from home (WFH), it is more difficult to monitor and manage the various systems. Systems need patching, must be monitored for disk space, and have hardware and applications installed—and these are just a few of the tasks.

This is where a *Remote Monitoring and Management (RMM)* solution can help IT across your enterprise or multiple enterprises and give you a holistic view of your enterprise. There are several different RMM solutions on the market today. Among the most popular solutions are managed service providers (MSPs) that manage your enterprise for a contracted price. These service providers ultimately use RMM software to monitor and maintain the enterprise. The MSP will require your organization to install an agent that is configured to report back to the MSP's RMM software.

You can also purchase a cloud-based or on-premises solution for maintaining your organization with your own IT department. Every vendor of RMM solutions has their own variation of features, which makes up the product's secret sauce. These solutions also require the installation of an agent that reports back to the RMM solutions. Regardless of which product you choose, there are two main features to RMM: the remote monitoring feature and the management feature.

Remote Monitoring The remote monitoring feature of an RMM system can monitor a number of different components, such as security, hardware, applications, and even activity on the operating system. These are just a few of the components of the remote monitoring feature found in RMM systems. The list grows depending on the RMM vendor. The most common monitoring is the security of the various systems across your enterprise, such as patch levels, antimalware status, and exploits. Another monitored component is the hardware and applications installed across the enterprise. Monitoring the hardware can identify your assets, as well as identify when upgrades are needed. Application monitoring can identify problems with a specific application or your vulnerability in the event the application needs to be patched. These are just a few monitored components—the list grows with every release of new RMM software by vendors.

Reporting is a major component of the remote monitoring capabilities of RMM systems. The reporting can be active or passive for most systems. In an active reporting system, the RMM software will compile a report periodically and alert you when a major change is discovered. As an example, if over 30 percent of your computers are vulnerable to a new exploit, the system can be configured to alert you. You may also set up a similar threshold alert for disk space. The passive reports can be run and give you

an overall picture of your network and are typically in the form of a drill-down report. A drill-down report allows you to view the overall health and drill down to specific areas of interest to a specific detail.

Remote Management The remote management feature of RMM solutions can also be quite diverse in the type of management they provide. Many RMM vendors offer the management of computers via a form of remote desktop control. This remote desktop control allows you to intervene when a user has a problem. RMM solutions also offer integrated patch management solutions to patch applications and the operating system. In addition to user intervention and security, you may find remote application installation, disk cleanup, and remote antimalware scan features, and these features just scrape the surface of RMM capabilities. Every vendor of RMM systems has their own unique management capabilities based on their product's focus, such as security, asset management, or user productivity.

Simple Protocol for Independent Computing Environments (SPICE)

The Simple Protocol for Independent Computing Environments (SPICE) is an open remote computing solution that provides high-quality and rich user experience for remote access to virtual desktops. The solution was originally developed by Qumranet in 2007 and later purchased by Red Hat. The original goal of SPICE was to enhance the interaction between users and virtual machines by offering a seamless and efficient protocol. SPICE provides a framework that enables users to connect to their virtual desktops securely and efficiently from any location.

SPICE is integrated with various virtualization platforms, such as Proxmox VE. This integration allows users to connect to the virtual desktop infrastructure (VDI) environment Proxmox VE hosts. The user can use keyboards, mice, audio, and video, with the virtual desktop as if these devices were connected to their local device connected to the virtual desktop. The experience is the same if the user is connected on a LAN or WAN connection.

SPICE operates by establishing a connection between the client device and the virtual desktop hosted on a server. The connection process involves several components, including the SPICE server, SPICE client, and SPICE protocol. The SPICE server resides on the host machine (usually a virtualization server) and handles the virtual machine's display and input/output operations. The SPICE client, installed on the user's device, is responsible for establishing the connection to the SPICE server and rendering the virtual desktop's display locally.

One of the features of SPICE is its robust security framework. The protocol supports multiple data channels, each of which can operate in either clear-text or TLS modes. This flexibility allows administrators to balance security and performance based on their specific needs. Additionally, SPICE offers a range of authentication schemes, including Kerberos.

Windows Remote Management (WinRM)

Windows Remote Management (WinRM) is a powerful tool used in Windows operating systems to facilitate remote management and automation of tasks. It allows administrators to securely execute commands, manage configurations, and retrieve data from remote computers. The initial WinRM protocol version 1.1 was originally shipped with Windows Vista and Windows Server 2008. Currently the WinRM protocol has matured to version 3.0 and is an integral part of the current Windows operating systems. WinRM is built on the WS-Management protocol and uses the Simple Object Access Protocol (SOAP). SOAP is a protocol used for exchanging information that is formatted in an XML format. WinRM also incorporates security protocols TLS, Kerberos, and NTLM (NT LAN Manager) to secure the WS-Management protocol. The WinRM protocol operates over standard HTTP port 80 and HTTPS port 443.

One of the most powerful features of WinRM is its ability to execute commands and scripts remotely. Administrators can use WinRM to run PowerShell scripts on remote systems, enabling them to perform a wide range of administrative tasks without needing physical access to the machines. This capability is particularly useful for automating repetitive tasks, such as software deployment, system updates, and configuration changes. By leveraging WinRM, administrators can write scripts that perform complex operations across multiple systems simultaneously, significantly reducing the time and effort required for system management.

WinRM also plays a crucial role in monitoring and troubleshooting remote systems. Administrators can use WinRM to gather performance data, check system logs, and diagnose issues without interrupting the end user's activities. This remote diagnostic capability helps in quickly identifying and resolving problems, minimizing downtime and maintaining system reliability. For example, an administrator can use WinRM to remotely check the event logs of a server to identify the cause of a system crash or performance issue.

Remote Access Tools

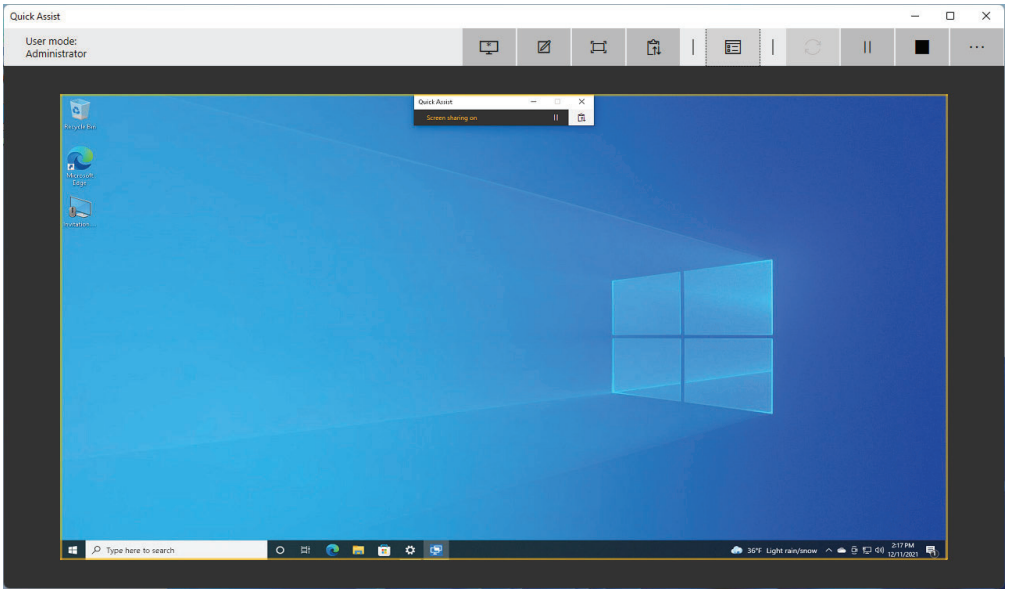
Since the release of Windows XP, Microsoft has included various tools to allow remote assistance to the Windows operating system. In addition to Microsoft's proprietary remote assistance tools, many vendors have entered the market. As a result, there are a number of third-party remote assistance tools freely available and out on the market, with varying costs. Let's explore some of the built-in capabilities of the Windows product and some of the features of third-party products.

Quick Assist

Microsoft includes a remote access tool in Windows 11 for soliciting help called *Quick Assist*, shown in Figure 8.11. Quick Assist operates similarly to third-party services such as Splashtop, GoToMeeting, and Join.me. These tools allow for remote desktop sharing and assistance for a user through screen sharing. Unfortunately, Quick Assist does not support

file transfers. Quick Assist does work behind routers and firewalls, unlike prior built-in applications. Quick Assist is becoming a tool of choice for many helpdesk tasks. Quick Assist even has its own shortcuts for easy access by the user.

FIGURE 8.11 Windows Quick Assist tool



You can launch Quick Assist on Windows 11 by clicking the Start menu and typing **Quick Assist**. Once the Quick Assist tool is started on your computer (the assistant), choose Give Assistance. You will then be prompted to log in with a Microsoft account. Once you are signed in, you will be presented with a six-digit number, which will be valid for 10 minutes. On the end user's (person being helped) computer, the end user will launch Quick Assist and select Get Assistance. The end user will need to obtain your six-digit code and enter it into the dialog box when prompted. Once the number is entered on the end user's side, both computers will connect, and you will be asked what type of control you want to provide for the end user: View Only or Full Control. Of course, the end user has to allow the control, but you can provide remote assistance without installing any third-party software.

Quick Assist offers chat functionality to the assistant and the end user in a chat window. You can launch this window by clicking Toggle Instruction Channel on the Quick Assist toolbar. You can also launch Task Manager automatically in later versions of the tool. By clicking the Task Manager item on the Quick Assist toolbar, you can remotely launch Task Manager on the end user's operating system. An annotation tool allows you to draw on the end user's screen to guide them. Select this tool by clicking the Annotate option on the Quick Assist toolbar. However, Quick Assist does not have a file transfer utility. Any file transfers

must be done with a file sharing service, where the user downloads the file from the Internet link you provide.

Third-Party Tools

Third-party tools such as Splashtop, GoToMeeting, and Join.me have their own unique features. Each tool fits into a category of screen sharing, videoconferencing, file transfer, or desktop management. When choosing a third-party tool, you should evaluate the main requirements and the category the software excels at. Let's examine the various categories to help you better identify the best software for your needs.

Screen Sharing Both the MSRA tool and Quick Assist fit into the category of *screen-sharing* software. Screen-sharing software will share your screen with another user, and it is most often used to troubleshoot problems. Two popular screen-sharing software utilities are Splashtop and TeamViewer. Both allow you to share your screen with another person, similar to the MSRA tool and Quick Assist. However, these tools have other built-in features that the MSRA tool and Quick Assist do not, such as file sharing and screen blanking.

Videoconferencing The *videoconferencing* category includes any screen-sharing software that allows more than one person to screen-share. The software also has some additional functionality, such as calendar invite, call-in numbers, and attendee management. Microsoft Teams, Webex, Zoom, and GoToMeeting are all examples of videoconferencing software. These tools are often used to present ideas and collaborate between groups of people.

File Transfer The ability to file transfer between two computers is not really a category of software; it is a feature of screen-sharing and/or videoconferencing software. *File transfer software* can, however, be a category of remote access software if you are solely looking for that feature. Popular file transfer software includes OneDrive, Dropbox, and Google Drive. These file transfer utilities allow the sharing of files between users and serve a specific purpose for the distribution of files bidirectionally.

Desktop Management The last category is an extension of some screen-sharing software packages such as Splashtop and TeamViewer. Both of these products can be used in an ad hoc fashion for screen sharing. They can also be installed on the operating system and configured to allow an inbound connection. When these tools are installed to await an incoming connection, they are considered to be *desktop management software* solutions. Many of these tools also allow integration with other management tools like Intune and Jamf. This integration gives you complete device management, and a lot of these tools will even show when a device is online and connectable.

Security Considerations

Each of the remote access technologies discussed in this chapter have security considerations. Before implementing a remote access technology, you should determine what type of data is

going to be exchanged and whether the level of encryption is sufficient. Telnet, which is not encrypted, might be fine if simple data is being transmitted, such as temperature or humidity readings. However, if passwords, configuration, or any type of sensitive information will be transmitted, then a more secure protocol, such as Secure Shell, should be used.

Beyond the data in transit and the method that provides the transit, there are other security considerations. One of the biggest concerns is any agent that awaits connection and is exposed to the Internet. A threat agent can exploit these software packages and compromise a host via these software packages. To combat this problem, keep your software package up to date, and if multifactor authentication is available, use it.

Videoconferencing packages should be secured so that a password is required to join a meeting. Setting a password will thwart conference bombing, also known as *Zoombombing*. This is the act of a threat agent guessing the meeting ID and joining an otherwise private conversation. This type of security concern isn't really your typical threat, but it is a liability to your organization if not avoided.



Zoombombing became an annoyance on the Internet during the COVID pandemic. Organizations were disbanded from their campuses and employees were sent home to work. Because Zoom conferencing software is a dominant software product for videoconferencing, many organizations used the software to continue doing business. With the higher usage and the relaxation of security by organizations, Internet trolls guessed meeting IDs and joined otherwise private conferences. The trolls mainly caused disruption to the ongoing conferences, but many exposed the participants to unsavory behavior.



Exam Objective 4.10, "Explain basic concepts related to artificial intelligence (AI)," requires you to know AI concepts. You should know the basic components of AI systems, how they are trained, application, policies, limitations, and private versus public datasets.

Artificial Intelligence

The goal of artificial intelligence (AI) is to allow computers to simulate or imitate human intelligence. Although there are certain parallels between AI and human intelligence, their nature and capacity are essentially distinct. Consciousness, self-awareness, and emotional experience are hallmarks of human intellect, which is a biological evolution. Human intelligence enables people to think abstractly, be creative, adapt to new circumstances, and learn from their experiences. Human culture and social interactions are closely linked to our intelligence; this type of intelligence is sometimes referred to as emotional intelligence.

For the majority of us, it allows us to comprehend and react to intricate social signs and emotional subtleties.

AI is a product of human ingenuity and is intended to emulate specific cognitive processes like learning, thinking, and problem-solving. AI systems are excellent at processing large volumes of data fast and precisely, especially those driven by machine learning and deep learning. In some areas, they are more accurate and efficient than humans at recognizing patterns, making predictions, and carrying out tasks. AI can, for instance, create text that appears human, streamline supply networks, and identify ailments by analyzing medical photos.

The subject of AI has become a hot topic as it has become integral in our lives and complementing human intelligence. Math teachers often said, "You must learn math because you are not always going to have a calculator in your pocket." Today, we not only have calculators as one of the many mobile apps, but we also have advanced AI that can help us solve problems. In this section, we will explore the components of AI, the characteristics of data AI uses, as well as the use of AI and considerations.

Evolution

Believe it or not, AI is not a new technology. In fact, the idea of neural networks and machine learning dates all the way back to the 1950s. John McCarthy and Alan Turing are claimed to be the original pioneers who introduced the concept of AI. In Turing's groundbreaking work "Computing Machinery and Intelligence," he raised the question of whether machines were capable of thought. The phrase *artificial intelligence* was first used by McCarthy, who also arranged the Dartmouth Conference in 1956. The Dartmouth Conference is regarded as the beginning of AI as a field of study.

In the preceding years, AI research helped fuel the subject and produced the first artificial neural network called the Perceptron. It was designed by psychologist Frank Rosenblatt in 1958. The Perceptron design was based on 1s and 0s and was a very simple proof of concept. The Perceptron would take input, weigh the input (which adjusted the input's importance), use a bias to make a prediction, and create output based on the weighted sum of the bias and prediction. This research is rudimentary but similar to how AI works today. It introduced the concept of machine learning to produce the bias and prediction.

The computer boom of the 1960s and 1970s helped continue the progression of AI research. The first working form of AI on computers of this era was something called expert systems. Expert systems were nothing more than a computer program that mimics human experts. It did this by using data and basic programming that mimicked a thought pattern. As an example, some of these expert systems allowed for the diagnosis of patients and a treatment plan. These expert systems were far from the AI we know today, but they were an important stepping-stone to where we are now. The early decades eventually paved the way for development of foundational concepts and algorithms, such as symbolic reasoning and early neural networks. However, progress was slow due to limited computational power and data.

The continued work on expert systems and the growth of machine learning methods in the 1980s and 1990s helped the expansion of AI. However, AI was limited to rudimentary chatbots and less well-known systems—hardly considered AI. The real breakthrough came in the 21st century with the availability of large datasets and advancements in neural network architecture. This gave new breath to machine learning algorithms, because the data was there to feed these systems and thus better decision models could be made. Of course, machine learning algorithms got better as a result, but the large dataset was the catalyst. The best example of this generation of AI is IBM's Deep Blue. In 1997 Deep Blue went head-to-head with the reigning chess world champion Garry Kasparov. Deep Blue won the title of chess master after a six-game marathon! This was a milestone for AI since its inception in the 1950s.

The true revolution in AI came in the 2010s with the advent of deep learning, a subset of machine learning that utilizes multilayered neural networks. In the decade that followed, we have seen the advancement of several foundation models. These foundation models allow us to perform image recognition, word recognition, voice recognition, and ultimately natural language processing. Each of these models was a milestone in AI.

As early as 2006, Google introduced language translation, or simple text-to-text translation from one language to another. In 2016, Google introduced the Google Neural Machine Translation (GNMT) system, an advancement that allowed the language translation service to take images with words and translate them to any language. In 2022, Google launched Translation Hub, which provides document translations. In the following years, Google added more languages and larger datasets. This is just one example of the advancement of AI to date. There are many other examples, such as ChatGPT from OpenAI, Microsoft Copilot, Apple Intelligence, and Google Gemini. These are just a few popular AI models that use generative AI. AI just keeps getting better and better with each refinement of these milestones. Just imagine what the next 70 years will look like.

Components

The subject of AI, like many other complex topics, requires us to know everything before we can learn something. This section explains the various components of AI we discussed in the previous section. As you progress through this section, how AI works should become clearer.

Machine Learning (ML)

Machine learning (ML) is a broad field of AI that focuses on developing algorithms that allow computers to learn from and make predictions or decisions based on data. ML works by using algorithms to identify patterns and relationships within large datasets. These algorithms can then make predictions or decisions based on new data inputs. For example, if you were to memorize a series of numbers such as 11364, 90210, 89049 and were presented with the first three digits of 902, you could predict the last two, or 10. This is a type of ML called supervised learning.

There are three different types of learning displayed in learning models. The supervised learning model allows AI to predict or decide, based on historical data. In an

unsupervised learning model, the AI analyzes data to derive a new outcome. And the reinforcement learning model allows AI to properly sequence events and learn by reward. All of these models make up some of the human intelligence that we use daily as human beings. Neural networks, deep learning, and tokenization are other examples.

Supervised Learning

With supervised learning, the model is trained on a labeled dataset, meaning that each training example is paired with an output label. The model learns to map inputs to the correct outputs. Let's build off the prior example of 5 digit numbers being trained as sequences such as 90210. The label for these numbers would be ZIP codes and of course the output would be Beverly Hills, California. Another example is that it can be used for email spam detection. The model is trained on a dataset of emails labeled as "spam" or "not spam" and learns to classify new emails accordingly.

Unsupervised Learning

In a supervised model, the dataset contains labeled data, like spam or not spam. In an unsupervised model, the dataset does not contain labeled data. With unsupervised learning, the model is given data without explicit instructions as to what to do with it. This model is often used to identify hidden patterns or groupings within the data. As an example, if you owned a retail store and had a large volume of transactions, you could use this model to identify segments of customers. You could do this by grouping the customers into clusters of transactions and then labeling the output as frequent shopper, bargain hunter, or high-value customer based on their transactions.

Reinforcement Learning

This type of learning model is based on a sequence of decisions to a desirable outcome or rewards. The model learns to achieve a goal by maximizing cumulative rewards. This type of learning model is similar to a computer playing chess against a human. The sequence of decisions leading up to checkmate is analyzed for the cumulative reward. Another example is a self-learning robot navigating a maze, where each movement is calculated against the previous with the goal of solving the maze.

Neural Networks

Neural networks are a class of machine learning model that mimic the human brain's structure and function. Neural networks are made up of several layers of interconnected nodes that process and send data, similar to real neurons. Each neuron receives input and applies a transformation to the data using weights and biases; then it passes the output to the next layer. This process allows neural networks to learn complex patterns and relationships within a dataset and makes them highly effective for tasks such as image recognition, natural language processing, and game playing.

Neural networks play a crucial role by enabling machines to perform tasks that typically require human intelligence. There are several different types of neural networks depending on the task. For example, convolutional neural networks (CNNs) are specialized for image and video analysis. These neural networks allow AI systems to identify objects, faces, and

even emotions in visual data. Recurrent neural networks (RNNs) are good at processing sequential data, making them ideal for applications like language translation, speech recognition, and time-series prediction. These are just a few examples, and several books have been dedicated to this topic.

Deep Learning

A branch of machine learning called *deep learning* uses artificial neural networks to mimic how the human brain interprets data. These neural networks are made up of layers of connected neurons, or nodes, that can do basic calculations. Deep learning's strength is its capacity to learn from experience and get better without explicit programming. Deep learning is used in various applications, such as image and speech recognition, natural language processing, and autonomous driving.

Large volumes of data are used to train deep learning models. For example, millions of labeled photos may be loaded into a deep learning model for image recognition. The model will begin with simple features like edges and textures at the early levels and progress to more intricate structures like shapes and objects in the later layers. The model progressively extracts details from the photos as it passes the photos deeper into its layers. This hierarchical feature extraction enables the model to recognize and classify images with high accuracy.

Tokenization

Tokenization is the method of dividing text into smaller parts known as tokens. Depending on the tokenization technique employed, these tokens may be words, subwords, or even single characters. Tokenizing the text `"Hello, world!"` might look like (‘Hello’ - ‘,’ - ‘world’ - ‘!’); this is just one example. This process is essential because it converts unprocessed text into a format that artificial intelligence models can train on.

Tokens are the basic units of input for AI models. Each token is numerically represented, often using embedding techniques that capture the token's semantic meaning in a high-dimensional space. This numerical representation is what the model uses to discover patterns and relationships in the data.

Tokenization helps manage the variability in language. By breaking down text into tokens, models can handle different forms of words, such as plurals or conjugations, more effectively. For example, the word *running* might be tokenized into (`"run" - "ning"`), allowing the model to recognize the root word `"run"` across different contexts. The following is a short list of the common tokens used to train AI models.

Word Tokens Each word is treated as a separate token. This is the simplest form of tokenization and can be less effective for languages with complex structure.

Subword Tokens Words are broken down into smaller meaningful units. This method helps handle out-of-vocabulary words better. For example, `"unbelievable"` might be tokenized into (`"un" - "believe" - "able"`).

Character Tokens Each character is treated as a token. This method is useful for languages with a large number of unique characters or for tasks requiring fine-grained text analysis.

Byte-Pair Encoding (BPE) This is a hybrid approach that merges the most frequently occurring character pairs in a text body. BPE is commonly used in modern NLP models to balance the granularity of tokens.

Tokenization also improves the efficiency of training AI models. Smaller tokens mean that the model can process text in manageable chunks, reducing computational complexity. This is particularly important for large-scale models like transformers, which need to handle vast amounts of data.

Natural Language Processing (NLP)

Natural language processing (NLP) is a component of artificial intelligence that focuses on the interaction between computers and humans through natural language. NLP enables machines to comprehend, interpret, and produce meaningful and practical human language. These tasks include text summarization, sentiment analysis, language translation, and speech recognition. Large volumes of natural language data can be processed and analyzed by AI systems using NLP. This allows AI to automate and enhance a variety of applications such as chatbots, virtual assistants, and language translation services.

Human language is complex and diverse, and this creates problems for NLP. The human language is frequently imprecise, context-dependent, and nuanced, in contrast to structured data. NLP systems combine machine learning algorithms with linguistic rules to overcome these difficulties. Deep learning models assist in using recurrent neural networks (RNNs) to discover patterns and relationships within the data. NLP also uses methods like tokenization, part-of-speech tagging, and named entity identification to aid in the breakdown and classification of text. These developments have greatly increased the precision and effectiveness of NLP applications, improving their dependability and potency.

Another exciting aspect of NLP is its potential to bridge language barriers and facilitate global communication. With the development of sophisticated language models, AI can now translate text and speech between multiple languages with impressive accuracy. This has profound implications for international business, education, and cross-cultural interactions. Additionally, NLP can be used to analyze social media and online content, providing valuable insights into public opinion, trends, and emerging issues.

Foundation Models

Foundation models are large-scale, pretrained models. These models leverage the transformer architecture that allows the model to handle sequential data through mechanisms like self-attention, allowing them to understand and generate complex patterns in text, audio, and video.

There are several different types of foundation models, such as Generative Pretrained Transformer (GPT) and OpenAI's DALL-E, among others. GPT is trained on a large amount

of text-based data, which in turn enables it to generate coherent and contextual information. OpenAI's DALL-E is trained on text-based and image-based data to create AI-generated images. Others like Microsoft Copilot are trained on a mixture of public and private data. These are just a few of the many foundation models available today.

Let's examine the components of a foundation model. GPT stands for Generative Pretrained Transformer. The generative part is the ability for the model to generate new content from its vast amount of pretrained data. The pretrained component is the data that is fed into the algorithm through machine learning or deep learning. The transformer component of GPT is what allows the foundation model to transform words into serialized data. Transformers can also be used to change serialized data into another form of data, such as images, audio, or even another language. The following are some of the more common transformers used today with AI.

Large Language Models (LLMs)

Large language models (LLMs) are the text-based models that understand written words and that can generate human-like responses. These models have been trained on very diverse datasets from books, articles, websites, literature, and more. This enables the LLM to understand context, nuance, and subtleties in language. They can mimic human communication, and this makes them a valuable tool in applications such as customer service, content creation, quality management, and a vast array of other applications.

They can even be tailored to have personas based on attitude, gender, behavior, and title, just to name a few factors. You could tailor an AI customer service representative to be flexible or rigid or even have the authority of a manager. Another way that LLMs are used is for quality management of calls. Since LLMs have a great grasp on the human language, they can pick up when someone is pleased or agitated, based on language. This sentiment analysis helps call centers with the disposition of calls.

Audio Transformers

The lines blur with audio transformers, because they work in conjunction with LLMs, but they are not specifically tied to text-based models. Audio transformers are used for audio-related tasks, such as speech-to-text and text-to-speech. They can understand spoken words and convert them to sequential data for further processing through LLMs. They can also transform sequential data from the LLMs back to speech with the right inflection, tone, pronunciation, and cadence. One great example of this is Apple's Siri; in recent years it went from a personal assistant to an integral part of Apple's AI ecosystem.

Audio transformers can also perform many other tasks, such as language translation, noise reduction, music generation, and audio classification, and it can even alter human speech. This short list is just a few of the tasks that AI performs for us today.

Video Transformers

Video transformers might seem like a real stretch from audio or text, but they are no different than image transformers. Video transformers either concentrate in an area of the video or they generate an image 30 times a second to create a moving picture. The most impressive and intriguing application of video transformers is deep fakes. They are so impressive and

advanced that the Screen Actors Guild-American Federation of Television and Radio Artists (SAG-AFTRA) went on strike to protest the use of AI in Hollywood.

Although deep fakes are interesting, video transformers are used for a wide array of other applications. Content creation, video editing, surveillance and security, healthcare, sports, and even video-based social media platforms are just a few of the many applications used with video transformers.

Datasets

Data is an essential component of any AI system. The more data that can be pretrained into a model, the better. LLMs and other foundation models need to be pretrained on insane amounts of data. These data sources can be terabytes or even petabytes in size. Remember, pretrained is GPT's middle name.

The old adage for computers, garbage in, garbage out, is true for AI as well. The data source for any AI model needs diversity and quality. The data can be sourced from any of the following channels:

Structured Data This type of data is highly organized and easily searchable. It is typically in the form of relational databases and spreadsheets. Examples include customer information, transaction records, and inventory data.

Unstructured Data Unlike structured data, unstructured data lacks a predefined format. The lack of structure makes it more challenging to process. This category includes text documents, emails, social media posts, images, and videos.

Semi-Structured Data Semi-structured data falls between structured and unstructured data. It includes the elements of both, such as JSON and XML files, which have a loose structure but do not fit neatly into relational databases.

Real-Time Data Real-time data is generated continuously by various sources, such as sensors, IoT devices, and streaming services. This data is crucial for applications that require up-to-date information, such as autonomous vehicles, financial trading systems, and real-time monitoring solutions.

External Data Sources These include publicly available datasets, third-party data providers, and open data platforms. Examples include weather data, economic indicators, and demographic information.

The content of these sources can and should be vast as well. The content can be provided from movie databases, legal databases, scientific journals, research papers, public datasets, government databases, books, music, and the list goes on. Just as a child learns from the time they are born to form structured meaning of their world, AI systems do the same as they are fed diversity and quality of knowledge.

Privacy vs. Security

As AI models are fed with data, they get better with their analysis, prediction, and generation of response. As we use these systems, we contribute to the learning process in a few different

ways. The first is referred to as continuous learning, or online learning, which is when content from the user's input is tokenized and fed into the AI model for pretraining. A great example of this is movie and music streaming services. They present you with the next thing to watch or listen to, based on your likes in comparison to others. Netflix, YouTube, Pandora, and Spotify are just a few content providers that use AI.

If you don't pay for the service, you are the service. That is a great sentiment to live by in today's age of technology. We surrender some, or all of our privacy, when we use services that ingest our participation. Sometimes we voluntarily give up our privacy, and other times we don't even realize we are doing so. Alexa, Apple Home, and Google Home listen to your conversations and spring to attention when you speak to these devices. The AI cloud behind the scenes learns from your input.

There are less obvious ways that information is used and privacy is surrendered, such as your use of your smartwatch or cell phone. These devices can track every breath you take and where you take it. By examining cell phone location data, relationships can be learned. For example, your phone is next to your significant other's between 11 p.m. and 7 a.m. Your phone is in proximity to your colleagues between 8 a.m. and 5 p.m., and from 7 p.m. to 10 p.m. next to your friends. If location services are turned on, patterns of travel can be learned. All this data is potentially available to providers and can be used for AI. There are of course privacy laws that should prevent breaches of your privacy. These are just a few ways we surrender our privacy.

Now that this section made you grab for the tinfoil hat, let's discuss security, which goes hand in hand with privacy. Data security focuses on protecting data from unauthorized access, breaches, and cyberattacks. You might wonder what that has to do with AI? A lot, because AI is pretrained on both material found online and material you enter into it. Threat actors are leveraging AI to uncover and exploit leaked data that may have been inadvertently included in the pretraining datasets of large language models (LLMs).

Human error is often the cause of data leakage where private or proprietary information is inadvertently leaked into LLMs. In a recent incident in 2023, Samsung experienced a data breach when employees inadvertently leaked sensitive information while using ChatGPT. The breach occurred as engineers at Samsung's semiconductor division input confidential data, including source code and internal meeting notes, into ChatGPT to optimize processes and create presentations. This data, once entered into ChatGPT, became part of the AI's training data, making it accessible to OpenAI, the company behind ChatGPT. Samsung has since banned the use of generative AI tools like ChatGPT and is developing its own in-house AI solutions to prevent future leaks. This incident highlights the risks associated with using AI tools in handling sensitive information and underscores the need for stringent data security measures.

Protecting yourself from these security concerns is simple. As an administrator you can create policies that restrict the usage of AI. Microsoft Defender Security supports creating policies that will deter employees from using AI and alert you to the concern. The other tact is to embrace AI, as it is not going away and seems to be entangled in everything. Microsoft offers an AI model called Copilot that keeps your data private by creating a private dataset and segmenting it from public data. In addition, Copilot abides by the security in your

organization. If you ask Copilot a question about the average salary at your organization, it won't dip into Human Resources files, unless you have access to them. This type of AI has been dubbed protected AI or enterprise AI; every vendor has their own buzzword.

Unfortunately, data leakage is not the only inherent problem with AI. Threat actors are using AI tools to help forge emails, research organizations, and even create malware. A threat actor can now formulate an email with these tools and custom-tailor it to your organization. They don't even need to speak your native language! Threat actors can also use tools like ChatGPT to create malware. Typically, mainstream AI tools like ChatGPT, Gemini, and Copilot will censor your requests for illegal activity. However, not all AI tools restrict content, and some sites even promote their use for illegal activity.

Integration

If your organization has not used AI or entertained incorporating it, your organization will not evolve in this new digital age. Competing organizations that do integrate AI will have a competitive advantage. Whether it is creating a sales presentation quicker than your organization, or the allure of AI for your customers, it creates a new competitive race.

So how will your organization stay ahead of the pack? It depends on how your organization chooses to integrate AI. In the early 2010s, business intelligence (BI) became a booming topic for organizations. BI allowed organizations to see relationships in their data, such as top customers, based upon location, time of year, or even the phases of the moon. Today we can use private AI to pretrain our data and extrapolate relationships in data we could not normally see.

Outside of business strategy and AI, Microsoft Copilot allows the summary of Teams meetings, the creation of presentations from Word documents, and meaningful searches of private data with natural language. The marriage of Copilot with the Microsoft 365 suite of products has become a productivity booster.

From an information technology aspect, AI can help you write code, debug code, formulate SQL statements, analyze packet captures, just about anything you can dream up! AI is to network administrators today as the first search engines were to network administrators 25 years ago. We can now use natural language to ask questions and get meaningful answers. One of the coolest tricks with AI is the transformation of information with the use of transformers. You can ask AI to translate pseudocode to PowerShell, Python, or even machine language for an 8080 processor! In Exercise 8.3 you perform this transformation of code.

EXERCISE 8.3

Creating a Script with AI

In the following exercise you will ask ChatGPT to create a PowerShell script from pseudocode. You will then ask it to create a Windows batch script with the same code.

1. Launch your preferred web browser and navigate to ChatGPT by entering `https://chat.openai.com` in the address bar, and press Enter.

EXERCISE 8.3 *(continued)*

2. Log in to ChatGPT. If you already have an account, click the Log In button. Enter your email address and password, then click Log In. If you don't have an account, click Sign Up and follow the prompts to create one.
3. Compose your request by entering
   `Write a PowerShell script with the following pseudocode: Loop ( print 1, wait, print 0 )`.
4. Review the output and test it in a PowerShell terminal.
5. Now ask ChatGPT to create a Windows batch script from the same code.
6. Review the output and test the code in a batch file.
7. Tell ChatGPT to modify the code, adding counters, then have it transform the code back to PowerShell.

For a bonus, you can have ChatGPT transform the code into other languages, such as C++, Python, or even COBOL!

Limitations

With great power comes great responsibility! This responsibility starts with your use of results from AI. You need to be able to spot the limitations of AI by using some critical thinking applied to your results.

Biases

The answers you get can often exhibit various types of bias. This is typically reflected on the data they are trained on and the design choices made by developers. The choice of algorithm can also be biased due to flawed design or assumptions. The following are some of the biases that you could encounter:

Algorithmic Bias Algorithmic bias occurs when the design of an algorithm favors certain outcomes over other outcomes. This can happen due to the way the algorithm is structured or the assumptions made during its development. For example, if an algorithm is designed to prioritize efficiency over fairness, it might systematically disadvantage certain groups of people. This type of bias can lead to unfair treatment and create inequalities.

Historical Bias Historical bias arises when AI systems are trained on data that reflects past inequalities or prejudices. Since AI models learn from historical data, they can inadvertently inherit and amplify these biases. For instance, if a hiring algorithm is trained on data from a time when certain demographics were underrepresented in certain jobs, it might continue to favor candidates from historically dominant groups, thus reinforcing the status quo.

Data Bias Data bias occurs when the data used to train an AI model is not representative of the real-world population or scenarios it will encounter. This can happen due to sampling errors, data collection methods, or intentional exclusion of certain data points. As a result, the AI system may perform well on the training data but poorly in real-world applications, leading to biased outcomes. For example, facial recognition systems trained predominantly on lighter-skinned individuals may have higher error rates for darker-skinned individuals. This bias made the headlines in 2018, when researchers from MIT and Stanford found this bias in commercial AI systems.

Confirmation Bias Confirmation bias in AI refers to the tendency of algorithms to favor information that confirms preexisting beliefs or hypotheses. This can happen if the training data or the feedback loop reinforces certain patterns while ignoring contradictory evidence. For example, a recommendation system might keep suggesting similar content to users based on their past behavior, thereby limiting their exposure to diverse perspectives and reinforcing their existing preferences.

Emergent Bias Emergent bias arises when an AI system interacts with users or other systems in ways that were not anticipated by its designers. This type of bias can emerge over time as the system learns from new data and adapts to changing environments. For instance, a chatbot might develop biased responses based on the interactions it has with users, especially if it encounters a lot of biased or harmful input. There is more than one story of Internet trolls training a chatbot to say and act in deviant ways. In one instance the chatbot didn't last for more than a few hours.

Evaluation Bias Evaluation bias occurs when the metrics or benchmarks used to assess the performance of an AI system are themselves biased. This can lead to misleading conclusions about the system's effectiveness and fairness. For example, if an AI model is evaluated using a dataset that is not representative of the target population, its performance metrics might not accurately reflect its real-world capabilities, leading to biased decision-making.

Addressing these biases requires a concerted effort to ensure diverse and representative data. Do not trust AI results wholeheartedly and fact-check your results. Knowing that these biases exist and applying some basic critical thinking skills can help you identify bias results.

Hallucinations

AI hallucinations happen when a system produces false or nonsensical information, frequently presenting it with assurance as fact. The intricacy of the task, limits in the training data, or AI's attempt to fill in the blanks with facts that seem true but aren't can contribute to this issue of hallucinations. Consider asking AI to create a pizza recipe. If it has a hallucination, it may recommend an illogical way to adhere cheese on pizza, such as applying glue. This demonstrates how AI can generate strange or inaccurate results when it deviates from its training data or logical thinking, and it is obviously incorrect.

Another instance of hallucinations is when AI provides historical information inaccurately. When AI is questioned about a well-known event, it can make up information.

For example, it might respond that Napoleon Bonaparte won the Battle of Waterloo, which is not true historically true. These hallucinations can be problematic, particularly in situations where accuracy is crucial, such as in legal counsel or medical diagnosis. The implications of a hallucination are as follows:

Misinformation AI hallucinations can lead to the dissemination of false information. For instance, if an AI generates incorrect details about a developing emergency, it can mislead the public and hinder effective response efforts.

Healthcare Risks In medical applications, AI hallucinations can be particularly dangerous. An AI might misdiagnose a condition, such as incorrectly identifying a benign lesion as malignant, leading to unnecessary treatments and anxiety for patients.

Legal and Financial In the legal and financial sectors, AI hallucinations can result in incorrect advice or data, potentially causing significant financial losses or legal complications.

Erosion of Trust Frequent AI hallucinations can erode user trust in AI systems. If users repeatedly encounter incorrect or nonsensical outputs, they may become skeptical of the technology's reliability and hesitate to use it in critical applications.

Bias and Discrimination AI hallucinations can also reflect and amplify biases present in the training data. This can lead to discriminatory outcomes, such as biased hiring recommendations or unfair treatment in customer service scenarios.

Operational Inefficiencies In industries like manufacturing or logistics, AI hallucinations can cause operational inefficiencies. For example, an AI might incorrectly predict demand, leading to overproduction or stock shortages.

Educational Impact In educational settings, AI hallucinations can misinform students. If an AI tutor provides incorrect explanations or answers, it can hinder learning and perpetuate misunderstandings.

Security Vulnerabilities AI hallucinations can create security vulnerabilities. For instance, an AI system responsible for monitoring network security might incorrectly identify benign activities as threats or fail to detect actual threats, compromising the security of the system.

Creative Industries In creative fields like writing, art, or music, AI hallucinations can produce outputs that are nonsensical or culturally insensitive, potentially offending audiences or misrepresenting artistic intent.

Customer Service Issues AI-powered customer service bots might provide incorrect information or solutions, frustrating customers and damaging the company's reputation.

Mitigation Strategies

A variety of techniques are used to mitigate AI hallucinations, in turn enhancing the precision and dependability of the AI system. Making sure the training data is of high quality is one of the best strategies. Pretraining the AI system with high-quality data lowers the

possibility of producing erroneous or nonsensical results. It also allows the AI model to learn precise patterns. It is also essential to update the training data on a regular basis to represent correct and most up-to-date information.

Improving the AI model itself is another crucial tactic, but typically it is out of our reach as end users of commercial systems. Other tactics include applying organized data templates, employing sophisticated algorithms that are better able to handle intricate data patterns, and fine-tuning the model with more training on certain tasks or domains. Hallucinations can also be considerably decreased by implementing human-in-the-loop procedures, in which human specialists examine and verify the AI's results. Before any created content is shown to users, this method guarantees that it is correct and trustworthy.

Continuous monitoring and evaluation of AI systems is an essential step for mitigating hallucinations. Implementing validation processes and regularly assessing the AI's performance can help identify and address any issues promptly. By combining these strategies, organizations can significantly reduce the risk of AI hallucinations and ensure that their AI systems are reliable and trustworthy.

Accuracy

Despite AI's immense versatility and power, AI is inherently inaccurate for several reasons. These include the quality and scope of the training data, which may create biases or knowledge gaps as previously discussed. AI models might also have trouble grasping the subtleties and context of the human language, which could result in misunderstandings or mistakes. It is difficult for AI to give accurate responses in every circumstance since real-world scenarios are frequently more complex than the training data. Also, when presented with new or unclear inputs, AI systems may make mistakes because they lack actual knowledge and rely on patterns rather than comprehension. This is evident, for example, if you ever ask AI to create a picture of a cat looking at the moon. You will find that the cat is looking through a telescope like an astronomer, because that is what AI knows when it associates moon and the act of looking or examining.

Considerations

AI has become wildly popular over the past 3 to 5 years. When planning the integration of AI into various applications and processes, be certain to address the considerations of AI adoption by your organization. Appropriate use of AI involves leveraging its capabilities to enhance productivity while respecting privacy and ethics. In this section, we will discuss the considerations for AI in organizations.

Policy

Organizations should establish a comprehensive AI policy that prioritizes ethical considerations, transparency, and accountability. Regular audits and assessments should be conducted to monitor and mitigate potential risks that AI could introduce. Additionally, organizations must plan for continuous learning and improvement in AI, encouraging employees to stay informed about the latest advancements. This should include reviewing the ethical standards of the AI in place.

Some of the typical policies that an organization might adopt are as follows:

Ethical Framework Organizations should develop an ethical framework that guides the development and deployment of AI systems. This framework should include principles such as fairness, accountability, transparency, and respect for user privacy. It should also address issues like bias, discrimination, and the potential for misuse.

Data Governance Establishing robust data governance policies is critical. Organizations must ensure that the data used to train AI models is accurate, relevant, and free from biases. Policies should also cover data security, privacy, and compliance with regulations such as national regulations like General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA), just to name a few.

Transparency and Explainability AI systems should be transparent, and their results should be explainable. Organizations need to provide clear information about how AI decisions are made and ensure that users can understand and trust these decisions. This includes documenting the AI models, their training data, and the decision-making processes.

Accountability and Oversight There should be clear lines of accountability for AI systems. Organizations should designate responsible individuals or teams to oversee AI initiatives, conduct regular audits, and ensure compliance with ethical and legal standards. This also involves setting up mechanisms for reporting and addressing any issues or concerns related to AI use.

Continuous Monitoring and Improvement AI systems should be continuously monitored to ensure they perform as intended and do not cause harm. Organizations should implement feedback loops to gather user input and make necessary adjustments. Regular updates and retraining of AI models are essential to maintain their accuracy and relevance.

Employee Training and Awareness Organizations should invest in training programs to educate employees about AI technologies, their potential impacts, and ethical considerations. This helps build a culture of responsible AI use. This training and awareness also empower employees to make informed decisions.

Stakeholder Engagement Engaging with stakeholders, including customers, employees, and regulators, is important for building trust and ensuring that AI policies align with the organization's values and culture. Organizations should seek feedback and collaborate with external experts to refine their AI strategies.

Appropriate Use

Promoting the appropriate use of AI within an organization is important to retain the culture and respect from employees. It is essential to recognize that AI can significantly transform various aspects of business operations. However, it is paramount that the overarching reason to adopt AI is to raise the level of service and not replace people. As an example, a customer service AI-powered chatbot can handle routine inquiries, freeing up human agents to tackle

more complex issues. In finance, AI algorithms can detect fraudulent activities by analyzing transaction patterns in real time, enhancing security and trust.

The appropriate use of AI does not only rest on the shoulders of the organization. The employees of the organization should be responsible for their actions and use of AI for their daily tasks. It's okay to use AI to speed up repetitive tasks, help compile reports, and lend guidance with writing an email. However, the employee engaging in AI needs to treat anything derived from AI as a first draft. The employee should always proof and refine the final product. This collaborative approach ensures that AI complements human capabilities rather than replacing them.

Plagiarism

Plagiarism is a big concern for organizations that publish information, as well as organizations that require end users to provide original works. Academia is one such place that requires students to provide original work to show that they have learned an objective and can synthesize the knowledge. This is similar to instructors requiring you to show mathematics work in math class, rather than just the answer. They want to see that you have learned the concepts and can apply them.

Outside of education, plagiarism is a consideration for corporate organizations. These organizations could suffer legal ramifications if work has been plagiarized by an AI system. The outcome could create an erosion of trust for the organization and carry many other undesirable consequences.

Summary

This chapter focused on the basics of scripting and introduced programming concepts such as scripting types, commenting, branch logic, loops, and use cases for scripting. Scripting is a somewhat newer objective for the CompTIA A+ objectives. It is a worthy skill for an A+ technician.

We also introduced you to the various remote access technologies, highlighting their advantages and weaknesses. In addition, we covered remote assistance tools, which allow a remote user to share their screen with you for support purposes.

Finally we concluded the chapter by discussing artificial intelligence, covering the evolution, components, potential integration, limitations, and considerations for organizations.

Exam Essentials

Understand the basics of scripting. You should understand that a programming language is compiled and a scripting language is interpreted. You should also understand the various elements of a script, such as the comment, interaction of environment variables, branch logic, and loops.

Know the various scripting languages. You should know the advantages and disadvantages of each scripting language and the use cases for scripting, as well as other security and system considerations. In addition, you should know which operating systems each of the various scripting languages operates on. Be familiar with the various script file types that correspond to each scripting language.

Know the various scripting file extensions. The `.bat` and `.cmd` extensions are associated with Windows Batch scripts. The `.ps1` is associated with PowerShell scripts. The `.vbs` extension is associated with VBScript scripts. The `.sh` extension is associated with Bash shell scripts. The `.js` extension is associated with JavaScript scripts. The `.py` extension is associated with Python scripts.

Know the various uses for scripts and their considerations. The various uses for implementing scripts are basic automation, restarting machines, mapping and remapping network drives, installation of applications, automated backups, gathering of information/data, and initiating updates. The considerations are unintentionally introducing malware, inadvertently changing system settings, and browser or system crashes due to mishandling of resources.

Know the various remote access technologies. You should know that RDP is a Microsoft protocol used for remote access. Virtual Network Computing (VNC) is an open standard similar to Microsoft RDP. SSH is an encrypted, secure console protocol. Remote Monitoring and Management (RMM) is a Microsoft protocol for remote administration. Simple Protocol for Independent Computing Environments (SPICE) is an open source protocol used for VDI. In addition, there are third-party screen-sharing and desktop management tools that function similarly and include file-sharing capabilities, as well as videoconferencing software.

Know the various components of artificial intelligence. You should know the various components of AI and understand the training of AI, neural networks, large language models (LLMs), natural language processing, and foundation models.

Understand the limitations and considerations for artificial intelligence. You should understand how biases can flaw the results of AI, as well as what you can do to mitigate the risks of bias. You should also understand how and why AI hallucinations occur and what you can do to prevent propagating wrong information. In addition, you should understand the considerations of policy, appropriate use, and plagiarism with AI.

Review Questions

The answers to the chapter review questions can be found in Appendix A.

1. Which statement about scripting languages is true?
   - **A.** Scripting languages require a compiler.
   - **B.** Scripting languages are strongly typed.
   - **C.** Scripting languages are interpreted.
   - **D.** Scripting languages have good memory management.
2. What level are scripting languages considered?
   - **A.** High
   - **B.** Mid
   - **C.** Intermediate
   - **D.** Low
3. Which type of variable will allow decimal math?
   - **A.** Boolean
   - **B.** Integer
   - **C.** Floating-point
   - **D.** String
4. Which environment variable is not inherited?
   - **A.** System variable
   - **B.** User variable
   - **C.** Program variable
   - **D.** String variable
5. Which statement will load a PowerShell variable `xvar` with a value of 2?
   - **A.** `xvar = 2`
   - **B.** `$xvar = 2`
   - **C.** `xvar = 2;`
   - **D.** `set /a xvar=2`
6. Which type of loop has a defined beginning and end, and steps from beginning to end?
   - **A.** `do while` loop
   - **B.** `while` loop
   - **C.** `if` statement
   - **D.** `for` loop

7. Which extension is used with the Windows batch scripting language?
   - **A.** `.vbs`
   - **B.** `.js`
   - **C.** `.bat`
   - **D.** `.py`

8. Which scripting language allows for the use of the Component Object Model (COM)?
   - **A.** PowerShell
   - **B.** VBScript
   - **C.** Windows batch script
   - **D.** JavaScript

9. Which extension is used with the Python scripting language?
   - **A.** `.vbs`
   - **B.** `.js`
   - **C.** `.bat`
   - **D.** `.py`

10. Which scripting language is used with Microsoft Azure and Microsoft 365?
    - **A.** PowerShell
    - **B.** VBScript
    - **C.** Windows batch script
    - **D.** JavaScript

11. Which scripting language is used within web pages to allow for interactive content?
    - **A.** PowerShell
    - **B.** Bash
    - **C.** Windows batch script
    - **D.** JavaScript

12. Which extension is used with the Bash scripting language?
    - **A.** `.vbs`
    - **B.** `.sh`
    - **C.** `.bat`
    - **D.** `.py`

13. What must be done before a Bash script can be executed?
    - **A.** `chown` permissions must be set.
    - **B.** The `execute` attribute must be set.
    - **C.** `chmod` permissions must be set.
    - **D.** An `.sh` must be added to the end of the script.

14. Which statement will load a JavaScript variable `mvar` with a value of 8?
    A. `$mvar = 8`
    B. `mvar = 8`
    C. `mvar = 8;`
    D. `set /a mvar=8`

15. Which scripting language has its own preinstalled Integrated Scripting Environment (ISE)?
    A. VBScript
    B. Bash
    C. Python
    D. PowerShell

16. Which line would be used to comment JavaScript code?
    A. `//comment`
    B. `'comment`
    C. `REM comment`
    D. `# comment`

17. Which extension is used with the JavaScript scripting language?
    A. `.js`
    B. `.sh`
    C. `.bat`
    D. `.py`

18. What is the primary goal of artificial intelligence (AI)?
    A. To replace human intelligence
    B. To simulate or imitate human intelligence
    C. To create emotional experiences
    D. To enhance human physical abilities

19. What is the primary function of convolutional neural networks (CNNs)?
    A. Language translation
    B. Image and video analysis
    C. Speech recognition
    D. Game playing

20. What is tokenization in the context of AI?
    A. Dividing text into smaller parts known as tokens
    B. Converting images into text
    C. Translating text into another language
    D. Analyzing audio data

Performance-Based Question

You will encounter performance-based questions on the A+ exams. The questions on the exam require you to perform a specific task, and you will be graded on whether or not you were able to complete the task. The following requires you to think creatively in order to measure how well you understand this chapter's topics. You may or may not see similar questions on the actual A+ exams. To see how your answers compare to the authors', refer to Appendix B.

You have been assigned to write a PowerShell script that will find other scripts in a user profile directory and all its subdirectories. Which PowerShell variable should you use, since `%UserProfile%` is an environment variable and will not run in PowerShell?

Chapter

9



Safety and Environmental Concerns

THE FOLLOWING COMPTIA A+ 220-1202 EXAM OBJECTIVES ARE COVERED IN THIS CHAPTER:

✓ **4.0 Operational Procedures**

✓ **4.4 Given a scenario, use common safety procedures.**

- Electrostatic discharge (ESD) straps
- ESD mats
- Electrical safety
  - Equipment grounding
- Proper component handling and storage
- Cable management
- Antistatic bags
- Compliance with government regulations
- Personal safety
  - Disconnect power before repairing PC
  - Lifting techniques
  - Fire safety
  - Safety goggles
  - Air filtration mask

✓ **4.5 Summarize environmental impacts and local environment controls.**

- Material safety data sheet (MSDS)/documentation for handling and disposal
  - Proper battery disposal



- Proper toner disposal
- Proper disposal of other devices and assets

- Temperature, humidity-level awareness, and proper ventilation
  - Location/equipment placement
  - Dust cleanup
  - Compressed air/vacuums
- Power surges, brownouts, and blackouts
  - Uninterruptible power supply (UPS)
  - Surge suppressor

✓ **4.6 Explain the importance of prohibited content/ activity and privacy, licensing, and policy concepts.**

- Incident response
  - Chain of custody
  - Inform management/law enforcement as necessary
  - Copy of drive (data integrity and preservation)
  - Incident documentation
  - Order of volatility
- Licensing/digital rights management (DRM)/end-user license agreement (EULA)
  - Valid licenses
  - Perpetual license agreement
  - Personal-use license vs. corporate-use license
  - Open-source license
- Non-disclosure agreement (NDA)/mutual non-disclosure agreement (MNDA)
- Regulated data
  - Credit card payment information
  - Personal government-issued information



    - PII
    - Healthcare data
    - Data retention requirements
- Acceptable use policy (AUP)
- Regulatory and business compliance requirements
    - Splash screens



In this chapter, we start by talking about safety, which includes your safety and the safety of your coworkers, as well as environmental concerns. Observing proper safety procedures can help prevent injury to you or to others.

Our discussion about the environment is two-sided. The environment affects computers (via things like dust, sunlight, and water), but computers can also potentially harm the environment. We'll consider both sides as we move through this chapter.

Next, we will cover some legal aspects of operational procedures. These include licensing of software, protection of personally identifiable information, and incident response.



Exam Objective 4.4, "Given a scenario, use common safety procedures," requires you to know the various safety procedures. You should know how electrostatic discharge (ESD) can be controlled, proper component handling and storage, compliance with government regulations, and personal safety.

Understanding Safety Procedures

The proliferation of computers in today's society has created numerous jobs for technicians. Presumably that's why you're reading this book: you want to get your CompTIA A+ certification. Many others, who don't fix computers professionally, like to tinker with them as a hobby. Years ago, only the most expert users dared to crack the case on a computer. Often, repairing the system meant using a soldering iron. Today, thanks to the cheap parts, computer repair is not quite as involved. Regardless of your skill or intent, if you're going to be inside a computer, you always need to be aware of safety issues. There's no sense in getting yourself hurt or killed.

As a provider of a hands-on service (repairing, maintaining, or upgrading someone's computer), you need to be aware of some general safety tips, because if you are not careful, you could harm yourself or the equipment. Clients expect you to solve their problems, not make them worse by injuring yourself or those around you. In this section, we'll talk about identifying safety hazards and creating a safe working environment.

Identifying Potential Safety Hazards

Anything can be a potential safety hazard, right? Okay, maybe that statement is a bit too paranoid, but there *are* many things, both human-made and environmental, that can cause safety problems when you're working with and around computers.

Perhaps the most important aspect of computers you should be aware of is that not only do they *use* electricity, but they also *store* electrical charge after they're turned off. This makes the power supply and older *cathode-ray tube (CRT)* monitor pretty much off-limits to anyone but a repairperson trained specifically for those devices. In addition, the computer's processor and various parts of the printer run at extremely high temperatures, and you can get burned if you try to handle them immediately after they've been in operation.

Those are just two general safety measures that should concern you. There are plenty more. When discussing safety issues with regard to PCs, let's break them down into four general areas:

- Computer components
- Electrostatic discharge
- Electromagnetic interference
- Natural elements

Computer Components

As mentioned earlier, computers use electricity. And as you're probably aware, electricity can hurt or kill you. The first rule when working inside a computer is always to make sure that it's powered off. If you have to open the computer to inspect or replace parts (as you will with most repairs), be sure to turn off the machine before you begin. Leaving it plugged in is usually fine, and many times it is actually preferred because it grounds the equipment and can help prevent electrostatic discharge.

There's one exception to the power-off rule: You don't have to power off the computer when working with hot-swappable parts, which are designed to be unplugged and plugged back in when the computer is on. Most of these components have an externally accessible interface (such as USB devices or hot-swappable hard drives), so you don't need to crack the computer case.

The Power Supply

Do not take the issue of safety and electricity lightly. Removing the power supply from its external casing can be dangerous. The current flowing through the power supply normally follows a complete circuit; when your body breaks the circuit, your body becomes part of that circuit. Getting inside the power supply is the most dangerous thing you can do as an untrained technician.

The two biggest dangers with power supplies are burning or electrocuting yourself. These risks usually go hand in hand. If you touch a bare wire that is carrying current, you could get electrocuted. A large-enough current passing through you can cause severe burns. It can also cause your heart to stop, your muscles to seize, and your brain to stop functioning. In short, it can kill you. Electricity always finds the best path to ground. And because the human body is basically a bag of saltwater (an excellent conductor of electricity), electricity will use us as a conductor if we are grounded.

Although it is possible to open a power supply to work on it, doing so is *not* recommended. Power supplies contain several capacitors that can hold *lethal* charges *long after they have been unplugged!* It is extremely dangerous to open the case of a power supply.

Besides, power supplies are relatively inexpensive and are considered *field replaceable units (FRUs)*. It would probably cost less to replace one than to try to fix it—and much safer.

In the late 1990s, a few mass computer manufacturers experimented with using open power supplies in their computers to save money. We don't know if any deaths occurred because of such incompetence, but it was definitely a very bad idea.

Current vs. Voltage: Which Is More Dangerous?

When talking about power and safety, you will almost always hear the saying, "It's not the volts that kill you; it's the amps." That's mostly true. However, an explanation is in order.

The number of volts in a power source represents its potential to do work. But volts don't do anything by themselves. Current (amperage, or amps) is the force behind the work done by electricity. Here's an analogy to help explain this concept: Say you have two boulders. One weighs 10 pounds and another weighs 100 pounds, and each is 100 feet off the ground. If you drop them, which one will do more work? The obvious answer is the 100-pound boulder. They both have the same potential to do work (100 feet of travel), but the 100-pound boulder has more mass and thus more force. Voltage is analogous to the distance the boulder is from the ground, and amperage is analogous to the mass of the boulder.

This is why you can produce static electricity on the order of 50,000 volts and not electrocute yourself. Even though this electricity has a great potential for work, it does very little work because the amperage is so low. This also explains why you can weld metal with a 110- or 220-volt welder. Welders use only 17 to 45 volts at the tip of the welder, but they also use anywhere from 55 to 590 amps!

If you ever have to work on a power supply, for safety's sake you should discharge all capacitors within it. To do this, connect a resistor across the leads of the capacitor with a rating of 3 watts or more and a resistance of 100 ohms (Ω) per volt. For example, to discharge a 225-volt capacitor, you would use a 22.5kΩ resistor (225V × 100Ω = 22,500Ω, or 22.5kΩ). You can also purchase a tool, known as a *grounding pen* or *discharge pen*, that will discharge any size capacitor. The tools look similar to a pen and have a wire that is connected to ground.

The Monitor

Other than the power supply, the most dangerous component to try to repair is an older style *cathode-ray tube (CRT)* computer monitor—specifically. In fact, we recommend that you *not* try to repair monitors of any kind.

The CRT monitor is a lift hazard due to the weight of these monitors. A CRT monitor can retain a high-voltage charge for hours after it's been turned off. The CRT monitor is also under a constant vacuum, which makes them implode if the glass is broken. This could cause sharp glass to scatter in all directions.

Even though we recommend not repairing monitors, the A+ exam may test your knowledge of the safety practices to use if you ever need to do so. If you have to open a CRT monitor, you must first discharge the high-voltage charge on it by using a *high-voltage probe*. This probe has a very large needle, a gauge that indicates volts, and a wire with an alligator clip. Attach the alligator clip to a ground (usually the round pin on the power cord). Slip the probe needle underneath the high-voltage cup on the monitor. You will see the gauge spike to around 15,000 volts and slowly reduce to 0 (zero). When it reaches 0, you may remove the high-voltage probe and service the high-voltage components of the monitor.



Do *not* use an ESD strap when discharging the CRT monitor. Doing so can lead to a fatal electric shock.

Working with *liquid crystal display (LCD)* monitors—or any device with a fluorescent or LCD backlight—presents a unique safety challenge. These types of devices require an *inverter*, which provides the high-voltage, high-frequency energy needed to power the backlight.

The inverter is a small circuit board installed behind the LCD panel that takes DC power and converts (inverts) it for the backlight. If you've ever seen a laptop or handheld device with a flickering screen or perpetual dimness, it was likely an inverter problem. Inverters store energy even when their power source is cut off, so they have the potential to discharge that energy if you mess with them. Be careful! Newer LED displays use LED light to backlight the display, but there is no clear way to distinguish between LCD backlit and LED backlit monitors, so use caution.

The Case

One component that people frequently overlook is the case. Cases are generally made of metal, and some computer cases have very sharp edges inside, so be careful when handling them. You can cut yourself by jamming your fingers between the case and the frame when you try to force the case back on. Also of particular interest are drive bays. Countless technicians have scraped or cut their hands on drive bays when trying in vain to plug a drive cable into the motherboard. You can cover particularly sharp edges with duct tape. Just make sure that you're covering only metal and nothing with electrical components on it.

The Printer

If you've ever attempted to repair a printer, you might have thought that a little monster was inside, hiding all the screws from you. Besides missing screws, here are some things to watch out for when repairing printers:

- When handling a toner cartridge from a laser printer or page printer, do not turn it upside down. You will find yourself spending more time cleaning the printer and the surrounding area than fixing the printer.
- Do not put any objects into the feeding system (in an attempt to clear the path) when the printer is running.

- Laser printers generate a laser that is hazardous to your eyes. Do not look directly into the source of the laser.
- If it's an inkjet printer, do not try to blow in the ink cartridge to clear a clogged opening—that is, unless you like the taste of ink.
- Some parts of a laser printer (such as the *EP cartridge*, which is an *image drum*) will be damaged if you touch them. Your skin produces oils and has a small surface layer of dead skin cells. These substances can collect on the delicate surface of the image drum (EP cartridge) and cause malfunctions. Bottom line: keep your fingers out of places where they don't belong.
- Laser printers use very high-voltage power sources to charge internal components, which can cause severe injuries.
- Laser printers can get extremely hot in order to fuse the toner to the page. Don't burn yourself on internal components. The fuser can get extremely hot!



Using an egg carton (or other container with small compartments) is a great way to store and keep track of screws that you take out of a device when you're working on it. If it's an impromptu repair and you don't have a container with compartments, you can use a piece of paper with tape that is looped. This allows you to also write down what the screws were removed from.

When working with printers, we follow some pretty simple guidelines. If there's a messed-up setting, paper jam, or ink or toner problem, we will fix it. If it's something other than that, we call a certified printer repairperson. The inner workings of printers can get pretty complex, and it's best to call someone trained to make those types of repairs.

The Keyboard and Mouse

Okay, we know that you're thinking, "What danger could a keyboard or mouse pose?" We admit that not much danger is associated with these components, but there are a couple of safety concerns that you should always keep in mind:

- If your mouse has a cord, it can catch on something, causing items to fall off a desk, or it might even be long enough that someone can trip over it. So, make sure that it's safely out of the way.
- You could short-circuit your keyboard if you accidentally spill liquid into it. Keyboards generally don't function well with half a can of cola in their innards.

Electrostatic Discharge

So far, we've talked about how electricity can hurt people, but it can also pose safety issues for computer components. One of the biggest concerns for components is *electrostatic discharge (ESD)*. For the most part, ESD won't do serious damage to a person other than

provide a little shock. But little amounts of ESD can cause serious damage to computer components, and that damage can manifest itself by causing computers to hang, reboot, or fail to boot at all. ESD happens when two objects of dissimilar charge come into contact with each other. The two objects exchange electrons in order to standardize the electrostatic charge between them. This charge can, and often does, damage electronic components.



CPU chips and memory chips are particularly sensitive to ESD. Be extremely cautious when handling them.

When you shuffle your feet across the floor and shock your best friend on the ear, you are discharging static electricity into their ear. The lowest static voltage transfer that you can feel is around 3,000 volts; it doesn't electrocute you because there is extremely little current. A static transfer that you can *see* is at least 10,000 volts! Just by sitting in a chair, you can generate around 100 volts of static electricity. Walking around wearing synthetic materials can generate around 1,000 volts. You can easily generate around 20,000 volts simply by dragging your smooth-soled shoes across a carpeted floor in the winter. (Actually, it doesn't have to be winter. This voltage can occur in any room with very low humidity—like a heated room in wintertime.)



Relative humidity has a significant impact on the electricity you generate. Walking around can generate 1,500 volts at 65–90 percent relative humidity, but it can produce 35,000 volts if the relative humidity is in the 10–25 percent range.

It makes sense that these thousands of volts can damage computer components. However, a component can be damaged with less than 300 volts! This means that if a small charge is built up in your body, you could damage a component without realizing it.



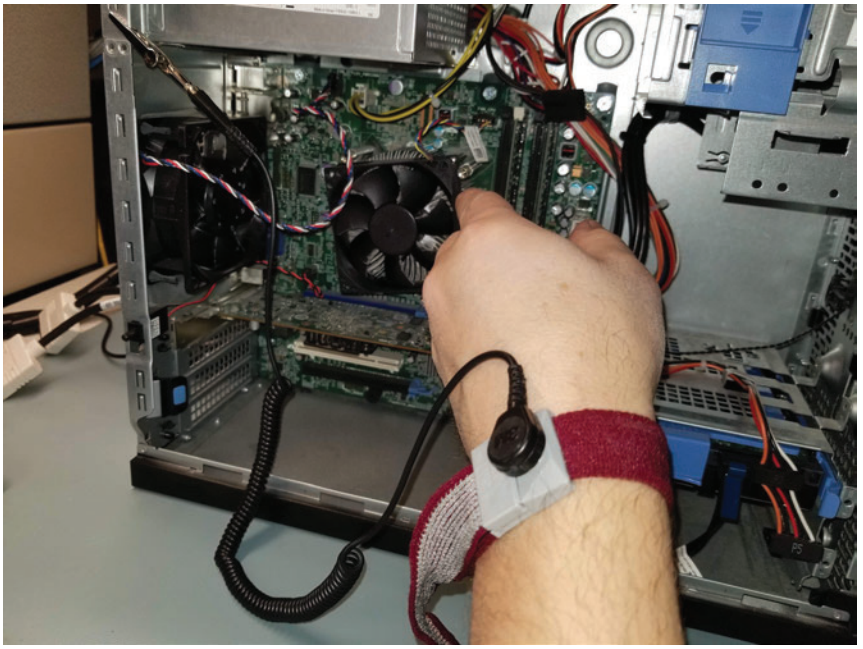
Do you have long hair or (gasp!) have to wear a tie when fixing computers? Tie it back. Letting long hair or dangling cloth get inside an open computer case is asking for trouble, because both are notorious for carrying and conducting static electricity.

The good news is that there are measures you can implement to help contain the effects of ESD. The first and easiest item to implement is the antistatic wrist strap, also referred to as an ESD strap. We will look at the antistatic wrist strap, as well as other ESD prevention tools, in the following sections.

Antistatic Wrist Straps

To use an ESD *antistatic strap*, you attach one end to an earth ground (typically, the ground pin on an extension cord) or the computer case and wrap the other end around your wrist. This strap grounds your body and keeps it at a zero charge. Figure 9.1 shows the proper way to attach an antistatic strap. Several varieties of wrist straps are available. The strap shown in Figure 9.1 uses an alligator clip and is attached to the computer case itself, whereas others use a banana clip that attaches to a grounding coupler.

FIGURE 9.1 ESD strap using an alligator clip



Real World Scenario

ESD Symptoms

Symptoms of ESD damage may be subtle, but they can be detected. One of the authors relates this experience:

"When I think of ESD, I always think of the same instance. Several years ago, I was working on an Apple Macintosh. This computer seemed to have a mind of its own. I would troubleshoot it, find the defective component, and replace it. The problem was that as soon as I replaced the component, it failed. I thought maybe the power supply was frying the boards, so I replaced both at the same time, but to no avail.

"I was about to send the computer off to Apple when I realized that it was winter. Normally this would not be a factor, but winters where I live are extremely dry. Dry air promotes static electricity. At first I thought my problem couldn't be that simple, but I was at the end of my rope. So, when I received my next set of new parts, I grounded myself with an antistatic strap for the time it took to install the components, and prayed while I turned on the power. Success! The components worked as they should, and a new advocate of ESD prevention was born."

In order for an antistatic wrist strap to work properly, the computer must be plugged in but turned off. When the computer is plugged in, it is grounded through the power cord. When you attach yourself to it with the wrist strap, you are grounded through the power cord as well. If the computer is not plugged in, there is no ground, and any excess electricity on you will just discharge into the case, which is not good.



An ESD strap is a device that is specially designed to bleed electrical charges away *safely*. It uses a 1 megohm resistor to bleed the charge away slowly. A simple wire wrapped around your wrist will not work correctly, and you could be electrocuted!

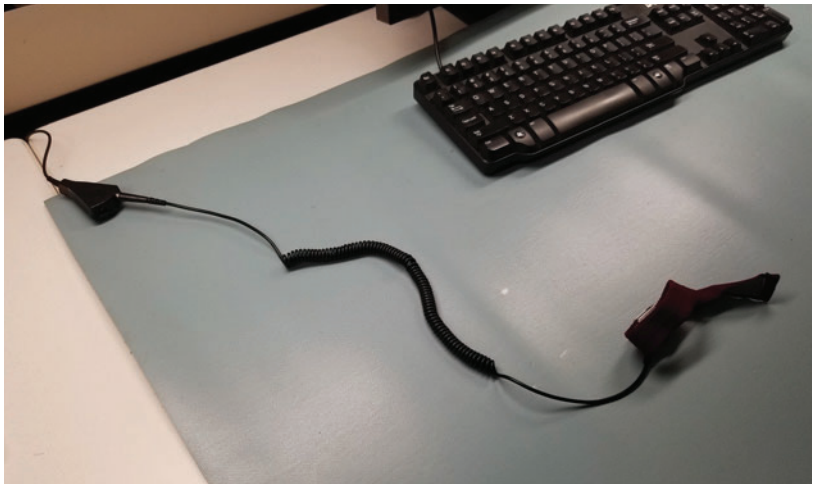


Never wear an ESD strap if you're working inside a monitor or inside a power supply. If you wear one while working on the inside of these components, you increase the chance of getting a lethal shock.

ESD Antistatic Mats

It is possible to damage a device by simply laying it on a benchtop. Therefore, you should have an ESD *antistatic mat* in addition to an ESD strap. An ESD mat drains excess charge away from any item coming in contact with it (see Figure 9.2). ESD mats are also sold as mouse/keyboard pads to prevent ESD charges from interfering with the operation of the computer. Many wrist straps can be connected to the mat, thus causing the technician and any equipment in contact with the mat to be at the same electrical potential and eliminating ESD. There are even ESD bootstraps and ESD floor mats, which are used to keep the technician's entire body at the same potential.

FIGURE 9.2 Proper use of an ESD antistatic mat



Antistatic Bags for Parts

Antistatic bags, shown in Figure 9.3, are important tools to have at your disposal when servicing electronic components because they protect the sensitive electronic devices from stray static charges. By design, the static charges collect on the outside of these silver or pink bags, rather than on the electronic components.

FIGURE 9.3 An antistatic component bag





Unlike antistatic mats, antistatic bags do not "drain" the charges away, and they should never be used in place of an antistatic mat. These bags conduct a uniform charge, because the surface is somewhat conductive.

You can obtain the bags from several sources. The most direct way to acquire antistatic bags is to go to an electronics supply store and purchase them in bulk. Most supply stores have several sizes available. Perhaps the easiest way to obtain them, however, is simply to hold onto the ones that come your way. That is, when you purchase any new component, it usually comes in an antistatic bag. After you have installed the component, keep the bag. It may take you a while to gather a collection of bags if you take this approach, but eventually you will have a fairly large assortment.

Self-Grounding

We recommend that you include a grounding strap in your toolkit so that you're never without it. But we also realize that things happen and you might find yourself in a situation where you don't have your strap or an ESD mat. In such cases, you should *self-ground*.

Self-grounding is not as effective as using proper anti-ESD gear, but it makes up for that with its simplicity. To self-ground, make sure the computer is turned off but plugged in. Then touch an exposed (but not hot or sharp) metal part of the case. This will drain electrical charge from you. Better yet is if you can maintain constant contact with that metal part.

That should keep you at the same bias as the case. Yes, it can be rather challenging to work inside a computer one-handed, but it can be done.

Additional Methods

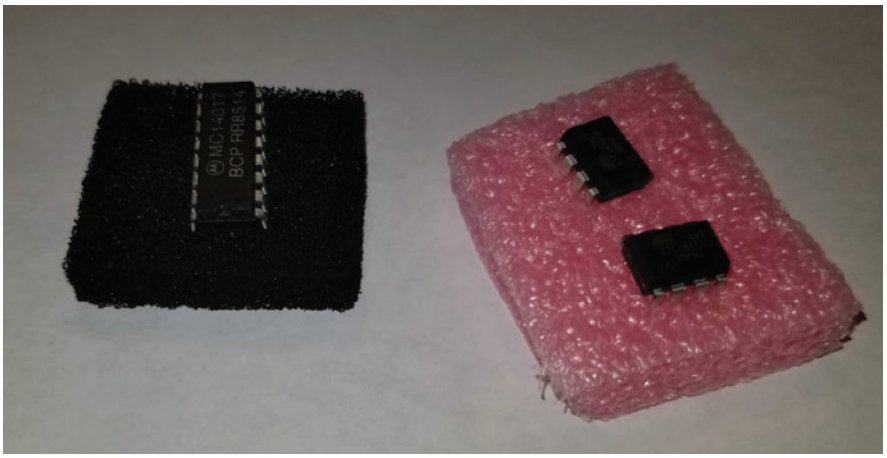
Another preventive measure that you can take is to maintain the relative humidity at around 50 percent. Don't increase the humidity too far—to the point where moisture begins to condense on the equipment. It is best to check with the manufacturer of the equipment that you are protecting to find the optimal humidity. Also, use antistatic spray, which is available commercially, to reduce static buildup on clothing and carpets.



If you don't have any antistatic spray, you can always use the "Downy solution." In a spray bottle, combine one part water with one part liquid fabric softener. Mist areas such as carpet and clothing that cause problems. If used regularly, it will keep static away—and keep your office smelling nice.

Vendors have methods of protecting components in transit from manufacture to installation. They press the pins of integrated circuits (ICs) into antistatic foam to keep all the pins at the same potential, as shown in Figure 9.4. In addition, most circuit boards are shipped in antistatic bags, as discussed earlier.

FIGURE 9.4 Antistatic foam





Antistatic foam looks a lot like Styrofoam. However, there are huge differences between the two. While antistatic foam helps reduce the transfer of electricity, Styrofoam does not. Styrofoam holds a charge on its surface quite easily. Have you ever tried to get some of those small packing "peanuts" off your hands? Be careful to not mix the two up, lest you fry your components.

At the very least, you should be mindful of the dangers of ESD and take steps to reduce its effects. Beyond that, you should educate yourself about those effects so that you know when ESD is becoming a major problem.

Electromagnetic Interference

When compared to the other dangers that we've discussed in this chapter, *electromagnetic interference (EMI)*, also known as *radio-frequency interference (RFI)* when it's in the same frequency range as radio waves, is by far the least dangerous. EMI really poses no threats to you in terms of bodily harm. What it can do is make your equipment or network malfunction.

EMI is an unwanted disturbance caused by electromagnetic radiation generated by another source. In other words, some of your electrical equipment may interfere with other equipment. Here are some common sources of interference:

Network Devices The popularity of wireless networking devices has introduced the possibility of interference. Some of the most popular wireless networking standards, 802.11b/g/n/ax/be, use the 2.4 GHz range for transmissions. Bluetooth devices happen to use the same frequency. In theory, they won't interfere with each other because they use different modulation techniques. In practice, however, interference between the two types of devices can happen.

Magnets Magnets work by generating an electromagnetic field. It might make sense, then, that this field could cause electromagnetic interference. For the most part, you don't need to worry about this unless you have huge magnets at work. Do note, however, that many motors use magnets, which can cause interference. For example, one of our friends used to have his computer on the opposite side of a wall from his refrigerator. Whenever the compressor kicked in, his monitor display would become wavy and unreadable. It was time to move his home office. Another common culprit is a desk fan. Put a desk fan next to a monitor and turn on the fan. What happens to the display? It will become wavy. This is another example of EMI.

Cordless Phones Cordless phones can operate at a variety of frequencies. Some of the more common frequencies are 900 MHz, 1.9 GHz, 2.4 GHz, and 5.8 GHz. Many of these are common ranges for computer equipment to operate in as well.

Microwave Ovens Microwave ovens are convenient devices for heating food and beverages. The radiation they generate is typically in the 2.45 GHz range, although it can vary slightly. If a microwave is being used near your computer, you'll often see a distorted display, just as if a fan or motor were being run next to your computer. You may also experience interference with wireless network communications.



Copper wires are susceptible to EMI. Fiber-optic cables, which use light to transmit data, are not susceptible to EMI.

Natural Elements

Computers should always be operated in cool environments and away from direct sunlight and water sources. This is also true when you're working on computers. We know that heat is an enemy of electrical components. Dirt and dust act as great insulators, trapping heat inside components. When components run hotter than they should, they have a greater chance of breaking down faster.

It pretty much should go without saying, but we'll say it anyway: water and electricity don't mix. Keep liquids away from computers. If you need your morning coffee while fixing a PC, make sure that the coffee cup has a tight and secure lid.



Real World Scenario

Water and Servers Don't Mix

This situation happened at a company where one of the authors used to work. The building needed some roof repairs. Repairs went on for several days, and then the weekend came. It just so happened that the area on which they were working was over the server room. That weekend was a particularly rainy one, and, of course, no one was in the office over the weekend.

Monday morning came, and the IT staff arrived to find that the server room was partially flooded. Rain had come in through weaknesses in the roof caused by the maintenance and had flooded through the drop ceiling and into the server room. Nearly half a million dollars' worth of equipment was ruined.

Although this is an extreme example, it illustrates an important point: always be aware of the environment in which you're working, and be alert to potential sources of problems for your computer equipment.

Creating a Safe Workplace

Benjamin Franklin was quoted as saying, "An ounce of prevention is worth a pound of cure." That sage advice applies to a lot in life and certainly to computer safety. Knowing how to work with and handle computer equipment properly is a good start. It's also important to institutionalize and spread the knowledge, and to make sure that your company has the proper policies and procedures in place to ensure everyone's safety.

Moving Computer Equipment

We've already talked about some of the hazards posed by computer parts. Many times it's the more mundane tasks that get us, though, such as moving stuff around. One of the most

common ways that IT employees get hurt is by moving equipment in an improper way. Changing the location of computers is a task often completed by IT personnel. You can avoid injury by moving things the right way.

To ensure your personal safety, here are some important techniques to consider before moving equipment:

- The first thing always to check for is that it's unplugged. There's nothing worse (and potentially more dangerous) than getting yanked because you're still tethered.
- Securely tie the power cord to the device or remove it altogether, if possible.
- Remove any loose jewelry and secure long hair or neckties.
- Lift with your legs, not your back. Bend at the knees when picking something up, not at the waist.
- Do not twist when lifting.
- Maintain the natural curves of the back and spine when lifting.
- Keep objects close to your body and at waist level.
- Push rather than pull, if possible.

The muscles in the lower back aren't nearly as strong as those in the legs or other parts of the body. Whenever lifting, you want to reduce the strain on those lower-back muscles as much as possible. If you want, use a back belt or brace to help you maintain the proper position while lifting.



CRT monitors can be heavy. (Thank goodness for flat screens!) When lifting and carrying any type of monitor, always keep the glass face toward your body. The front of the monitor is the heaviest part, and you want the heavy part closest to your body in order to reduce strain on your muscles.

If you believe that the load is too much for you to carry, don't try to pick it up. Get assistance from a coworker. Another great idea is to use a cart. It will save you trips if you have multiple items to move, and it saves you the stress of carrying components.

If you do use a cart to move the equipment, make sure that you do not overload the cart. Know the cart's weight limitation and estimate the weight of the equipment you will be hauling. Most small, commercial service carts will hold around 100–200 pounds. If you're moving a battery backup unit that requires two people to lift, you may be pushing the limitations of the cart. Also make sure the load is not top heavy. Always place the heaviest items on the lower shelves of a cart.

When moving loads, always be aware of your surrounding environment. Before you move, scout out the path to see whether there are any *trip hazards* or other safety concerns, such as spills, stairs, uneven floors (or ripped carpet), tight turns, or narrow doorways.

Using Appropriate Repair Tools

A big part of creating a safe working environment is having the right tools available for the job. There's no sense implementing a sledgehammer solution to a ball-peen hammer

problem. Using the wrong tool might not help fix the problem, and it could very possibly hurt you or the computer in the process.

Most of the time, computers can be opened and devices removed with nothing more than a simple screwdriver. But if you do a lot of work on PCs, you'll definitely want to have additional tools on hand.

Computer toolkits are readily available on the Internet or at any electronics store. They come in versions from inexpensive (under $10) kits that have around 10 pieces to kits that cost several hundred dollars and have more tools than you will probably ever need. Figure 9.5 shows an example of a basic 13-piece *PC toolkit*. All of these tools come in a handy zippered case, so it's hard to lose them.

FIGURE 9.5 PC toolkit



Figure 9.5 shows the following tools, from left to right:

- Sockets, ranging from 1/4″ and 1/2″
- Screwdriver bits: #1 and #1 Phillips tips, #1 and #2 flat tip, T10, T15, T20 Torx tips, and a 1/4″ drive adapter
- Hex keys
- ESD strap
- Screw container

- Screwdriver
- 1/8″ flat screwdriver
- Tweezers
- Long-nose pliers
- A three-claw retriever
- A #2 Phillips extension
- An integrated circuit (IC) extractor
- Wire strippers

A favorite of ours is the three-claw retriever, because screws like to fall and hide in tiny places. While most of these tools are incredibly useful, an IC extractor probably won't be. In today's environment, it's rare to find an IC that you can extract, much less find a reason to extract one.

The following sections look at some of the tools of the PC troubleshooting trade.

Screwdrivers

Every PC technician worth their weight in pocket protectors needs to have a *screwdriver*—at least one. There are three major categories of screwdrivers: flat-blade, Phillips, and Torx. In addition, there are devices that look like screwdrivers, except that they have a hex-shaped indented head on them. They're called *hex drivers*, and they belong to the screwdriver family.

When picking a screwdriver, always keep in mind that you want to match the size of the screwdriver head to the size of the screw. Using a screwdriver that's too small will cause it to spin inside the head of the screw, stripping the head of the screw and making it useless. If the screwdriver is too large, on the other hand, you won't be able to get the head in far enough to generate any torque to loosen the screw. Of course, if the screwdriver is way too big, it won't even fit inside the screw head at all. Common sizes for Phillips-head screws are 000, 00, 0, 1, 2, and 3. When you are dealing with Torx screws, the two most common sizes are T-10 and T-15.



When tightening screws, you don't need to make them so tight that they could survive the vibrations of an atmospheric reentry. Snug is fine. Making them too tight can cause problems when loosening them, which could cause you (or someone else not so strong) to strip the head. Using an electric screwdriver is fine, if you have one. The only problem with them is that they tend to be larger than manual screwdrivers and can be difficult to get inside a case.



Using magnetic-tipped screwdrivers is not recommended. Many computer disks contain magnetically coded information, and the magnetic tip of a screwdriver could cause a problem. Keep a retrieving tool handy instead, just in case you drop a screw.

Antistatic Wrist Straps

We've already talked about these, but they are important, so we'll mention them again. An antistatic wrist strap is essential to any PC technician's arsenal. They don't typically come with smaller PC toolkits, but you should always have one or two handy.

Other Useful Tools

PC techs also commonly carry the following tools:

Pliers Pliers are useful for a variety of tasks, especially gripping something. Long-nose or needle-nose pliers extend your reach.

Wire Cutters Wire cutters come in a variety of forms but are primarily used for cutting cables. It's not likely that you'll need any sort of heavy-duty metal cutters.

Wire Strippers If you are making your own network cables or fixing them, having a cable stripper (and crimper) is essential.

Mirrors Mirrors are handy inside tight spaces. Many techs like to use a dentist-style mirror because of its compact size and good reach.

Flashlight Never underestimate the utility of a good flashlight. You never know what your lighting situation will be like when you're at a repair site. Smaller flashlights with good output are great to have because they can fit into tight spaces.

Compressed Air For as much as computers and dust don't get along, it sure seems like they are attracted to each other. In all seriousness, computer components are powered by electricity, which causes the components to have a slight electrical charge. Dust is also electrically charged, so it's attracted to computer components. Compressed air can help you clean off components, especially in hard-to-reach places.



Be judicious about your use of compressed air. Often, you will find yourself just blowing the dust from one part of a computer to another.

Multimeter If you're having power issues, a multimeter is an invaluable tool. It measures electrical current, voltage, and resistance. You'll also hear of voltmeters, and while the two have somewhat different functions, both of them can be used to troubleshoot power problems. You can use a voltmeter to determine whether a computer's power supply is producing the right amount of current for the devices that depend on it.

Air Filter Masks We've mentioned dust, and you will occasionally encounter large amounts of it when working inside computers. If you're using compressed air to blow dust out, it could easily get into your eyes or lungs. You don't want to breathe these airborne particles, and in some environments (like machine shops), you don't know what

types of hazardous materials are mixed in with the airborne particles. You don't need a big army-grade gas mask, but a small respirator will help considerably.

Safety Goggles Your eyesight should be a primary concern when working with any type of tools, because accidents happen. You should always wear protective glasses that cover all sides of your eyes. When you think of *safety goggles*, you probably think of the science lab rubberized goggles from high school chemistry class. Safety goggles can be as comfortable as a pair of glasses, if you purchase approved safety glasses.

Fire Safety

Repairing a computer isn't often the cause of an electrical fire. However, you should know how to extinguish such a fire properly. Four major classes of fire extinguishers are available, one for each type of flammable substance: A, for wood and paper fires; B, for flammable liquids; C, for electrical fires; and D (metal powder or NaCl [salt]), for flammable metals, such as phosphorus and sodium.

The most popular type of fire extinguisher today is the multipurpose, or ABC-rated, extinguisher. It contains a dry chemical powder (e.g., sodium bicarbonate, monoammonium phosphate) that smothers the fire and cools it at the same time. For electrical fires (which may be related to a shorted-out wire in a power supply), make sure the fire extinguisher will work for Class C fires. If you don't have an extinguisher that is specifically rated for electrical fires (Class C), you can use an ABC-rated extinguisher.

The topic of electrical fire safety is a very broad subject. The best prevention method for electrical fires is to follow building codes. If building codes are not followed, your organization could be fined. Every state and locality has a different building code that you should reference with new structures and alterations to existing structures. In addition to state and local building codes, you also need to reference national building codes related to fire prevention. One organization that supplies reference to these codes is the National Fire Protection Association (NFPA) at `www.nfpa.org/for-professionals/codes-and-standards`. The U.S. Fire Administration (`www.usfa.fema.gov/prevention`) is a government organization that provides general training and prevention material.

Creating a Safe Work Environment

We've already talked about some work environment issues. For example, don't put a computer next to the breakroom sink, and keep computers out of direct sunlight (even if the desk location is great). A few other things to watch out for are trip hazards, atmospheric conditions, and high-voltage areas.

Cables are a common cause of tripping. If at all possible, run cables through drop ceilings or through conduits to keep them out of the way. If you need to lay a cable through a trafficked area, use a floor cable guard to keep the cables in place and safe from crushing. Floor guards come in a variety of lengths and sizes (for just a few cables or for a lot of cables). Figure 9.6 shows a cable guard.

FIGURE 9.6 Floor cable guard



Another useful tool to keep cables under control is a *cable tie* (see Figure 9.7). It's simply a plastic tie that holds two or more cables together. Cable ties come in different sizes and colors, so you're bound to find one that suits your needs.

FIGURE 9.7 Cable ties





In a pinch, and without a floor cable guard, you can use tape, such as gaffing tape, to secure your cables to the floor. This is recommended only as a temporary fix for two reasons. First, it's not much less of a trip hazard than just having the cables run across the floor. Second, gaffing tape doesn't protect the cables from being crushed if people step on them or heavy objects are moved over them.

Exercise 9.1 is a simple procedure that you can modify and use as needed. Its purpose is to illustrate common office hazards that you may not have realized were there.

EXERCISE 9.1

Finding Trip Hazards

1. Walk around the server room and count how many cables are lying on the floor.
2. Walk around the client areas and see how many cables are lying on the floor or are exposed underneath cubicles.

Maybe you're fortunate and don't find any, but odds are that you found at least one area with exposed cables that should not be exposed. You can reapply this exercise for other dangerous items, such as exposed wires and exposed sharp edges.

Atmospheric conditions you need to be aware of include areas with high static electricity or excessive humidity. Being aware of these conditions is especially important for preventing electrostatic discharge, as we've already discussed.

Finally, be aware of high-voltage areas. Computers do need electricity to run, but only in measured amounts. Running or fixing computers in high-voltage areas can cause problems for the electrical components and problems for you if something should go wrong.

Implementing Safety Policies and Procedures

The *Occupational Safety and Health Act* states that every working American has the right to a safe and healthy work environment. To enforce the act, the *Occupational Safety and Health Administration (OSHA)* was formed. OSHA covers all private-sector employees and U.S. Postal Service workers. Public-sector employees are covered by state programs, and federal employees are covered under a presidential executive order. In a nutshell, OSHA requires employers to "provide a workplace that is free of recognized dangers and hazards."

There are three overarching criteria to a safe work environment:

- The company and its employees have identified all significant hazards in the work setting.
- Preventive measures have been taken to address each significant hazard.
- The company and its employees understand how to respond to accidents or near-miss accidents if or when they occur.

Next, let's explore specific responsibilities and how to create a safe work environment plan.



Always ensure that your company's safety policies and procedures comply with all government regulations.

Employer and Employee Responsibilities

Maintaining workplace safety is the responsibility of employers as well as employees. Here are some of the important responsibilities of employers:

- Provide properly maintained tools and equipment.
- Provide a warning system, such as codes or labels, to warn employees of potential hazards or dangerous chemicals.
- Post the OSHA poster in a prominent location.
- Keep records of workplace injuries or illnesses.
- Continuously examine workplace conditions to ensure OSHA compliance.

It's also the responsibility of the employee to help maintain a safe work environment. Specifically, employees are charged with the following tasks:

- Read and understand OSHA posters.
- Follow all employer-implemented health and safety rules and safe work practices.
- Use all required protective gear and equipment.
- Report hazardous conditions to the employer.
- Report hazardous conditions that the employer does not correct to OSHA.

As you can see, employers and employees need to work together to keep the workplace safe. It is illegal for an employee to be punished in any way for exercising their rights under the Occupational Safety and Health Act.



Real World Scenario

Play It Safe with Common Sense

When you're repairing a PC, do not leave it unattended. Someone could walk into the room and inadvertently bump the machine, causing a failure. Worse, they could step on pieces that may be lying around and get hurt. It is also not a good idea to work on the PC alone. If you're injured, someone should be around to help. Finally, if you're fatigued, you may find it difficult to concentrate and focus on what you are doing. The most important safety measure to remember when you are repairing PCs is to pay close attention to what you are doing.

Safety Plans

We recommend that your company create and follow a workplace safety plan. Having a safety plan can help avoid accidents that result in lost productivity, equipment damage, and employee injury or death.

A good safety plan should include the following elements:

- A written document that states, among other things, who is responsible for implementing and managing the plan
- Systematic periodic inspections to identify workplace hazards
- Procedures for eliminating hazards once identified
- Processes for investigating the cause of accidents, injuries, or illnesses
- A safety and health training program specific to the job duties performed
- A system for employees to communicate safety or health concerns without fear of reprisal
- A system to ensure that employees comply with safety and health rules
- A system to maintain safety and health records, including steps taken to implement accident-prevention initiatives

It might seem like a laundry list of items to consider, but a good safety program needs to be holistic in nature to be effective.

Many companies also incorporate rules against drug or alcohol use in their safety and health plans. Specifically, employees are not allowed to come to work if under the influence of alcohol or illegal drugs. Employees who do come to work under the influence may be subject to disciplinary action up to and including termination of employment.

After your safety plan has been created, you need to ensure that all employees receive necessary training. Have each employee sign a form at the end of the training to signify that they attended, and keep the forms in a central location (such as with or near the official safety policy). In addition to the training record, you should make available and keep records of the following:

- Safety improvement suggestion form
- Accident and near-accident reporting form
- Injury and illness log
- Safety inspection checklist
- Hazard removal form
- Material safety data sheets

Safety rules and regulations will work only if they have the broad support of management from the top down. Everyone in the organization needs to buy into the plan; otherwise, it won't be a success. Make sure that everyone understands the importance of a safe work environment, and make sure that the culture of the company supports safety in the workplace.

Incident Management

Accidents happen. Hopefully, they don't happen too often, but we know that they do. Details on how to handle accidents are a key part of any safety plan so that when an accident does happen, you and your coworkers know what to do. Good plans should include steps for handling a situation as well as reporting an incident. We will cover *incident response* in more detail later in this chapter. Two major classifications of accidents are environmental and human.

When related to computers, environmental accidents typically come in one of two forms: electricity or water. Too much electricity is bad for computer components. If lightning is striking in your area, you run a major risk of frying computer parts. Even if you have a surge protector, you could still be at risk.

The best bet in a lightning storm is to power off your equipment and unplug it from outlets. Make the lightning have to come inside a window and hit your computer directly in order to fry it.



Those cheap $10 surge suppressors will fry right along with your computer. And don't be fooled—most power strips do *not* protect against power surges.

Water is obviously also bad for computer components. If there is water in the area and you believe that it will come in contact with your computers, it's best to get the machines powered off as quickly as possible. If components are not powered on but get wet, they may still work after thoroughly drying out. But if they're on when they get wet, they're likely cooked. Water + electronic components = bad. Water + electronic components + electricity = *really* bad.

Many server rooms have raised floors. Although this serves several purposes, one is that equipment stored on the raised floor is less susceptible to water damage if flooding occurs.

Human nature dictates that we are not infallible, so, of course, we're going to make mistakes and have accidents. The key is to minimize the damage caused when an accident happens.

If a chemical spill occurs, make sure that the area gets cordoned off as soon as possible. Then clean up the spill. The specific procedure on how to do that depends on the chemical, and that information can be found on material safety data sheets (MSDSs). Depending on the severity of the spill or the chemical released, you may also need to contact the local authorities. Again, the MSDS should have related information. We cover MSDSs in more detail later in the chapter.

Physical accidents are more worrisome. People can trip on wires and fall, cut or burn themselves repairing computers, and incur a variety of other injuries. Computer components can be replaced, but that's not always true of human parts (and it's certainly not true of lives). The first thing to keep in mind is always to be careful and use common sense. If you're trying to work inside a computer case and you see sharp metal edges inside the case, see whether the metal (or component on which you are working) can be moved to another location until you finish. Before you stick your hand into an area, make sure that nothing is hot or could cut you.

When an accident does happen (or almost happens), be sure to report it. Many companies pay for workers' compensation insurance. If you're injured on the job, you're required to

report the incident, and you might also get temporary payments if you are unable to work because of the accident. Also, if the accident was anything but minor, seek medical attention. Just as victims in auto accidents might not feel pain for a day or two, victims in other physical accidents might be in the same position. If you never report the accident, insurance companies may find it less plausible that your suffering was work related.



Exam Objective 4.5, "Summarize environmental impacts and local environment controls," requires you to know the impact of the physical environment on electronic equipment. This includes power issues and how you can protect your equipment from common power issues. In addition, you should know how to properly dispose of electronics.

Understanding Environmental Controls

It is estimated that more than 25 percent of all the lead (a poisonous substance) in landfills today comes from consumer electronics components. Because consumer electronics (televisions, cell phones, laptops, stereos) contain hazardous substances, many states require that they be disposed of as *hazardous waste*. Computers are no exception. CRT monitors contain several carcinogens and phosphors as well as mercury and lead. The computer itself may contain several lubricants and chemicals as well as lead. Printers contain plastics and chemicals, such as those in toners and inks, which are also hazardous. All of these items should be disposed of properly.

Remember all those 386 and 486 computers that came out in the late 1980s and are now considered antiques? Maybe you don't, but there were millions of them. Where did they all go? Is there an Old Computers Home somewhere that is using these computer systems for good purposes, or are they lying in a junkyard somewhere? Or could it be that some folks just cannot let go and have a stash of old computer systems and computer parts in the dark depths of their basements? Regardless of where they are today, all of those old components have one thing in common: They are hazardous to the environment.

On the flip side, the environment is also hazardous to our computers. We've already talked about how water and computers don't mix well, and that's just the beginning. Temperature, humidity, and air quality can have dramatic effects on a computer's performance. And we know that computers require electricity; too much or too little can be a problem.

With all these potential issues, you might find yourself wondering, "Can't we all just get along?" In this section, we will talk about how to make our computers and the environment coexist as peacefully as possible.

Managing the Physical Environment

Some of our computers sit in the same dark, dusty corner for their entire lives. Other computers are carried around, thrown into bags, and occasionally dropped. Either way, the

physical environment in which our computers exist can have a major effect on how long they last. It's smart to inspect the physical environment periodically in order to ensure that there are no working hazards. Routinely cleaning components will also extend their useful life, and so will ensuring that the power supplying them is maintained.

Maintaining Power

As electronics, computers need a power source. Laptops can free you from your power cord leash for a while, but only temporarily. Power is something that we often take for granted until we lose it, and then we twiddle our thumbs and wonder what people did before the Internet. Most people realize that having too much power (a *power surge*) is a bad thing because it can fry electronic components. Having too little power, such as when a *blackout* occurs, can also wreak havoc on electrical circuits.

Obviously, if we lose power, the equipment stops working. We have all experienced blackouts, but there are many other electrical problems we can encounter that will affect our network equipment and interrupt operations:

Sags An electrical sag is created when the line voltage dips below the nominal operating voltage. Normally the voltage should stay consistent at 90 percent of the voltage specification. An electrical sag is considered an event if the voltage drops within 1/2 second to 3 seconds of a power cycle.

Brownout An electrical sag (undervoltage event) where the line voltage dips below the nominal operating voltage for more than 3 seconds of a power cycle. Brownouts are commonly caused by large upstream draws on electrical demand.

Spike Power spikes are created when the line voltage increases over the nominal operating voltage. A voltage spike can be tens of thousands of volts, but the event only lasts 1 to 2 nanoseconds.

Surges Power surges are also created when the line voltage spikes over the nominal operating voltage. However, the difference between a spike and a surge is that a surge can last over 3 nanoseconds.

Power strips come in all shapes and sizes and are convenient for plugging multiple devices into one wall outlet. Most of them even have an on/off switch so that you can turn all the devices on or off at the same time. Figure 9.8 shows a simple power strip.

Don't make the mistake of thinking that power strips will protect you from electrical surges, though. If you get a strong power surge through one of these $10 devices, the strip and everything plugged into it can be fried. Some people like to call power strips "surge protectors" or "surge suppressors," but power strips do nothing to protect against or suppress surges.

Devices that actually attempt to keep power surges at bay are called *surge protectors*. They often look similar to a power strip, so it's easy to mistake them for each other, but protectors are more expensive, usually starting in the $25 range. Surge protectors have a fuse inside them that is designed to blow if it receives too much current and not to transfer the current to the devices

plugged into it. Surge protectors may also have plug-ins for RJ-11 (phone), RJ-45 (Ethernet), Universal Serial Bus (USB), and F-connector (coaxial cable) connectors.

FIGURE 9.8 A simple power strip

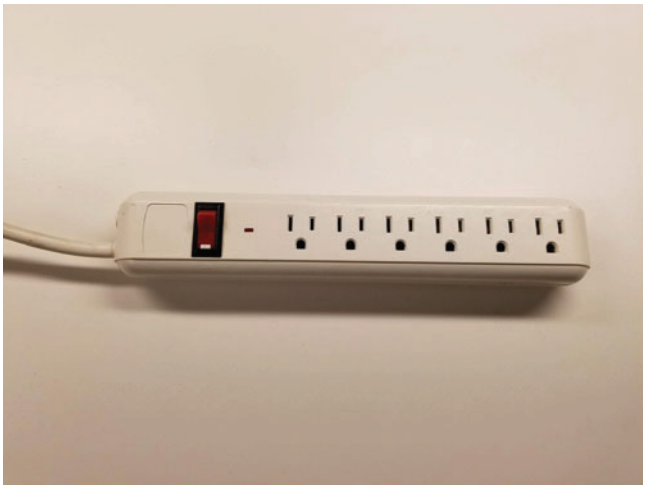


Figure 9.9 shows a surge protector, which doesn't look too different from a simple power strip. The key is to read the packaging and the labels on the product. Make sure that the device will protect your electronics from electrical surges. There is usually a printed specification of 200V to 500V. This indicates how much of a surge in voltage the surge protector can handle.

FIGURE 9.9 A surge protector



The best device for power protection is called an *uninterruptible power supply (UPS)*. These devices can be as small as a brick, like the one shown in Figure 9.10, or as large as an entire server rack. Some just have a few indicator lights, while others have LCD displays that show status and menus and that come with their own management software.

FIGURE 9.10 An uninterruptible power supply



Inside the UPS are one or more batteries and fuses. Much like a surge suppressor, a UPS is designed to protect everything that's plugged into it from power surges. UPSs are also designed to protect against *power sags* and even power outages. Energy is stored in the batteries, and if the power fails, the batteries can power the computer for a period of time so that the administrator can then safely power it down. Many UPSs and operating systems will also work together to power down a system automatically (and safely) or switch it to UPS power. These types of devices may be overkill for Uncle Bob's machine at home, but they're critically important fixtures in server rooms.

UPSs can accommodate several different devices; the number depends on the size and power rating. The model shown in Figure 9.11 has four plugs for battery backup and surge protection, and another four outlets for surge protection only. Two of each of the four outlets are controlled by a master switch on the unit.

The UPS should be checked periodically as part of the preventive maintenance routine to make sure its battery is operational. Most UPSs have a test button you can press to simulate a power outage. You will find that batteries wear out over time, and you should replace the battery in the UPS every couple of years to keep the UPS dependable.

FIGURE 9.11 The back of a UPS





Power blackouts are generally easy to detect. Power sags without a complete loss, called a *brownout,* are also very damaging to electrical components but often go unnoticed. Uninterruptable power supplies can prevent this type of damage. A UPS provides enough time to safely shut down equipment during a power outage. It protects electronic devices and offers immediate emergency power in case of failure.

Managing the Environment

Sometimes we can't help how clean—or unclean—our environments are. A computer in an auto body shop is going to face dangers that one in a receptionist's office won't. Still, there are things that you can do to help keep your systems clean and working well. We're going to break these concepts down into two parts. First, we'll look at common issues you should be aware of, and then we'll discuss proper cleaning methods.

Avoiding Common Problems

In a nutshell, water and other liquids, dirt, dust, unreliable power sources, and heat and humidity aren't good for electronic components. Inspect your environment to eliminate as many of these risks as possible. Leaving your laptop running outside in a rainstorm? Not such a good idea. (Been there, done that.)

Computers in manufacturing plants are particularly susceptible to environmental hazards. One technician reported a situation with a computer that had been used on the manufacturing floor of a large equipment manufacturer. The computer and keyboard were covered with a black substance that would not come off. (It was later revealed to be a combination

of paint mist and molybdenum grease.) There was so much diesel fume residue in the power supply fan that it would barely turn. The insides and components were covered with a thin, greasy layer of muck. To top it all off, the computer smelled terrible!

Despite all this, the computer still functioned. However, it was prone to reboot itself every now and again. The solution was (as you may have guessed) to clean every component thoroughly and replace the power supply. The muck on the components was able to conduct a small current. Sometimes, that current would go where it wasn't wanted and zap—a reboot. In addition, the power supply fan is supposed to partially cool the inside of the computer. In this computer, the fan was detrimental to the computer because it got its cooling air from the shop floor, which contained diesel fumes, paint fumes, and other chemical fumes. Needless to say, those fumes aren't good for computer components.

Computers and humans have similar tolerances to heat and cold, although computers like the cold better than we do. In general, anything comfortable to us is comfortable to a computer. They don't, however, require food or drink (except maybe a few RAM chips now and again)—keep those away from the computer.



It's bad practice to eat, drink, or smoke around your computer. Smoke particles contain tar that can get inside the computer and cause problems similar to those just described.

Computers need lots of clean moving air to keep them functioning. One way to ensure that the environment has the least possible effect on your computer is always to leave the blanks in the empty slots on the back of your box. These pieces of metal are designed to keep dirt, dust, and other foreign matter out of the inside of the computer. They also maintain proper airflow within the case to ensure that the computer does not overheat.

You can also purchase computer enclosures to keep the dust out—just make sure that they allow for proper air ventilation. Many times these devices use air filters in much the same way a furnace or a car engine does.

Cleaning Systems

The cleanliness of a computer is extremely important. Buildup of dust, dirt, and oils can prevent the various mechanical parts of a computer from operating. Cleaning them with the right cleaning compounds is equally important. Using the wrong compounds can leave residue behind that is more harmful than the dirt that you are trying to remove.

Most computer cases and monitor cases can be cleaned by using mild soapy water on a clean, lint-free cloth. Do *not* use any kind of solvent-based cleaner on monitor screens, because doing so can cause discoloration and damage to the screen surface. Most often, a simple dusting with a damp cloth (moistened with water) will suffice. Make sure that the power is off before you put anything wet near a computer. Dampen (don't soak) a cloth in mild soap solution and wipe the dirt and dust from the case. Then wipe the moisture from the case with a dry, lint-free cloth. Anything with a plastic or metal case can be cleaned in this manner.



Don't drip liquid into any vent holes on equipment. Monitors in particular often have vent holes in their tops.

Additionally, if you spill anything on a keyboard, you can clean it by soaking it in distilled, *demineralized water* and then drying it off. The extra minerals and impurities have been removed from this type of water, so it will not leave any traces of residue that might interfere with the proper operation of the keyboard after cleaning. The same holds true for the keyboard's cable and its connector.

The electronic connectors of computer equipment, on the other hand, should never touch water. Instead, use a swab moistened in distilled, *denatured isopropyl alcohol* (also known as electronics or contact cleaner and found in electronics stores) to clean contacts. Doing so will take oxidation off the copper contacts.

Finally, the best way to remove dust and debris from the inside of the computer is to use compressed air (not a vacuum). Compressed air can be more easily directed and doesn't easily produce ESD damage, as a vacuum could. Simply blow the dust from inside the computer by using a stream of compressed air. Make sure to do this outside so that you don't blow dust all over your work area or yourself. Also be sure to wear safety goggles and use an air mask. If you need to use a vacuum, a nonstatic *computer vacuum* that is specially made for cleaning computer components is recommended. Their nozzles are grounded to prevent ESD from damaging the components of the computer.

One unique challenge when cleaning printers is spilled toner. It sticks to everything and should not be inhaled—it's a carcinogen. Use an electronics vacuum that is designed specifically to pick up toner. A typical vacuum's filter isn't fine enough to catch all the particles, so the toner may be circulated into the air. Normal electronics vacuums may melt the toner instead of picking it up.



If you get toner on your clothes, use a magnet to get it out. (Toner is half iron.)

Table 9.1 summarizes the most common cleaning tools and their uses.

TABLE 9.1 Computer cleaning tools

Tool	Purpose
Computer vacuum	Sucking up dust and small particles.
Mild soap and water	Cleaning external computer and monitor cases.
Demineralized water	Cleaning keyboards or other devices that have contact points that are not metal.
Denatured isopropyl alcohol	Cleaning metal contacts, such as those on expansion cards.

Tool	Purpose
Monitor wipes	Cleaning monitor screens. Do *not* use window cleaner.
Lint-free cloth	Wiping down anything. Don't use a cloth that will leave lint or other residue behind.
Compressed air	Blowing dust or other particles out of hard-to-reach areas.

Periodically cleaning equipment is one of the easiest ways to prevent costly repairs, but it's also one of the most overlooked tasks. We're often too busy solving urgent crises to deal with these types of tasks. If possible, block out some time every week for the sole purpose of cleaning your equipment.

Handling and Disposing of Computer Equipment

Each piece of computer equipment that you purchase offers a manual, usually found online. Detailed instructions on the proper handling and use of that component can be found in the manual. In addition, many manuals give information on how to open the device for maintenance or on whether you should even open the device at all.

If you have the luxury of having paper manuals, don't throw them away. Keep a drawer of a file cabinet specifically for hardware manuals (and keep it organized). You can always look up information on the Internet as well, but having paper manuals on hand is useful for two reasons:

- You may need to fix something when Internet access isn't readily available. (Router problems, anyone?)
- Some companies are required to keep hardware documentation, in the event of an audit (such as for ISO 9000–compliant organizations).

In the following sections, we'll cover two topics: using safety documentation and following safety and disposal procedures.



There are several different ISO specifications that an organization can be certified for by the International Organization for Standardization (ISO). Each different ISO certification corresponds to a specific set of policies and procedures that an organization must adhere to. When a company is certified, it means they meet or exceed following the standardized policies and procedures for their organization.

Using Safety Documentation

In addition to your product manuals, another place to find safety information is in material safety data sheets (MSDSs). MSDSs include information such as physical product data (boiling point, melting point, flash point, and so forth), potential health risks, storage and disposal recommendations, and spill/leak procedures. With this information, technicians and emergency personnel know how to handle the product as well as respond in the event of an emergency.

MSDSs are typically associated with hazardous chemicals. Indeed, chemicals do not ship without them. MSDSs are not intended for consumer use; rather, they're made for employees or emergency workers who are consistently exposed to the risks of the particular product.

The U.S. *Occupational Safety and Health Administration (OSHA)* mandates MSDSs only for the following products:

- Products that meet OSHA's definition of *hazardous* (they pose a physical or health hazard)
- Products that are "known to be present in the workplace in such a manner that employees may be exposed under normal conditions of use or in a foreseeable emergency"

One of the interesting things about MSDSs is that OSHA does not require companies to distribute them to consumers. Most companies will be happy to distribute one for their products, but they are under no obligation to do so.

If employees are working with materials that have MSDSs, those employees are required by OSHA to have "ready access" to MSDSs. This means that employees need to be able to get to the sheets without having to fetch a key, contact a supervisor, or submit a procedure request. Remember the file cabinet drawer that you have for the hardware manuals? MSDSs should also be kept readily accessible. Exercise 9.2 helps you find your MSDSs and get familiar enough with them to find critical information.

EXERCISE 9.2

Finding MSDSs

1. Locate the MSDSs in your workplace. You might have to ask a manager. (Do you even have them?)
2. Find an MSDS for a product you're interested in.
3. Does the MSDS list any potential health effects for this item? What are they?
4. What is the proper disposal procedure for this item?

It's not likely that you're going to memorize or need to memorize everything on an MSDS. The key things are to know where to find the MSDS and how to find information in it quickly. If you have a spill of a potentially dangerous chemical, the last thing you need to do is spend your time figuring out how to handle the spill without causing injury to yourself or others.

At this point, you might stop to think for a second, "Do computers really come with hazardous chemicals? Do I really need an MSDS?" Consider this as an example: oxygen. Hardly a dangerous chemical, considering we need to breathe it to live, right? In the atmosphere, oxygen is at 21 percent concentration. At 100 percent concentration, oxygen is highly flammable and can even spontaneously ignite some organic materials. In that sense, and in the eyes of OSHA, nearly everything can be a dangerous chemical.



If you are interested in searching for free MSDSs, several websites are available, such as www.msds.com. Many manufacturers of components also provide MSDSs on their websites.

The sections within an MSDS are the same regardless of the product, but the information inside each section changes. Here is a truncated sample MSDS for ammonium hydrogen sulfate:

```
**** MATERIAL SAFETY DATA SHEET ****
Ammonium Hydrogen Sulfate
90009
**** SECTION 1—CHEMICAL PRODUCT AND COMPANY IDENTIFICATION ****
MSDS Name: Ammonium Hydrogen Sulfate
Catalog Numbers:
A/5400
Synonyms:
Sulfuric acid, monoammonium salt; Acid ammonium sulfate; Ammonium
acid sulfate.
**** SECTION 2—COMPOSITION, INFORMATION ON INGREDIENTS ****
CAS# Chemical Name % EINECS#
7803-63-6 Ammonium hydrogen sulfate 100 % 232-265-5
Hazard Symbols: C
Risk Phrases: 34
**** SECTION 3—HAZARDS IDENTIFICATION ****
EMERGENCY OVERVIEW
Causes burns. Corrosive. Hygroscopic (absorbs moisture from the air).
Potential Health Effects
Skin:
Causes skin burns.
Ingestion:
May cause severe gastrointestinal tract irritation with nausea, vomiting,
and possible burns.
Inhalation:
Causes severe irritation of upper respiratory tract with coughing, burns,
breathing difficulty, and possible coma.
```

**** SECTION 4—FIRST-AID MEASURES ****
Skin:
Get medical aid immediately. Immediately flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes.
Ingestion:
Do not induce vomiting. If victim is conscious and alert, give 2–4 cupfuls of milk or water. Never give anything by mouth to an unconscious person. Get medical aid immediately.
Inhalation:
Get medical aid immediately. Remove from exposure and move to fresh air immediately.
If not breathing, give artificial respiration. If breathing is difficult, give oxygen.
**** SECTION 5—FIREFIGHTING MEASURES ****
**** SECTION 6—ACCIDENTAL RELEASE MEASURES ****
General Information: Use proper personal protective equipment as indicated in Section 8.
**** SECTION 7—HANDLING and STORAGE ****
Handling:
Wash thoroughly after handling. Wash hands before eating. Use only in a well-ventilated area. Do not get in eyes, on skin, or on clothing. Do not ingest or inhale.
Storage:
Store in a cool, dry place. Keep container closed when not in use.
**** SECTION 8—EXPOSURE CONTROLS, PERSONAL PROTECTION ****
Engineering Controls:
Use adequate general or local exhaust ventilation to keep airborne concentrations below the permissible exposure limits.
Respirators:
Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Always use a NIOSH or European Standard EN 149 approved respirator when necessary.
**** SECTION 9—PHYSICAL AND CHEMICAL PROPERTIES ****
Physical State: Solid
Color: White
Odor: Not available
**** SECTION 10—STABILITY AND REACTIVITY ****
Chemical Stability:
Stable under normal temperatures and pressures.
Conditions to Avoid:

```
Incompatible materials, dust generation, exposure to moist air or water.
**** SECTION 11—TOXICOLOGICAL INFORMATION ****
RTECS#:
CAS# 7803-63-6: BS4400500
**** SECTION 12—ECOLOGICAL INFORMATION ****
**** SECTION 13—DISPOSAL CONSIDERATIONS ****
Products which are considered hazardous for supply are classified as Special
Waste, and the disposal of such chemicals is covered by regulations which may
vary according to location. Contact a specialist disposal company or the local
waste regulator for advice. Empty containers must be decontaminated before
returning for recycling.
**** SECTION 14—TRANSPORT INFORMATION ****
**** SECTION 15—REGULATORY INFORMATION ****
European/International Regulations
European Labeling in Accordance with EC Directives
Hazard Symbols: C
Risk Phrases:
R 34 Causes burns.
Safety Phrases:
S 26 In case of contact with eyes, rinse immediately with plenty of water and
seek medical advice. S 28 After contact with skin, wash immediately with...
**** SECTION 16—ADDITIONAL INFORMATION ****
MSDS Creation Date: 6/23/2004 Revision #0 Date: Original.
```

Following Proper Disposal Procedures

It is relatively easy to put old components away, thinking that you might be able to put them to good use again someday, but doing so is not realistic. Most computers are obsolete as soon as you buy them. And if you have not used them recently, your old computer components will more than likely never be used again.

We recycle cans, plastic, and newspaper, so why not recycle computer equipment? The problem is that most computers contain small amounts of hazardous substances. Some countries are exploring the option of recycling electrical machines, but not all have enacted appropriate measures to enforce their proper disposal.



Some countries are ahead of others on the recycling issue. For example, in 2013, the United Kingdom introduced the Waste Electrical and Electronic Equipment (WEEE) Regulations, which obligates manufacturers to provide disposal methods for items they manufacture. Consumers can insist that manufacturers take back dangerous waste and electrical and electronic equipment. Many local governments also provide e-waste programs and disposal centers.

Regardless of manufacturer or community programs, we can take proactive steps, as consumers and caretakers of our environment, to promote the proper disposal of computer equipment:

- Check with the manufacturer. Some manufacturers will take back outdated equipment for parts (and may even pay you for them).
- Properly dispose of solvents or cleaners (as well as their containers) used with computers at a local hazardous waste disposal facility.
- Disassemble the machine and reuse the parts that are good.
- Check out businesses that can melt down the components for the lead or gold plating.
- Contact the Environmental Protection Agency (EPA) for a list of local or regional waste disposal sites that accept used computer equipment. The EPA's web address is `www.epa.gov`.
- Check with the EPA or at `www.msds.com` to see if what you are disposing has an MSDS. These sheets contain information about the toxicity of a product and whether it can simply be disposed of as trash. They also contain lethal-dose information.
- Check with local nonprofit or education organizations that may be interested in using the equipment.

Check out the Internet for possible waste disposal sites. Table 9.2 lists a few websites that we came across that deal with the disposal of used computer equipment. A quick web search will likely locate some in your area.

TABLE 9.2 Computer recycling websites

Site name	Web address
The Home Depot	`www.homedepot.com/c/electronics_recycling_programs`
Staples	`https://www.staples.com/stores/recycling`
U.S. EPA	`www.epa.gov/recycle/electronics-donation-and-recycling`
Repowered	`https://getrepowered.org`



The National Conference of State Legislatures details each state's legislation regarding electronics waste. The site also details state programs to collect electronics waste. For more information, visit:

`www.ncsl.org/environment-and-natural-resources/electronic-waste-recycling`

Following the general rule of thumb of recycling your computer components and consumables is a good way to go. In the following sections, we'll look at four classifications of computer-related components and the proper disposal procedures for each.

Batteries

The EPA estimates that more than 350 million batteries are purchased annually in the United States. One can only imagine what the worldwide figure is. Batteries contain several heavy metals and other toxic ingredients, including alkaline, mercury, lead acid, nickel cadmium, and nickel metal hydride.



Never burn a battery to destroy it. Doing so will cause the battery to explode and could result in serious injury.

When batteries are thrown away and deposited into landfills, the heavy metals inside them will find their way into the ground. From there, they can pollute water sources and eventually find their way into the supply of drinking water. In 1996, the United States passed the *Mercury-Containing and Rechargeable Battery Management Act* (aka the *Battery Act*) with two goals: to phase out the use of mercury in disposable batteries and to provide collection methods and recycling procedures for batteries.



Several countries around the world have battery-recycling programs. Information on battery recycling in the United States can be found at `www.call2recycle.org`. You can find information on the Battery Act at

`www.epa.gov/rcra/mercury-containing-and-rechargeable-battery-management-act-public-law-104-142`

Five types of batteries are most commonly associated with computers and handheld electronic devices: alkaline, nickel-cadmium (NiCd), nickel-metal hydride (NiMH), lithium-ion (Li-ion), and *button cell*.

Alkaline Batteries *Alkaline batteries* have been incredibly popular portable batteries for several decades. Before 1984, one of the major ingredients in this type of battery was mercury, which is highly toxic to the environment. In 1984, battery companies began reducing the mercury levels in batteries, and in 1996, mercury was outlawed in alkaline batteries in the United States. Still, it's strongly recommended that you recycle these batteries at a recycling center. Although newer alkaline batteries contain less mercury than their predecessors, they are still made of metals and other toxins that contaminate the air and soil.

Nickel-Cadmium *Nickel-cadmium (NiCd)* is a popular format for rechargeable batteries. As their name indicates, they contain high levels of nickel and cadmium. Although nickel is only semi-toxic, cadmium is highly toxic. These types of batteries are categorized by the EPA as hazardous waste and should be recycled.

Nickel-Metal Hydride (NiMH) and Lithium-Ion (Li-ion) Laptop batteries are commonly made with *Nickel-Metal Hydride (NiMH)* and *lithium ion (Li-ion)*. Unlike the previous types of batteries that we have discussed, these are not considered hazardous waste, and there are no regulations on recycling them. However, these batteries do contain elements that can be recycled, so it's still a good idea to go that route.

Button Cell These batteries are so named because they look like a button. They're commonly used in calculators and watches as well as portable computers. They often contain mercury and silver (and are environmental hazards due to the mercury) and need to be recycled.

You may have noticed a theme regarding the disposal of batteries: recycling. Many people just throw batteries in the trash and don't think twice about it. However, there are several laws in the United States that require the recycling of many types of batteries, and recycling does indeed help keep the environment clean. For a list of recycling centers in your area search the Internet.



If you're ever exposed to the electrolyte (the inside "juice") of the battery, immediately flush the exposed area with water. If it gets on your eye, immediately contact a physician and wash the eye for 15 minutes.

Display Devices

Computer monitors (CRT monitors, not LCDs) are big and bulky, so what do you do when it's time to get rid of them? As previously mentioned, monitors contain capacitors that are capable of retaining a lethal electric charge after the monitors have been unplugged. You wouldn't want anyone to set off the charge accidentally and die. But what we didn't mention earlier, which is important now, is that most CRT monitors contain high amounts of lead. Most monitors contain several pounds of lead, in fact. Lead is very dangerous to humans and the environment and must be dealt with carefully. Other harmful elements found in CRTs include arsenic, beryllium, cadmium, chromium, mercury, nickel, and zinc.

If you have to dispose of a monitor, contact a computer-recycling firm. It's best to let professional recyclers handle the monitor for you.



Real World Scenario

How Not to Dispose of Your Monitors

This story comes from the technical support division of a now-defunct major computer manufacturer, which used a lot of computers at its own facility. At one time, the company had as many as 500 technicians working the phones. So you can imagine that they burned out a lot of equipment.

Here's how they disposed of dead monitors. An IT staff member would take the monitor out to the dumpster and bring along a sledgehammer. Setting the monitor on its back, he

would take one good swing at the glass panel with the hammer to shatter the screen. (This was done, by policy, to ensure that no one would want to go out to the dumpster and try to salvage the dead monitor.) After spraying glass everywhere, he picked up the monitor and threw it in the dumpster.

One employee observed that it probably wasn't good to be spreading glass all over the parking lot by shattering monitors. That advice was taken, and the sledgehammer was retired. Instead, an IT staff member would use a permanent black marker and draw all over the screen (again, so no one would want to try to salvage it), and again, it was thrown in the dumpster.

In our enlightened state today (as opposed to the mid-1990s), we can see how this was not a good plan for disposing of broken monitors. In fact, many states today have laws prohibiting the disposal of computer monitors in trash bins. This is a good law because, with the amount of harmful elements in monitors, they're every bit the environmental hazard that batteries are.

Laser Printer Toner Disposal

Toner cartridges should be recycled as well. PC recycling centers will take old toner cartridges and properly dispose of them. The toner itself is a carcinogen, and the cartridges can contain heavy metals that are bad for the environment.

Toner cartridges are valuable to companies that refurbish and refill these cartridges. It's actually big business to refill these expensive cartridges, and it's environmentally responsible. Most toner is the same for all types and models of laser printers, so a toner refurbishing center will refill and test these cartridges. Then these companies sell the cartridges for a fraction of the price of new toner cartridges. If a new toner cartridge is installed, the old toner cartridge is boxed up and sent back. Even if your organization doesn't contract with one of these services, they would be happy to take the old cartridges off your hands and keep them out of the trash.

Mobile Devices

Cell phones and tablets are considered disposable units, with the average life expectancy of two to four years. Their popularity has outpaced mobile computing and the desktop computer market. These mobile devices are extremely small and fit neatly into the trash. However, mobile devices contain the same toxic metals and chemical compositions as their larger cousins. Every mobile device has a circuit board that contains lead, batteries that contain other heavy metals, and bodies that contain plastics.

These devices should be recycled responsibly. Many big-box retailers have an anonymous recycling drop bin, where you can recycle a mobile device. There are even automated kiosks where you can get money for a defective device. You simply tell it what is wrong with the device, the device's make and model, and its condition. The kiosk will then offer you a few

dollars to recycle it. The company that owns and operates the kiosk refurbishes the device and resells it for a fraction of the cost of a new one.



Before recycling a mobile device, make sure the device is wiped of your personal data with the factory reset function. You never know where the device will end up. It could be refurbished and resold locally or sold for parts to a nefarious market.

Chemical Solvents and Cans

Nearly every chemical solvent that you encounter will have a corresponding MSDS. On the MSDS for a chemical, you will find a section detailing the proper methods for disposing of it. Chemical solvents were not designed to be released into the environment, because they could cause significant harm to living organisms if they're ingested. If in doubt, contact a local hazardous materials handler to find out the best way to dispose of a particular chemical solvent.

Cans are generally made from aluminum or other metals, which are not biodegradable. It's best always to recycle these materials. If the cans were used to hold a chemical solvent or otherwise hazardous material, contact a hazardous materials disposal center instead of a recycling center.



Always be sure that you are following all applicable laws and regulations when disposing of computer equipment.



Exam Objective 4.6, "Explain the importance of prohibited content/activity and privacy, licensing, and policy concepts," requires you to know how to perform incident response, the various licensing/digital rights management (DRM), end-user license agreement (EULA), non-disclosure agreement (NDA)/mutual nondisclosure agreement (MNDA), acceptable use policy (AUP), and regulatory and business compliance requirements.

Understanding Policies, Licensing, and Privacy

Many of the operational procedures that we've discussed up to this point have been about safety—yours, your computer equipment's, and the environment's. We've also touched on regulations, as in always be sure to comply with local government regulations. In this section, we focus more on the legal side of things. Not understanding legal requirements is not

a justifiable defense in a court of law. Considering that IT professionals often deal with software licensing and personally identifiable information, or sometimes encounter prohibited activity or have to deal with a security incident, you should understand the general principles related to these concepts.

Dealing with Prohibited Content/Activity

This is a situation that no one really wants to deal with, but it happens more often than we would care to admit: a computer you are fixing has content on it that is inappropriate or illegal, or you see someone on your network performing an action that is against policy or laws. How you respond in such a situation can have a significant bearing on your career, the other people involved, and, depending on the situation, the well-being of your organization. The key to dealing with *prohibited content* or activity is to have a comprehensive policy in place that covers appropriate behavior. After that, it's a matter of executing the proper steps per the plan when something happens.

Situations involving prohibited content or activities are not easy to address. The accused person might get angry or confrontational, so it's important always to have the right people there to help manage and defuse the situation. If you feel that the situation is severe enough to worry about your own personal safety, don't be afraid to involve the police. While the situation needs to be handled, there's no sense in putting yourself in direct danger to do so.

Creating a Prohibited Content Policy

Creating a policy is the most important part of dealing with prohibited content or actions. Without a policy in place that specifically defines what is and what isn't allowed, and what actions will be taken when a violation of the policy occurs, you don't really have a leg to stand on when a situation happens.

What is contained in the policy depends on the organization for which you work. Generally speaking, if something violates an existing federal or local law, it probably isn't appropriate for your network either. Many organizations also have strict policies against the possession of pornographic or hate-related materials on the organization's property. Some go further than that, banning personal files such as downloaded music or movies on work computers. Regardless of what is on your policy, always ensure that you have buy-in from very senior management so that the policy will be considered valid.

Here are some specific examples of content that might be prohibited:

- Adult content
- Content that advocates violence against an individual, group, or organization
- Unlicensed copyrighted material
- Content related to drugs, alcohol, tobacco, or gambling
- Content about hacking, cracking, or other illegal computer activity
- Violent or weapons-related content

A good policy will also contain the action steps to be taken if prohibited content or activity is spotted. For example, what should you do if you find porn on someone's work laptop?

The policy should explicitly outline the punishment for performing specific actions or possessing specific content. The appropriate penalty may very well be based on the type of content found. Something that is deemed mildly offensive might result in a verbal or written warning for the first offense and a more severe sentence for the second offense. If your company has a zero-tolerance policy, then employees may be terminated and possibly subject to legal action.

Finally, after the policy has been established, it's critical to ensure that all employees are aware of it and have proper training. In fact, it's highly recommended that you have all employees sign a disclosure saying they have read and understand the policy, and that the signed document be kept in their human resources file. Many organizations also require that employees review the policy yearly and re-sign the affidavit as well.

Incident Response

If you have your policy in place, then your *incident response* plan should be relatively scripted. It might not be easy to deal with, but the steps you should take should be outlined for you. Professionalism should be maintained during the incident. This is a good time to remind you that people will be looking at your reaction as well as your actions, and professionalism will define how you dealt with the incident. If you see prohibited content and start giggling and walk away, that probably doesn't reflect well on you. Always remember that others are watching you. The specific steps that you take will depend on your policy. The following sections describe the best practices for dealing with security incidents.

Detecting the Incident

An incident can be detected in several different ways. The preceding section used the example of pornographic content on a laptop or mobile device. This is a great illustration of an example of passive detection. You were not looking for this material, but you found it, and now you must respond. A full list of incident detections is detailed in the following:

Passive *Passive detection* is used to clue in on what happened and how it happened. Passive detection can be when you are reviewing logs, content, or events and discover an incident. Another example of passive detection in a physical security context is the use of a video camera. The video camera will show you an event in the past, but it is normally not actively looking for an event. An example of passive detection in a digital context is the use of file hashes. If a file hash does not match the original, then someone or something has tampered with the original content.

Active *Active detection* is used to actively look for security incidents. This type of detection in a physical security context is similar to having a security guard patrol your premises. A digital example might be the use of an *intrusion detection system (IDS)*

running on your firewall that alerts you to an attempted break-in. This type of detection is used to catch a perpetrator in the act of a security incident.

Proactive *Proactive detection* is the act of proactively securing possible holes in security. *Penetration testing*, also known as *pentesting*, involves someone trying to penetrate either the physical or software defenses of the organization. The exercise is done to strengthen the organization's security for either the physical or digital defenses. Proactive detection normally does not create a security incident, unless the potential for the loss of information was discovered during the process.

Responding to the Incident

Once an incident is detected using the methods mentioned in the preceding section (passive, active, or proactive), or it's detected through dumb luck, it's time to spring into action and respond to the incident. The person responding to the incident, called the *first responder*, should be versed in how to collect evidence in the order of volatility. If the evidence is not collected, for example, from a computer's RAM, and the computer reboots, the evidence will be gone. The *order of volatility* is as follows:

Memory Contents Evidence of the incident located in the RAM of the computer or system. This evidence can be a file that has not been saved to the filesystem. It can also be a process running in RAM that requires a memory dump for further analysis.

Swap Files/Virtual Memory Evidence that has been saved out to a swap file/virtual memory, because the process has been backgrounded or is no longer in foreground RAM. The swap file/paging file should be copied to removable media.

Network Processes Evidence that is part of a network process, such as a browser redirection or spam or active network communications. This is where most of the incidents occur on a network. This information should be saved to removable media.

System Processes Evidence that is part of a system, such as an exploit that has been rooted into the operating system. This is also a common place for incidents to occur. The information should be separated out of the system and saved to removable media.

Filesystem Information This evidence is in the form of files on a hard drive. An example of filesystem information is illegal content that has been saved to the filesystem. This information must be preserved in its original form but also copied for evidence purposes.

Raw Disk Blocks This evidence is in the form of a block-level copy of the data. Once an incident is discovered and all the other volatile evidence has been collected, the last step is to create an image of the affected system. This can be done with a tool called a *write-blocker* (which makes the media read-only) and specialized software that copies data at the block level.

The preceding evidence can be collected with the sophisticated tools that only a highly trained first responder might have on hand. However, not all the tools need to be complex tools. A simple camera can preserve information. For example, if you walk up to a system that displays a ransomware screen, your first reaction should be to take a photo of the screen. If you press a key, the ransomware could crash and disappear. Photo evidence of data and processes that are loaded in RAM is a good alternative to not having any evidence, but having a digital copy of the evidence is preferred.

The act of photographing the scene should not be limited to just the computer screen; anything relevant to the incident should be photographed as evidence. Time and date stamps should be overlaid onto the image. This is normally a function of any camera. However, if you can't digitally record the time stamps, a simple alternative is to include a watch in the frame of the photo.

You should take notes with a pad and pen, recording the initial scene, including time and date. Create a chronology of the discovery and collection of the evidence. Remember, any of this could potentially be used in a court of law. The underlying premise is to record as much evidence as possible before the crime scene is tainted by others.

Following the Right Chain of Custody

The removed materials should be secured and turned over to the proper authorities. Depending on the situation, materials may be held in a safe, locked location at the office, or they may need to be turned over to local authorities. Have a documented procedure in place to follow, given a situation.

The materials that are deemed as evidence should be well documented as to why they are considered evidence. The *chain of custody* documentation should define the following:

- Who obtained the evidence
- Who secured the evidence
- Who controlled the evidence during the entire process

If the evidence is moved, the chain of custody documentation should reflect the following:

- The reasons it was moved
- Who moved it
- How it was secured
- Who controlled it

The chain of custody must be maintained at all times. If a chain of custody of the evidence is not maintained, the evidence may not be admissible in a court of law.

Reporting the Incident

Once you've collected the initial set of evidence and in the order of volatility, it's time to report the incident. The incident should be reported to management and a decision should be made whether to involve law enforcement. Involving law enforcement is, of course, dependent on the severity of the incident. A piece of malware that has infected a

single machine is a bit different than malware that has infected an entire network. Your management might elect not to involve law enforcement at all.

Regardless of the direction management takes, the evidence collected, as well as notes taken during the collection, will help an escalation team or law enforcement to proceed in building a case. The goal of the first responder is to collect evidence that answers the following questions:

- What happened?
- When did it happen?
- How did it happen?
- Who made it happen?
- Why did it happen?

An escalation team or law enforcement's job is to fill in the blanks by using the evidence. If the evidence is complete and concise, it will be used to build a case against the threat agent. The ultimate goal is to stop a future incident from happening.

Recovering from the Incident

Your most important task is to recover from the incident. If your critical ordering system was affected during the incident, it's your job to get it back online. If the incident involved one computer that is used by a task worker, then it's your job to get it back up and running. You might notice a common theme here: it's your job to get things back to normal after the incident. Once you can get the flow of information flowing again, you can move on to remediating the incident.

During the recovery from the incident, you may have to make changes to the network or systems that support the clients. Any changes during the recovery process should be documented thoroughly. Documenting these changes is important if you are submitting claims to insurance, assessing damages, or looking for future reparations.

Remediating the Incident

All components affected by the incident should be remediated to ensure that all traces of the incident have been removed. Remediation can be as simple as adding firewall rules to the firewall, or it can involve formatting a server and reloading it. The steps to remediation should also include steps to prevent the incident from happening in the future.

Documenting the Incident

Before, during, and after the incident, the documentation process should begin. You should collect as much information as possible, as a formal incident report will eventually be formulated defining the following key elements.

- Date and time
- Summary
- Root cause
- Actions taken

- Remediation
- Services impacted
- Recommendations

It really doesn't matter how you collect information for documentation purposes. It can be pad and pen or something more elaborate. The only stipulation is that the documentation should not be on a system that can be affected by the incident. An offline laptop is fine, as long as the laptop is never introduced to the network affected by the ongoing incident. This could jeopardize all of the documentation efforts and hinder the outcome.

Reviewing the Incident

The final step for incident response is to review all the documentation and findings of the incident—a process often called a hot-wash meeting. During a hot-wash meeting, the incident response team should talk about what has been done properly during the incident and what procedures should be changed for future incidents. These meetings should be constructive and support standards of excellence for the incident response team.

Another key goal of the review process is to identify threats similar to the characteristics of the incident. If an employee entered credentials into a phishing page, what measures are in place to prevent this from happening to others in the organization? You may have rules in place for this particular phishing email, but are your employees trained for future incidents similar to this? If not, end-user training may be required.

Best Practices for Incident Response

Now that you have a good understanding of the process involved with incident response, let's look at some of the best practices. The following best practices should be applied to all elements of the incident response process:

Follow your policies exactly as they are written. Yes, we've already said this several times. It's crucial that you do this. Not following the policies and procedures can derail your case against the offender and possibly set you up for problems as well.

If you are the first responder, get a verifier. Your first priority as the first responder is to identify the improper activity or content. Then you should always get someone else to verify the material or action so that it doesn't turn into a situation of your word against someone else's. Report the situation immediately through the proper channels.

Preserve the data or device. The data or device should immediately be removed from the possession of the offending party and preserved. This will ensure that the data doesn't mysteriously disappear before the proper parties are notified.

Use documentation. Document everything that could be relevant to the situation. Many companies have standard documentation that is used in incident response in order to be sure that the responder captures important information and does not forget to ask critical questions or look for vital clues.



Always be sure to follow government regulations as well as corporate end-user policies and security best practices when dealing with sensitive information.

Common Agreements

There are several agreements that you will need to be familiar with for the CompTIA exam. This short list does not constitute all of the agreements you might have to abide by as a network professional. These are just the most common that you might need to follow or create.

Nondisclosure Agreement (NDA)

When an organization hires a new employee, the employee may be privilege to certain proprietary information. This information could be a company's secret sauce, passwords to administrative systems, or customer information. A nondisclosure agreement (NDA) legally protects an organization from an employee disclosing this information to a person outside the organization. The NDA is a document that defines the consequences of disclosure by termination and legal recourse. The NDA should be developed by the organization's legal counsel, HR department, and stakeholders of proprietary information in the organization. These documents are typically signed by the employee during the onboarding process.

Mutual Nondisclosure Agreement (MNDA)

A mutual nondisclosure agreement (MNDA) is a legal agreement in which both parties agree to protect each other's confidential information. This type of agreement is often used in situations where both parties need to share sensitive information with one another. This is commonly found with business partnership, joint venture, or collaboration. The mutual NDA ensures that both parties have a legal obligation to keep the shared information confidential and not disclose it to a third party.

An MNDA differs from a standard NDA, because an NDA typically involves only one party disclosing confidential information to the other. In this case, the receiving party agrees to keep the information secret and not use it for any unauthorized purposes. While a standard NDA imposes confidentiality obligations on just one party, an MNDA imposes these obligations on both parties equally. The key difference between the two types of NDAs lies in the reciprocity of the mutual NDA.

Acceptable Use Policy (AUP)

There can be many different resources employees use that the organization supplies to conduct its business. An *acceptable use policy (AUP)* helps protect the organization from employee abuse of these resources. Email is an example of one such resource. It is generally not acceptable for an employee to use an organization's email for religious, political, or personal causes; illegal activities; or commercial use outside of the organization's interest.

The AUP should be developed by the organization's legal counsel, HR department, and IT department. The systems that should be included in the AUP are telephone, Internet, email, and subscription services. The AUP might not be exclusive to electronic resources; the organization might also include postage and other nonelectronic resources that could be abused. The AUP is typically signed or electronically acknowledged during the employee's onboarding process, or when a resource is accessed that could be misused.

Regulations and Compliance

Regulations are rules imposed on your organization by an outside agency, like a certifying board or a government entity, and they're usually totally rigid and immutable. The list of possible regulations that your organization could be subjected to is so exhaustively long, there's no way I can include them all in this book. Different regulations exist for different types of organizations, depending on whether they're corporate, nonprofit, scientific, educational, legal, governmental, and so on, and they also vary by where the organization is located.

For instance, U.S. governmental regulations vary by county and state, federal regulations are piled on top of those, and many other countries have multiple regulatory bodies as well. The Sarbanes–Oxley Act of 2002 (SOX) is an example of a regulation system imposed on all publicly traded companies in the United States. Its main goal was to ensure corporate responsibility and sound accounting practices, and although that may not sound like it would have much of an effect on your IT department, it does because a lot of the provisions in this act target the retention and protection of data. Believe me, something as innocent sounding as deleting old emails could get you in trouble—if any of them could've remotely had a material impact on the organization's financial disclosures, deleting them could actually be breaking the law. All good to know, so be aware, and be careful!

The penalties of noncompliance to regulations can be in the form of financial penalties, reputation damage, loss of services, loss of licensing, and ultimately disruptions to operations. Therefore, it is paramount to the continued operation of our organizations that we comply with safeguarding the data entrusted.

Splash Screens

Splash screens are used by organizations to communicate compliance-related information. These screens can be customized to display various types of content, such as legal disclaimers, privacy policies, or terms of service (ToS), which users must acknowledge before accessing the system. This ensures that users are consistently reminded of their obligations and the organization's policies. The organization will typically use these splash screens on a captive portal, before an employee or guest is granted access to a network. The splash screen can also include an AUP reminding the end-user of the acceptable use.

For example, a healthcare organization that needs to comply with healthcare regulations, might implement a splash screen that appears each time a user logs into their system. This will help remind them of the importance of protecting patient information and adhering to data privacy standards. The splash screen could include a brief summary of key compliance points and a prompt for the user to acknowledge their understanding before proceeding.

This not only reinforces the organization's compliance policies, but also helps to create a culture of accountability and awareness among the organization's employees.

Healthcare organizations are not the only organizations that can benefit from splash screens. This approach can be particularly effective in industries with stringent regulatory requirements, such as finance, healthcare, and legal sectors, just to name a few.

Managing Software Licenses

When you buy an application, you aren't actually buying the application. Instead, you're buying the right to use the application in a limited way, as prescribed by its licensing agreement. Most people don't read these licensing agreements closely, but suffice it to say, they're pretty slanted in favor of the software manufacturer.

Don't like the terms? Too bad. No negotiation is allowed. If you don't accept the *end-user license agreement (EULA)*, your only recourse is to return the software for a refund. (Most vendors will refuse to take back an opened box. Still, the software manufacturer is required to take it back and refund your money if you reject the licensing. This is true of programs purchased online as well.)



Many companies rely on *digital rights management (DRM)* to protect digital assets, such as online photos or videos. DRM is not as established as licensing agreements are, but you should still respect the property of the owners of digital content.

Although the majority of the applications that you acquire will probably be commercial products, there are a number of alternatives to commercial software sales. Here are some of the license types that you may encounter:

Freeware *Freeware* is software that is completely free. On a small scale, you can get such software from download sites such as `https://download.cnet.com` or from the creator's personal website. Large companies like Google and Microsoft also sometimes offer products for free, because it serves the company's interests to have a lot of people using their software. Examples include Google Chrome and Microsoft Internet Explorer. Freeware doesn't include source code, and users aren't allowed to modify the application.

Open Source Open source software is freer than free: not only is the application free, but the source code (code used by programmers) is also shared to encourage others to contribute to the future development and improvement of the application. OSs such as Linux and applications such as OpenOffice fit this category. Open source software can't be sold, although it can be bundled with commercial products that are sold.

Shareware Shareware is software that provides a free trial, with the expectation that you'll pay for it if you like it and decide to keep it. In some cases, a shareware version isn't the full product; in other cases, it expires after a certain amount of time. Some shareware provides a full and unlimited version, with payment requested on the honor system.

Multiuser This is commercial software that you're allowed to install on more than one computer. For example, some versions of Microsoft Office allow you to install the same copy on two or three PCs.

Single User This is commercial software for which the license restricts installation to a single PC. A common misconception is that a single-user license allows you to install the software on more than one computer as long as you use only one instance at a time, but that's not accurate. Commercial products sometimes have activation systems that lock the software to a specific PC once it's installed, so you can't install it elsewhere. Microsoft Office is a good example of commercial software.

Concurrent This license allows the software to be installed on many PCs but used concurrently by a smaller number. For example, you may have 1,000 computers with the application installed, but only 100 users can use it simultaneously. This is useful in situations in which everyone needs to have an application available but the application gets very little actual use.

Perpetual A perpetual or non-expiring license is a license that can be purchased for a software product. Many video games still use this license model. You purchase the software and the right to use the software for life. Software vendors that use perpetual licensing will also offer software upgrades or patches for the first year, as customary. However, any upgrades after the first year are generally covered under a support or maintenance fee.

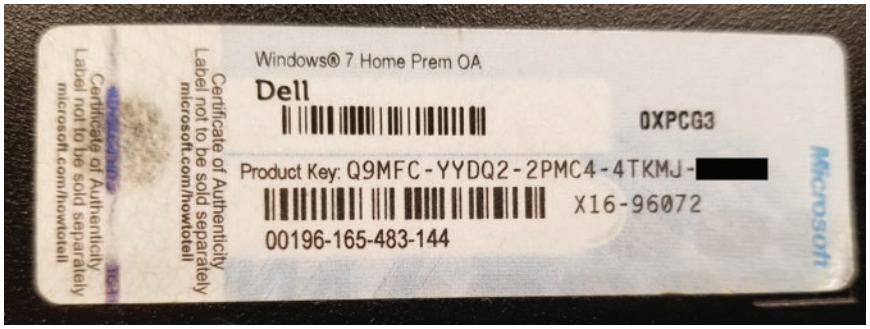
Corporate, Campus, or Site These are enterprise licenses, which permit an organization to install the application on an agreed-upon number of PCs. For example, a school may buy a site license of an antivirus program and allow all students to download and install it freely to ensure that the school's network remains virus free. Microsoft calls this a volume license key.

Valid License A valid license is any software license that is current in its term and not counterfeit. As long as the license is not used already, it is valid for activation for the software product it is associated with.

If you buy any sort of commercial software, you will receive a *product key*, which you will need to enter during installation or the first time the application is opened. The product key might be emailed to you, or it could be located on the physical media if you got an installation CD-ROM, DVD, or thumb drive. Figure 9.12 shows an example of a product key.

In a corporate environment, license management is a critical responsibility. The company may spend thousands or even millions of dollars on software licenses. Money could be wasted on unused licenses, or if the company's computers have unlicensed software, it could result in huge fines. Ignorance is not a legal excuse in this area.

FIGURE 9.12 A Microsoft product key



To avoid these problems, it may be best for your company to purchase a software asset management tool, such as Microsoft's Software Asset Management (SAM) guide (`www.microsoft.com/en-us/download/details.aspx?id=31382`), License Manager by License Dashboard (`www.licensedashboard.com`), or FlexNet Manager by Flexera Software (`www.flexera.com/products/flexnet-manager`). In general, here are the steps to take for proper license management:

1. Build a database of all licenses owned by your company. This includes what type of license it is, license numbers, and expiration dates, if any.
2. Perform an inventory of all licensed software installed on your computers. (This can be quite an effort.)
3. Compare the license list to the installation base.
4. For any gaps where you have unlicensed software, either remove the software or procure licenses.

Because of the potential for heavy fines, many companies prohibit the installation of software on client computers unless specifically authorized by a manager or the IT department.



Real World Scenario

How Do I Buy the Right Licenses?

Consider this situation: After talking to your boss about software licenses, you decide to investigate the office productivity software on your department's computers. You are unable to find proper documentation that the correct software licenses were purchased, leading you to wonder whether the company has the right licenses. Your boss wants to avoid any

potential legal issues and asks you to go buy enough copies of the latest version of Microsoft Office for the 20 users in your department. What do you do?

The first question to ask yourself is, do all 20 people use their computers at the same time? If not, then you might be able to purchase a concurrent license for fewer than 20 users. If there's the possibility of all 20 users needing Office at once, then you'll definitely need licenses for everyone. You might purchase a corporate license as well. Now that you have figured out what to buy, how do you do it?

One option is to go to the local computer store, load up 20 boxes of Office (if they have that many in stock), and trudge up to the cashier. A second option is to go to Microsoft's volume licensing site, at `www.microsoft.com/en-us/licensing`, to learn about purchasing multiple licenses. Microsoft will direct you to an authorized reseller so that you can purchase the licenses. The authorized reseller will register the list of license numbers with Microsoft's licensing center. Now you need just one physical (or downloaded) copy to perform the installation.

Managing Sensitive Information

As an IT manager, you will very likely have access to information that you will need to keep closely guarded. For example, you might have access to username and/or password lists, medical or educational records, addresses and phone numbers, or employee records. It's your responsibility to ensure that sensitive information does not get released into the wrong hands. On the flip side, you may encounter information that's sensitive because it's prohibited or illegal. You need to know how to react in those situations as well.

Much of the data you come into contact with could be regulated outside of the organization's internal policies. Regulated data must be identified as it enters your network, and the proper operating procedures should be followed. The operating procedures for such data should be constructed so that you can adhere to the data's regulatory compliance rules. The regulatory rules for compliance can be at any local, state, or federal level. Many of the regulations in the following sections are at a federal level of compliance.

Personally Identifiable Information

Personally identifiable information (PII) is anything that can be used to identify an individual person on its own or in context with other information. This includes someone's name, address, other contact information; the names of family members; and other details that people would consider private.

PII should always be kept confidential and secure. It seems like every few months or so we see news stories of data breaches at big companies resulting in stolen credit card data or username and contact lists. This information finds its way into hackers' hands and causes millions of people grief and monetary damages. Be sure that this information is properly secured and can be accessed only by authorized personnel.

Personal Government-Issued Information

Any personal information contained in a document issued by a government or state is considered personal government-issued information. Examples of government-issued documents are defined as a birth certificate, Social Security card, identification card, driver's license, resident card, taxpayer ID number, or password, just to name a few. The category of information is very broad and can overlap with other types of protected data, such as protected health information or other PII.

The threat to personal government-issued information being compromised is that the information is how a person is defined by the government. The theft of a Social Security number is a direct theft of identity in the eyes of the government. Tax records to credit information is tied to a Social Security number. Therefore, like any PII, this data should also be kept confidential and secure.

Payment Card Industry Data Security Standard

Payment Card Industry Data Security Standard (PCI DSS) is a standard of processes and procedures used to handle data related to transactions using payment cards. A payment card is any card that allows the transfer of money for goods or services. Types of payment cards include credit cards, debit cards, or even store gift cards.

PCI DSS compliance is not enforced by government entities. PCI DSS compliance is actually enforced by banks and creditors. Merchants must comply with the PCI DSS standard to maintain payment card services. If a merchant does not comply with PCI DSS standards and a breach occurs, the merchant can be fined by the banks. Once a breach of PCI data occurs, then local, state, and federal laws can apply to the merchant. For example, some laws require the merchant to pay for credit-monitoring services for victims after a breach.

General Data Protection Regulation

The *General Data Protection Regulation (GDPR)* is a European Union (EU) law governing how consumer data can be used and protected. The GDPR was created primarily to protect citizens of the European Union. It applies to anyone involved in the processing of data based on the citizens of the European Union, regardless of where the organization is located.

The GDPR recommends that organizations hire a *data protection officer (DPO)*. This person is the point of contact for all compliance with GDPR, as well as any other compliances your organization falls under. The underlying goal is to achieve consent from the end user of your product or service. Consent to collect information must be proven by an organization beyond a shadow of doubt. This means that if someone visits your website from the European Union, you must receive consent in clear language to even place a cookie in their web browser. The DPO is responsible for coordinating this language, as well as the life cycle of any data that is collected.

Protected Health Information

Protected health information (PHI), also known as *personal health information*, refers to any information used in the health care industry to describe a patient or ailment. This information can be considered "the patient chart" you always see on television. However, *electronic health records (EHRs)* go way beyond the current condition of a patient; they describe a person from the cradle to the grave.

Electronic health records (EHRs) are used to record a patient's vitals every time the patient visits a doctor's office. They represent historical information about patients, as well as billing information used by health care providers. This makes the EHR extremely valuable to a hacker and represents a large makeup of identity theft.

This type of identity theft is really dangerous! Your diagnosis could be determined based upon vitals, allergies, or conditions that are recorded from a person who assumes your identity. Try to explain to the insurance company that your gallbladder needs to be removed, but their records show they paid to have it taken out already.

PHI can also be used to track the statistics of larger groups of people. However, the data must be anonymized first, before it is put into a publicly addressable database for these statistics and studies. Information such as names, Social Security numbers, phone numbers, health insurance information, account numbers, and specific geographical information must be removed. Interesting enough, geographical information can include only the first three digits of a zip code.

Data Retention Requirements

As you've learned, there are all different forms of data and they are regulated for sensitivity and confidentiality purposes. The storage of data is also regulated depending on the type of data. The data to be retained is usually transactional in nature, such as a financial transaction, but it can also be data that has touched your system.

Regulations are not the only governing factor for data retention—your company can also have an internal requirement. An example of an internal retention requirement might be for memorandums, access control records, or even video footage. There are different reasons for adhering to an internal data retention requirement, but they all shield the organization from liability in some way. This is very apparent in legal situations. If you keep all data indefinitely on old backups and there is a lawsuit, you will be liable to restore records in a timely fashion, even if those old systems are no longer supported. If you don't produce the data in a timely fashion, you could default a lawsuit in favor of the plaintiff. These discoveries of information are considered e-discovery by court proceedings.

As an IT professional, it is your responsibility to work with legal counsel in your organization to define data retention periods. However, before you can define the data retention periods, you will need to identify the data to be protected. A document profile should be created that clearly defines the types of data you deal with on a daily basis. This stage of the process is where you can also identify outside governing requirements for retention. Once the document profile is created, you should start tagging data in your environment with the

data types. This may be as simple as naming the email backup job email data. You should then set up a hard policy to control how long the data is kept. Keep in mind that holding the data too long is just as bad as not holding the data long enough.

Summary

This chapter covered three areas of operational procedures that you should integrate into your work:

- Safety procedures
- Environmental impacts and controls
- Licensing, policies, and sensitive materials

First, we looked at the importance of safety procedures. Safety is about protecting you from harm as well as protecting your computer components from getting damaged. Then, we outlined some methods to apply the policies and procedures for a safe working environment, and identified potential safety hazards. Included were preventing *electrostatic discharge (ESD)* and *electromagnetic interference (EMI)*, creating a safe work environment, and properly handling computer equipment.

Safety involves you and your coworkers, but it also includes environmental issues. The environment can have a harmful effect on computers, but computers can also greatly harm the environment. You need to be familiar with the importance of material safety data sheets (MSDSs) as well as the proper disposal procedures for batteries, display devices, and chemical solvents and cans. These items need to be kept out of the environment because of the damage that they can cause.

Finally, we looked at potential legal issues. Failure to follow certain procedures can expose you or your company to legal proceedings. Make sure that all the software on your computers is legal and licensed and that the computers contain no illegal or prohibited materials. You may also need to protect personally identifiable information, depending on the type of data you have. When incidents happen, you need to know how to respond properly to mitigate the issue.

Exam Essentials

Know which computer components are particularly dangerous to technicians. The most dangerous components are the power supply and the monitor. Both are capable of storing lethal charges of electricity, even when unplugged. You also need to be aware of parts that get incredibly hot, such as the processor, which can cause severe burns if touched.

Understand where to find safety information regarding chemicals. You can find this information in a material safety data sheet (MSDS). An MSDS might not have come with your purchase, but most suppliers will gladly supply one.

Know which tool to use for which job. The majority of computer repair jobs can be handled with nothing more than a Phillips-head screwdriver. However, you might need cutters, extra light, or a mirror for some jobs. Avoid using magnetic-tipped tools.

Understand methods to help prevent ESD. One of the biggest and most common dangers of electronic components is electrostatic discharge (ESD). You can employ several methods to help avoid ESD-related problems, such as grounding yourself; using an antistatic wrist strap, bag, or mat; and controlling the humidity levels.

Know the proper disposal procedures for used computer parts, batteries, and chemical solvents. The specific disposal procedure depends on the item. However, the safe answer is always to recycle the component, not to throw it in the trash.

Know the differences between license types. Open source applications don't require a purchased license, whereas commercial applications do require a purchased license. Some licenses are personal licenses, meaning the applications are for use by one person on one computer, whereas others are enterprise licenses, meaning the applications can be used on multiple computers at the same time.

Understand how to handle prohibited content or activity. Always have policies and procedures in place to deal with prohibited content or activity. When an incident happens, follow the procedures, report through proper channels, preserve the data or device using the order or volatility, and follow the chain of custody.

Know the various types of sensitive information. You should know that personal identifiable information (PII) is any information used to identify an individual. The Payment Card Industry Data Security Standard (PCI DSS) is a standard of policies and procedures used by the payment industry. General Data Protection Regulation (GDPR) is a European Union (EU) law governing how EU citizen data is used. Protected health information (PHI) is any information used to describe a patient or a patient's ailment.

Know the common business agreements. These agreements include nondisclosure agreements (NDA), mutual nondisclosure agreements (MNDA), and acceptable use policy (AUP). You should understand how they are used to protect organizations and individuals.

Review Questions

The answers to the chapter review questions can be found in Appendix A.

1. You have dropped a screw into a tight location of a computer. How should you retrieve it?
   - **A.** Use a magnetic-tipped screwdriver.
   - **B.** Use a magnetic grabber.
   - **C.** Use a three-pronged grabber.
   - **D.** Shake the computer until it falls out.

2. Your company's AUP states that employees must not use company resources for personal gain. You have a side business and want to use your work computer to manage it. What should you do?
   - **A.** Use your work computer to manage your side business.
   - **B.** Use your personal computer to manage your side business.
   - **C.** Ask your manager for permission to use your work computer.
   - **D.** Ignore the AUP and use your work computer anyway.

3. One of your coworkers just spilled a chemical solvent in a warehouse, and you have been asked to help clean it up. Which of the following must contain information about a chemical solvent's emergency cleanup procedures?
   - **A.** OSHA
   - **B.** MSDS
   - **C.** Product label
   - **D.** CRT

4. You are purchasing an inkjet printer cartridge for home use that you know has an MSDS. How do you obtain the MSDS for this product?
   - **A.** The store is required to give you one at the time of purchase.
   - **B.** It's contained in the packaging of the printer cartridge.
   - **C.** You are not legally allowed to have an MSDS for this product.
   - **D.** You should visit the printer cartridge manufacturer's website.

5. In the interest of a safe work environment, which of the following should you report? (Choose two.)
   - **A.** An accident
   - **B.** A near-accident
   - **C.** Spills on the floor inside a building
   - **D.** Rain forecasted for a workday

6. What is the approximate minimum level of static charge for humans to feel a shock?

- **A.** 300 volts
- **B.** 3,000 volts
- **C.** 30,000 volts
- **D.** 300,000 volts

7. Your work environment has been unusually dry lately, and several components have been damaged by ESD. Your team has been asked to be extra careful about ESD damage. Which of the following measures can be implemented to reduce the risk of ESD? (Choose two.)

- **A.** Use an antistatic wrist strap.
- **B.** Use an antistatic bag.
- **C.** Spray disinfectant spray.
- **D.** Shuffle your feet.

8. Which of the following are OSHA requirements for a safe work environment that must be followed by employers? (Choose two.)

- **A.** Attend yearly OSHA safe work environment seminars.
- **B.** Provide properly maintained tools and equipment.
- **C.** Have an OSHA employee stationed within 5 miles of the facility.
- **D.** Display an OSHA poster in a prominent location.

9. Your office just added 20 new workstations, and your manager has put you in charge of configuring them. The users need to have Microsoft Office installed. What should you do to install Microsoft Office properly on these computers?

- **A.** Ensure that the company has the proper licenses to install 20 additional copies.
- **B.** Agree with the open source license agreement during installation.
- **C.** Use the personal license key from an existing system to install Office on the new computers.
- **D.** Follow normal installation procedures; nothing else needs to be done.

10. Your office is moving from one floor of a building to another, and you are part of the moving crew. When moving computer equipment, which of the following are good procedures to follow? (Choose two.)

- **A.** Lift by bending over at the waist.
- **B.** Carry CRT monitors with the glass face away from your body.
- **C.** Use a cart for heavy objects.
- **D.** Ensure that there are no safety hazards in your path.

11. You just removed four AA alkaline batteries from a remote-control device. What is the recommended way to dispose of these batteries?

- **A.** Throw them in the trash.
- **B.** Incinerate them.
- **C.** Take them to a recycling center.
- **D.** Flush them down the toilet.

12. When replacing a hard drive, you discover prohibited material on a user's laptop. What should you do first? (Choose two.)

A. Destroy the prohibited material.

B. Confiscate and preserve the prohibited material.

C. Confront the user about the material.

D. Report the prohibited material through the proper channels.

13. You need to investigate how to protect credit card data on your network. Which information should you research?

A. PCI DSS

B. GDPR

C. PHI

D. PII

14. Which class of fire extinguisher is recommended for use in a wood and paper fire?

A. A

B. B

C. C

D. D

15. Which of the following are common types of screwdrivers? (Choose two.)

A. Circular

B. Phillips

C. Torx

D. Helix

16. Which of the following are elements of a good workplace safety plan? (Choose two.)

A. Periodic workplace inspections

B. A safety and health training program

C. Retribution for employees who report violations

D. An independent third-party auditor of the safety plan

17. What is the recommended use policy on magnetic-tipped screwdrivers inside computers?

A. Do not use them.

B. It's okay to use them, but keep them away from the processor.

C. It's okay to use them, but keep them away from the RAM.

D. It's okay to use them, but only if they're of the powered variety.

18. Which of the following are usually contained on an MSDS? (Choose two.)

A. Freezing point

B. Handling and storage instructions

C. Personal protection instructions

D. Weight

19. Which of the following are OSHA requirements for a safe work environment that must be followed by employees? (Choose two.)

- **A.** Immediately report all accidents to OSHA.
- **B.** Use protective gear and equipment.
- **C.** Attend safety training.
- **D.** Follow all employer-implemented health and safety rules.

20. Which of the following types of batteries are not considered environmental hazards?

- **A.** Alkaline
- **B.** Nickel-metal hydride (NiMH)
- **C.** Nickel-cadmium (NiCd)
- **D.** Button cell

Performance-Based Question

You will encounter performance-based questions on the A+ exams. The questions on the exam require you to perform a specific task, and you will be graded on whether or not you were able to complete the task. The following requires you to think creatively in order to measure how well you understand this chapter's topics. You may or may not see similar questions on the actual A+ exams. To see how your answers compare to the authors', refer to Appendix B.

One of your office coworkers recently tripped on a power cord and injured himself. What should you do to find potential trip hazards in your office? Once the hazards are identified, what actions should you take?



Chapter 10

Documentation and Professionalism

THE FOLLOWING COMPTIA A+ 220-1202 EXAM OBJECTIVES ARE COVERED IN THIS CHAPTER:

✓ **4.0 Operational Procedures**

✓ **4.1 Given a scenario, implement best practices associated with documentation and support systems information management.**

- Ticketing systems
  - User information
  - Device information
  - Description of issues
  - Categories
  - Severity
  - Escalation levels
  - Clear, concise written communication
    - Issue description
    - Progress notes
    - Issue resolution
- Asset management
  - Inventory lists
  - Configuration management database (CMDB)
  - Asset tags and IDs
  - Procurement life cycle
  - Warranty and licensing
  - Assigned users
- Types of documents
  - Incident reports
  - Standard operating procedures (SOPs)



        - Software package custom installation procedure
    - New user/onboarding setup checklist
    - User off-boarding checklist
    - Service-level agreements (SLAs)
        - Internal
        - External/third-party
    - Knowledge base/articles

✓ **4.2 Given a scenario, apply change management procedures.**

- Documented business processes
    - Rollback plan
    - Backup plan
    - Sandbox testing
    - Responsible staff members
- Change management
    - Request forms
    - Purpose of the change
    - Scope of the change
    - Change type
        - Standard change
        - Normal change
        - Emergency change
    - Date and time of change
        - Change freeze
        - Maintenance windows
    - Affected systems/impact
    - Risk analysis
        - Risk level
    - Change board approvals
    - Implementation



- Peer review
- End-user acceptance

✓ **4.3 Given a scenario, implement workstation backup and recovery methods.**

- Backup
  - Full
  - Incremental
  - Differential
  - Synthetic full
- Recovery
  - In-place/overwrite
  - Alternative location
- Backup testing
  - Frequency
- Backup rotation schemes
  - On site vs. off site
  - Grandfather-father-son (GFS)
  - 3-2-1 backup rule

✓ **4.7 Given a scenario, use proper communication techniques and professionalism.**

- Present a professional appearance and wear appropriate attire.
  - Match the required attire of the given environment
    - Formal
    - Business casual
- Use proper language and avoid jargon, acronyms, and slang, when applicable
- Maintain a positive attitude/project confidence
- Actively listen, take notes, and avoid interrupting the customer
- Be culturally sensitive
  - Use appropriate professional titles, when applicable
- Be on time (if late, contact the customer)



- Avoid distractions
  - Personal calls
  - Texting/social media sites
  - Personal interruptions
- Appropriately deal with difficult customers or situations
  - Do not argue with customers and/or be defensive
  - Avoid dismissing customer problems
  - Avoid being judgmental
  - Clarify customer statements (i.e., ask open-ended questions to narrow the scope of the problem, restate the issue, or question to verify understanding)
  - Use discretion and professionalism when discussing experiences/encounters.
- Set and meet expectations/timeline and communicate status with the customer
  - Offer repair/replacement options, as needed
  - Provide proper documentation on the services provided
  - Follow up with customer/user at a later date to verify satisfaction
- Appropriately handle customers' confidential and private materials
  - Located on a computer, desktop, printer, etc.



Every day at work, we do what it takes to get the job done. As IT professionals, we have millions of facts crammed into our heads about how various hardware components work and what software configuration settings work best for our systems. We know which servers or workstations give us constant trouble and which end users need more help than others. We are counted on to be experts in knowing *what* to do to keep computers and networks running smoothly. And even though we don't tend to think about it overtly, we need to be experts in *how* we get things done as well. Even though the *how* might not be at the top of your mind every day (hopefully you don't go to work every day thinking "Okay, don't get killed by a monitor" or "Let's see if I can be nice to someone today"), but it should be integrated into your work processes. Operational procedures define the *how*, and they provide guidance on the proper ways to get various tasks accomplished.

In this chapter, we will describe documentation used to document the network and the policies that you will encounter and need to enforce. We will then look at change management practices used to make changes to the network that will affect other business units in your organization.

We will also look at disaster prevention and recovery. In the process, we will cover topics such as backup and recovery, power resiliency, and cloud storage versus local storage.

In the final part of this chapter, we'll switch to discussing professionalism and communication and focus on topics that you need to know for your exam study. Applying the skills learned here will help you pass the exam, but on a more practical level, it will help you become a better technician and possibly further advance your career.



Exam Objective 4.1, "Given a scenario, implement best practices associated with documentation and support systems information management," requires you to know ticketing systems and the various elements they contain. You must know the various elements of asset management and the types of documents you will use daily.

Documentation and Support

Creating documentation of the network and systems you work on is the last step to many projects. However, it is the most overlooked, or it is considered secondary to fixing new problems or upgrading to the next system. The documentation produced has positive effects

on production and problem solving. It is used so that you don't need to redo the step of discovery the next time you have a problem. Documentation also allows you to have others work on a problem with the same distinctive view you had at the time of documentation.

Some of the documentation you create will help to develop policies and procedures that others will need to follow. Throughout this book, we have discussed the hard controls of these policies. For example, when you implement a password policy, you can dictate that a password be complex and of a certain length. A written policy is a soft control that might detail how to create a complex password. In this chapter, we will look at several different policies that you will come across as a technician.

Ticketing Systems

One of the most important functions of the information technology (IT) department is to solve problems and fulfill end-user requests. The core of this function is the ability to receive incoming requests, track progress, and assure that problems and requests are solved and completed. It is common to find a ticketing system at the heart of this function called help desk services or support services. A ticketing system adds other benefits to the function of the help desk, such as accountability, reporting, collaboration, and escalation, just to name some of the important benefits.

There are hundreds of ticketing system vendors on the market today, and each one has some unique features. When choosing a ticketing system, the first decision you must make is to host it on-premises, also known as on-prem, or host it through a cloud option from the vendor. Many ticketing systems can be purchased as a software-as-a-service (SaaS) cloud model. You are charged a monthly fee based on the number of tickets, storage, or the number of agents (help desk personnel), whereas on-prem systems are licensed either per agent or per supported users, and storage is not a concern. There are a number of considerations when choosing between on-premises and cloud-based. The SaaS option is a great option for a number of reasons, mainly for availability. However, if all of your employees are local to your site, then on-premises might be the best solution. If you are conscientious about keeping the system up to date for security patches, having an on-premises system is another box to patch (so to speak). But if you elect SaaS hosting, the vendor is responsible for patching and keeping the system secure.

Besides the initial hosting options, there are several features that are somewhat standard with any ticketing system. These features include escalation, automated routing, knowledge base, email to open a ticket, ticket management, and real-time reporting, and those are just a few of the features. We will cover many of them in this section.

Entering Tickets

Once you have a ticketing system in place, it's just a matter of having tickets entered into the system. Your users should have no problem finding network issues to create tickets for. There are three main entry methods of generating a ticket: email, portal, and manually.

Email is probably the easiest way to enter a ticket. It is customary when setting up a ticket system to dedicate an email address for entering tickets, such as `helpdesk@wiley.com` or `support@wiley.com`. When a user has a problem, they simply have to email the ticketing system, and a ticket is automatically generated. Their email address will become the identity for the ticket related to the problem, and IT can converse with them via email. This method of ticket entry is commonly used for external customers, such as product support.

Every ticketing system will have a portal for the users and help desk personnel to log into. This of course means that the users and the help desk personnel need an existing account, or users can sign up for an account when they enter the ticket. These types of ticketing systems are often established for internal issues inside the organization. Once signed into the portal, a user has several fields to fill out in order to submit the ticket. This type of setup is nice for users if they have multiple tickets and want to track them simultaneously.

The manual entry of tickets is a catch-all entry method. A help desk support person is responsible for entering the ticket information manually to create a ticket. This is a common practice when someone calls into the help desk. The entry of the ticket serves two main purposes: The first is that it allows for follow-up or escalation of the problem. The second purpose is that entering a ticket can identify a problem common to your network—for example, if a system has just been upgraded and calls come into the help desk because of a problem with the upgraded system. The user or even your help desk personnel might not know of the recent upgrade. However, the accumulation of tickets for similar issues will allow help desk personnel to identify a systemwide problem. The problem(s) can then be escalated to the party responsible for the problematic upgraded system. There are many other reasons for manually entering a ticket, such as scheduling of help desk staff, identifying specific skills required by help desk staff, and the overall volume of work. These are just a few; each organization has its own reasons.

These are the most common ticket entry methods for ticketing systems. There are many other methods, such as application-triggered entry and interactive voice response (IVR) automation.

Ticket Details

The devil is in the details, as the saying goes, and that is also true when entering a ticket into the ticketing system. There are several elements that need to be entered correctly for the successful resolution of an issue. If a user is self-entering the ticket, then you will have no control over this initial process.

When a user enters a ticket using the portal, it is common for them to be very brief with their issue description. This is usually because the description window is one field of many and the user may feel they have to be brief. Issues that are submitted via email might have more elaborate explanations, because users may feel more comfortable communicating in email. However, other fields may not be filled out, such as severity, contact information, or issue category. The ticketing system generally cannot decipher these elements from a simple email.

When acting as a help desk support person answering the help desk line, we have the most control over manually entering a ticket into the ticketing system. We also directly

interface with the user. So, there should be a standardized process for collecting the necessary information, as well as some best practices to be followed. Always keep in mind that your interactions with the user make an impression on the entire department. You should have good interpersonal skills, display empathy, and above all have patience. All of these skills will help both you and the user reach a resolution more quickly.

Another important skill that you should display is actively listening to the user. This skill is often related to the skill that detectives have when interviewing a person. When a user calls into the help desk, the critical information to collect is who they are, how to contact them, and a description of the issue. In addition, you should ask the urgency of the issue.

If you are able to enter information into the ticket entry form as the person is talking to you, do so. If you are unable to type and listen, or you must allow for long awkward pauses as you type, you should use the trusted method of pad and paper. You can always enter the information after the person is off the phone and moves on to their next task, awaiting a response for the issue. Or, if you solved the issue, then you can enter the ticket with the resolution after they are off the phone to act as a follow-up. Always exercise speed and accuracy in obtaining the information, especially if you are not able to help the person and need to escalate the ticket to someone else.

The following is a list of information you should obtain on the initial call with the user as you exercise all the best practices:

User Information The first information you want to obtain is the user's name. This is important for your communication with the user. Always use their name throughout the interview about the issue. In addition, you should obtain the best method to contact them, such as their phone number, cell phone number, email address, or physical location. This information is important if you must escalate the issue to another technician.

Description of the Issue A good description of the issue is the most important information you must gather. During your communications with the user, ask them to describe the exact issue they are having. If the user offers multiple issues, ask to review one at a time. Also, ask the user how you can reproduce the issue, then review the steps to reproduce the issue with the user. After you learn about the issue and the steps to reproduce it, repeat the steps to the user in a calm way to verify that you have clear and concise information.

Device Information Although the device being used by the user might not be apparent to you as being part of the issue, if the ticket needs to be escalated it may help in solving the issue more quickly. Information to be obtained is the type of device, operating system version, hardware service tag, and method of network access. Another technician might find that a model or version of operating system is a key to the issue.

Severity of the Issue You may pick up the urgency in the user's voice, and that can serve as an indicator of the severity of the issue. You must communicate this in entering the ticket—often there is a drop-down menu for low, mid, or high severity. Always keep in mind that not everything can be of the utmost urgency; although we would like to make everything an emergency, we often don't have the staff to do so. Communicate with the user about the severity of the issue and agree upon a severity level. Always use patience, empathy, and professionalism in your communications.

Category of the Issue The category of an issue is a critical element of the ticket. In some cases, the category may automatically route a ticket to a group of people in your support staff. For example, if a user calls in and has an issue with the organization's database and reporting, the ticket may need to be routed to the database administrators for the organization. Likewise, if there is an issue with a phone, the ticket may be routed to the phone administrators for the organization. In addition to automatic routing to a group of administrators, categories are useful for reporting. They help identify categories of issues that see the most tickets, which can be an indication you need more staff or need to adopt a quality control process.

Levels of Support

Every IT department has a structure, and they vary from organization to organization. On a very high level there are typically two main groups of support personnel: network administrators and application/database administrators. However, your organization might have security administrators, application developers, storage administrators, virtualization administrators—and these just scratch the surface.

In each group of support personnel, there are varied levels of experience, support, and responsibilities. The simplest structure is front-office personnel, who interface with the users, and back-office personnel, who interface with the front-office personnel and make systemwide changes. Depending on the size of your organization, you may also have an intermediate level of personnel who interface with the backend and frontend personnel. This intermediate level serves as a buffer to keep engineers separated from the day-to-day issues. These levels are often numbered from basic knowledge to expert knowledge, level 1 through level 3.

Regardless of your structure, these various levels are considered escalation points. As an IT technician or even an IT administrator, you are not always expected to have the answer, but you are expected to be able to get the answer. When you don't have the answer and must ask someone more knowledgeable, this process is called escalation. With a ticketing system, if all the information is properly obtained, you can escalate the ticket to a higher level of technician or administrator. The next level up the support chain should be able to read through the ticket notes and work on a resolution. When a ticket is escalated to another person, it is common for that person to own the ticket (issue) and be responsible for communicating with the user.

You may also find that you are the most knowledgeable person about a particular system in your organization, yet you still do not have an answer. This is fine and it happens all the time, but always keep in mind you that are expected to be able to find the answer. This requires an escalation of the issue to a third party, potentially outside your organization. It is also the reason support contracts should be kept current so that you have an escalation point outside of your organization. When an escalation is made outside of your organization, the point of contact (POC) inside your organization will be the owner of the ticket.

Clear Communication

The requirement and benefits of clear communications cannot be overstated enough—not just verbal communications, but also written communications between technicians and users, as well as technicians and their escalation points. Clear, concise written communications can also break down verbal communications issues between regions of the world. There are four stages to any issue where clear, concise written communications are required:

Issue Description Not only should the issue description be clear and concise to you and the user, but it should communicate the issue to someone not part of the initial communications. Don't use jargon, acronyms, or references to anything that is not clearly defined in the issue description. Always keep in mind that the issue may have to be escalated.

Work-Arounds When an issue is identified and a solution requires time, a work-around should be communicated to the user. A work-around serves as a temporary way of allowing the user to continue their work by working around the issue. Always communicate that it is a temporary solution and that you will work on a permanent solution.

Progress Notes The most important part of the ticketing process is the progress of solving the issue. You should adopt the rule of daily communications with the user and communicate where you are in the process of obtaining a resolution. Be as transparent with the user as possible—always keep in mind that the user might ask for a supervisor to review the progress. If the user stops communicating with you on the progress of a resolution, give them a drop-dead date when the ticket will be suspended or closed.

Issue Resolution The last stage to any issue is the resolution, and you should strive to have a resolution to every issue. The resolution or steps to the resolution should be clear and concise written communications. This also serves as a reference to other technicians with the same issue.

Follow-Up

Although the issue resolution should be where the communication with the user ceases, it is important to follow up with the user to make sure the issue is resolved. This important step should be done right before the ticket is closed. You may learn that the final resolution presented does not work for the user and that they continue to use the work-around that you presented as a temporary fix. This often happens when you must escalate the issue and communications are not clear and concise between the next technician and the user.

The ticket management process should include the steps of entry, resolution, solution, follow-up, and knowledge base. Each ticket that is completed strengthens your technical support for future issues. It also strengthens the faith the users have that the IT department can solve their issues. When you follow up with the user, you verify that all of your efforts and your escalation point's efforts are justified by the solution.

Many ticket systems allow for the entry of a knowledge base article for future users and technicians to self-service their issues. Writing the draft of a knowledge base article should happen after you have identified a successful follow-up with the user.

Asset Management

Asset management is an important part of the IT department's responsibilities, because the IT assets are considered *fixed tangible assets*. Some other examples of fixed tangible assets are land, furniture, and office equipment. When equipment is initially purchased, the accounting department records it as an asset on the company's general ledger, because it adds to the value of the company. Over time, however, the asset will lose its initial value. The accounting department will depreciate the value of the asset based on its perceived lifespan.

The management of these assets benefits the organization in defining the organization's worth. The management of assets also helps the IT department in forecasting upgrades and future expenditures for growth. In this section we will cover the various elements of asset management as it applies to the IT department.

Asset Management Systems

There are a number of ways to manage assets for the organization. Choosing a way to manage assets depends on what needs to be done with the information. Asset management at an organizationwide level is often a module of an accounting package used by the company. This software allows an asset (equipment) to be tracked by associating a number on the asset tag with the condition, business unit, and perceived value of the equipment. Examples of this equipment are desks, land, and even computer equipment. These types of databases work well for reporting on the value of equipment that the organization owns to calculate a net worth for an organization, but they do very little in helping an IT department plan upgrades.

Laptops, desktops, and other devices have variables such as storage, RAM, operating system versions, and other unique variables to the hardware and software of the device. Asset management systems are databases that collect data from the operating system through the use of an agent. This type of asset management is more detailed than a purchasing record from the accounting department. Once the information is collected, reports can be drawn when upgrades are required. For example, a report you may compile in the asset management system might be all operating systems that match Windows 10 and that have less than 4 GB of RAM and hard drives smaller than 100 GB. You then have a report of what needs to be upgraded in terms of hardware for an upgrade of the operating system to Windows 11. When using an asset management for an organization that spans a large geographic area, this is invaluable information that otherwise would have taken days to collect.

Software and Licensing

Asset management systems don't stop at hardware; software packages and their accompanying licensing are considered assets as well. Many asset management systems can also collect a list of the software installed on the devices in your organization. They can also include

detailed licensing usage information so that you can gauge where licensing is being used efficiently and where it is not based on usage.

Inventory Lists

Not all asset management requires databases and asset management systems. When managing a small amount of equipment, an inventory list is more than sufficient. The list can be a simple Microsoft Excel sheet detailing the types of equipment and their associated quantities. These inventory lists work really well when trying to control consumable electronics like mice, keyboards, and monitors. Once the rotating stock of equipment becomes too large in quantity and value, it's time to look at an asset management system.

Configuration Management Database (CMDB)

The configuration of devices and systems can also be assets. The fact is that your organization didn't pay for goods—they paid you for services to configure the device. The resultant configuration is considered an asset to the organization.

A configuration management database (CMDB) is a centralized repository that stores information about the configuration of items within an IT organization. These items, known as configuration items (CIs), can include hardware, software, systems, facilities, and even personnel. This centralized information can also be used for making business decisions and running efficient IT operations. Organizations can better manage changes, incidents, and compliance requirements when they know the configuration items.

The use of a CMDB enhances visibility and control over the IT environment. The CMDB helps with better incident and change management. When an incident occurs, the CMDB can be used to quickly identify the affected CIs and their relationships. The access to this information facilitates faster root cause analysis and resolution. Similarly, during change management, the CMDB provides a clear picture of the potential impact of proposed changes, helping to minimize risks and ensure successful implementation. We will cover incident and change management later in this chapter.

Asset Tags

All computer and network equipment should be tracked, from the cradle to the grave, by the IT department. When equipment enters the company, it should be labeled with an *asset tag*, as shown in Figure 10.1. The asset tag is often a permanent metallic sticker or metallic plate that is riveted to the equipment. The asset tag often has a *barcode*, which defines the numeric number that identifies the asset. This asset tag should then be entered into the asset management software, by either typing in the number or scanning the bar code.

FIGURE 10.1 An asset tag

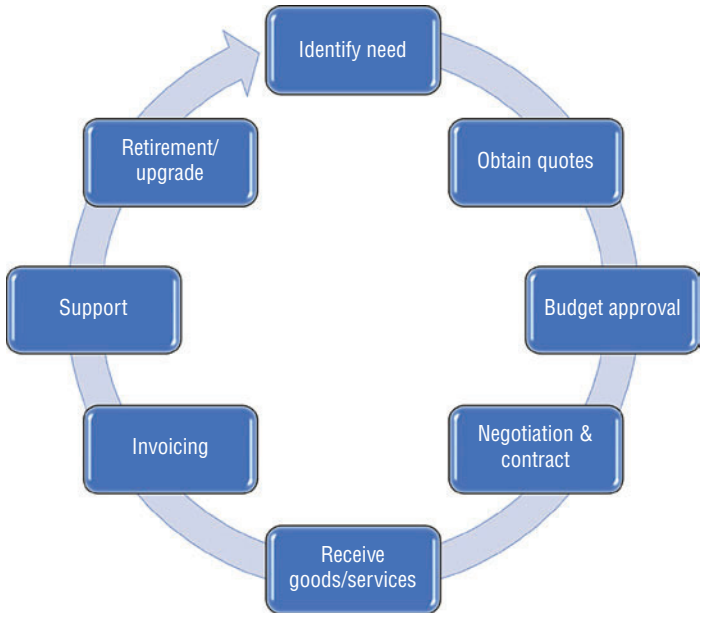


Procurement Life Cycle

The entire life cycle for any IT system is cyclical and differs slightly depending on the assets. The typical life cycle for IT systems consists of purchasing, deploying, managing, and retiring. The exact procedures for the IT life cycle will depend on your organization and the goods or services.

The first step is the procurement of the goods or services. This process is typically standardized by your organization's business affairs department or finance department. Just like any process, the procurement process differs from organization to organization. An example of a procurement life cycle is shown in Figure 10.2; yours may differ slightly. Regardless, the procurement life cycle will always start by identifying the need for the goods or services being requested. If it costs money for the organization, then it has to fill a need or solve an issue. This is probably the most important step in the procurement process. If it does not solve an apparent issue or need, then it might not get approved or may be sidelined for another budget cycle.

FIGURE 10.2 An example of a procurement life cycle



After the need is identified, it's time to obtain quotes for the goods or services. You should have three comparative quotes for the goods or services being requested. However, depending on the goods or services, achieving that may not be feasible. Examples where comparable quotes are unobtainable are direct purchasing from the vendor, vendor-registered value-added resellers (VARs), and custom goods or services.

The budget approval process will be dependent on the reported needs of the goods or services, as previously explained. The goods or services are submitted for approval for the operational budget (OpX) or the capital budget (CapX). Items being submitted for the CapX budget will depend on their value and utility. The test is always, can the item be depreciated over the expected life of the product? Examples are servers, workstations, and other equipment. Anything that cannot be depreciated, such as services, will fall into the OpX budget. It is likely that your organization has a standard characterization of goods and services and which budget is applicable. The outcome of the budget approval process will either be approved or denied, but it can also be conditionally approved based on meeting goals or other conditions.

Once the goods or services are approved, then your business affairs department will work with the vendor or reseller to negotiate money, terms and conditions, and the overall contract/scope of work (SOW). Once the purchase is completed, you will receive the goods or services. At this point, the contract or SOW is important, because it will define when the vendor is to be paid. If all the goods are not received or the services are not complete, then the vendor is not entitled to send an invoice for payment or the invoice can be held. Although this sounds like a simple part of the procurement process, it is often overlooked. You should never begin payment until the goods are received or the contract/SOW is satisfied.

During the initial phase of obtaining quotes, you should identify the cost for ongoing support, maintenance, or licensing of the goods or services. These costs should be submitted to the operational expense budget as an ongoing/recurring cost, since these costs are usually a monthly or annual cost. The vendor or VAR might also include 3–5 years of support, maintenance, or licensing in the quote so that it can be submitted to a capital budget. This will be based on your organization's processes. Typically, the vendor will include an initial warranty or license with the original purchase of the product or service.

Every product or service outlives its usefulness. This is where we identify retiring or upgrading the product or service. This retirement or upgrade will then start the procurement process all over again. This time around, identifying the needs is easier, unless the retirement does not necessitate replacing the product.

Assigned Users

When assets are acquired by the organization, they must be managed throughout their life cycle. This typically requires assigning a person to manage the group of assets, such as laptops, servers, and hotspots. This is a critical step in the management of the asset. The person who manages the assets is responsible for identifying users who are assigned to the devices in the event of termination. The responsible person is also required to forecast upgrades and perform accounting for all assets over their life cycles. Each organization has its own requirements, but these are the top requirements for asset management.

Common Documentation

Documentation is extremely important to an IT department, not to mention the entire organization. It serves many different purposes, such as educating new IT workers, recording work performed, highlighting issues, and describing normal functionality. However,

documentation is usually one of the functions that suffer the most when projects are hurried and there is a push to start the next project.

In the following sections, we will cover the most common documents that help support the IT department and day-to-day operations. Some of these documents are prepared by collecting information in a specific manner and then detailing the results; examples are incident and standard operating procedures.

Incident Documentation

An *incident* is any event that is unusual or outside of the normal processes. You may encounter many different types of incidents as a technician: network security incidents, network outage incidents, and even customer service incidents. Regardless of which type of incident transpires, an incident document should be completed so that there is a record of the event. A record of the incident allows for further review after the incident has subsided so that it is not repeated.

The incident document should be completed as soon as possible so that key details are not forgotten. This document is often used as an executive brief for key stakeholders in the company, such as C-level people—for example, the chief information officer (CIO). The incident document can also be public-facing and used to inform customers of the incident. When used in this fashion, the incident document allows the organization to communicate with transparency about a major incident they may have experienced. Volume 2 Chapter 9, "Safety and Environmental Concerns," covered the processes and procedures for incident response in further detail. Here are common elements of a network incident document:

Date and Time The date and time of the incident is probably the most important element, because several incidents could happen on the same day. The date and time allow you to distinguish between various incidents. The date and time should be referenced using the *Universal Time Code (UTC)* so that someone anywhere in the world does not need to calculate the time offset.

Summary The incident summary is another important piece of information that should be documented. It will often be the first thing that is looked at because it describes what happened during the incident.

Root Cause The root cause is the reason the incident occurred. Every incident needs to have a root cause defined; otherwise, the incident could happen over and over again. The network team's first responsibility is to identify the root cause.

Actions Taken The actions taken are the actions that transpired during the incident to rectify the situation. Most of the time, they are temporary actions so that business can resume.

Remediation The remediation is the ultimate fix that will repair the root cause of the incident. This could be as simple as installing a patch, or it could be as complex as redesigning a network component.

Services Impacted The services impacted section details all the network systems and business units that were affected during the incident. This section defines the severity of the incident.

Recommendations The recommendations section details the next steps to take to remediate the issue. It also explains how to avoid similar incidents in the future.

Although these are the most common elements of an incident document, the document is not limited to these elements. Each organization has different needs for the process of reviewing network incidents. A template should be created so that there is consistency in the reporting of incidents.

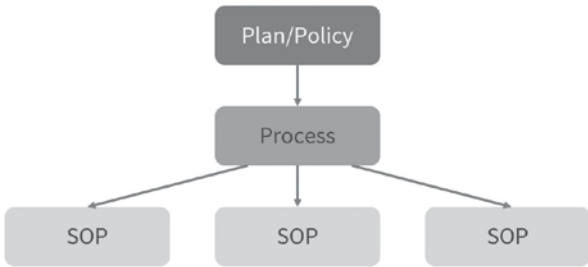
Standard Operating Procedures

When organizations create policies, they will outline specific processes to adhere to the policies. Throughout this discussion you may see the words *policy* and *plan*. Policies are also considered plans for the organization; once a plan is ratified, it becomes a policy. Put another way, a policy is a mature and enforced plan of action. Each process derived from the plan or policy contains a list of steps that are called *standard operating procedures (SOPs)*, as shown in Figure 10.3. All of these components are part of the documentation process for a *quality management system (QMS)*. QMSs are created to meet certain International Organization for Standardization (ISO) requirements. A common ISO certification is ISO 9001. When a company is ISO certified, it means that it adheres to strict quality standards for consistent outcomes and has a detailed operation plan. There are many information technology ISO standards that your organization can be certified with. You have probably seen these certifications in a sales manual at some point.



ISO 9001 is a globally recognized standard for Quality Management Systems (QMS), published by the International Organization for Standardization (ISO). It helps organizations ensure they meet customer and regulatory requirements while improving customer satisfaction. The latest version, ISO 9001:2015, focuses on a process-oriented approach and risk-based thinking. Adopting ISO 9001 demonstrates a commitment to quality and continuous improvement.

FIGURE 10.3 Standard operating procedures



A process in a QMS is just that—a process. A process is defined as taking an input and creating an output. Here's an example: a specific server needs to be decommissioned, which is the input to the process of decommissioning a server, and the output of the process is that the server is wiped clean of corporate information. The process in this example is the decommissioning of a server. (We are oversimplifying the process in this example.) Most processes have an input specification and an output specification.

The SOP in this example outlines how to perform the process of decommissioning the server. The SOP should clearly explain who is responsible and the standard they must achieve for the process. In the example of a decommissioned hard drive, the SOP would define the following:

- Who is responsible for removal of the server from the rack
- Who is responsible for wiping the drives in the server
- What standard should be used to wipe the drives
- Who is responsible for removing DNS entries
- Who is responsible for removing firewall rules
- Who is responsible for clearing the BIOS/UEFI
- In which order the tasks are performed

Several tasks are created from the SOP document. Each task is defined as part of the procedure to achieve the decommissioning of the server. The work instructions serve two primary purposes. The first is to detail how each task should be performed. The exact steps are listed in the work instructions, and for the previous example, they may include the following:

- How to remove the server from the rack
- How to wipe the drives in the server
- How the drives should be wiped
- How to remove the DNS entries
- How to remove the firewall rules
- How to clear the BIOS/UEFI

The second purpose of the work instructions is to provide a training process for new employees. The work instructions are a more detailed portion of the procedure, so it becomes a training item for new employees on how to perform their job.

Procedures for Installing Software Packages

If something is worth doing the first time, then it's worth documenting and chances are it will need to be done again. This is often the case when installing custom software packages. When a new software package is purchased, a technician is often assigned to assist and install the application for the users. It is that technician's responsibility to document the software installation process. This ensures that other technicians will not have to start from scratch to help the next user who needs the software package installed.

Documenting the software installation will save time for consecutive installations. It also allows the initial technician to pass the knowledge on to other teammates. This is obviously important for the rest of the IT team, and it's also important for the initial technician. It is often the case that the original technician will be assigned the consecutive installation, mainly because they are the only ones who know how to install the software. Therefore, documenting the process of installing the software serves two purposes of saving time and passing information on to other teammates.

Onboarding/Offboarding Procedures

As employees are hired in your organization, a certain amount of initial interaction with IT is required. This interaction is called the *onboarding procedure* and is often coordinated with the HR department in your organization. During the onboarding procedure a *new-user setup checklist* should be followed. Examples of the items on the checklist include showing the user how to log in the first time and changing their password. The password policy is often the first policy discussed with the user. Other policies such as bring your own device (BYOD), acceptable use policies (AUPs), and information assurance should also be discussed during the onboarding procedure. Email, file storage, and policies should be covered as well. Each organization has a different set of criteria that make up the onboarding procedures.

Eventually, employees will leave your organization. The offboarding procedure ensures that information access is terminated when the user is terminated. A *user termination checklist* should be followed during the offboarding procedure. The process will be initiated by the HR department and should be immediately performed by the IT department. This process can be automated by using the organization's employee management system. The procedure can also be manually performed if the employee management system is not automated. However, the procedure must be performed promptly, since access to the company's information systems is the responsibility of the IT department. During the offboarding procedure, email access or BYOD access is removed through the use of the mobile device management (MDM) software; the user account is disabled; and IT should make sure the user is not connected to the IT systems remotely. The offboarding procedure may also specify that a supervisor must assume ownership of the terminated employee's voicemail, email, and files.

SLA Agreements

A *service-level agreement (SLA)* is an expected level of service that a service provider will adhere to for uptime. The SLA is detailed in the service contract as a percentage of uptime per year; it is often called "the nines." When a service provider exceeds the percentage of uptime defined in the SLA, the provider is in breach of the contract. The contract will often identify what is considered in the SLA requirement as well as the terms if they breach the SLA requirement.

You should be careful with SLAs because being up 99 percent of the time means the service provider can be down 3 days and 15 hours over the course of a year before breaching the contract. A service provider with 5 nines (99.999 percent) can be down 5 minutes and 15

seconds over the course of a year before breaching the contract. Table 10.1 details the different nines of an SLA for uptime.

TABLE 10.1 Uptime for nines of an SLA

SLA %	Downtime per year	Downtime per month	Downtime per day
99% (two nines)	3.65 days	7.2 hours	14.4 minutes
99.9% (three nines)	8.76 hours	43.8 minutes	1.44 minutes
99.99% (four nines)	52.56 minutes	4.38 minutes	8.64 seconds
99.999% (five nines)	5.26 minutes	25.9 seconds	864.3 milliseconds
99.9999% (six nines)	31.5 seconds	2.59 seconds	86.4 milliseconds

Internal

Internal service level agreements (SLAs) are used to ensure that teams within an organization maintain a specific performance goal. These agreements outline specific performance metrics, such as response times, resolution times, and quality standards, which are expected to be met by the team. By doing so, they create a clear understanding of what is required and help to manage expectations.

One of the key benefits of internal SLAs is that they promote accountability. Each team knows exactly what is expected of them and can be held responsible for meeting those expectations. This accountability helps to drive performance improvements and ensures that any issues are addressed promptly.

Knowledge Base/Articles

If time is spent on an issue, it's worth documenting so that the same amount of time is not required by someone else or yourself in the future. A *knowledge base* is a collection of issues with solutions that both your internal customers (IT staff) and external customers (end users) can use to solve common issues. The Microsoft Knowledge Base is a great example of a knowledge base. It contains more than 200,000 public articles and just as many that are private and accessible only by partners. Most helpdesk software allows for the creation of a knowledge base article from the resolution of an issue. A knowledge base can become very large, so many knowledge bases allow keyword searches.

Fortunately, you don't need fancy helpdesk software to create a knowledge base. You can simply have a collection of articles that are accessible to either your colleagues or your end users. A knowledge base article should be clear and easy to understand. Make

sure that you define any terms or jargon used within the article so that you don't lose the audience it is intended for. Here are the common elements of a typical knowledge base article:

Title The title should represent the key symptom of the issue and be action-based. For example, if the issue is that the Num Lock key stays on after the system boots up, the title could be "Num Lock Enabled After Boot."

Introduction The introduction should explain, in plain language, what the rest of the article will cover. It should include the thesis of the issue as well as the systems affected by the issue.

Symptoms The symptoms section explains how to identify the symptoms of the issue. It assists the user in validating that the issue covered in the introduction is affecting their system. This section can include steps to reproduce the issue. For example, after a fresh bootup and login, your Num Lock key stays lit up.

Solution The solution section contains the steps to fix the issue outlined in the introduction. The solution should not be restricted to just one solution; many solutions can be detailed, as well as work-arounds, if the primary solutions cannot be implemented.

More Information The more information section contains a list of resources that were used to formulate the solution. This section is important because the symptoms might be similar to those of another issue or an unidentified issue. The resources listed will allow the technician to research the issue deeper.

Date The date is important because it allows the reader to see if the issue is an old issue or a newer one, to gauge how relevant the solution is to the current issue.

Author The author section allows for the technician reading the knowledge base to contact the original technician who created the article.

Although these are the most common elements for a knowledge base article, you are not constricted to only these elements. Whichever format you end up using, it is important to be consistent. A template should be drafted so that there is consistency. Consistency allows either the technician or the end user to expect the elements when they review the article.



Exam Objective 4.2, "Given a scenario, apply change management procedures," requires you to know the change management process and how it is applied in an organization to reduce risk of changes. In addition, you should know the various components of the documented business processes, such as the rollback plan, backup plan, sandbox testing, and responsible staff members.

Change Management Best Practices

When you implement a new system or change an existing system, you affect a lot of people. You also affect business processes and other business units with these changes. Don't underestimate the power of the documentation you produce as a technician. It can and often will be used by change management groups to review the impact of your proposed changes.

Change management is a process often found in large corporations, publicly held corporations, and industries such as financial services that have regulatory requirements. However, change management is not exclusive to these types of organizations. The main purpose of change management is to standardize the methods and procedures used to handle changes in the company. These changes can be soft changes of personnel or processes, or hard changes of network services and systems.

When changes are proposed to a process or set of processes, a document is drafted called the *change management plan document*. This document is used throughout the change management process to evaluate the impact to the business continuity of the organization. In the following section, we will discuss the elements of a change management plan document.

Documented Business Processes

The documented business process is incorporated into the change management plan document. It provides an overview of the business process that the changes are expected to affect. It allows everyone involved in the process both directly and indirectly to understand the entire process.

The documentation specifically defines who interacts with, how they interact with, why they interact with, and when they interact with the process. For example, if your company created widgets, your documentation might detail the process of manufacturing the widget. The document would describe the following:

- How the raw components enter the production line
- Who delivers the raw materials
- When they deliver the raw materials
- Who assembles the raw materials to create a finished product
- Who inspects the finished product
- When and how the product is shipped

Rollback Plan (Backout Plan)

The *rollback plan*, also called the backout plan, describes the steps to roll back from a failed primary plan. If it were determined that the primary plan could not be completed, you would either implement an alternate (secondary) plan or a rollback plan, depending on the changes proposed in the change management plan document. Like the primary and alternate

plans, the rollback plan should contain the steps to be taken in the event the rollback plan must be executed. The rollback plan should also document any changes to configuration so that it can be reverted back. Most of the rollback plan will consist of the original configuration, with any additional steps to revert it back.

Backup Plan

The *backup plan* is an essential component of the change management process, safeguarding against potential failures that may arise during implementation. This plan involves creating comprehensive backups of all critical systems and data before any changes are made. The backups should be performed systematically, ensuring that each element of the infrastructure and data is covered. Verification of the backup integrity is also a crucial part of the backup plan. This involves regular testing to confirm that the backups are complete and can be successfully restored.

In the event of an unforeseen issue or failure, this plan ensures that the organization can quickly restore its systems to their previous operational state with minimal disruption. The detailed steps for restoring systems from backups should include configurations, revalidation processes, and checks to ensure operational integrity after restoration. The backup plan should be created in conjunction with the rollback plan.

Sandbox Testing

Sandbox testing is extremely useful when you want to test a change before placing it into production. A sandbox can be constructed to match your environment; you can then implement the change and fine-tune your primary plan. The use of a sandbox testing environment allows you to hone your process for the proposed change while observing any potential issues.

The introduction of virtual machines makes it very easy to set up a sandbox for testing. You can clone production servers into an isolated network and then create snapshots on the server in the sandbox and test over and over again, until all the bugs are worked out of the primary plan.

Responsible Staff Members

Every process in the organization must have a person who is assigned to be the *responsible staff member*. This person oversees the process and can answer questions about the process. If there are any changes to the process or changes that can affect the process, this person acts as the main point of contact. They can then facilitate any changes to the process.

As an example, you may assign a person to be the responsible party for the electronics decommissioning process. Any questions about disposal of electronics should be directed to this person. If your organization is choosing a new e-waste company, it will affect the decommissioning process. Therefore, this person should be included in the decision as a stakeholder. Any changes can then be adjusted or integrated into the decommissioning process, and this person can facilitate the changes.

Request Forms

The change management process often begins with a *request form* that details the proposed change. The exact elements in the request form will differ slightly depending on your organization's requirements. The following lists the most common elements found on the change management request form. Some of the information found on the request form is preliminary; the information will be expanded upon as the request form transitions into the change control document.

Item to Be Changed This is the item that is being requested for a change. Examples are software, hardware, firmware, configuration, or documentation.

Reason The reason the item is being submitted for a change. Examples are legal, marketing, performance, software bug, or process problem.

Priority The urgency, or priority, of the change is documented in this section of the form, although this should also be conveyed in the reason for the change. The priority is often a separate field on the change control form. Examples are emergency, urgent, routine, or a specific date.

Change Description/Plan The description or plan for the change is documented in this section. For changes in configuration, you would detail the changes to the configuration and why each part of the configuration is being changed. Firmware changes would list the version being upgraded from and the version being upgraded to.

Change Rollback Plan The rollback plan describes the steps to roll back from a failed primary plan. If it was determined that the primary plan could not be completed, you would implement either an alternate plan or a rollback plan depending on the changes proposed.

Technical Evaluation In this section of the form, you document why the primary plan will succeed. The changes should be tested in a lab environment closest to the production environment and documented in this section. When you're creating the technical evaluation, outline specific objective goals along with the metrics with which they can be measured.

Duration of Changes Here, you document the estimated duration of the change process. Any service outages will be documented in this section.

Purpose of Change

The *purpose of change* is the reason the change management process starts. Either your business unit requires a change that will affect others, or another business unit requires a change that can affect your business unit indirectly. A change to any part of the process, such as the intake of raw materials, could affect the end result. Change is an essential component of a business, and it should be expected. If your company only created widgets and never evolved, you would eventually be out of business.

Unfortunately, not all changes support the company's product line directly. Some changes are imposed on the company, because IT systems are constantly changing. As a technician,

you are constantly upgrading and patching systems and equipment. These upgrades and patches are considered changes that can affect the entire business process.

This section of the change management plan document should explain why the change is necessary. It needs to include any vendor documentation explaining the change to the product. For example, if the proposed change were to install a Microsoft Windows security patch, the purpose of the change would be the security of the Windows operating system. The vendor documentation in this example would be the knowledge base article that normally accompanies Windows security patches. Other examples of purposes of change might be legal, marketing, performance, capacity, a software bug, or a process problem that requires a change.

Scope of Change

The *scope of change* details how many systems the proposed change will affect. The scope could involve only one system, or it could be all the systems in an entire enterprise. The scope of change is not limited to the number of systems that will be changed. The scope can also describe how many people a proposed change will affect. For example, if you propose to change lines of code in an ordering system, the change could affect your salespeople, customers, and delivery of the products. The scope of this change could impact the business continuity directly if something goes wrong during the change.

When creating this section of the change management plan documentation, be sure to document which systems the proposed change will affect, the number of systems the proposed change will affect, the number of people the proposed change will affect, as well as whether anyone will be directly or indirectly affected. In addition, you should include the proposed date and time of the change and how long the change will take. Keep in mind that this section allows the change management team to evaluate how big the proposed change is. The scope should answer the following questions:

- Who will the change affect?
  - IT resources
  - Other departments
  - Business units
  - Partners
  - Customers
- What will the change affect?
  - IT systems
  - Sites/locations
  - Processes/procedures
  - Availability of systems
  - Outage duration

- What will be impacted?
  - Other proposed changes
  - System performance
  - System capacity
  - Other resources (personnel, security, etc.)
- When is the proposed change?
  - Date and time of the change
  - Duration of the change

Change Type and Schedule

In addition to the purpose of the change, and the preceding questions about the change, the driving factor of the change is going to be the change type. The change type will ultimately affect the schedule of the change. The change type is the driver behind the change. The common change types are as follows:

Standard Change In the realm of change management, a standard change is a pre-approved, low-risk change that follows a predefined process and is frequently performed. These changes typically have minimal impact on the organization and require no additional approvals.

Normal Change A normal change is any change that does not fall under the standard or emergency categories. It requires a thorough review and approval process, including a risk assessment and a detailed implementation plan. This type ensures that all potential implications are considered before proceeding.

Emergency Change An emergency change is a high-priority change that must be implemented immediately to address critical issues that could cause significant disruption or harm to the organization. Due to its urgent nature, the usual review and approval processes are expedited, with the focus being on resolving the issue as quickly as possible to restore normal operations.

Each of these change types requires a tailored approach to ensure that they are managed effectively and align with the organization's overall change management strategy.

Maintenance Window

The change type will obviously dictate the date and time of the change or timing of the change. When deciding on the timing of the change, the maintenance window should be considered. The maintenance window is a scheduled period during which regular maintenance tasks, updates, and system enhancements are performed on an organization's IT infrastructure. This predefined time frame is carefully selected to minimize disruption to normal business operations, typically occurring during off-peak hours when system usage is at its lowest.

Change Freeze

A change freeze is a period during which no changes or very limited changes are allowed to be made to the system. This is typically enforced during critical business periods, major projects, or before and during the implementation of significant IT changes. A change freeze is a deliberate pause in the implementation of new changes within an IT environment. During the change freeze, only emergency changes or pre-approved exceptions are allowed, and these are closely monitored and controlled.

During a change freeze, the primary objective is to maintain the stability of the IT environment. With no new changes being introduced, the risk of unforeseen issues and disruptions decreases significantly. This is especially critical during high-stakes periods such as end-of-quarter financial reporting, major product launches, or other times when system availability and reliability are paramount.

Peer Review

Typically, we will not solely understand the impact of a change to other systems. This is why a peer review should be performed with peers in other departments. In some circumstances, the change might appear to be successful but inadvertently cause problems elsewhere in the business process. An example is a change made to a shipping system, which could affect your data analytics or accounting team's business process. This is where changes should be identified for risk and introduced into the risk analysis process.

Risk Analysis

Whenever a change is made to a system or equipment, there is the potential for the system or equipment to fail. The change could even cause another system or piece of equipment to fail. In some circumstances, the change might be successful but inadvertently cause problems elsewhere in the business process. For example, if a change to an ordering system causes confusion in the ordering process, sales might be inadvertently lost.

Risk analysis is the process of analyzing the proposed changes for the possibility of failure or undesirable consequences. Although you will include this section in the initial change management plan document, your sole risk analysis will be narrow in perspective, because you will focus on the process from the IT aspect. A change advisory board will perform a much larger risk analysis. This team will have a much larger perspective, since they come from various business units in the organization. From this analysis, a proper risk level to the organization can be determined. The risk level will dictate how much time is spent on the possibility of failure or undesirable consequences from the change.

Risk Levels

Risk levels are used to communicate our perceived risk for failure or a change. For most purposes the risk levels can be kept simple, such as a three-tier system of: low, moderate, or high. You can build risk level structures with more than three tiers of risk, such as the popular four-tier model of low, moderate, high, and extreme, or the six-tier model of very-low, low, medium, high, very-high, and extreme. The number of risk levels your organization

will use to convey risk will be based on several factors, such as the typical audience and the amount of quantitative data used to determine the perceived risk.

If you are analyzing risk for a development department, then you may want to use a granular system that has many tiers. Developers are accustomed to details, and they typically prioritize their work on a granular level. However, if you are presenting to a C-suite level of individuals on a daily basis, then a more simplistic system is in order, since executives drive their decisions based on the brevity of information and not always the finite details.

The three-tier model serves many organizations just fine when it comes to risk analysis. Here are the risk levels when we classify a change:

- Low Risk
  - The system is easily recoverable and reproducible.
  - The system provides an informational (non-critical) service.
- Moderate Risk
  - The system is integral with other systems.
  - The system provides a normal or important service.
- High Risk
  - The system is highly integrated with other systems.
  - The system provides a critical organizationwide service.

Implementation

The implementation section of the change management plan explains how the proposed change will be executed. Steps should be detailed on the changes and the order of the changes. If changes were to be made in configuration files, switches, or routers, you would document the changes to the configuration and why each part of the configuration is being changed. Firmware changes would list the version being upgraded from and the version being upgraded to. The idea is to provide as much detail as possible about the documented changes to be made to the systems or equipment.

When a change is implemented or planned, there is always the potential for problems, or you may identify a consideration in the execution of the plan. The plan for change section should detail those considerations. It's common for a primary plan to be drafted as well as an alternate plan in the event the primary plan cannot be executed. For example, if the primary plan is to move a server from one rack to another so that it can be connected to a particular switch, the alternate plan could be to leave it in the rack and use longer cables. Be sure to have multiple plans; once the change is approved, the plan(s) outlined in this document must be executed closely.

You should also document why the primary plan will succeed. The changes should be tested in a lab environment closest to the production environment (if possible) and documented in this section as well. When creating the plan, you should outline specific, objective goals, along with the metrics with which they can be measured. For example, if you are planning to make a change because there is a high error rate on an interface, then the metric measure to be

compared would be the error rate on the interface. You would document what you expect the error rate to be after the change is made so that you can measure the success of the change.

Change Board

The change board, also known as the *change advisory board*, is the body of users who will ultimately evaluate and then approve or deny the change you propose. This group of people often meets weekly to discuss the changes detailed in the change management plan documents. The goal of the change advisory board is to evaluate the proposed changes in order to reduce the impact on day-to-day operations of the organization.

It is common practice for the meetings of the change advisory board to be held via a conference call at a set time every week. This allows key stakeholders in an organization to be available regardless of where they are in the world. Because it's at a set time every week, there are no excuses for not being available during the change control meetings.

It is likely that if you are the technician proposing the change, you will be on the call for questions or clarification. The key to getting a change approved is to know your audience and communicate clearly in the change control plan document. Remember, the change advisory board is often composed of various stakeholders from the entire organization, not just IT. You should not assume that the change you are proposing is as clear to them as it is to you. Some change advisory boards are made up strictly of IT stakeholders, so you must understand who will review the proposed changes and choose your wording appropriately.

The change management document must be approved by the majority of change advisory board members or by specific board members. The approval of the proposed change should be documented in the change control policy for the organization. Only approved changes can be executed. If other changes need to be made outside of the original submission, additional approvals must be acquired.



Each organization is different, and each change management documentation process will be slightly different. However, the sections described here are the most common elements found on these documents.

End-User Acceptance

Although the CompTIA A+ exam does not focus on application development testing and approval, *user acceptance* is an objective on the exam as it pertains to the change management process. It should be noted that user acceptance is not solely used for application development; it is also used when there is a significant update to an interface or a process, such as a service pack or upgrade to an operating system.

When a change is to be made in which the user's interaction will be impacted, it is common practice to beta-test the change. This is also known as user testing or just plain application testing. You can achieve user acceptance two different ways:

In-Person Testing The person responsible for testing dedicates time and performs testing in person with the developer.

Self-Paced Testing This has its advantages, since the user will be relaxed in their own environment during testing.

Regardless of which method of testing you choose, a strict time frame must be communicated to the user testing the change.

Once user acceptance is obtained, it should be documented in the user acceptance section of the change management documentation. The methods of testing, the users and groups involved in testing, and the time invested in testing should be included in this section as well. Remember that the goal is the approval and successful implementation of the changes, so it is important that you are convincing and, more importantly, convinced that the change will succeed without repercussions.



Exam Objective 4.3, "Given a scenario, implement workstation backup and recovery methods," requires you to know how to implement backup and recovery methods, backup testing, and backup rotational schemes. This will ensure you can recover from information that is deleted or when you fall prey to a disaster.

Backup and Recovery

As a technician you're responsible for recovering from uncontrolled disasters. Luckily, you can prevent disasters by taking proper precautions, as we will discuss in this section. You'll find that you're prepared when disaster strikes and can restore business continuity that much more quickly. This section discusses data loss from the following types of disasters: data loss, human error, and data loss as the result of threat actors.

When we think of data backups, we usually relate them to disasters. However, data backups are not just used to restore from disaster; we often use data backups when a user inadvertently deletes files they shouldn't have deleted. Data backups are also used when users overwrite files or just plain forget where they put them in the first place. Regardless of how the data was lost, the underlying reason we create data backups is to recover from data loss.

Because you can't choose the disaster or situation that causes the loss of data, you should adopt a layered strategy, starting with the user and expanding outward to the infrastructure. The following sections cover several types of strategies that can protect you from data loss.

File-Level Backups

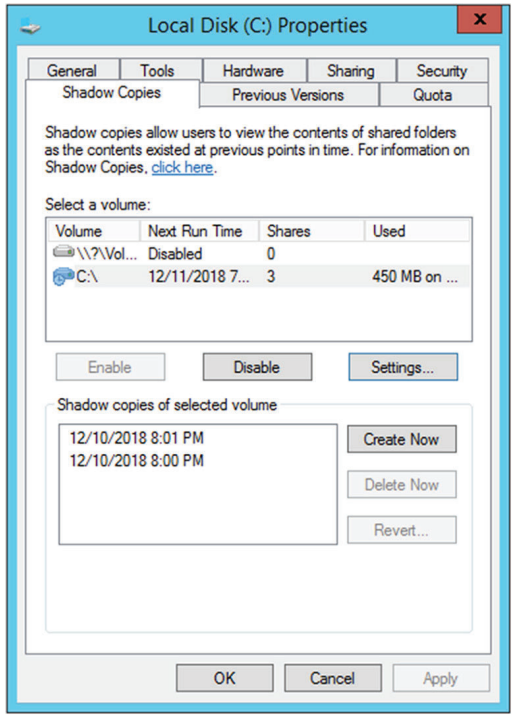
Most of the time, your users will need to restore a single file or perhaps a few files, but definitely not the entire server or server farm. Therefore, you should make sure that one of the layers of protection allows for the restoration of individual files. You can implement this type of strategy in several different ways. Depending on your resources, you should use them all.

Volume Shadow Copy

Volume Shadow Copy, also known as the *Volume Snapshot Service (VSS)*, has been an integral part of the Windows Server operating system since the release of Windows 2000.

Volume Shadow Copy can be enabled on a volume-by-volume basis. Once it's turned on, all the shares on the volume are protected. You can access Volume Shadow Copy by right-clicking a volume and selecting Properties. You can then configure it by using the Shadow Copies tab, as shown in Figure Figure 10.4.

FIGURE 10.4 The Shadow Copies tab

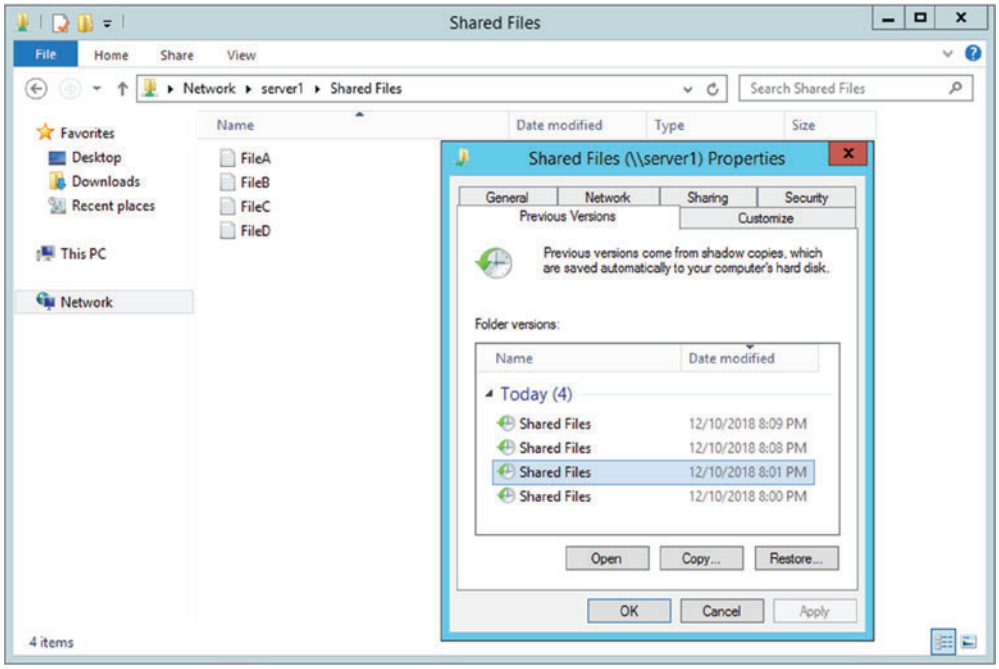


Volume Shadow Copy has one amazing advantage: it empowers the user to restore their own files. All the user needs to do is right-click the file or empty space in the shared folder, select Properties, and then in the Properties window, select the Previous Versions tab. This will open a list of snapshots, as shown in Figure 10.5. The user can then double-click the snapshots to open them as if they were currently on the filesystem. This allows the user to evaluate what they are looking for. Once they find what they are looking for, they can either click the Restore button or drag the files over to the current folder.

One limitation to Volume Shadow Copies is the number of snapshots that can be active. Only 64 snapshots can be active at one time. The oldest snapshot is deleted when a new snapshot is created to maintain a running total of 64 snapshots. By default, Volume Shadow Copy is not enabled. When it is enabled, the default schedule creates a snapshot twice a day, at 7 a.m. and 12 p.m. It's advisable to set a schedule that creates a snapshot every hour

during normal business hours. This will give the user the last 64 hours of work, which could be well over a week and a half, if you were open 9–5.

FIGURE 10.5 The Previous Versions tab

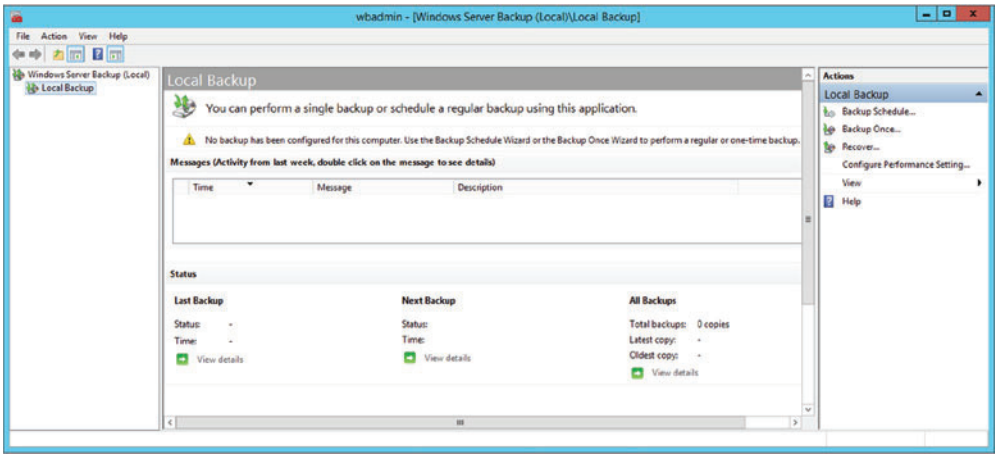


File-Based Backups

File-based backups are a common type of backup in organizations today and have been since the introduction of backup software. The Windows Server operating system includes a backup program capable of protecting the local server, as shown in Figure 10.6. It is somewhat limited, because it only supports a file-based destination and does not offer options for data tapes. It also only allows for the management of the local server. However, the product is free and is included with the Server operating system, so there is no reason not to have some type of backup.

Advanced backup software, such as Veeam Backup & Replication and Veritas Backup Exec, allows for the centralized management of all backups. Multiple backup jobs can be created for various groups of servers and can be directed to various destinations. For example, the accounting servers might back up to a tape library unit, whereas the sales servers back up to a disk storage unit. We'll discuss media types later in this chapter, but the key takeaway is that multiple jobs can be created and executed at the same time.

FIGURE 10.6 Windows Server Backup



Advanced backup software often requires a licensed agent to be installed on each server. Depending on the type of agent purchased, the agent might just allow for a simple backup of files, or it might allow for open files to be backed up while they are in use. Some agents even allow for the snapshot of all files so that a point-in-time image can be made of the filesystem. The backup is then created from the snapshot. This type of backup is common in financial institutions, where an end-of-day cutoff needs to be created.

Advanced backup software normally performs a pull of files from the selected source server and directs the information to the selected media. This is called the pull backup method, and it is probably the most common type of backup you will encounter. However, there are also push backup methods, in which the backup software directs the selected source server to push the files to the destination media using the backup server. This reduces the utilization on the backup server and speeds up the backup process, also known as the backup window.

Image-Based Backups

Image-based backups allow for a complete server to be backed up. This type of backup is also called a *bare-metal backup*. It's called a bare-metal backup because if the server hardware were to fail, you would restore the backup to a new server (bare-metal) and restore it completely. The inherent problem with these types of restorations is that they require administrator intervention. However, the technology is impressive and spares you from reinstalling the server from scratch.

Virtualized environments are where image-based backups really add value. Virtualization is changing the landscape of IT, and the area of backups is no different. When a server is virtualized, the guest virtual machine consists of configuration files, a virtual filesystem file, and other supporting files. When access is given to the underlying filesystem where the files can

be directly accessed, they can be backed up. This allows for an image to be created for the current state of an operating system—files and all.

Most enterprise backup software supports image-based backups for an additional license fee. It normally requires an agent to be installed on the host operating system, such as Microsoft Hyper-V. In VMware environments, a VMware Consolidated Backup (VCB) proxy is required. This application proxy allows the backup software to create snapshots for the guest virtual machines and assists in backing up the virtual machine files.

Critical Applications

So far, we've discussed how to use file server backups to protect an organization. However, an organization does not rely solely on file servers; there are many other types of servers in an organization. Examples include Microsoft SQL, for databases, and Microsoft Exchange, for email. In addition, there are several other types of applications that might be custom to an organization.

Just like file servers, Microsoft SQL and Microsoft Exchange have custom agents that are licensed. These agents allow for the data contained in the proprietary data stores to be backed up to your backup media. In addition to the backup of data, the agent starts a maintenance process at the end of a backup. This maintenance process checks the consistency of the current data store by replaying transaction logs, also called *tlogs*.

Critical applications for an organization do not have to be on-site. As organizations adopt a cloud-based approach to IT, they push critical applications out of the network and into the cloud. Providers such as Amazon Web Services (AWS) and Microsoft Azure offer not only the critical applications but also backup services that are contained in the cloud.

Backup Media

When creating a backup job, you choose what you want to back up (source) and a destination where it is to be stored. Depending on the backup software, you may have several different destinations to select from. Examples include iSCSI storage area networks (SANs), network-attached storage (NAS), tape library units (TLUs), or even cloud-based storage, such as Amazon S3. These are just some examples; there are many different media options for storing backups. Each backup media option uses a specific media type, and each media type has unique advantages and disadvantages. Here are the three media types commonly used for backups:

Disk-to-Tape Disk-to-tape backups have evolved quite a bit throughout the years. Today, *Linear Tape-Open (LTO)* technology has become the successor for backups. LTO can provide 6 TB of raw capacity per tape, with plans for 48 TB per tape in the near future. Tapes are portable enough to rotate off-site for safekeeping. However, they require time to record the data, resulting in lengthy overall backup time. Restore requires time to tension the tape, locate the data, and restore the data, making the RTO a lengthy process.

Disk-to-Disk Disk-to-disk backups have become a standard in data centers as well because of the proximity of the data and the short RTO. This type of media is usually based on-site and then it is used to create an off-site copy. It can record the data faster than traditional tape, thus shortening overall backup time. It also does not require tensioning and seeking for the data, like a tape requires.

The capacity of a disk, however, is much smaller than a tape because the drives remain in the backup unit. Data deduplication can provide a nominal 10:1 compression ratio, depending on the data. This means that 10 TB of data can be compressed on 1 TB of disk storage. So, a 10 TB storage unit could potentially back up 100 TB of data. Again, this depends on the types of files you are backing up. The more similar the data, the better the compression ratio.

Disk-to-Cloud Disk-to-cloud is another popular and emerging backup technology. It is often used with disk-to-disk backups to provide an off-site storage location for end-of-week backups or monthly backups. The two disadvantages of a disk-to-cloud backup are the ongoing cost and the lengthy RTO. Anytime you want to restore data, you must download the data from the cloud. Depending on your Internet connection speeds, this could require some time. The advantage is that expensive backup equipment does not need to be purchased along with the ongoing purchase of tapes.

Media Rotation

Administrators will adopt a rotation schedule for long-term archiving of data. The most popular backup rotation is *grandfather, father, son (GFS)*. The GFS rotation defines how tapes are rotated on a *first-in, first-out (FIFO)* basis. One of the daily backups will become the weekly backup on a FIFO basis. And lastly, one of the weekly backups will become the month-end backup. Policies should be created such as retaining 6 daily backups, 4 weekly backups, and 12 monthly backups. As you progress further away from the first six days, the RPO jumps to a weekly basis, then to a monthly basis. The benefit is that you can retain data over a longer period of time with the same number of tapes.

Backups are created for one of two main reasons: accidental deletion and disaster. Therefore, it makes sense that a disaster that could destroy your data center could also destroy the backup media. For this reason, media should be rotated off-site from the on-site presence of the original media.

The *3-2-1 backup rule* method is a common method for maintaining both on-site and off-site backups. The 3-2-1 method works like this: Three instances of the data should exist at all times. The original copy of the files and a backup of the files should be on-site, and the third copy of the data should be off-site in the event of tragedy at the site. Here's an example: You create a business proposal on your computer (first instance), and nightly your files are backed up (second instance). You now have two instances local to your immediate site (on-site) in the event of an accidental deletion. A second backup job then backs the file up to the cloud. This provides a third instance of the file, which is off-site.

There are a number of ways you can achieve this method of disaster recovery. For instance, you create the file, Volume Shadow Copy snapshots the drive on the hour, and a nightly backup copies the file to the cloud for off-site storage.

Backup Methods

There are several options for creating file-based backup jobs. Each backup method has advantages and disadvantages, depending on the media you are using and the amount of time in your backup window. The following are several of the backup methods you will find primarily with file-based backups:

Full A full backup is just that: a full backup of the entire data set. A full backup requires the longest backup window of all the methods, because it is the slowest. Therefore, full backups are generally performed on weekends, when you have a lengthy window. All files are backed up, regardless of the state of the archive bit. The archive bit is an attribute of each file; when a file is created or modified, the archive bit is turned on. When a full backup is performed, all the archive bits are reset on the files in the data set. A full backup is not sustainable through the week because it backs up all the files, regardless of whether they have been modified.

Incremental An incremental backup is used to speed up backups through the week when backup windows are short. An incremental backup backs up all files with the archive bit set. After the files are backed up, the archive bit is reset. Only the files that were created and modified from the last full or prior incremental backup are backed up, so backups are small. If you need to restore from an incremental backup, however, you will need to restore the full backup as well as all of the incremental backup files up to the RPO required. This type of restore creates a longer RTO because of the multiple backups that are required.

Differential A differential backup is also used to speed up backups through the week. It will back up all the files with the archive bit set as well, but it will not reset the archive bit after it has done so. A differential backup creates a gradually larger backup until a full backup is completed and the archive bits are reset again. This type of backup has a shorter RTO than incremental backups, because only the full and the last differential backups are needed to restore to the RPO required.

Copy A copy backup is used when you want to make an exact copy of the server. This backup method is identical to a full backup, with the exception that it does not reset archive bits. This backup method is often used when you want to make a copy of a server before a major upgrade. This backup method allows you to create a backup without affecting an ongoing nightly backup.

Synthetic Full A *synthetic full backup*, also known as just a synthetic backup, is a full backup without the overhead of performing a full backup on the entire data set. A synthetic backup is performed by the backup software. The software takes the latest full backups and applies the differential backups or incremental backups up to the desired date the job runs. For example, if you create a full backup once a month but want weekly full backups, a synthetic full backup can be created by taking the last full backup and applying the changes up to the current date. The synthetic full backup now serves as a full backup in the event a restore is required, thus lowering the RTO.

Data Restoration

When you have implemented the appropriate backup types for your organization, implemented the proper methods and media, and have a backup rotation, then you are prepared for when disaster strikes! If a user suffers data loss, someone deletes something inadvertently, or a threat actor ransoms data, you have the ability to restore the data. The question is: where to restore the data?

In-Place If the restore is being performed because a user inadvertently deleted a folder or folder structure, then the answer is pretty obvious, the data should be restored in-place. This may be the server share it was originally shared from. The location might also be a folder on the server that is used for application data, such as a database.

Overwrite There are certain situations where you may want to overwrite the existing data. These situations might be from ransomware, or just plain old people being people and corrupting their files. In either case, the restoration of data will need to be done over the existing data. The location could be to a share or file location.

Alternative Location When you restore data, most backup software will allow you to choose an alternative location. Using an alternative location is your best choice if a user is questioning the validity of their existing data. They might want to compare a group of files from a prior time period, against their current files. It is also a very low risk to restore files to an alternative location. However, the consideration should be the size of the files being restored and the potential for someone to start using the restored files in lieu of the current files. Therefore, communication is key with the user requesting the files to be restored. The alternative location should be labeled as a restore from a specific date.

Characteristics

When discussing the restoration of data, two characteristics dictate when you back up and how you back up. The concept of the *recovery point objective (RPO)* defines the point in time that you can restore to in the event of a disaster. The RPO is often the night before, since backup windows are often scheduled at night. The concept of the *recovery time objective (RTO)* defines how fast you can restore the data.

Backup Testing

Over the years, we've seen fellow administrators rely on their backups—up to the point when they try to restore them. It's a very different story when they fail during a critical moment. Fortunately, this only happens to you once, and then you adopt testing strategies. You should not consider data on a backup to be safe until you have proven that it has been restored successfully. There are so many things that can go wrong with a restore, the most common being media failure.

We recommend that the frequency of performing a restore of your backup is at least once a month. This will allow you to verify that you actually have data that is restorable in the event of an emergency. Many backup products restore the data to a temporary location and compares the restore to what was originally backed up on the media. This functionality allow you to schedule a test restore. When it's done testing the restore, it deletes the restored data and notifies you of any discrepancies.



Exam Objective 4.7, "Given a scenario, use proper communication techniques and professionalism," requires you to know how to demonstrate professionalism. This is to include your proper attire, language, and attitude. You should also know how to deal with customers, handle confidentiality, and meet expectations of the customer.

Demonstrating Professionalism

As a professional technician, you need to possess a certain level of technical competence, or you'll quickly find yourself looking for work. Technical ability alone isn't enough, though; there are many people out there with skills similar to yours. One thing that can set you apart is acting like a true professional and building a solid reputation. As the noted investor Warren Buffett said, "It takes 20 years to build a reputation and 5 minutes to ruin it. If you think about that, you'll do things differently."

You could probably break down professionalism 100 different ways. For the A+ 220-1202 exam, and for the purposes of this chapter, we're going to break it down into two critical parts: communication and behavior.

Good communication includes listening to what the user or manager or developer is telling you and making certain that you understand completely. Approximately half of all communication should be listening. That a user or customer may not fully understand the terminology or concepts doesn't mean they don't have a real problem that needs to be addressed. Therefore, you must be skilled not only at listening but also at translating.

Professional behavior encompasses politeness, guidance, punctuality, and accountability. Always treat the customer with the same respect and empathy that you would expect if the situation were reversed. Likewise, guide the customer through the problem and the explanation. Tell them what has caused the problem they are currently experiencing and the best solution for preventing it from recurring in the future.

Professional Appearance

Demonstrating professionalism begins with your appearance and attire. You should always dress for the respect and professionalism that you deserve. It's an easy element of being professional, but it is often an overlooked element.

If you dress down, then you will be judged as being less professional than your intellect or position deserves. The customer might not communicate with you the way you'd expect, because the customer will make assumptions about your intellect or position based on your appearance. The opposite can also happen if you overdress. The customer might not give you the same respect that a technician of your caliber might expect. You could be viewed as overqualified for the customer's needs or nontechnical.

These judgments shouldn't happen from the customer's perspective, but they often do because of the impression that appearance makes. Therefore, you should always attempt to match the required attire of the environment you are working in. For example, a technician on a construction site should have the expected attire, such as safety equipment and rugged clothes. Showing up in a suit is not what the customer would expect. Conversely, if you showed up at an office environment wearing jeans and a t-shirt you'd be looked at as being unprofessional.

Luckily for IT professionals, there are two norms of appearance and attire that are expected, depending on the environment and workplace:

Formal Business formal, also known as business professional, typically requires that male employees be dressed in a suit and tie. The male attire will consist of a long-sleeve dress shirt, tie, dark or striped pants, a suit jacket, and dress shoes. Female employees should be dressed in a business suit or business skirt. Women's clothes can be lighter in color than men's clothes, and long sleeves are preferable. Women should wear heeled shoes or dress shoes.

Business Casual Business casual is often interpreted as casual and therefore many professional organizations have not embraced it. The standard definition is dress pants or khakis, collared shirt, blazer, and professional-looking shoes. For male employees a tie is not required and the color palette can be lighter for both pants and blazer. Female employees can wear a collared or noncollared blouse, dress or skirt (typically below the knee), along with heels, dress shoes, dress boots, or flats.

The dress attire of the organization will differ slightly from the preceding definitions based on season, organization type, and even the day of the week. It is popular now for organizations to have a casual Friday that differs in definition based on the norm of business formal or business casual the rest of the week. Therefore, it is always best to check with your supervisor or coworkers about what is appropriate and what is not.

Communicating with Customers

The act of diagnosis starts with the art of customer relations. Go to the customer with an attitude of trust. Believe what the customer is saying. At the same time, retain an attitude of hidden skepticism; *don't* believe that the customer has told you everything. This attitude of hidden skepticism is not the same as distrust. Just remember that what you hear isn't always the whole story; customers may inadvertently forget to provide some crucial details.



One of the best ways to become proficient in communicating with customers is to put yourself in the shoes of the novice user. None of us are experts in every field, so think of an area where you are weak—auto repair or home repair, for example—and imagine how you would want a professional in that area to communicate with you.

For example, a customer may complain that their optical drive doesn't work. What they fail to mention is that it has never worked and that they installed it. On examining the machine, you realize that they mounted it with screws that are too long, preventing the tray from ejecting properly.

Here are a few suggestions for making your communication with the customer easier:

Use proper language. End users are likely to be less computer literate than you—that's why you are there to fix the problem. Therefore, they might not know computer acronyms, slang, or jargon. In fact, the excessive use of computer terms might make them feel stupid (and make the situation uncomfortable) or make you seem aloof. It's always best to start off with basic terms, such as "Where does it plug into?" If the user answers, "Oh, it's an eSATA SSD," then you know you have the freedom to use more technical terms. Users with computer knowledge will often let you know. Users with little knowledge may be afraid to tell you that they don't understand something because they don't want to look dumb.

Have the customer reproduce the error. The most important part of this step is to have the customer show you what the problem is. The best method we've seen of doing this is to ask, "Show me what 'not working' looks like." That way, you see the conditions and methods under which the problem occurs. The problem may be a simple matter of doing an operation incorrectly or performing the operation in the wrong order. During this step, you have the opportunity to observe how the problem occurs, so pay attention.

Identify recent changes. The user can give you vital information. The most important question is, "What changed?" Problems don't usually come out of nowhere. Was a new piece of hardware or software added? Did the user drop some equipment? Was there a power outage or a storm? These are the types of questions that you can ask a user when trying to find out what is different. If nothing changed, at least outwardly, then what was going on at the time of the failure? Can the problem be reproduced? Is there a work-around? The point here is to raise as many questions as you need to ask in order to pinpoint the source of the trouble.

Use the collected information. Once the problem or problems have been clearly identified, your next step is to isolate possible causes. If the problem cannot be clearly identified, then further tests will be necessary. A common technique for hardware and software problems alike is to strip the system down to bare-bones basics. In a hardware situation, this could mean removing all interface cards except those absolutely required for the system to operate. In a software situation, this may mean disabling elements within Device Manager.

Generally, then you can gradually rebuild the system toward the point where the trouble started. When you reintroduce a component and the problem reappears, you know that component is the one causing the problem.

Customer satisfaction goes a long way toward generating repeat business. If you can *meet* the customer's expectations, you will most likely hear from them again when another problem arises. However, if you can *exceed* the customer's expectations, you can almost guarantee that they will call you the next time a problem arises.



Real World Scenario

Communication Is Key

Marriages disintegrate when couples do not communicate effectively—or so many experts proclaim. Communication is ranked as one of the most important skills needed to make a marriage work. The same can be said for business partnerships. It is important to make certain that you are listening to your customers, whether they are truly customers in the traditional sense of the word or internal users that you support. You also need to listen carefully to managers and vendors and make sure that you understand them before beginning a project.

Similarly, you need to make certain that the parties in question understand what you are saying to them. It isn't acceptable to resort to the "But I told you . . ." excuse when customers or partners aren't pleased with the results. Making certain that they understand what you are telling them is as important as making certain that you understand what they are telling you.

Customer satisfaction is important in all communications media—whether you are on-site, providing phone support, or communicating through email or other correspondence. If you are on-site, follow these rules:

- When you arrive, immediately look for the person (user, manager, administrator, and so on) who is affected by the problem. Make sure that the user knows you are there and assure that person that you will do all you can. Project a positive attitude and confidence that you can resolve their problem.
- Listen intently to what your customer is saying and avoid interrupting them. Make it obvious that you are listening and respecting what they are telling you.
- If there is a problem with understanding the client, go to whatever lengths you need to in order to remedy the situation. Look for verbal and nonverbal cues that can help you isolate the problem. Taking notes is also helpful. Make sure to say something to the user like, "I'm listening, and I am going to write down a few notes just to make sure I remember everything."

- Share the customer's sense of urgency. What may seem like a small problem to you can appear to your customer as if the whole world is collapsing around them.
- Be honest and fair with the customer and try to establish a personal rapport. Explain what the problem is, what you believe the cause is, and what can be done in the future to prevent the problem from recurring.
- Handle complaints as professionally as possible. Accept responsibility for errors that may have occurred on your part and never try to pass the blame elsewhere. Avoid arguing with a customer; it serves no purpose. Resolve the customer's anger with as little conflict as possible. Remember, the goal is to keep the customer, not to win an argument.

When you finish a job, notify the user that you have finished. Make every attempt to find the user and inform them of the resolution. If you cannot find the customer, leave a note explaining the resolution.

You should also leave a means by which the customer can contact you if they have any questions about the resolution or a related problem. In most cases, you should leave your business number and, if applicable, your cell phone number, in case the customer needs to contact you after hours.

You should also notify both your manager and the user's manager that the job has been completed.

If you are providing phone support, keep the following guidelines in mind:

- Always answer the telephone in a professional manner, announcing the name of the company and yourself.
- Using the customer's name can help build rapport. Using it in every sentence can sound condescending, but using it once in a while can make you seem more personable.
- Make a concentrated effort to ascertain the customer's technical level and communicate at that level, not above or below it.

  The most important skill that you can have is the ability to listen. You have to rely on the customer to describe the problem accurately. They cannot do that if you are second-guessing or jumping to conclusions before the whole story is told. Ask broad questions to begin, and then narrow them down to help isolate the problem.

  It is your job to help extract the description of the problem from the user. For example, you might ask the following questions:

  - Is the printer plugged in?
  - Is it online?
  - Are there any lights flashing on it?

  Complaints should be handled in the same manner in which they would be handled if you were on-site. Make your best effort to resolve the problem and not argue. Again, your primary goal is to keep the customer.

  Close the incident only when the customer is satisfied that the solution is the correct one and the problem has gone away.

  End the phone call in a courteous manner. Thanking the customer for the opportunity to serve them is often the best way.

Talking to the user is an important first step in the troubleshooting process. Your first contact with a computer that has a problem is usually through the customer, either directly or by way of a work order that contains the user's complaint. Often, the complaint is something straightforward, such as, "There's smoke coming from the back of my monitor." At other times, the problem is complex, and the customer does not mention everything that has been going wrong. Regardless of the situation, always approach it calmly and professionally, and remember that you get only one chance to make a good first impression.



Real World Scenario

Communication Is Everywhere

Almost every profession stresses the importance of good communication, and the IT industry is no different. Jamie Walters, founder and chief vision and strategy officer for Ivy Sea, Inc., and Sarah Fenson, Ivy Sea's guide to client services, wrote an article for `Inc.com` on steps to smooth conversations (`www.inc.com/articles/2000/08/20000.html`) that included the following advice:

- Don't take things personally. If someone acts inappropriately toward you, just react in a calm manner. They are likely responding that way because of outside factors.
- Admit when you don't know the answer to something. It's okay to defer to somebody else or tell the user or customer that you'll have to look into their complaint and will get back with them as soon as possible.
- It is better to validate someone's feeling or respond to the information that they have given to you than to react to them. For instance, if somebody complains that a help ticket has not been handled in a timely manner, tell them that you understand how they feel and will look into it, instead of reacting in a defensive manner.
- Don't let your personal opinions or feelings get in the way of the real complaint. Try to put yourself in the user's or customer's shoes.
- Be sympathetic. If you need a user to leave their laptop with you overnight, tell them that you realize it's frustrating and apologize.
- Try to provide a solution from which you both can benefit. Look for commonalities between you and the client, and work to find a solution that is agreeable to both of you.
- Try to be as informative as possible when discussing a solution to their problem. Most people are uncomfortable with change, so explaining the benefits of a particular solution might help ease this discomfort.
- Try to keep a positive attitude and be optimistic.
- Always work on your listening skills!

Using Appropriate Behavior

Critical to appropriate behavior is to treat the customer, or user, the way you would want to be treated. Much has been made of the Golden Rule—treating others the way you would have them treat you. Six key elements to this, from a business perspective, are punctuality, accountability, flexibility, confidentiality, respect, and privacy. The following sections discuss these elements in detail.

Punctuality

Punctuality is important and should be a part of your planning process. If you tell the customer that you will be there at 10:30 a.m., you need to make every attempt to be there at that time. If you arrive late, you have given them false hope that the problem will be solved by a set time. That can lead to anger because it can appear that you are not taking the problem seriously. Punctuality continues to be important throughout the service call and does not end with your arrival. If you need to leave to get parts and return, tell the customer when you will be back, and be there at that time. If for some reason you cannot return at the expected time, alert the customer and tell them when you can return.

Along those same lines, if a user asks how much longer the server will be down and you respond that it will up in five minutes only to have it down for five more hours, the result can be resentment and possibly anger. When estimating downtime, always allow for more time than you think you will need, just in case other problems occur. If you greatly underestimate the time, always inform the affected parties and give them a new time estimate. To use an analogy that will put it in perspective: if you take your car to get an oil change and the counter clerk tells you it will be "about 15 minutes," the last thing you want is to be still sitting there four hours later. If you ever feel that you won't be able to meet the timeline you proposed, communicate that as quickly as possible. It's better to overcommunicate than to have users wondering where you are.

Exercise 10.1 is a simple drill that you can modify as needed. Its purpose is to illustrate the importance of punctuality.

EXERCISE 10.1

Understanding Punctuality

1. Consider this scenario: You call someone important in your life—your spouse, a parent, an in-law, or a close friend—and tell them that you have something very important that you need to discuss. You give that person no other details but ask them to meet you in exactly one hour at a location familiar to both of you.
2. Now imagine that you waited two hours before showing up.
3. What would be that person's reaction? How would that person feel about having to wait for you? What kind of an impact would it have on the person's mood and behavior?

This is an interaction with someone who matters in your life. Imagine how it would affect a customer who does not know you. Punctuality can go a long way toward keeping dialogue pleasant between two parties.

Accountability

Accountability is a trait that every technician should possess. When problems occur, you need to be accountable for them and not attempt to pass the buck to someone else. For example, suppose you are called to a site to put a larger hard drive into a server. While performing this operation, you inadvertently scrape your feet across the carpeted floor, build up energy, and zap the memory in the server. Some technicians would pretend the electrostatic discharge (ESD) never happened, put the new hard drive in, and then act completely baffled by the fact that problems unrelated to the hard drive are occurring. An accountable technician would explain to the customer exactly what happened and suggest ways of proceeding from that point—addressing and solving the problem as quickly and efficiently as possible.

Accountability also means that you do what you say you're going to do, ensure that expectations are set and met, and communicate the status with the customer. Here are some examples of ways to be accountable:

- Offer different repair or replacement options, if they're available.
- Provide proper documentation on the services that you provided.
- Follow up with the customer at a later date to ensure satisfaction.

The last one is the most overlooked, yet it can be the most important. Some technicians fix a problem and then develop an "I hope that worked and I never hear from them again" attitude. Calling your customer back (or dropping by their desk) to ensure that everything is still working right is an amazing way to build credibility and rapport quickly.

Flexibility

Flexibility is another trait that's as important as the others for a service technician. You should respond to service calls promptly and close them (solve them) as quickly as you can, but you must also be flexible. If a customer cannot have you on-site until the afternoon, you must make your best effort to work them into your schedule around the time most convenient for them. Likewise, if you are called to a site to solve a problem and the customer brings another problem to your attention while you are there, you should make every attempt to address that problem as well. Under no circumstances should you give a customer the cold shoulder or not respond to additional problems because they were not on an initial incident report.



You should always follow the express guidelines of the company you work for as they relate to flexibility, empowerment, and other issues.

It's also important that you remain flexible in dealing with challenging or difficult situations. When someone's computer has failed, they likely aren't going to be in a good mood, which can make them a "difficult customer" to deal with. In situations like these, keep in mind the following principles:

Avoid arguing or being defensive. Arguing with the customer—about anything—is only going to make the situation worse. The customer may be mad and may be yelling

at you, but don't argue back, act defensive, or take their comments personally. Try to defuse the situation by calmly reminding them that you're there to help and you want to understand what's going on so you can do that. They may need to vent for a bit, so let them do that. Just focus on doing what you need to do to resolve the problem.

Don't minimize their problems. Although the customer's problem might seem trivial to you, it isn't to them. Treat the problem as seriously as they're treating it. Keep in mind that facial expressions and body language are also important. If someone tells you their problem and you look at them like they're delusional, they're probably going to pick up on that, which can make the situation worse.

Avoid being judgmental. Don't blame or criticize. As previously mentioned, focus on what needs to happen to fix the problem. Accusing the user of causing the problem does not build rapport. Even seemingly innocent statements such as "What did you do to the computer?" can be considered judgmental and put the customer on the offensive. Stick with more neutral language, such as, "Can you help me understand what happened?"

Focus on your communication skills. If you have a difficult customer, treat it as an opportunity to see how good a communicator you really are. (Maybe your next job will be a foreign ambassador.) Ask nonconfrontational, open-ended questions. "When was the last time it worked?" is more helpful than "Did it work yesterday?" or "Did you break it this morning?" These can help you narrow down the scope of the problem.

Another good tactic here is to restate the issue or question to verify that you understand. Starting with "I understand that the problem is . . ." and then repeating what the customer said can show empathy and proves that you were listening. If you have it wrong, it's also a good opportunity to let your customer correct you so that you're on track to solve the right problem.

Confidentiality

The goal of confidentiality is to prevent or minimize unauthorized access to files and folders and disclosure of data and information. In many instances, laws and regulations require confidentiality for specific information. For example, Social Security records, payroll and employee records, medical records, and corporate information are high-value assets. This information could create liability issues or embarrassment if it fell into the wrong hands.

Over the last few years, there have been a number of cases in which bank account and credit card numbers were published on the Internet. The loss of confidence by consumers due to these types of breaches of confidentiality can far exceed the actual monetary losses from the misuse of this information.



Confidentiality entails ensuring that data expected to remain private is seen only by those who should see it. Confidentiality may be implemented through authentication and access controls.

As a computer professional, you are expected to uphold a high level of confidentiality. Should a user approach you with a sensitive issue—telling you their password, asking for assistance obtaining access to medical forms, and so on—it is your obligation as a part of your job to make certain that information goes no further.



Confidential materials on workspaces and printers should always be protected.

As part of confidentiality, don't ever disclose work-related experiences via social media. You might have had a terrible day and really want to say something like, "Wow, the people at XYZ Company sure are insufferable morons," but just don't do it. It's not professional, and it could expose you to legal action.

Respect

Much of the discussion in this chapter is focused on respecting the customer as an individual. However, you must also respect the tangibles that are important to the customer. While you may look at a monitor that they are using as an outdated piece of equipment that should be scrapped, the business owners may see it as a gift from their children when they first started their business.

Treat the customer's property as if it had value, and you will win their respect. Their property includes the system you are working on (laptop/desktop computer, monitor, peripherals, and the like) as well as other items associated with their business. Avoid using the customer's equipment, such as telephones or printers, unless it is associated with the problem you've been summoned to fix.

Another way to show respect is to focus on the task at hand and avoid distractions. For example, you should avoid the following:

- Personal calls
- Texting or social media sites
- Talking to coworkers while interacting with customers
- Personal interruptions

As for texting or talking to coworkers, there may be times when it's appropriate for you to do so based on the situation. The key is to find the right time to do it and, if appropriate, tell the customer what you are doing. For example, after gathering information, you might say something like, "Do you mind if I give my coworker Jen a quick call? The other day she told me about a situation she had that sounded exactly like this, and I want to see if her fix worked well." But then make the call quick and business-focused.



The American Customer Satisfaction Index (`https://theacsi.org`) measures the behavior of corporations and the respect they give to customers.

Respecting the customer is not rocket science. All you need to do—for this exam and in the real world—is think of how you would want someone to treat you. Exercise 10.2 explores this topic further. Like Exercise 10.1, it can be modified to fit your purpose or constraints. Its goal is to illustrate the positive power of the unexpected.

EXERCISE 10.2

Surprise Someone

1. Pick a random, toll-free number used for business solicitation and call it.
2. Chat with the operator for a few moments about the company's product or service, and then ask to speak to the supervisor.
3. When the supervisor comes on, commend the operator you were talking to for the job that they have done.

It is likely the operator became confused when you asked to speak to the supervisor; this almost always occurs only in a negative situation. How did the operator handle the request? Did it change the tone of the communication that was taking place? Did they fulfill your request even though they feared they could lose from it? Did the supervisor respond by expecting negative comments? How was the positive information you offered accepted?

Ideally, this illustrated the importance of staying professional and keeping the channel of communication open even in a tough situation. You should be able to adapt this to the workplace when a customer asks to speak to your superior or has another request that is difficult for you to fulfill.

One last area to consider that directly relates to this topic is that of ethics. *Ethics* is the application of morality to situations. While there are different schools of thought, one of the most popular areas of study is known as *normative ethics*, which focuses on what is normal or practical (right versus wrong and so on). Regardless of religion, culture, and other influences, there are generally accepted beliefs that some things are wrong (stealing, murder, and the like) and some things are right (e.g., the Golden Rule). You should always attempt to be ethical in everything you do, because it reflects not only on your character but also on your employer.

Privacy

Although there is some overlap between confidentiality and privacy, privacy is an area of computing that is becoming considerably more regulated. As a computing professional, you must stay current with applicable laws because you're often one of the primary agents expected to ensure compliance.

Although the laws provide a minimal level of privacy, you should go out of your way to respect the privacy of your users beyond what the law establishes. If you discover information about a user that you should not be privy to, you should not share it with anyone, and you should alert the customer that their data is accessible and encourage them—if applicable—to remedy the situation. This includes information that you see on their computer, on their desk, on printers, or anywhere else in their facility.



Real World Scenario

A Little Goes a Long Way

The following examples of respecting and disrespecting the customer come from one of the authors' own experiences:

"My wife and I were in an unfamiliar part of Chicago without ready access to a vehicle when we started to get hungry. I am a meat-and-potatoes man, and I rarely take a chance on anything else. There were no restaurants of that type around, however, and we wound up at an Asian grill. Expecting not to like the buffet, we ordered a side of lettuce wraps and then two buffets and drinks. As it turned out, I liked the buffet a great deal, and I went back through the line many times. We also liked the drinks and got several of those.

"Everything was great, except the waiter forgot to bring the lettuce wraps. I dismissed it and made a mental note to inform the waiter when he brought the bill and have him deduct them from our tab. Instead, the manager brought the bill over when we were finished eating, and he had scribbled on it 'no charge.' When I asked him why, he apologized that no one brought the wraps and said he hoped we would come back another time. I was beside myself with disbelief and thanked him profusely, and since then I have told many people about that restaurant, describing it as the best place in Chicago I know of to eat.

"In a very different situation, while driving home one night, the 'low tire pressure' dashboard light came on. Upon inspection, I could hear the right-rear tire hissing. I drove to a tire store and explained the situation. I had used this same tire store over the past 14 years for tires, oil changes, exhaust, maintenance, and a number of other things on the vehicles I've owned. The manager came out and said that they found a nail in the tire. They removed the nail, patched the tire, and charged me $13. I was delighted, expecting it to cost much more, and so I paid the bill and went on my way.

"The next morning, I woke up to find the right-rear tire completely flat. I canceled the morning's appointment, filled the tire with an air compressor, and drove back to the tire store. The manager came out and told me that they found another nail in that tire; they were going to eat the $13 on this one, but it had better not happen again. I could not believe the insinuation—that I was driving about looking for nails to hit with that one tire just so I could spend my morning taking them for $13! Instead of offering the possibility that they had

overlooked a nail the previous night, apologizing for the inconvenience, or anything of that sort, he shifted the responsibility to me. Needless to say, I have not been back since, and all of my repair business is now done elsewhere."

These two examples illustrate two different approaches to treating the customer. In the first example, the customer is well respected and treated better than expected. In the second example, the customer is disrespected and is treated as an inconvenience. Given the lifetime value of customers, it is always better to respect them—and retain them—than to dismiss them offhandedly.

Putting It All in Perspective

Whether you are dealing with customers in person or on the phone, there are rules to which you should adhere. These were implied and discussed in the previous sections, but you must understand them and remember them for the exam.

- Use standard language and avoid using jargon, abbreviations, and acronyms. Every field has its own language, and outsiders feel lost when they start hearing it. Put yourself in the position of someone not in the field and explain what is going on by using words they understand.
- Maintain a positive attitude and tone of voice, and project confidence. The customer is counting on you to fix their problem. The last thing they want is for you to sound defeated when you hear about the problem.

  Listen to your customers and take notes. Allow them to complete their statements and avoid interrupting them. People like to know that they are being heard, and as simple an act as it is, this can make all of the difference in making them feel at ease with your work.

  Everyone has been in a situation where they have not been able to explain their problem fully without being interrupted or ignored. It is not enjoyable in a social setting, and it is intolerable in a business setting.
- Be culturally sensitive. Some people may have a language barrier that makes it difficult to explain their problem. (Think about how much computer language you learned in your high school language courses.) Others may have different habits or practices in their workplace. Be respectful of their world. In some cases, using the appropriate professional titles is a sign of respect, and not using them is an insult.
- Be on time. If you're going to be late, be sure to contact your customer. Not doing so indicates that you think their problem isn't important.
- Avoid distraction and/or interruptions when talking with customers. You need to make them feel that their problem is important and that it has your full attention. Distractions can include personal calls, texting or social media, talking to co-workers, and other personal interruptions.

- Exercise patience with difficult customers and situations.
- Avoid arguing with customers and/or becoming defensive.
- Do not minimize a customer's problem. While it may be a situation you see every day, it is a crisis to them.
- Avoid being judgmental and/or insulting or calling the customer names.
- Clarify the customer's statements and ask pertinent questions. The questions you ask should help to guide you toward isolating the problem and identifying possible solutions. Don't be afraid to nod, ask questions, and repeat to the customer what you think they are saying to make sure that you understand it correctly.
- Don't vent about customers on social media.

Set and meet—or exceed—expectations and communicate timelines and status. Customers want to know what is going on. They want to know that you understand the problem and can deal with it. Being honest and direct is almost always appreciated.

Deal appropriately with confidential materials. Don't look at files or printouts that you have no business looking at. Make sure the customer's confidential materials stay that way.

Summary

In this chapter, we covered ticketing systems and how they are implemented for an organization. We also covered the various documentation types that you will encounter as a technician throughout your career. Some of the documentation is created from external regulations. This, in turn, creates policies, which dictate processes and procedures. You'll create some of the documentation as you are troubleshooting a problem and some of the documentation after fixing a problem. We also covered documentation often used with the change management process.

Next, we looked at disaster prevention and recovery. The two main areas are data loss and equipment failure due to power issues. Data loss can be prevented with data backups and other user-facing strategies, such as Volume Shadow Copy. Power problems can be prevented with the appropriate use of uninterruptable power supplies and surge protection equipment.

Finally, we moved on to professionalism and communication. You should treat your customers as you would want to be treated. Your actions and behavior should let them know that you respect them and their business.

Exam Essentials

Understand the elements of a ticketing system. You should have a clear understanding of the importance of clear communications in the support process, as well as the important elements in a ticket, such as user information, device information, category of problem, severity of the problem, and most important, the description of the problem.

Know the common documentation in a network. Incident documentation is created when a computer, network, or security incident has happened. You should be familiar with various elements of this documentation. The acceptable use policy (AUP) is a soft policy that protects an organization's resources from employee abuse. The password policy is also a soft policy that describes best practices for password creation and expectations of the end user.

Know the elements of change management documentation. Change management is adopted by organizations to standardize the methods and procedures for any changes in the organization. Change management documentation is an integral part of this process. It is used to document a change so that it does not impede the organization's operations.

Understand how to prevent disasters and recover from data loss. Data loss is the most common disaster an organization can suffer from on a day-to-day basis. You should understand how to prevent data loss with the implementation of data backups and other prevention methods. You should also understand how to protect an organization from power-related problems using uninterruptable power supplies and surge protection.

Know the various backup methods to prevent data loss. There are three types of backups: full, incremental, and differential. Full backups copy all the data in a data set to media and reset the archive bit. An incremental backup copies only the data to media that has the archive bit set; it then resets all of the archive bits. A differential backup copies all the data in the data set to media that has the archive bit set, but it does not reset the archive bit.

Know proper communication techniques. Listen to your customers. Let them tell you what they understand the problem to be, and then interpret the problem and see if you can get them to agree to what you are hearing them say. Treat your customers with respect, whether end users or colleagues, and take their issues and problems seriously. Use professionalism when working with customers, such as minimizing distractions, speaking in plain English, being punctual, and following up.

Review Questions

The answers to the chapter review questions can be found in Appendix A.

1. What is the first step to consider when entering a ticket manually?
   - **A.** Determine the urgency of the issue.
   - **B.** Collect user information.
   - **C.** Identify recent changes.
   - **D.** Verify the issue description.

2. You are executing the primary plan in the change management documentation and realize that you cannot proceed. Which section details the original configuration?
   - **A.** Purpose
   - **B.** Risk analysis
   - **C.** Rollback
   - **D.** Plan for change

3. Which section of the change management documentation contains whom the change will affect?
   - **A.** Business processes
   - **B.** Scope of change
   - **C.** User acceptance
   - **D.** Plan for change

4. You just had an outage of Internet connectivity. Which document should you complete so that stakeholders understand the reason for the outage?
   - **A.** Change management documentation
   - **B.** Knowledge base article
   - **C.** Acceptable use policy
   - **D.** Incident documentation

5. Which type of backup allows users to restore their own files without IT intervention?
   - **A.** Image-based backup
   - **B.** Volume Shadow Copy
   - **C.** Incremental backup
   - **D.** Differential backup

6. Why is it important to document the root cause in an incident document?
   - **A.** To satisfy legal requirements
   - **B.** To ensure accurate billing
   - **C.** To prevent the recurrence of the incident
   - **D.** To train new employees

7. End users are abusing the email system by selling personal items. Which policy would detail the proper use of the email system for business purposes?
   - **A.** MDM
   - **B.** Password policy
   - **C.** AUP
   - **D.** Incident management

8. Which backup media is the fastest from which to recover?
   - **A.** Disk-to-tape
   - **B.** Disk-to-disk
   - **C.** Disk-to-flash
   - **D.** Disk-to-cloud

9. You need to upgrade a server and want to make a backup of the data before you begin. Which backup method should you choose so that your normal backups are not affected?
   - **A.** Full
   - **B.** Copy
   - **C.** Incremental
   - **D.** Differential

10. What is the primary benefit of having a knowledge base in an IT environment?
    - **A.** Reduces the number of help desk staff needed
    - **B.** Ensures compliance with company policies
    - **C.** Provides a quick reference for resolving common issues
    - **D.** Improves employee morale

11. You promised a customer that you would be out to service their problem before the end of the day but have been tied up at another site. As it now becomes apparent that you will not be able to make it, what should you do? (Choose the best answer.)
    - **A.** Arrive first thing in the morning.
    - **B.** Wait until after hours and then leave a message that you were there.
    - **C.** Call the customer and inform them of the situation.
    - **D.** Email the customer to let them know that you will be late.

12. A user reports that a workstation has two significant problems that do not seem related. How should you approach these problems?
    - **A.** Look for what the two problems would have in common.
    - **B.** Assume that a virus is involved.
    - **C.** Deal with each issue separately.
    - **D.** Order a new machine.

13. A customer is trying to explain a problem with their system. Unfortunately, the customer has such a thick accent that you are unable to understand their problem. What should you do? (Choose the best answer.)

A. Just start working on the system and look for obvious errors.

B. Call your supervisor.

C. Ask that another technician be sent in your place.

D. Apologize and find another user or manager who can help you translate.

14. You have been trying to troubleshoot a user's system all day when it suddenly becomes clear that the data is irretrievably lost. When you inform the customer, they become so angry that they shove you against a wall. What should you do?

A. Shove the user back, only a little harder than they shoved you.

B. Shove the user back, only a little easier than they shoved you.

C. Try to calm the user down.

D. Yell for everyone in the area to come quickly.

15. A customer tells you that a technician from your company spent three hours on the phone making personal calls. What should you do with this information?

A. Nothing.

B. Inform your manager.

C. Talk to the technician personally.

D. Ask the customer to prove it.

16. You arrive at the site of a failed server to find the vice president nervously pacing and worrying about lost data. What should you do?

A. Offer a joke to lighten things up.

B. Downplay the situation and tell him that customers lose data every day.

C. Keep your head down and keep looking at manuals to let him know that you are serious.

D. Inform him that you've dealt with similar situations and will let him know what needs to be done as soon as possible.

17. You are temporarily filling in on phone support when a caller tells you that they are sick and tired of being bounced from one hold queue to another. They want their problem fixed, and they want it fixed now. What should you do?

A. Inform them up front that you are only filling in temporarily and won't be of much help.

B. Transfer them to another technician who handles phone calls more often.

C. Try to solve their problem without putting them on hold or transferring them elsewhere.

D. Suggest that they call back at another time when you are not there.

18. At the end of the day, you finish a job only to find that the user you were doing it for had to leave. What should you do? (Choose two.)

A. Clean up and leave no evidence that you were there.

B. Leave a note for the user detailing what was done and how to contact you.

C. Notify the user's manager and your own manager that you have finished.

D. Put the system back to its original state.

19. A user on the phone does not seem to be able to explain their problem to you without using profanity. That profanity is making you unable to understand their problem. What should you do? (Choose the best answer.)

A. Ask the user to refrain from the offensive language.

B. Overlook the profanity.

C. Hang up.

D. Show them that you know just as many expletives as they do.

20. Which of the following is not a benefit of implementing asset tags for inventory management?

A. Tracking of the equipment

B. Scheduling the depreciation of the equipment

C. Identifying assets

D. Providing ownership of the equipment

Performance-Based Question

You will encounter performance-based questions on the A+ exams. The questions on the exam require you to perform a specific task, and you will be graded on whether or not you were able to complete the task. The following requires you to think creatively in order to measure how well you understand this chapter's topics. You may or may not see similar questions on the actual A+ exams. To see how your answers compare to the authors', refer to Appendix B.

A user has called in and explained they accidentally overwrote a file and need to retrieve the freshest copy of the file. Luckily, you have Volume Shadow Copy configured on the share where the file was overwritten. What are the steps to recover the file?

Appendix

A



Answers to Review Questions

Chapter 1: Operating System Basics

1. B, C. You can open a command prompt by typing `cmd` or `command` in the Start menu. The command prompt utility will pop up in the search results. Run is not a command; it is a dialog box. Open is not a command; it is an operating system action.

2. D. A driver is specifically written to instruct the operating system to communicate with a piece of hardware. Source code is the original code the software is written in. An application is a program written to perform a specific purpose to interact with the user. The kernel is the core of the operating system.

3. B. OneDrive is a cloud-based file synchronization application; it is similar to Google Drive. Excel is a spreadsheet application that is part of the Microsoft 365 suite of products. Microsoft Exchange is an on-premises email server; its cloud-based counterpart is Microsoft Exchange Online, and it is similar to Google Gmail. Outlook is the email client used with Microsoft Exchange.

4. A. If you wanted to install a program on a virtual machine, the preferred method would be to mount an ISO image of the application. USB and optical discs are not common installation methods for virtual machines. Although ZIP files could be used, a ZIP file is only a compression method and not a delivery method.

5. B, C. The Windows File Explorer program can be used to copy and move files. Windows File Explorer can also be used to change file attributes. Windows File Explorer cannot be used to browse the Internet or to create backup jobs.

6. D. The maximum allowable length for a filename is 255 characters.

7. B. The shell is a program that runs on top of the OS and allows user interaction with the operating system. The Taskbar is an element of the Desktop. The Desktop does not allow for commands to be executed. Source is not related to the graphical interface.

8. B. Typing the program name in the Start box allows you to launch the program. The Shut Down command will shut down the operating system. Run is not a command; it is a dialog. The `cmd` command starts the command-prompt application.

9. D. When a program is multithreaded, it is written to allow for multiple requests into the processor at one time. Multiuser mode, dystopia, and preemption are not correct answers.

10. B. The Recycle Bin is a temporary spot that files are deleted to but from where they can be retrieved if deleted mistakenly. The My Computer icon is another older way to start the File Explorer. Control Panel and the Settings app are used to customize the operating system.

11. D. In Windows 11, when you click the Start button, a power icon is displayed on the lower left. From the power icon, you can sleep, reboot, or shut down the operating system. There is no command named turnoff. Turning the power switch off and unplugging the machine are not graceful ways to power off the operating system. Pressing Ctrl+Alt+Del alone will not power down the computer.

12. A. A Trusted Platform Module (TPM) 2.0 is required for the installation of Windows 11. 64 GB or greater of storage is required for the installation of Windows 11. A Microsoft DirectX 12 graphics device is required for the installation of Windows 11. The minimum amount of memory required for the installation of the 64-bit Windows 11 operating system is 4 GB. More RAM is always recommended, but only 4 GB is required.

13. D. The minimum amount of free hard drive space required for Windows 11 64-bit is 64 GB.

14. D. The minimum amount of memory required for the 64-bit installation of the Windows 11 is 4 GB.

15. C. A quick method of accessing Help is to press the F1key. The Windows key on the keyboard launches the Start menu. The F12 key and the Alt key perform various tasks, depending on the application.

16. A. The Windows key+R will open a run box, where you can launch applications by typing in their executable name. Ctrl+R is used to reload web pages in web browsers. The Windows key alone will launch the Start menu. The F1 key will launch help in applications and on the desktop it will launch a web browser.

17. C. The notification area is the area on the rightmost portion of the Taskbar. It is used to display the date and time and running background applications. The pinned apps and Start menus are located on the center-left portion of the Taskbar. Shutdown options are found inside the Start menu in Windows 11.

18. A. Clicking the Settings gear in the Start menu will open the Settings app. From there, the System category will allow you to change the Display Properties. Clicking the System icon under Control Panel allows you to change system settings. Pressing Ctrl+Alt+Esc launches the Task Manager. Pressing Ctrl+Alt+Tab launches the Task View in Windows 11.

19. B. The File Explorer allows you to navigate the file/folder structure in Windows 11. The Start menu allows you to launch applications. KDE and GNOME are Linux desktop shells.

20. C. The minimum processor speed required for the 64-bit installation of the Windows 11 is 1GHz.

Chapter 2: Windows Configuration

1. B. Task Scheduler (`taskschd.msc`) will allow you to start a program based on a condition met, such as the computer starting up. The Programs and Features applet is not an MMC snap-in. The Disk Management MMC snap-in allows you to manage the disks attached to the computer. The Group Policy Editor lets you edit the local Group Policy.

2. C. FAT32 is a filesystem that was popular with Windows 9x operating systems and is also commonly used for USB flash drives for compatibility between platforms. NTFS is the native filesystem for Windows. EFS is a technology that encrypts files. The CDFS filesystem is used for CD media.

3. C. The `shutdown` utility can be used to schedule a remote shutdown—for example, `shutdown /t 60 /m \\computer`. The `taskmgr` utility is used to view tasks, `kill` is used to kill processes, and `netstat` is used to view network statistics and activity.

4. D. The System event log would contain an entry when the operating system reboots. The Application log is used to log applications installed on the system. The boot log is not a valid log. The Security log contains log information pertaining to the security of objects and files.

5. D. The exFAT filesystem is a Microsoft proprietary filesystem created for large flash drives. GPT is a partition table type, NFS is a network filesystem used with Linux/UNIX, and ext3 is a filesystem used with Linux.

6. D. The Accounts settings screen allows you to access the Backup screen, where you can configure backups for the user under the Windows Backup menu item. The System section allows you to configure a multitude of options for Windows itself. The Devices section lets you configure devices, and the Apps section allows you to uninstall and configure applications.

7. C. The Apple File System (APFS) is a proprietary filesystem created for SSD support; it can be found on macOS Sierra 10.12.4, iOS 10.3, and tvOS 10.2 and later operating systems. Hierarchical File System (HFS) is a proprietary filesystem that was developed by Apple for the Mac OS back in 1985, ext4 (Fourth Extended File System) is used with Android and other Linux-based operating systems, and exFAT is a Microsoft proprietary filesystem created for large flash drives.

8. A. The System Control Panel applet is used to change the operating system computer name and change its membership from a workgroup to a domain, or domain to workgroup. The User Accounts applet is used to make changes to your user account or others on the operating system. The Windows Tools option in Control Panel opens up a menu of tools, none of which join the operating system to a domain. The Programs And Features applet is used to uninstall and repair installed programs, as well as manage Windows features.

9. D. When a program is not responding to any commands, the process must be terminated via the Task Manager. You can end the process by opening the Task Manager, right-clicking the application, and selecting End Task. Viewing the performance will not make the process respond. Adding more memory is not possible without shutting down the computer. Pressing Ctrl+Alt+Delete without any further action will not reboot the computer.

10. B. The command `eventvwr.msc` will start the Event Viewer snap-in. The command `eventviewer.exe` is not a valid command. The command `lusrmgr.msc` will start the Local Users and Group snap-in. The command `devmgmt.msc` will start the Device Manager snap-in.

11. C. The iSCSI Initiator utility can be found in Control Panel, in Windows Tools. The Storage Spaces applet is used to configure storage spaces of already connected storage. The Disk Management snap-in is used to configure disks already connected to the operating system. Device Manager is used to configure hardware attached to the computer.

12. A. Hardware resources are configured in the Device Manager console. Files and folders and applications are configured in the operating system GUI. Memory is configured in System Properties.

13. C. The mechanical hard drive should be periodically defragmented to arrange concurrent blocks of storage of files so that the hard drive arm does not need to move much when reading and writing files. Solid-state drives (SSDs) do not suffer poor performance from fragmentation, because they have no moving parts. Freeing up space and emptying the Recycle Bin are good ideas but do not contribute to the hard drive's performance as much as defragmenting. Trim is a feature used with SSD drives, where the operating system notifies the SSD device that data blocks have been freed up.

14. A. The Indexing Options applet allows you to configure custom extensions so that metadata can be searched. The File Explorer Options applet is used to view and configure the File Explorer. The Windows Tools option in Control Panel is not really an applet; it opens up a menu of tools. The Ease of Access Center applet is used to configure options related to user disabilities to make the operating system easier to use.

15. A. The operating system must boot from an active primary partition if MBR style partitioning is used. Extended partitions are used to contain logical partitions. There is no such thing as a dynamic partition.

16. B. The `HKEY_LOCAL_MACHINE` Registry hive contains information about the computer's hardware. It is also known as `HKLM`. `HKEY_CURRENT_MACHINE` and `HKEY_MACHINE` are not valid Registry hives. `HKEY_RESOURCES` was used with Windows 9*x* operating systems but is no longer used.

17. C. GUID [globally unique identifier] Partition Table (GPT) allocates a 64-bit logical block addressing entry to partition tables to accommodate larger drives. GPT has rapidly replaced the MBR-style partition tables. LILO and GRUB are boot managers for Linux. NFS is a network filesystem used with Linux/UNIX operating systems.

18. A. Disk Defragmenter will rearrange blocks of files on a hard drive to create contiguous files and space. Windows File Explorer is used to navigate the filesystem. Scandisk is not a command. Windows Backup allows for file backups.

19. A. The hibernate power mode will write the contents of the RAM to the disk so that the computer can enter a low power state. The standby power mode will turn off the monitor and lower power consumption by the computer. The sleep and suspend power mode will lower power consumption, but it will not write the contents of the RAM to the disk.

20. C. The Dynamic Disk partition style allows up to 2,000 volumes to be created on a physical disk. GUID [globally unique identifier] Partition Table (GPT) allow up to 128 volumes to be created on a physical disk. The Master Boot Record (MBR) and Basic Disk partitioning scheme are one and the same and allow up to 26 partitions to be created on a physical disk.

Chapter 3: Windows Administration

1. D. Windows 11 requires a minimum of 4 GB of RAM and 64 GB of storage. Therefore, to meet these requirements, you need to upgrade both the RAM and the storage. Simply proceeding with the installation would not work because the current specifications do not meet

the minimum requirements. Upgrading only the RAM or only the storage would still leave one of the requirements unmet, making both upgrades necessary.

2. A. The Generalize pass is where the operating system detects hardware and installs the appropriate drivers. The Out-Of-Box Experience (OOBE) is the pass that is responsible for configuring and creating the user environment during the setup process. The Specialization pass is where the operating system is configured during setup. The WinPE pass starts the procedure of installing the operating system.

3. D. The WinRE partition is where the Windows Recovery Environment is located. The BCD is a term used with the boot configuration data to direct the operating system how to boot. The system partition contains the kernel of the operating system. The EFI system partition (ESP) contains the boot configuration data (BCD) that is used to boot the rest of the operating system on a UEFI system.

4. A. The Windows Deployment Services (WDS) is a server role that is used to create a Windows image for mass deployment. The Microsoft Assessment and Planning (MAP) toolkit is used to create reports on system hardware for image deployment. The User State Migration Toolkit (USMT) is used in a corporate environment to migrate user data. The `sysprep` utility is used to prepare an operating system for imaging.

5. D. The `robocopy` command will copy all data and will include NTFS permission to remain intact. The `xcopy` and `copy` commands will copy files from a source folder to a destination folder but will not copy NTFS permissions. The `chkdsk` command is used to check the integrity of the NTFS filesystem.

6. B. The General Availability branch will install updates as they are released to the general public. The Semi-Annual Channel is no longer a channel supported by Windows 11. The Long-Term Servicing Channel never installs new features during the life of the version of Windows. The Insider Program allows for the installation of brand-new features before they are publicly released.

7. A. Windows 11 Home only allows for the General Availability branch releases. Windows 11 Pro allows for the use of the Insider Program branch releases. Windows 11 Education allows for the use of the Insider Program branch releases. Windows 11 Enterprise allows for the use of Insider Program branch releases.

8. B. Unified Extensible Firmware Interface (UEFI) firmware is required for Secure Boot, which enhances security by ensuring that only trusted software is loaded during the boot process. Basic Input/Output System (BIOS) and Legacy BIOS do not support Secure Boot. Extensible Firmware Interface (EFI) is an older term; UEFI is the correct modern term.

9. C. The `pathping` command will measure the packet loss at each router as the packet travels to the destination address; it combines the `ping` and `tracert` commands. The `ping` command will return a single destination's response time. The `nslookup` command is used to resolve DNS addresses. The `tracert` command allows you to see how a packet travels to its destination.

10. B. The `msinfo32.exe` tool allows for the remote reporting of a computer's hardware. `regedit.exe` is used to edit the Registry. `msconfig.exe` is used to change the startup of services and change the boot process. `dxdiag.exe` is used to diagnose DirectX problems.

11. C. The `chkdsk` command is used to check a volume for corruption, as well as attempt to repair the corruption. The `diskpart` command allows you to create, modify, and view volumes on a disk. The `format` command allows you to format a filesystem on a volume. The `sfc` command is used to fix corrupted files but not volume corruption.

12. C. Users will not be able to change the wallpaper or personalize the operating system until it is active. Applications will still launch fine. Windows Updates will continue to function to keep Windows up-to-date. Browsing the Internet will not be affected.

13. B. The Out-Of-Box Experience (OOBE) is the pass that is responsible for configuring and creating the user environment during the setup process. The Generalization pass is where the operating system detects hardware. The Specialization pass is where the operating system is configured during setup. The WinPE pass starts the process of installing the operating system.

14. A. When installing Windows, you can control telemetry data in the Privacy Settings dialog box. The Locale options allow you to control how the language and local notations of money, weight, date, and several other measurements for a particular region of the world. Disk partitioning is configured during the initial installation in order to direct Windows where to install. Account creation is performed to create the first user on the operating system.

15. B. The `tracert` command shows the path packets take to a destination, helping to identify where delays or failures occur. The `ping` command checks connectivity but not the path. `nslookup` resolves DNS names but does not trace the path. `netstat` shows network connections and ports but not the path.

16. D. There is no way to directly convert a 32-bit installation of Windows to 64-bit Windows without performing a clean installation. Upgrading directly to Windows 11 Pro 64-bit is not possible. Upgrading to Windows 11 Pro 32-bit is possible, but you cannot upgrade to a 64-bit edition. Upgrading to Windows 11 64-bit is not possible.

17. A. The BCD is stored in the EFI System Partition on an EFI installation of Windows. The WinRE partition is used for the Windows Recovery Environment. Secure Boot is a feature of an EFI installation and does not contain its own partition. The `C:\WINDOWS` folder is where the installation of Windows exists.

18. A. The `netstat` command can be used to view ports in use by the operating system and network applications communicating with the network. The `ipconfig` command allows you to see the current IP address and DNS information for the operating system. `pathping` allows you to view packet loss along the path to a destination IP address. The `nslookup` command is used to resolve DNS records.

19. C. The Wireless Wide Area Network (WWAN) connection is a cellular connection and is configured on the Cellular screen in the Settings app. A wired connection is configured on the

Ethernet screen in the Settings app. A wireless connection is configured on the Wi-Fi screen in the Settings app. A virtual private network (VPN) connection is configured on the VPN screen in the Settings app.

20. C. The maximum number of connections that can simultaneously be made to a Windows workstation is 20. All other answers are incorrect.

Chapter 4: Working with macOS and Linux

1. C. The command `ls -la` will list all the files in a long format. The command `ls -a | ls-l` will not work. `ls -s; ls -a` will show two listings: one with the size and the other with all the files. The `ls -a\ls -l` command will show two listings—one with all the files and the other in a long format—but it will not show all the files in a long format.

2. B. Mission Control is a quick way on macOS to view what is currently running. Keychain or keychain access is a tool that saves credentials so that the user does not need to be prompted. The Finder is the macOS equivalent to the Windows File Explorer. Force Quit is a function of the macOS for killing a process.

3. A. Time Machine is the Apple application that can be used to create backups on a regular basis. The Finder is the macOS equivalent to Windows File Explorer. VSS is a Windows technology that is used to create snapshots of a volume. Keychain is a tool that saves credentials so that the user does not need to be prompted.

4. B. The `nano` command is used to edit files. The `ps` command lists processes running. The `rm` command removes files or directories. The `ls` command lists files and folders in the filesystem.

5. C. The `ip` command can be used to edit an Ethernet connection's configuration settings. The `dd` command is used to duplicate disks. The `apt` command is used with the APT package management system for downloading packages. The `pwd` command will show the current working directory.

6. A. The Passwords feature has replaced the Keychain tool in the latest version of macOS 15.2 (Sequoia). The Passwords feature saves credentials so that the user does not need to be prompted. Spotlight is a utility in macOS that allows a search of the operating system and Internet for search terms. The Dock is the macOS application menu that is located on the bottom of the screen.

7. A. The shell is the interpreter in Linux between the operating system and the user. The translator is a wrong answer. Login is something you perform to access the shell. The graphical user interface (GUI) does not interpret commands and therefore is not considered an interpreter.

8. D. Working copies are backups that are kept on-site at the computer center for immediate recovery purposes. An on-path attack is an attack where an eavesdropper listens between two people or devices. Cloud copies are backups that are sent to cloud storage and cannot be immediately recovered. Journal copies is a wrong answer.

9. C. The `apt` utility can be used to download and apply patches to a Linux installation. The `update` command is not a utility. Shell/terminal is an interface for interacting with the operating system with the command line. The `patch` command is not a utility.

10. C. The `chown` command is used to change ownership of a file. The `cd` command changes the working directory. The `chmod` command changes permissions on files. The `pwd` command displays the current working directory.

11. A. The `fsck` Linux utility is used to check and repair disks. The `chkdsk` utility is a Windows utility used to check and repair disks. The `du` utility is used to show the current disk usage. `dumgr` is not a utility and is a wrong answer.

12. B. By pressing the Home button twice on an iPad, you can Force Quit an application that will not stop running. The `kill` utility can be used only at the command line of Linux/macOS. The Task Manager is a Windows utility. Close Quit is not a feature and therefore a wrong answer.

13. C. Bash is the most common command-line shell used with Linux. Tcl/Tk is a shell used with Linux, but it is very uncommon. Terminal is a utility used to access the command-line shell. SSH, or Secure Shell, is a secure protocol used to access the shell over a network.

14. D. The Dock is the area at the bottom of the macOS screen that is used to launch applications. The footer is not a valid answer. Mission Control is a macOS feature that allows you to see all the applications running. The Taskbar is a Windows Desktop element that is used to track running applications and quickly launch them.

15. C. The Command+spacebar key combination will bring up the Spotlight utility. Control+Shift, Option+Tab, and Alt+Home are wrong answers.

16. A. The `sudo` command can be used to run a single command as another user. The `su` command allows you to change user logins at the command line. The `passwd` command changes the user's password. The `ifconfig` command allows you to view and modify the wired network interface.

17. C. The `ps` command will display a snapshot of the current running processes on a Linux operating system. The `ls` command will display a listing of files from the working directory. The `cat` command will display the contents of a file. The `su` command allows you to change user logins at the command line.

18. B. The command `cd ..` will take you one level back from the current working directory. The command `cd .` will do nothing, because the period signifies the current working directory. The command `cd ...` is not a valid command. The command `cd ~` will change directories to the home directory of the user.

19. B. The effective permissions are read and write. From left to right, the permissions are `rwx` for the user, `rw-` for the group, and `r--` for everyone else. Since the user is only a member of the group applied to the file, they will have read and write permissions.

20. C. The `-p` option on the `mkdir` command allows subfolders to be created as well as the target folder. All other answers are incorrect.

Chapter 5: Security Concepts

1. A. Fences are intended to delay or deter entrance into a facility. Access control vestibules are used for mid-layer access control to prevent tailgating. Multifactor authentication is used for mid- and inner-layer access control. Strong passwords are used for mid- and inner-layer access control.

2. B. A magnetometer, also known as a metal detector, can detect weapons on a person entering a facility. Biometrics are used to authenticate someone based on their face, retina, fingerprint, or some other method that biologically verifies their identity. A motion sensor is used to detect motion. A badge reader is used to scan security ID badges.

3. A. Social engineering uses the inherent trust in the human species, as opposed to technology, to gain access to your environment. IDSs are network-based systems that detect intrusions. Perimeter security describes physical security. Biometrics describes an authentication method based on human physical traits.

4. C. A fingerprint scanner, or any device that identifies a person by a physical trait, is considered a biometric security control. Access control is the system that controls access for users. Physical barriers are structures that limit physical access. Softening refers to weakening of security.

5. A. Although the end result of any of these attacks may be denying authorized users access to network resources, a denial-of-service (DoS) attack is specifically intended to prevent access to network resources by overwhelming or flooding a service or network. Worms reproduce and move throughout the network to infect other systems. Trojans are programs that enter a system or network under the guise of another program. Social engineering is a process in which an attacker attempts to acquire information about your network and system by social means, such as talking to people in the organization.

6. B. A distributed denial-of-service (DDoS) attack uses multiple computer systems to attack a server or host in the network. A denial-of-service (DoS) is a one-on-one attack to disrupt service. Worms reproduce and move throughout the network to infect other systems and therefore do not attack one victim. Rootkits are software programs that have the ability to hide themselves from the operating system.

7. D. Tokens are rotating numerical keys that you must physically have with you to gain access to the operating system or applications. Biometrics relies on a physical characteristic of the user to verify identity. Biometric devices typically use either a hand pattern or a retinal scan

to accomplish this. Smartcards contain a private certificate key and are protected with a password. Geofencing uses your GPS coordinates to ensure that the authentication happens when you are in a defined geographic area.

8. A. An on-path attack intercepts data and then sends the information to the server as if nothing were wrong, while collecting the information. Zero-day attacks are attacks in which a developer has not properly patched a hole yet and is unaware of the hole. A denial-of-service (DoS) attack is used to disrupt legitimate requests from being answered. A Smurf attack is a type of distributed denial-of-service (DDoS).

9. A. A spoofing attack is an attempt by someone or something to masquerade as someone else (IP address) and is often used to disrupt access. Social engineering is a process in which an attacker attempts to acquire information about your network and system by social means, such as talking to people in the organization. Worms reproduce and move throughout the network to infect other systems. Password attacks are used in an attempt to guess passwords.

10. A. A worm is different from a virus in that it can reproduce itself, is self-contained, and doesn't need a host application to be transported. A smurf attack is a type of distributed denial-of-service (DDoS). A phishing attack is an attempt to gain a user's credentials to a network resource. Trojan horses are programs that enter a system or network under the guise of another program.

11. D. The reflective attack is using a broadcast ping (ICMP) on a network. The return address of the ping may be that of a valid system in your network. The Transmission Control Protocol (TCP) is not typically used with a reflective attack. The Internet Protocol (IP) is a suite of protocols and solely used with a reflective attack. The User Datagram Protocol (UDP) is not described in this reflective attack.

12. C. A SQL injection attack is a method of passing a SQL query with a web request by using an escape code sequence. An insider threat is a threat from within your organization, such as a disgruntled employee. An evil twin attack involves a rogue access point with the same SSID as your organization. Tailgating is the act of walking behind someone who has swiped to get into an area so the attacker can gain entry.

13. B. A key fob is an example of authentication for something you have. A password is something you know. A fingerprint is something you are. A place is a geographical place in which you are.

14. B. End-user education is the best way to protect your users from the threat of phishing via email. Antivirus software is used to prevent viruses, not phishing attempts. SecureDNS can be useful in protecting your users, but not from phishing emails. The principle of least privilege assigns only the permissions that users need to do their work, and no more.

15. C. Spear phishing is a type of social engineering, where someone is trying to con your organization into revealing account and password information by pretending to be a high-level person. A spoofing attack is an attempt by someone or something to masquerade as someone else, with the intent of disrupting access. A replay attack is a form of an on-path attack, where packets are replayed at a critical time. Trojan horses are programs that enter a system or network under the guise of another program.

16. A. Some viruses won't damage a system in an attempt to spread into all the other systems in a network. These viruses use that system as the carrier of the virus. A password attack would not prompt your antivirus software to notify you. Your antivirus software could be malfunctioning, but it would not suggest the same virus is infecting you over and over again. A denial-of-service (DoS) attack would not prompt your antivirus to notify you.

17. B. A symptom of many viruses is unusual activity on the system disk. The virus spreading to other files on your system causes this. A disk failure will not create high disk activity. A spear phishing attack is a social engineering attack and will not create high disk activity; neither denial-of-service attacks nor TCP/IP hijacking attacks will create high disk activity.

18. A. A brute-force attack is a type of password attack in which a password is guessed over and over, until the right password is guessed. A backdoor attack is an embedded account that allows unauthorized access through an unpatched coding hole. A worm is different from a virus in that it can reproduce itself, is self-contained, and doesn't need a host application to be transported. A TCP/IP hijacking attempt is an attack that attempts to redirect the TCP/IP conversation to the threat agent.

19. A. The thesis of your presentation should outline the need of a firewall to isolate the external network from the internal network. Firewalls will not scan packets for viruses. Firewalls will not prevent password attacks or harden physical security.

20. C. A login script is used by Active Directory during login to map drives and printers. A home folder is a private network location in which the user can store their personal files. Organizational units (OUs) are used to group computers and users so that Group Policy can be applied. The MMC is used to manage various aspects of Active Directory and the local operating system.

Chapter 6: Securing Operating Systems

1. C. A bring your own device (BYOD) policy defines the minimum specifications for an employee's device used for work-related access. The mobile device management (MDM) software would usually police these specifications, but it would not define them. The acceptable use policy (AUP) is a code of conduct when dealing with organization resources. The nondisclosure agreement (NDA) is an agreement used when dealing with intellectual property.

2. D. Synchronizing a mobile device with a computer system allows you to mirror personal data between the devices, regardless of which one contains the most current data. Calibration refers to matching the device's and the user's perceptions of where the user is touching the screen. Remote wipes allow you to remove personal data from a lost or stolen device. Pairing is what must be done in Bluetooth for two Bluetooth devices to connect and communicate.

3. A. The Guest account should be disabled on the operating system, unless there is good reason to leave the account enabled. Print Operators is a group found on Windows servers. Power Users is a group found on both Windows workstations and servers. Userone is

obviously a user account. Unless the user has left the organization, there should be no reason to disable the account.

4. C. Locator apps can find the phone and then display a message that can be read even while the phone is locked. Locator apps also often give the user the option to make the phone blare a noise to aid in finding the device. Failed login restrictions will wipe a device if a specific number of wrong passwords are entered. Antivirus apps make sure that your phone does not get a virus, but they offer no way to message or locate the device. Remotely wiping a mobile device is a feature of locator apps, but it is not itself an app that displays messages and produces noises.

5. C. A screensaver should automatically start after a short period of idle time, and that screensaver should require a password before the user can begin the session again. The system should neither shut down nor restart, since work could still be open. For the same reason, the system should not log out the user.

6. C. User Account Control (UAC) helps protect against unauthorized changes to system files by prompting for administrative approval. BitLocker and EFS provide encryption. Windows Defender offers antivirus protection.

7. C. The goal of single sign-on (SSO) is to allow a username/password combination. Once the combination is entered, claims are used to access additional resources. Tokens are given to the operating system after a user successfully logs in; they allow a user to access rights on the operating system. Kerberos is used (along with Active Directory) to authenticate a user on the Windows operating system. Multifactor authentication is the use of two or more factors to authenticate a user.

8. B. The Windows Credential Manager is a password manager that is built into the operating system of Windows. Edge and Internet Explorer 11 work in conjunction with the Credential Manager, but they are not password managers. Active Directory is used to authenticate domain users, but it does not manage passwords for end users.

9. A. The advantage to assigning the IP addresses statically is that you can make certain which host is associated with which IP address, and then use filtering to limit network access to only those hosts. Disabling DNS will limit all hosts on a network from getting to the Internet and will not limit network access. A default subnet mask will not limit network access and will cause other problems. An empty default gateway will not limit network access.

10. C. Wi-Fi Protected Access 2 (WPA2) offers the Advanced Encryption Standard (AES) for encrypting wireless communications. Wired Equivalent Privacy (WEP) offers weak 64- or 128-bit encryption. Wi-Fi Protected Access (WPA) uses the RC4 encryption algorithm. Temporal Key Integrity Protocol (TKIP) is a part of the WPA encryption protocol.

11. B. An extension is a type of add-on that will extend the functionality of the web browser in a way it was never originally designed. A pop-up blocker changed the way a web page is rendered. A plug-in changes the way a web page is rendered. An ad blocker also changes the way a web page is rendered.

12. A. When an employee is offboarded, their user account is deleted or disabled. A user account is not created, nor is its password reset, during the offboarding process. A user account normally is created during the onboarding process.

13. B. By default, a Google account is required on Android devices; it is used to synchronize data and app purchases to the cloud. The device does not require email to be set up, but the account can be used for the setup. The account is not used for desktop backups. The device does not require registration.

14. A. Fingerprint locks are the most secure of all the lock methods, since fingerprints are hard to duplicate. Face locks have a high number of false positives, which can be used to gain access to the phone. Passcode locks can be cracked or shoulder surfed. Swipe locks are not really locks; they just allow the screen to be opened with a swipe.

15. A. The Encrypted File System (EFS) is a functionality of the Windows NTFS filesystem. EFS can encrypt individual files and folders. BitLocker is a full-device encryption technology. NTFS is a filesystem that supports encryption and security, among other functionality. BitLocker to Go is used for full-device encryption of removable drives.

16. B. Because the user is in both groups and the Sales group has full share permissions and the Sales group has modify NTFS permissions, the most restrictive of the two is Modify, so that will be the effective permission for the user. All of the other answers are incorrect.

17. A. The permissions will be the same as before the move, since you are just moving the files and not creating a new entity. The permissions will not be inherited from the parent folder. The permissions will not be configured the same as the root folder. The permissions will not be blank.

18. B. The user will have only the Modify permission when logged in locally to the computer, since the filesystem is not shared with the appropriate permissions. The user will not have the Modify permission when connecting from the network. The user will still have the Modify permission when logged in locally, because of the NTFS permissions. The user will not have read-only permissions when connecting from the network.

19. C. The file attributes are most likely set to Read-only, and thus you are not permitted to delete the file. The Modify permission would not hinder the file from being deleted. The share permissions would not have any bearing on the problem, since the local filesystem is being accessed. The file attributes being set to System would not restrict a file from being deleted.

20. D. Mobile device management (MDM) software enables you to enforce profile security requirements on mobile devices. The acceptable use policy (AUP) is a code of ethics your users should follow when dealing with organizational resources. A nondisclosure agreement (NDA) is an agreement between an employee and the organization to protect intellectual property. A bring your own device (BYOD) policy explains how devices should be secured but provides no enforcement.

Chapter 7: Troubleshooting Operating Systems and Security

1. D. Windows Update is responsible for downloading and installing Windows service packs, patches, and security updates. Device Manager is used to view devices installed on the operating system. The Microsoft Management Console is a console that allows snap-ins to be added for management. Download Manager is a component of Edge.

2. B. Enable Safe Boot with the `msconfig` utility allows you to boot with basic drivers and minimal startup of nonessential services. Enable Debugging is used by kernel developers. Disable Driver Signature Enforcement is used to allow an unsigned driver to load during boot. Enable Low-Resolution Video will boot the operating system into a VGA mode.

3. B. The `/REBUILDBCD` option can be used with the `bootrec` tool to rebuild the boot configuration data (BCD). The `/FIXBOOT` option writes a new boot sector to the system partition. The `/SCANOS` option scans all other partitions that are found to have Windows installations. The `/FIXMBR` writes a new master boot record (MBR) to the partition.

4. B. The most important first step is to identify and verify the malware symptoms. You should quarantine the infected system once you have verified it is infected. Remediating the infected system happens after you disable System Restore. Education of the end user is the last step to malware removal.

5. A. The Profile tab of the Advanced Systems Properties dialog box allows you to view the total size of a local or remote profile. Regedit and the Windows Recovery Environment will not aid in troubleshooting a slow-loading profile. Windows Preinstallation Environment is the mini-Windows version used for installation of Windows.

6. D. `winresume.exe` is used to load Windows from a suspended state. The Boot Configuration Data (BCD) is used to direct Windows to boot the proper installation. `ntoskrnl.exe` is the Windows kernel. `winload.efi` is used for the normal booting of the Windows operating system.

7. A. The System Restore option should be used first to restore the operating system to an earlier point before the problem. This will restore the device back to a previous state before the installation of the drivers. System Restore will not affect user data files. Reset This PC will reset the PC back to factory default before the tax application was installed. Reimaging the laptop will erase all programs and data files. Manually reinstalling Windows 10 will erase all programs and data files.

8. A. When you have UEFI firmware, you must have the disk set up with a GUID Partition Table (GPT) partitioning type. The standard master boot record (MBR) partitioning type can be used with BIOS. Power-on self-test (POST) is a routine the BIOS or firmware performs to test hardware before boot. The Boot Sector is contained on both MBR and GPT partitioning types.

9. B. Pop-up blockers are used to prevent pop-ups and pop-unders from appearing. Antimalware utilities will remove and prevent malware. Phishing sites are used to collect users credentials' by tricking users. Antivirus software is used to protect the operating system from viruses.

10. C. Antivirus definitions should be updated daily, because new viruses are identified by the minute. Updating antivirus definitions weekly or monthly will open you up to the possibility of infection.

11. A. Resource Monitor can be used to identify slow and sluggish performance, as well as identify the source of the problem. The `msconfig.exe` tool can be used to enable or disable services on startup and launch tools, but it cannot be used to diagnose performance issues. The Device Manager MMC can be used to view and modify devices, but it will not help diagnose performance problems. Reliability Monitor will display the reliability of the operating system, but it will not help diagnose problems with performance.

12. B. Windows Update Troubleshooter can assist in diagnosing problems with Windows Update. The `ntbtlog.txt` file is used to diagnose problems with bootup. Windows Recovery Environment is used to solve problems with Windows and is not typically used for problems with Windows Updates. Safe mode is a boot mode that loads minimal drivers and services.

13. A. Windows Defender Security is considered antimalware and antivirus protection for the Windows operating system. Mobile device management (MDM) software is used to manage mobile devices. Windows Action Center is a notification center for action to be taken in the operating system. VirusTotal is a third-party site that analyzes virus signatures, but it does not protect you from them.

14. B. The System File Checker (SFC) allows you to manually scan for modified operating system files and repair them. `regedit` is used to modify the Registry. `bootrec` is used to repair the boot records on an operating system installation. User Account Control (UAC) is used to control access to administrative credentials.

15. A. Establishing security policies and procedures will help eliminate security problems and guide employees on what to do if they arise. Optimizing drives will defragment drives and has no effect on security. Preventing booting into safe mode will only hinder diagnostics. Preventing booting into Windows Recovery Environment will also hinder diagnostics.

16. D. The device most likely has too many background applications open that are using RAM. That the device is not charged to capacity would not affect RAM. A digitizer not functioning properly would resemble inaccurate touchscreen responses. If the device were in Do Not Disturb (DND) mode, the speakers would not work.

17. B. If auto-reconnect is configured on an SSID, the device could be susceptible to an evil twin attack, in which the device connects to any device with the same SSID. The device will not reconnect to any SSID but only to the SSID configured as auto-reconnect. Battery life will remain unaffected with auto-reconnect. Exceeding limits can be avoided with auto-reconnect, because the wireless network is used for data usage.

18. B. Event Viewer will allow you to see more detailed information on why programs have crashed. The Event Viewer logs may not give the exact reason, but they will aid in understanding the root cause. Device Manager is used to view and manage devices connected to the operating system. The Windows Recovery Environment is used to repair the Windows operating system. The `msconfig.exe` tool is used to modify startup programs and launch other diagnostic tools.

19. C. The reason that the operating system would write out large amounts of RAM to the page file is that the system is running low on physical RAM and is attempting to free up physical RAM. The CPU might run high when paging occurs, but it will not trigger excessive paging. Although it is normal for the operating system to write out RAM to the page file, it is not normal for large amounts to be written out. The page file is not faster than conventional RAM.

20. B. Degraded battery life can be expected from an overheating mobile device if the problem persists for a long time. Higher RAM usage will not occur with overheating, but it could be a cause of overheating. Inaccurate touchscreen responses are not a symptom or a consequence of overheating. The inability to decrypt emails depends on having the proper certificate installed.

Chapter 8: Scripting, Remote Access, and Artificial Intelligence

1. C. Scripting languages are interpreted languages that run on top of a runtime environment. Programming languages, not scripting languages, require a compiler. Scripting languages are not strongly typed; programming languages are strongly typed. Scripting languages have bad memory management because of loosely typed variables.

2. A. Scripting languages are considered high-level languages because they do not directly access hardware and use an intermediary called the interpreter. Mid-level languages are Java and C/C++, not scripting languages. There is no such thing as an intermediate-level language. Low-level languages are machine language and assembly language, which are not scripting languages.

3. C. Floating-point variables allow for precision math, also known as decimal math. Boolean variables allow for true or false values. Integer variables allow for whole numbers values. String variables allow for text values.

4. C. A program variable is the least significant and it is not inherited. A system variable is defined for the entire system and is the most significant because it is inherited by all users and programs. A user variable is significant as well, since all applications will inherit the variable. A string variable is not inherited, but it is not an environment variable.

5. B. The statement `$xvar = 2` is a PowerShell statement that will load the variable `xvar` with a value of `2`. The statement `xvar = 2` is Bash syntax. The statement `xvar = 2;` is JavaScript syntax. The statement `set /a xvar=2` is Windows batch script syntax.

6. D. A `for` loop has a defined beginning and end, and steps from the beginning to the end. A `do while` loop is a type of `while` loop and has only a defined end. A `while` loop has only a defined end. An `if` statement is branch logic, not a loop.

7. C. The `.bat` extension is used with the Windows batch scripting language. The `.vbs` extension is used with VBScript language. The `.js` extension is used with the JavaScript scripting language. The `.py` extension is used with the Python scripting language.

8. B. The VBScript language allows for the use of the Component Object Model (COM). The PowerShell scripting language allows for the use of the .NET Framework. Windows batch scripts use existing applications. JavaScript is primarily web browser–based and does not allow for the use of external objects.

9. D. The `.py` extension is used with the Python scripting language. The `.vbs` extension is used with the VBScript language. The `.js` extension is used with the JavaScript scripting language. The `.bat` extension is used with the Windows batch scripting language.

10. A. The PowerShell scripting language allows for the use of the .NET Framework and is commonly used with Microsoft Azure and Microsoft 365. The VBScript language allows for the use of the Component Object Model (COM) and is not used for cloud services. A Windows batch script uses existing applications and is not used for cloud services. JavaScript is primarily web browser–based, does not allow for the use of external objects, and is not used for cloud services.

11. D. JavaScript is primarily web browser–based and allows for interactive content. The PowerShell scripting language is used to manage the operating system. The Bash scripting language is primarily used with Linux and UNIX systems. Windows batch scripts use existing applications and are used to manage the operating system.

12. B. The `.sh` extension is used with the Bash scripting language. The `.vbs` extension is used with the VBScript language. The `.bat` extension is used with the Windows batch scripting language. The `.py` extension is used with the Python scripting language.

13. C. Before a Bash script can be executed, you must use the `chmod` command to grant execute permissions. The `chown` command changes ownership. There is no such thing as an `execute` attribute. Adding `.sh` to the end of the script doesn't serve any purpose.

14. C. The statement `mvar = 8;` is JavaScript syntax to load a variable of `mvar` with a value of `8`. The statement `$mvar = 8` is PowerShell syntax. The statement `mvar = 8` is Bash syntax. The statement `set /a mvar=8` is Windows batch script syntax.

15. D. PowerShell is the only scripting language that has a preinstalled Integrated Scripting Environment (ISE), called the PowerShell ISE. The VBScript language requires the installation of Microsoft Visual Studio Code. The Bash scripting language requires a text editor or

other package to be installed. The Python scripting language requires a third-party integrated development environment (IDE), also known as an ISE.

16. A. The line `//comment` is used to comment JavaScript code. The line `'comment` is used to comment VBScript code. The line `REM comment` is used to comment Windows batch script code. The line `# comment` is used to comment Bash script code and PowerShell code.

17. A. The `.js` extension is used with the JavaScript scripting language. The `.sh` extension is used with the Bash scripting language. The `.bat` extension is used with the Windows batch scripting language. The `.py` extension is used with the Python scripting language.

18. B. The primary goal of AI is to simulate or imitate human intelligence. This involves processes like learning, thinking, and problem-solving. AI is not intended to replace human intelligence but to complement it. Creating emotional experiences is not the main goal of AI. AI focuses on cognitive processes, not physical abilities.

19. B. Convolutional neural networks (CNNs) are specialized for image and video analysis. Language translation is typically handled by recurrent neural networks (RNNs). Speech recognition is also handled by RNNs. Game playing can be handled by various types of neural networks.

20. A. Tokenization is the method of dividing text into smaller parts known as tokens. Tokenization does not involve converting images into text. Tokenization is not about translating text. Tokenization deals with text, not audio data. Translating text into another language is the function of transformers.

Chapter 9: Safety and Environmental Concerns

1. C. A three-pronged grabber should be used to retrieve the screw from the computer. Using a magnetic-tipped screwdriver is not advisable, because many components are sensitive to magnets. Using a magnetic grabber is not advisable, because many components are sensitive to magnets. Shaking the computer until it falls out can be dangerous and make the screw harder to get out.

2. B. Using your personal computer to manage your side business adheres to the AUP and prevents misuse of company resources. Using your work computer violates the AUP. Asking for permission is unnecessary as the AUP clearly prohibits such use.

3. B. The material safety data sheet (MSDS) will have the necessary information about the cleanup procedures. The Occupational Safety and Health Administration (OSHA) is an organization that oversees safety in the workplace. The product label may or may not have this information. A cathode-ray tube (CRT) is a type of monitor that uses a glass tube.

4. D. You can download the MSDS by visiting the printer cartridge manufacturer's website. The store is not required to furnish an MSDS at the time of purchase. The manufacturer is not required to furnish an MSDS with the packaging of the product. You are legally allowed to have an MSDS for products, as per OSHA. You should have them on hand in case of an emergency.

5. A, C. An accident should always be reported so that if there is a hazardous condition, it can be fixed. Report hazardous conditions, such a spill on the floor, to the employer before they are a problem. A near-accident does not need to be reported unless it is caused by a hazardous condition, like spills. Rain forecasted for a workday is not a hazardous condition.

6. B. You can feel an ESD shock of 3,000 volts or more. 300 volts is too low for a human to feel. A minimum of 30,000 volts can be felt by a human. 300,000 volts can be felt by humans and could be painful.

7. A, B. Using an antistatic wrist strap allows you to ground yourself to dissipate a static charge. Using antistatic bags allows for a uniform charge around a component. Spraying disinfectant spray will only kill germs, not reduce static. Shuffling your feet will build a charge of static.

8. B, D. Providing properly maintained tools and equipment is a requirement for an OSHA-compliant workplace. Displaying an OSHA poster in a prominent location is also a requirement for an OSHA-compliant workplace. Attending yearly OSHA safe work environment seminars is not a requirement. Having an OSHA employee stationed within five miles of the facility is also not a requirement.

9. A. You should ensure that the company has the proper licenses to install 20 additional copies to be compliant with licensing. Microsoft Office is not an open source licensed product. Using the personal license key from an existing system violates the license agreement. Following normal installation procedures will not ensure compliance of licensing.

10. C, D. You should use a cart to move heavy objects. You should also ensure that no safety hazards are in your path. Lifting by bending over at the waist can hurt your back. Carrying CRT monitors with the glass facing outward is not safe, as the weight will be farthest from your body.

11. C. Alkaline batteries should be taken to a recycling center. Throwing batteries in the trash is not environmentally responsible. Incinerating batteries is not advisable, since they can explode and will create pollution. Flushing batteries down the toilet is not an acceptable disposal method.

12. B, D. The first step is to confiscate and preserve the prohibited materials on the drive. The next step is to report the prohibited materials through the proper channels. Destroying the prohibited material will not remedy the situation. Confronting the user about the material is not suggested, as they could become hostile.

13. A. You should research information on the Payment Card Industry Data Security Standard (PCI DSS) standard. The General Data Protection Regulation (GDPR) is used for protecting EU citizens. Protected health information (PHI) is any data that defines a patient or an

ailment of a patient. Personally identifiable information (PII) is any information that can be used to identify a person.

14. A. Wood and paper fires can be put out by a Class A fire extinguisher. Class B fire extinguishers are used for flammable liquids. Class C fire extinguishers are used for electrical files. Class D fire extinguishers are used for flammable metals.

15. B, C. Phillips and Torx are two common types of screwdrivers. Neither circular nor helix is a type of screwdriver.

16. A, B. Good safety plans protect the interests of the workers and also help to keep company costs down. Periodic workplace inspections and a training program are good components to implement. Employees cannot be punished in any way for reporting safety violations. Third-party audits of safety programs are not necessary but can complement periodic inspections and safety and health training programs.

17. A. A good rule of thumb when it comes to magnetic-tipped screwdrivers is to avoid using them inside a computer case. Magnetic tools can damage data on disks that use a magnetic storage scheme. Magnetic-tipped screwdrivers should not be used inside a computer even if they are kept away from the processor and the RAM. Regardless of whether they are powered or not, magnetic-tipped screwdrivers should not be used inside computers.

18. B, C. Material safety data sheets (MSDSs) contain handling and storage instructions as well as personal protection instructions. MSDSs do not contain freezing point specifications or weight.

19. B, D. To maintain a safe work environment, all employees must follow certain protocols. These include using protective gear and equipment and following all health and safety rules. Accidents must be reported to the employer, not OSHA (unless action is not properly taken by the employer). There are no rules requiring safety training, although it is a good idea to have these sessions and for employees to attend them.

20. B. Nickel-metal hydride (NiMH) batteries are not considered environmental hazards. Alkaline batteries used to contain mercury, which is an environmental hazard, so they are assumed to be environmental hazards. Nickel-cadmium (NiCd) batteries and button cell batteries are considered environmental hazards.

Chapter 10: Documentation and Professionalism

1. B. Collecting user information is correct because it is essential to know who is reporting the issue and how to contact them. Determining the urgency of the issue is important, but communication to the customer is of higher importance. Identifying recent changes would happen after you enter the ticket. Verifying the issue description is an important, but secondary, step.

2. C. The rollback section contains the original configuration that can be used to revert the changes. The purpose section contains the reason for the proposed change. The risk analysis section explains the risks involved with the proposed changes. The plan for change section contains the primary change configuration and the alternate change configuration.

3. B. The scope of change section details whom the change will affect. The business processes section details the current business processes the change will affect. The end-user acceptance section details how the changes were tested and accepted by the users. The plan for change contains the primary and alternate plans for the proposed change.

4. D. Incident documentation should be completed so that key stakeholders can understand the reason for the outage. Change management documentation is used for proposed changes to the network. Knowledge base articles are used to document symptoms and solutions. An acceptable use policy (AUP) is used to protect an organization's resources from user abuse.

5. B. Volume Shadow Copy allows users to restore previous versions of their files themselves. Image-based restoration typically requires a desktop administrator. Incremental and differential backup and restores also require IT intervention for restoration.

6. C. Documenting the root cause helps in preventing the incident from recurring. There are typically no legal requirements around the documentation of the root cause, unless there is something specifically in a contract. Billing is also not related to documenting root cause, unless there is something specifically in a contract. Training new employees is important but is not a primary reason for documenting the root cause.

7. C. The acceptable use policy (AUP) details the acceptable use of the email system for business purposes. Mobile device management (MDM) is software that allows you to manage mobile devices in the workplace. A password policy details the appropriate handling and management of passwords. Incident management is how a network or security incident is handled.

8. B. Disk-to-disk is the fastest recovery method and backup method as well, because you are backing up from a disk to another disk attached via the network. Disk-to-tape is slower because you must re-tension the tape and then locate the data on the tape to recover it. Disk-to-flash is not a backup method, because of the price of flash. Disk-to-cloud is the slowest recovery method because you must recover from the cloud over a network connection.

9. B. You should use the copy backup method, since it will perform a full backup of the files without resetting the archive bits. A full backup makes a full backup and resets all the archive bits affecting the normal backups. An incremental backup copies only the files that have changed since the last backup and leaves the archive bits unchanged. A differential backup backs up only the files that have changed since the last backup and then resets all the archive bits.

10. C. Providing a quick reference for resolving common issues helps both users and technicians solve problems more efficiently. Creating a knowledge base rarely ever reduces help desk staff. Knowledges bases do not ensure compliance. Although knowledge bases are great tools, they do not improve morale.

11. C. Calling the customer and informing them of the situation is the best action that can be performed, since you are having direct communications. Arriving first thing in the morning should be done, but communication is the first action. Waiting until after hours and then leaving a message is not an appropriate action. Sending an email letting the customer know that you will be late is lying, since you will not make it to their site.

12. C. If the problems do not appear to be related, then deal with them separately. Looking for commonalities between the problems would just waste valuable time. No assumptions should be made, such as that a virus is causing the problem, unless there is proof. Ordering a new machine is disruptive to the customer, unless the problem dictates that this action be performed.

13. D. If you do not understand the customer, you should apologize, treat the customer with respect, and seek a manager to help translate. Ignoring the customer and starting to work on the system is not an appropriate response. Calling your supervisor will not solve the problem, since your supervisor will have the same issue. Asking that another technician be sent in your place is being insensitive to the customer and "passing the buck."

14. C. You should avoid confrontations with the user by attempting to calm them down. You should never shove the user back. Yelling for everyone in the area to come quickly will escalate the situation, which is the opposite of what you want to do.

15. B. You should take appropriate action and inform your manager, as well as notify the customer that you've done so. Doing nothing is not an appropriate response to this situation. Talking to the technician personally will create a confrontation and should be avoided. Asking the customer to prove it is an inappropriate action to this problem.

16. D. You should assure the vice president that you are optimistic and skilled to deal with these problems. Offering a joke is an inappropriate action. Downplaying the situation does not show respect to the customer's problem. Keeping your head down and ignoring the customer does not display appropriate communications.

17. C. You should try to solve the customer's problem without further escalating the frustration by putting them on hold or transferring them. Informing them that you won't be able to help is only going to anger the caller, further validating their frustration. Transferring them to another technician is not the appropriate action, because it will add to their frustration. Suggesting that they call back will create more frustration and is poor customer service.

18. B, C. Leaving a note for the user detailing what was done and how to contact you displays appropriate communications. Notifying the user's manager and your own manager that the problem is resolved is also appropriate communications. Cleaning up and leaving no evidence that you were there is an inappropriate action, as it is deceiving to the user. Putting the system back to its original state is an inappropriate action because it does not solve the problem.

19. A. While the user's profanity is likely linked to frustration, it hinders the communication and should be eliminated. You should ask the user to refrain from the offensive language. There may be circumstances where it is necessary to overlook the profanity, if your request is likely

to make the already upset customer even angrier, as long as you can understand what is being said. Hanging up is not an appropriate action and will further the user's frustration. Firing back with profanity is not an appropriate response, as it will escalate the situation further.

20. B. Scheduling of the depreciation of the equipment is performed in accounting software. Tracking of the equipment is a benefit to an asset tag. Identifying assets is a direct benefit of asset tags. An asset tag provides a proof of ownership.

Appendix

B



Answers to Performance-Based Questions

Chapter 1: Operating System Basics

Answer to Performance-Based Question

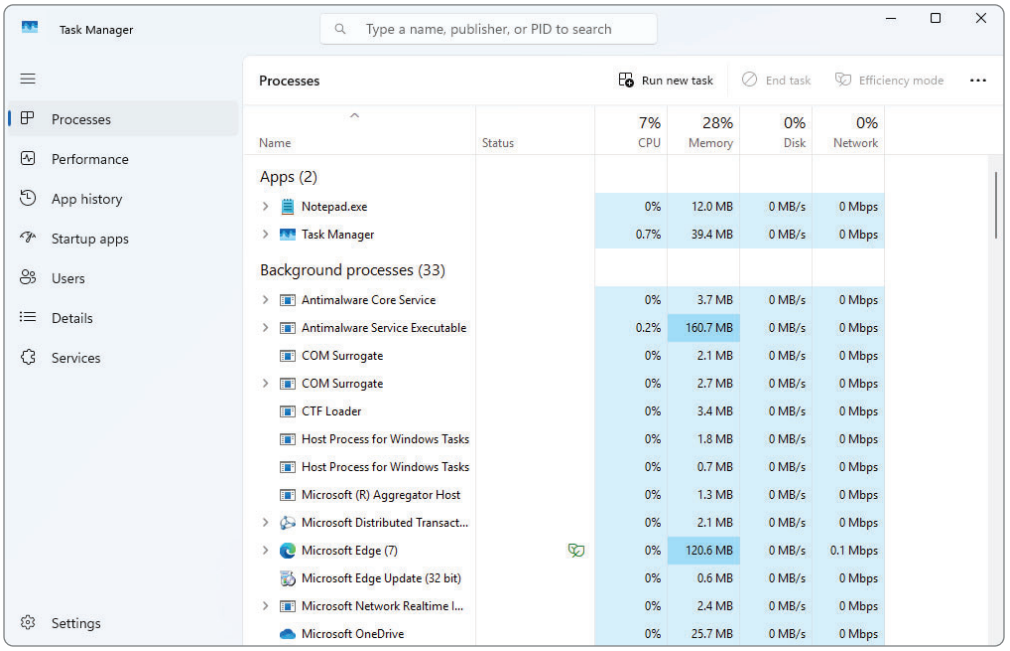
In order to accommodate the future requirement of BranchCache, your organization will need to purchase a volume license agreement with Microsoft. The BranchCache feature is only available in Windows 11 Enterprise. Windows 10 Pro is a retail operating system that can be upgraded to Windows 11 Enterprise. However, the upgrade will require a different 25-digit product key and activation of the Windows 11 Enterprise operating system.

Chapter 2: Windows Configuration

Answer to Performance-Based Question

To check to see which background processes are running and the resources they are using, open Task Manager. Do so by pressing Ctrl+Alt+Delete and selecting Task Manager. You can also press Ctrl+Shift+Esc.

Once in Task Manager, click the Processes tab, as shown here. If all the processes are not shown, then expand the More Details chevron in the lower left, if it is not already expanded. Click the CPU column header to sort by CPU usage. If a process is taking up a considerable amount of CPU time, you can highlight it and click End Process to shut it down. You can also sort by memory used and shut down processes that look to be using excessive amounts of memory. Note that shutting down critical processes may cause Windows to lock up or otherwise not work properly, so be careful what you choose to terminate. Note also that right-clicking one of the processes offers the End Process Tree option—a useful option when the process being killed is associated with others.



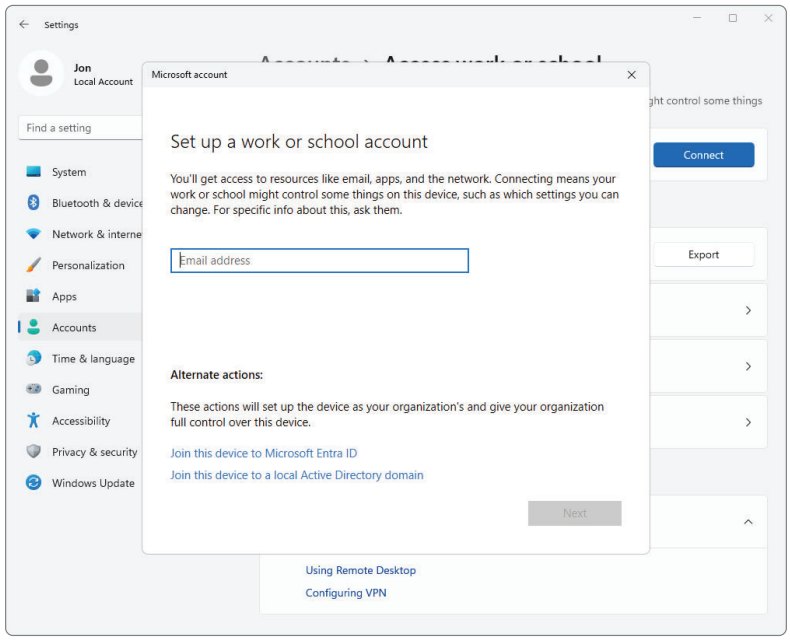
Chapter 3: Windows Administration

Answer to Performance-Based Question

Assuming you are on a version of Windows 11, the process is as follows:

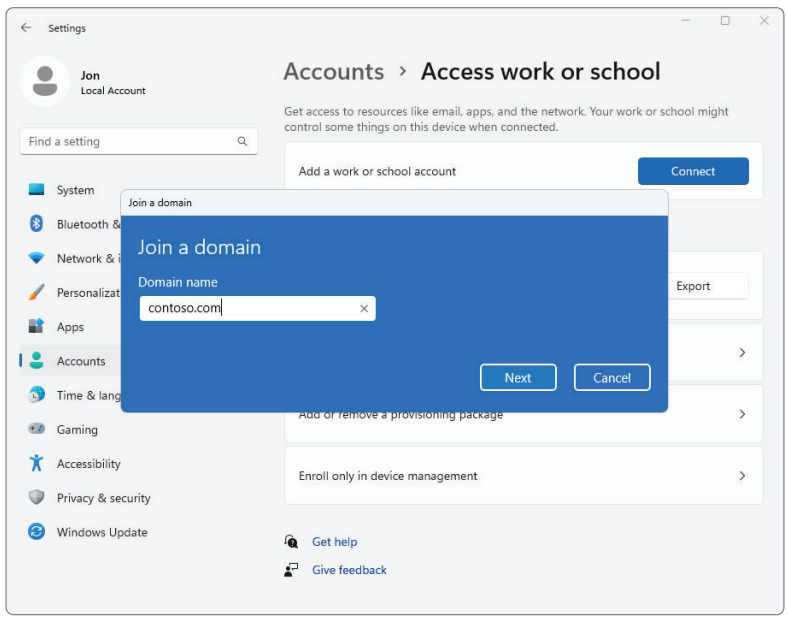
1. Click the Start menu.
2. Click Settings.
3. Select Accounts.
4. Click Access Work Or School.
5. Click Connect.
6. When the Set Up A Work Or School Account dialog box appears, click Join This Device To A Local Active Directory Domain as the alternate action, as shown in Figure 3.61.

FIGURE 3.61 The Set Up A Work Or School Account dialog box



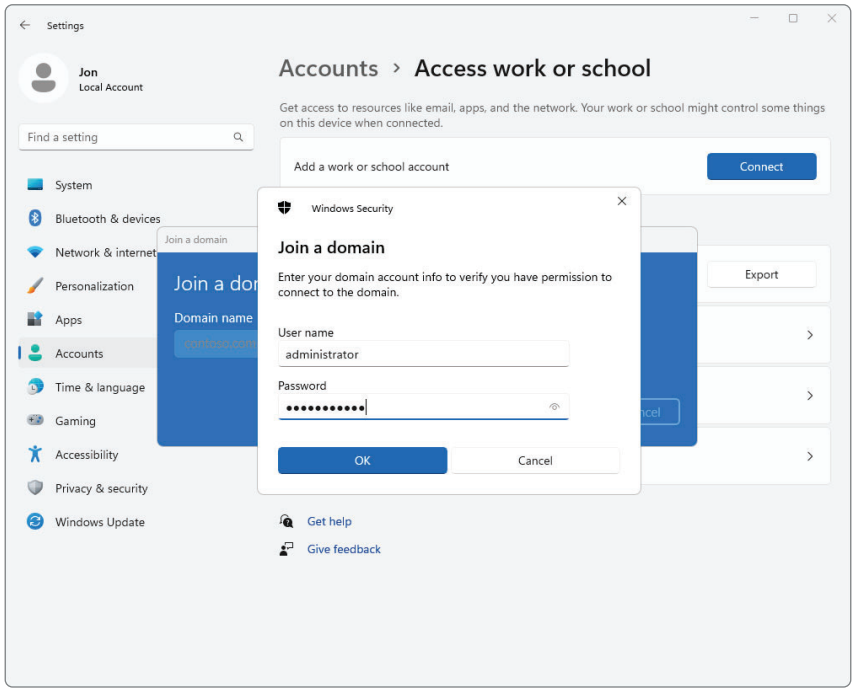
7. Enter the domain name in the dialog box, as shown in Figure 3.62.

FIGURE 3.62 Domain dialog box



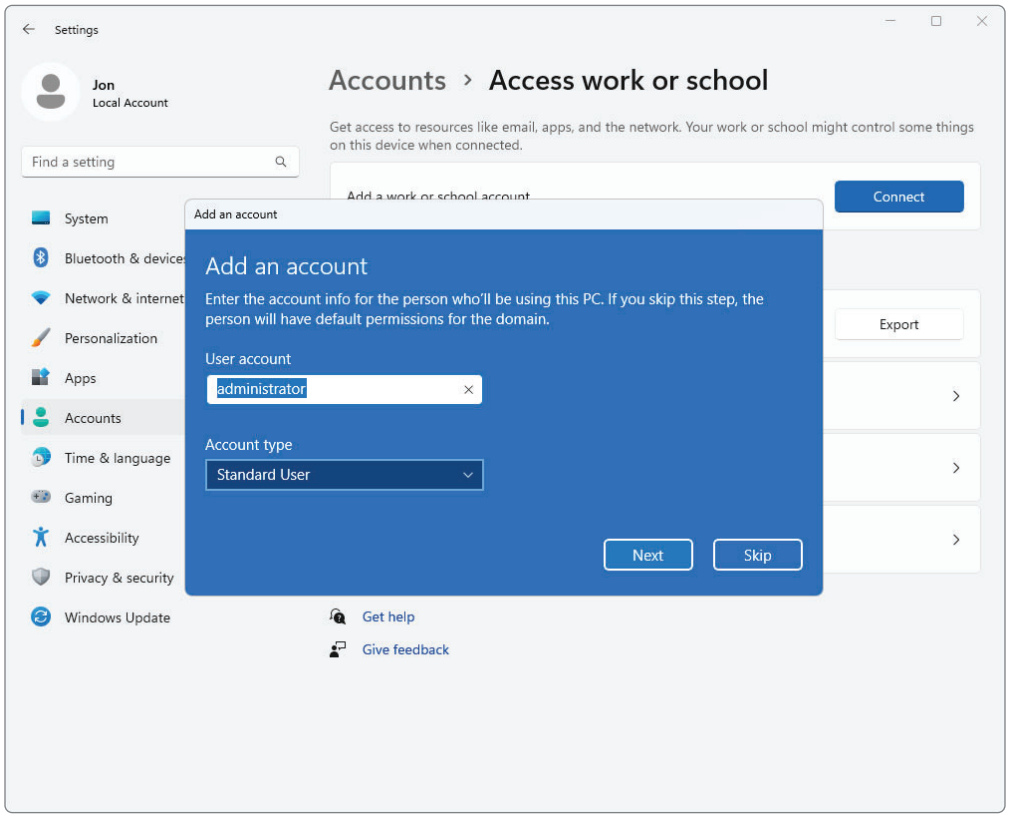
8. Enter the user account on the domain that has rights to join workstations, as shown in Figure 3.63.

FIGURE 3.63 Domain administrative credentials dialog box



9. Enter the domain user and their respective security level on this PC, as shown in Figure 3.64. Or you can click Skip, and the user will automatically have the same rights as the domain.

FIGURE 3.64 Local administrative rights dialog box

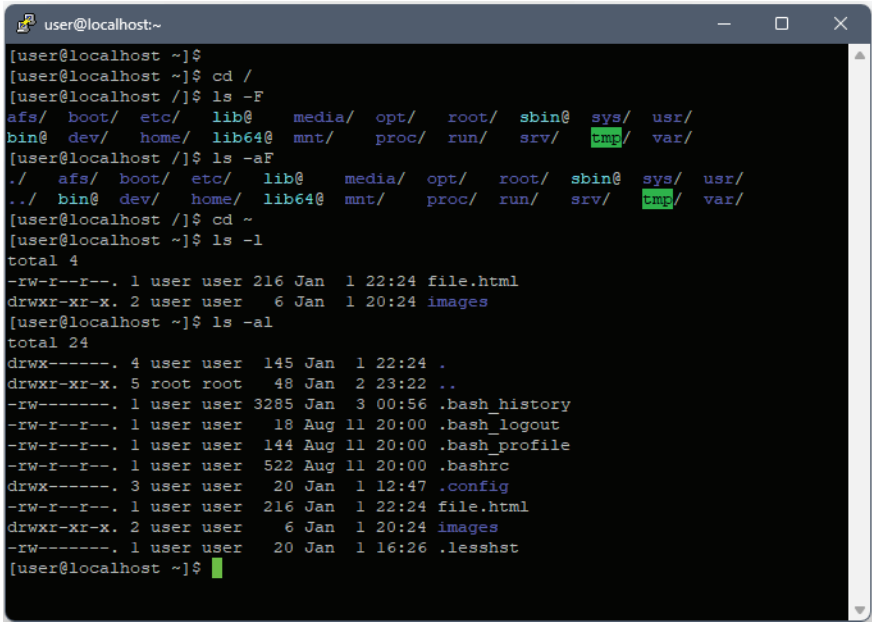


10. Click the Restart Now option and reboot.

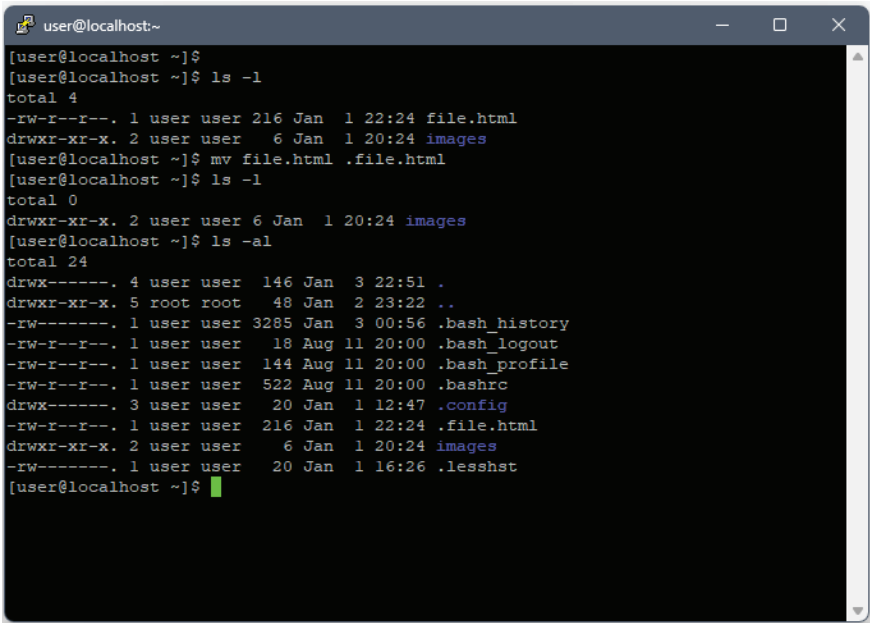
Chapter 4: Working with macOS and Linux

Answer to Performance-Based Question

The listing you see when typing these commands will differ based on such factors as the system, the directory, your permissions, and the files/subdirectories present, but in all cases, there will be entries present with the `-a` option that do not appear in the display without it. Among those listings that now appear are a single period (representing the present directory) and a double period (representing the parent directory), as shown here:



If there are any files or directories starting with a period, they will now appear where they did not before. The easiest way to "hide" a file or directory in Linux is to start the name of it with a period; thus, it will not show up in a listing unless the `-a` option is used. An example of this is shown here:



Chapter 5: Security Concepts

Answer to Performance-Based Question

A simple 8-character alpha numeric password contains 0–9 for a total of 10 characters, 26 uppercase and 26 lowercase characters. This gives you a total of 52 letters and 10 numbers, for a total of 62 combinations per character: 62 to the power of 8, or 62 × 62 × 62 × 62 × 62 × 62 × 62 × 62 = 218,340,105,584,896 combinations. A 25-character alphanumeric password with symbols contains 95 combinations per character; 95 to the power of 25 is 2.77×10^{54} combinations. If you are using a calculator, you might see 2.7738957e+49 as a result. Although the exact math is not significant, the deep understanding of combinations and complexity is the underlying lesson.

Chapter 6: Securing Operating Systems

Answer to Performance-Based Question

The following explains how you would achieve the goal:

1. Create three folders on the local filesystem of the server: Sales, Marketing, and R&D.
2. Set NTFS permissions on all folders for Everyone and the Modify permission.

3. Set the Sales folder to have the following share permissions:
   - **Sales:** Change
   - **Marketing:** Read
4. Set the Marketing folder to have the following share permissions:
   - **Marketing:** Change
   - **Sales:** Read
5. Set the R&D folder to have the following share permissions:
   - **R&D:** Change
   - **Marketing:** Read

Chapter 7: Troubleshooting Operating Systems and Security

Answer to Performance-Based Question

1. Investigate and verify malware symptoms.
2. Quarantine the infected systems.
3. Disable System Restore in Windows.
4. Remediate infected systems.
   a. Update antimalware software.
   b. Scan the removal techniques, such as booting into Safe Mode and using the Windows Preinstallation Environment.
   c. Reimage/reinstall.
5. Schedule scans and run updates.
6. Enable System Restore and create a restore point in Windows.
7. Educate the end user.

Chapter 8: Scripting, Remote Access, and Artificial Intelligence

Answer to Performance-Based Question

To write a PowerShell script to find other scripts in a user profile directory and all its subdirectories, you need the `$home` environment variable. There are a number of ways of writing this script to achieve the solution. The following is just one of the possibilities, using the parameters found on the Microsoft website:

```
Get-ChildItem -Path $home\* -Include *.ps1 -Recurse
```



PowerShell cmdlets and their associated parameters are well documented on Microsoft's website. For more information about the `Get-ChildItem` cmdlet, visit:

```
https://learn.microsoft.com/en-us/powershell/module/
microsoft.powershell.management/
get-childitem?view=powershell-7.5
```

To learn about other PowerShell cmdlets, visit:

```
https://learn.microsoft.com/en-us/powershell/scripting/
overview?view=powershell-7.5
```

Chapter 9: Safety and Environmental Concerns

Answer to Performance-Based Question

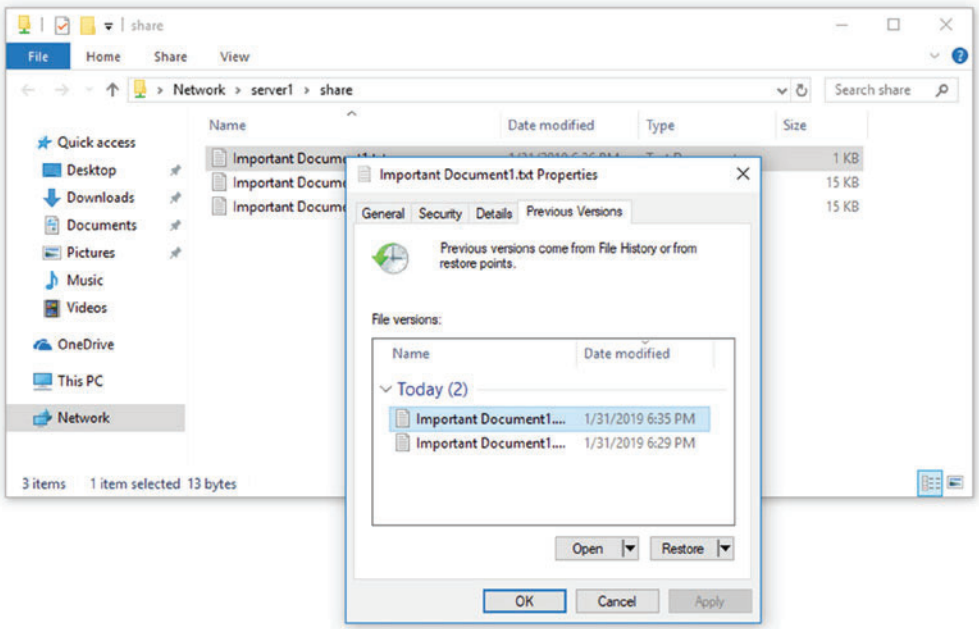
Here are some steps to take to look for trip hazards and eliminate them:

1. Walk around the server room and count how many cables are lying on the floor.
2. Walk around the client areas and see how many cables are lying on the floor or are exposed underneath cubicles.
3. Devise a plan to secure the cables and prevent them from being hazards. For example, purchase and install floor guards, use *cable ties*, or install conduits, as necessary, to secure all loose cables.

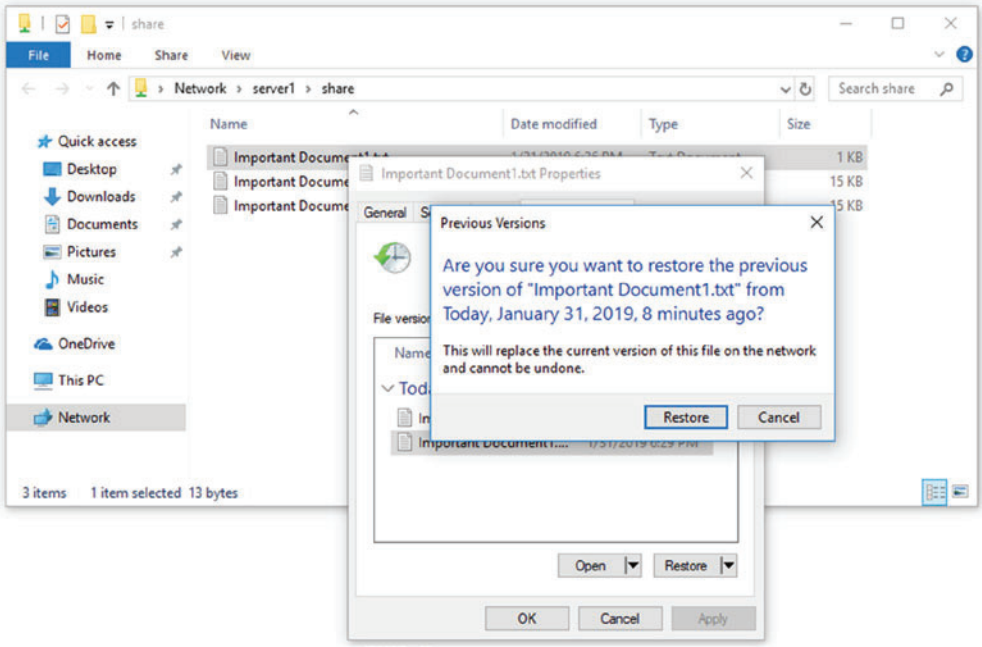
Chapter 10: Documentation and Professionalism

Answer to Performance-Based Question

1. Browse to the folder containing the overwritten file.
2. Right-click the file and select Properties.
3. Select the Previous Versions tab.
4. Find the latest version of the file and click the Restore button, as shown here:



5. In the confirmation box, select Restore again, as shown here:



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